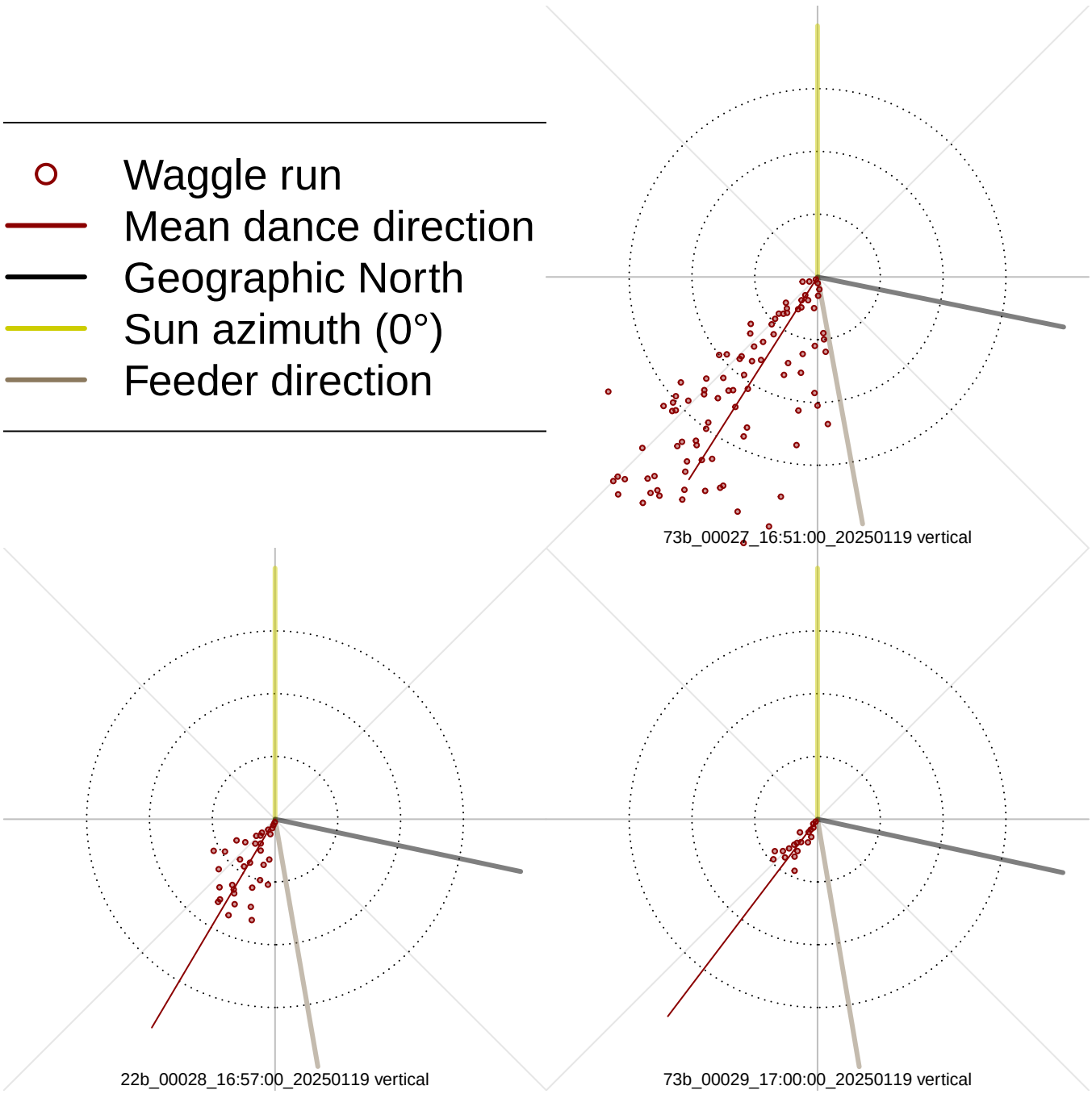
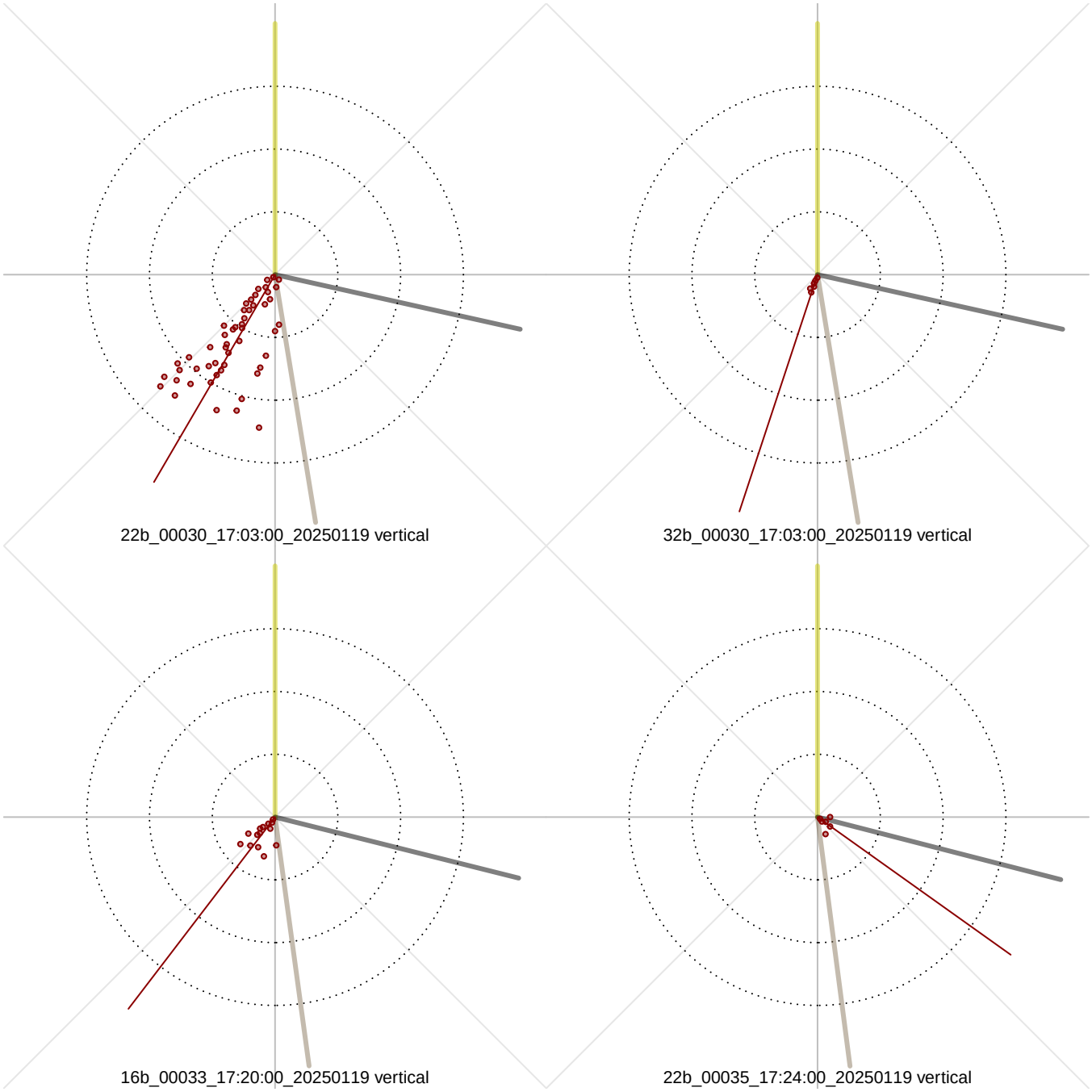
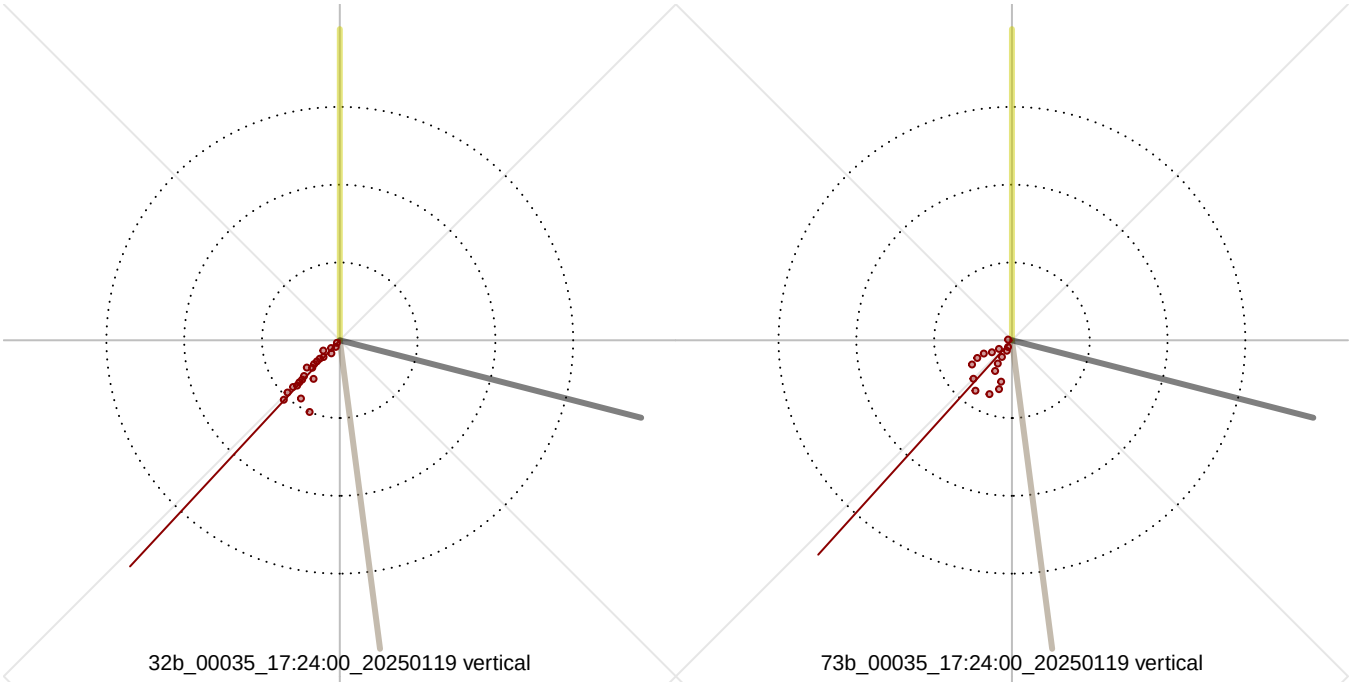
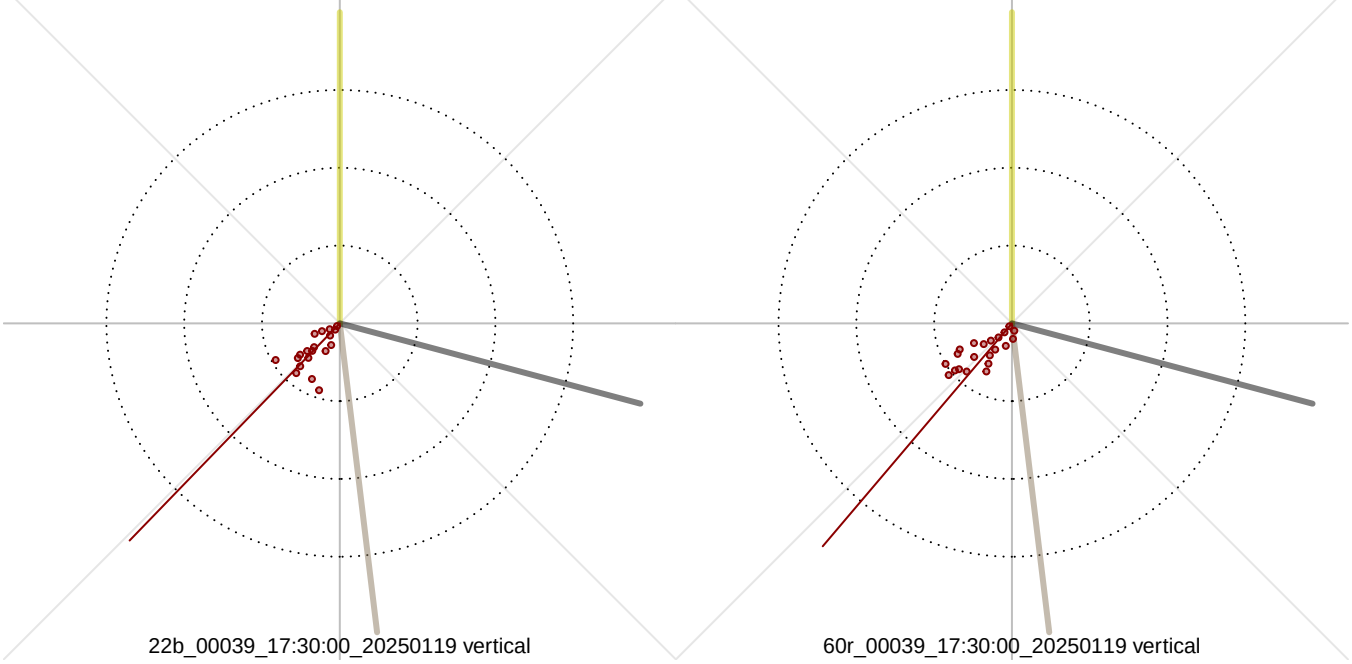

Waggle run
Mean dance direction
Geographic North
Sun azimuth (0°)
Feeder direction
73b_00027_16:51:00_20250119 vertical
22b_00028_16:57:00_20250119 vertical
73b_00029_17:00:00_20250119 vertical


22b_00030_17:03:00_20250119 vertical
32b_00030_17:03:00_20250119 vertical
16b_00033_17:20:00_20250119 vertical
22b_00035_17:24:00_20250119 vertical



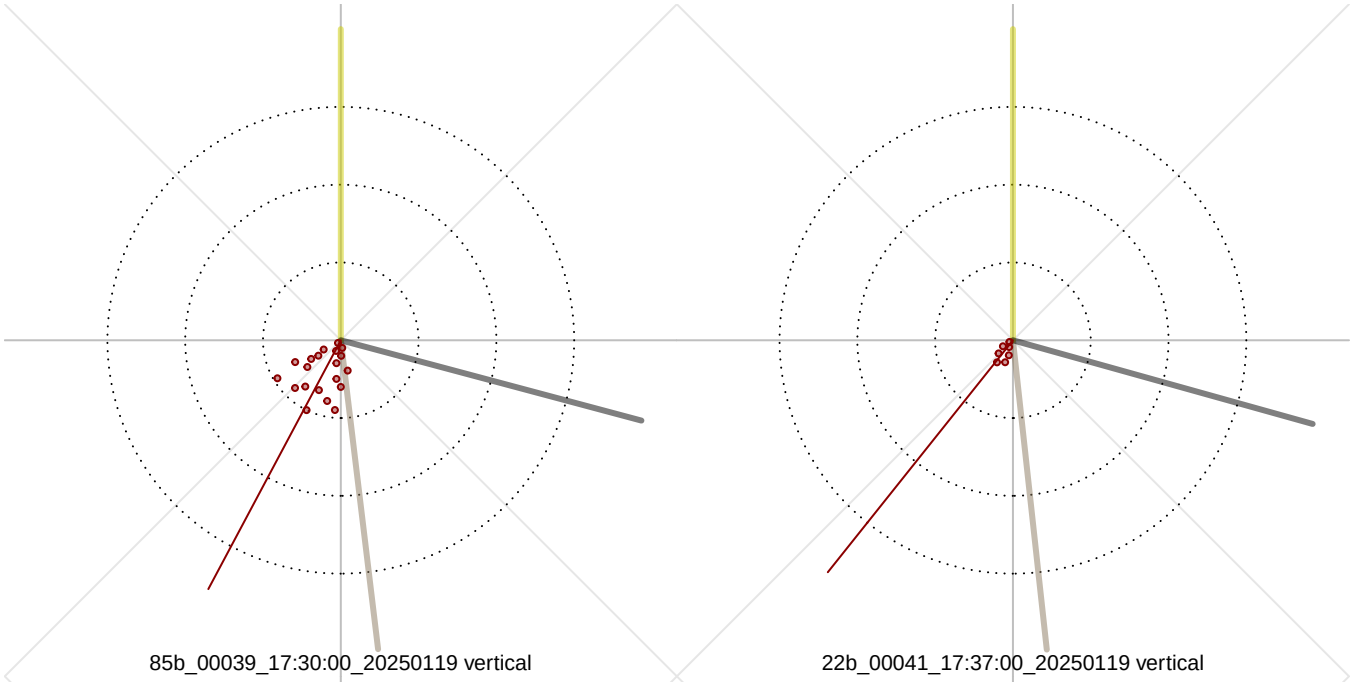
32b_00035_17:24:00_20250119 vertical

73b_00035_17:24:00_20250119 vertical

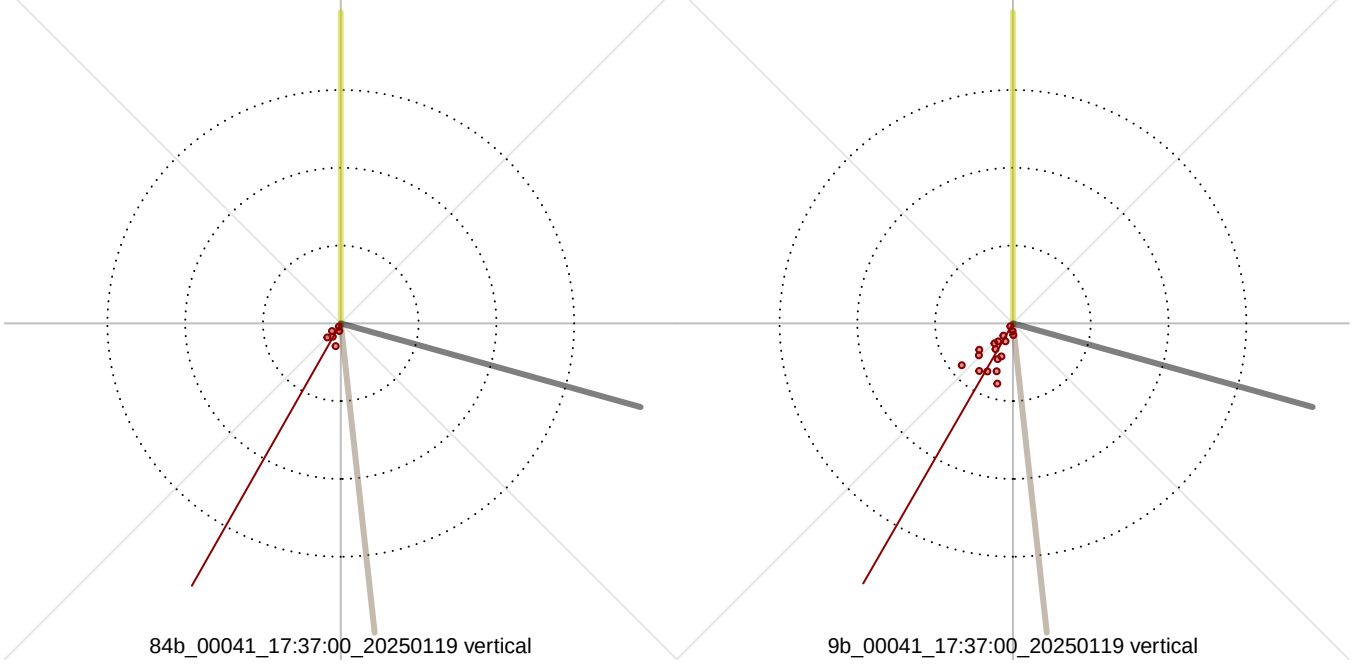


22b_00039_17:30:00_20250119 vertical

60r_00039_17:30:00_20250119 vertical



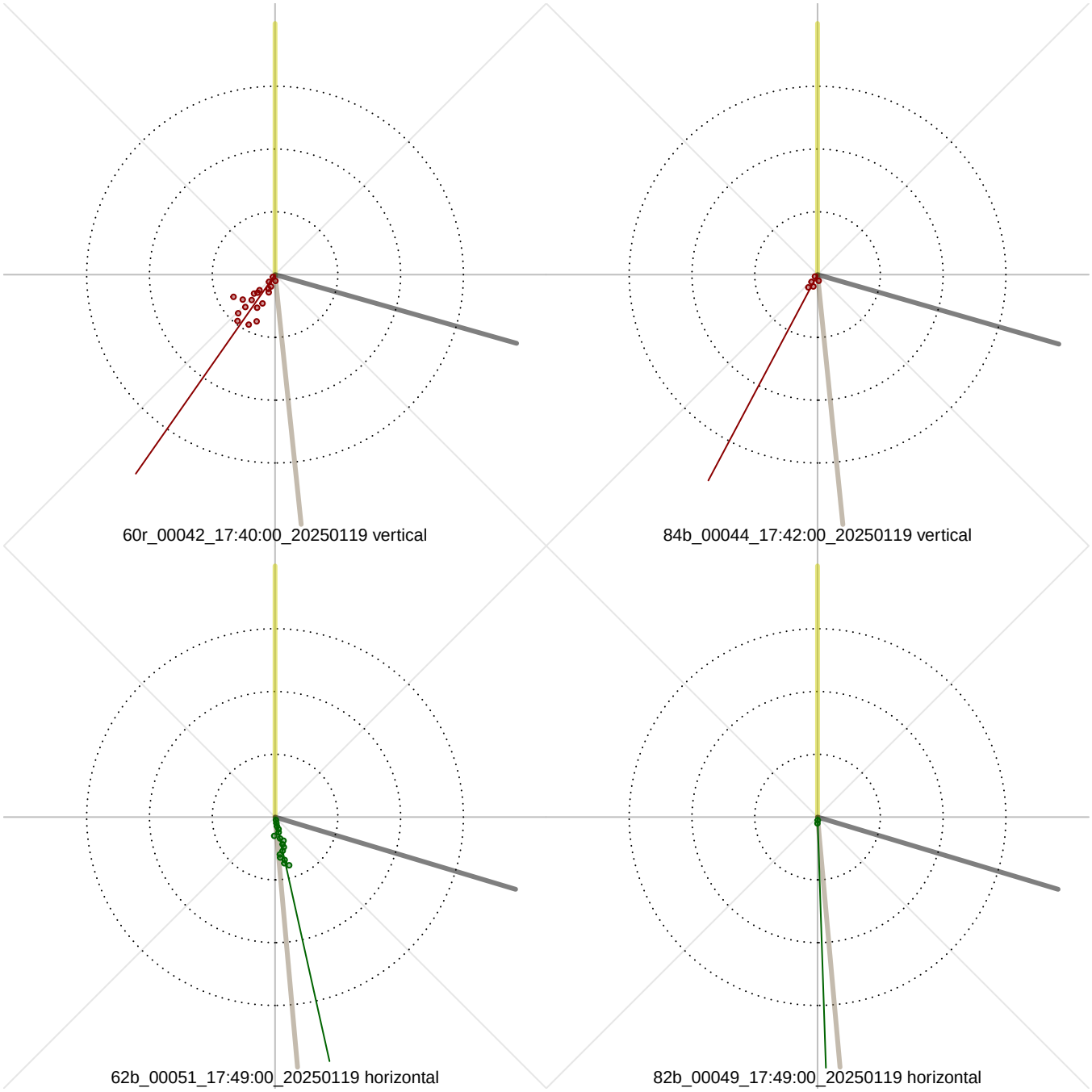
85b_00039_17:30:00_20250119 vertical

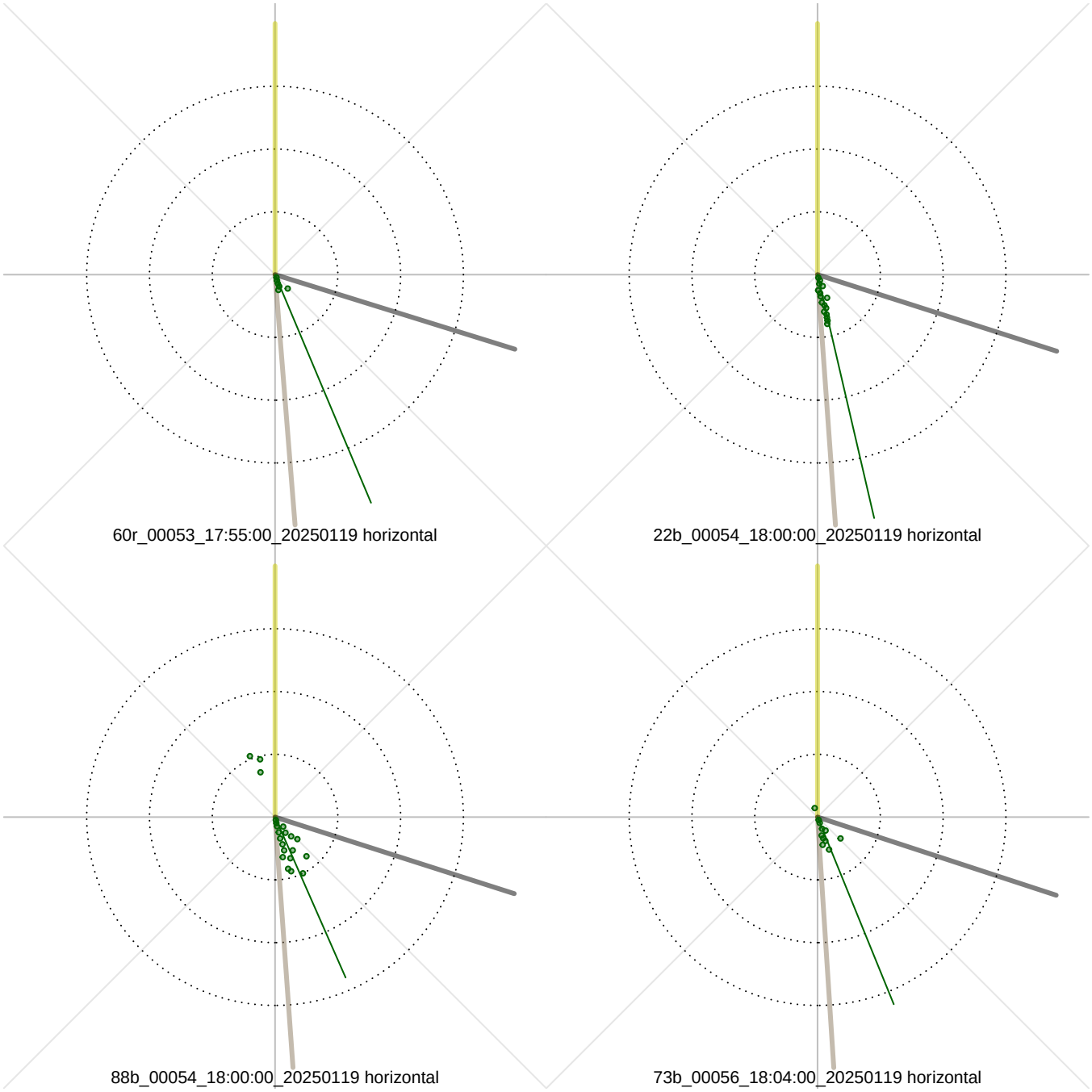


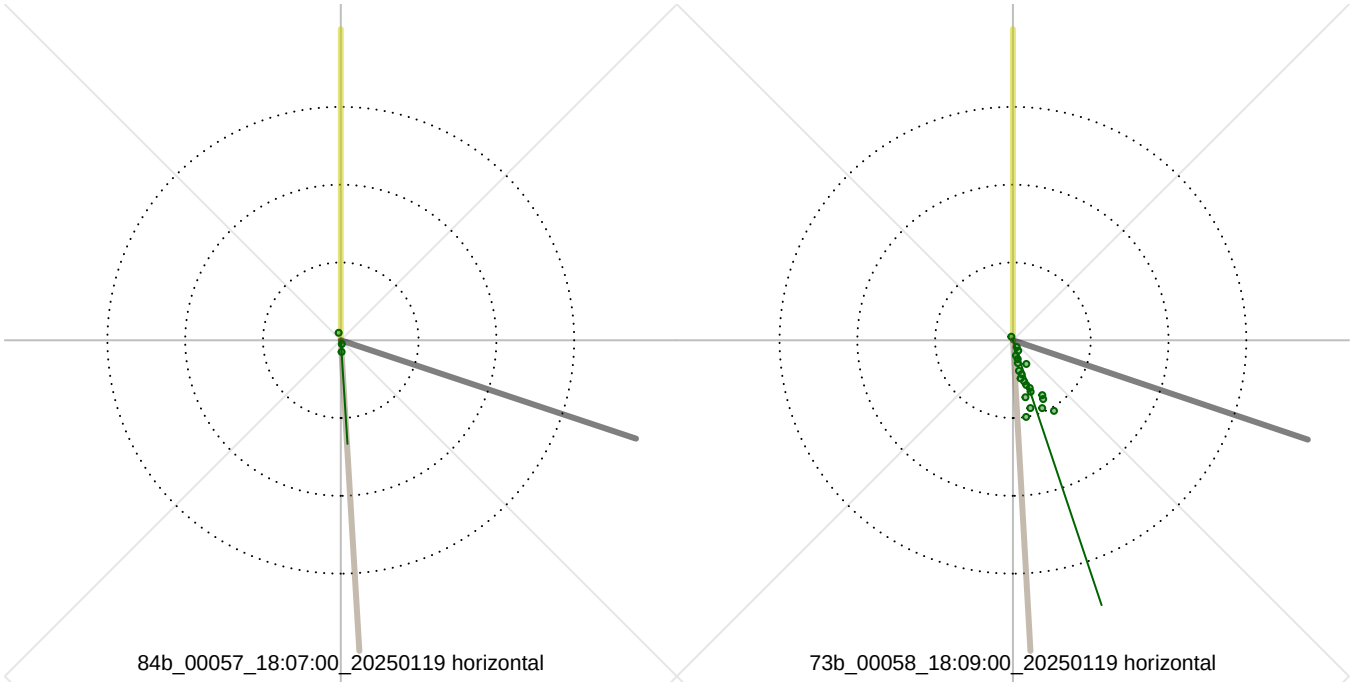
84b_00041_17:37:00_20250119 vertical

22b_00041_17:37:00_20250119 vertical

9b_00041_17:37:00_20250119 vertical

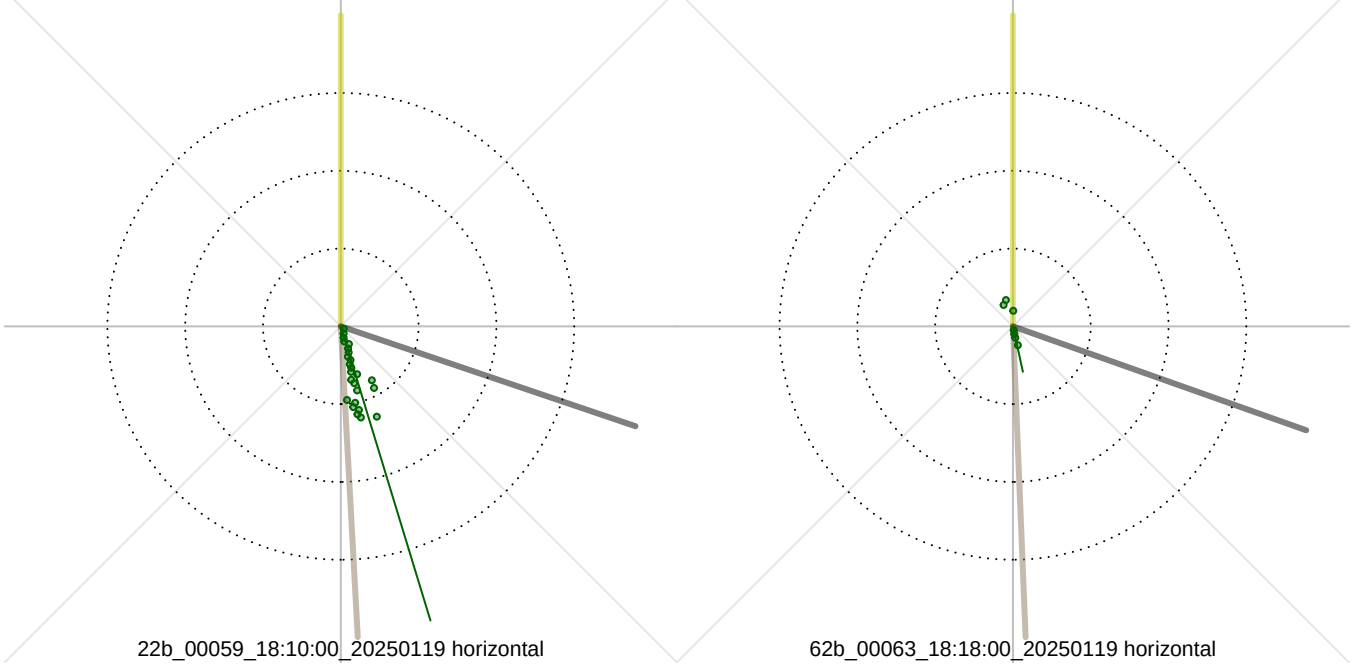

60r_00042_17:40:00_20250119 vertical
84b_00044_17:42:00_20250119 vertical
62b_00051_17:49:00_20250119 horizontal
82b_00049_17:49:00_20250119 horizontal


60r_00053_17:55:00_20250119 horizontal
22b_00054_18:00:00_20250119 horizontal
88b_00054_18:00:00_20250119 horizontal
73b_00056_18:04:00_20250119 horizontal



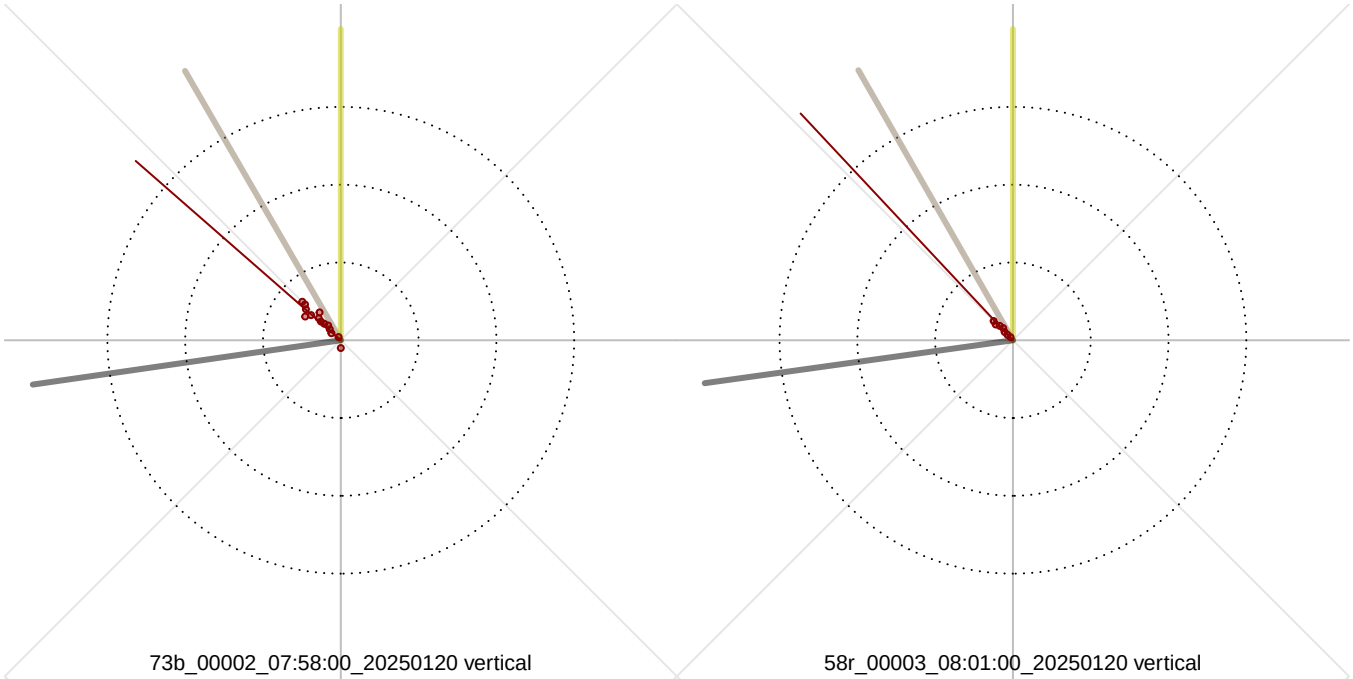
84b_00057_18:07:00_20250119 horizontal

73b_00058_18:09:00_20250119 horizontal

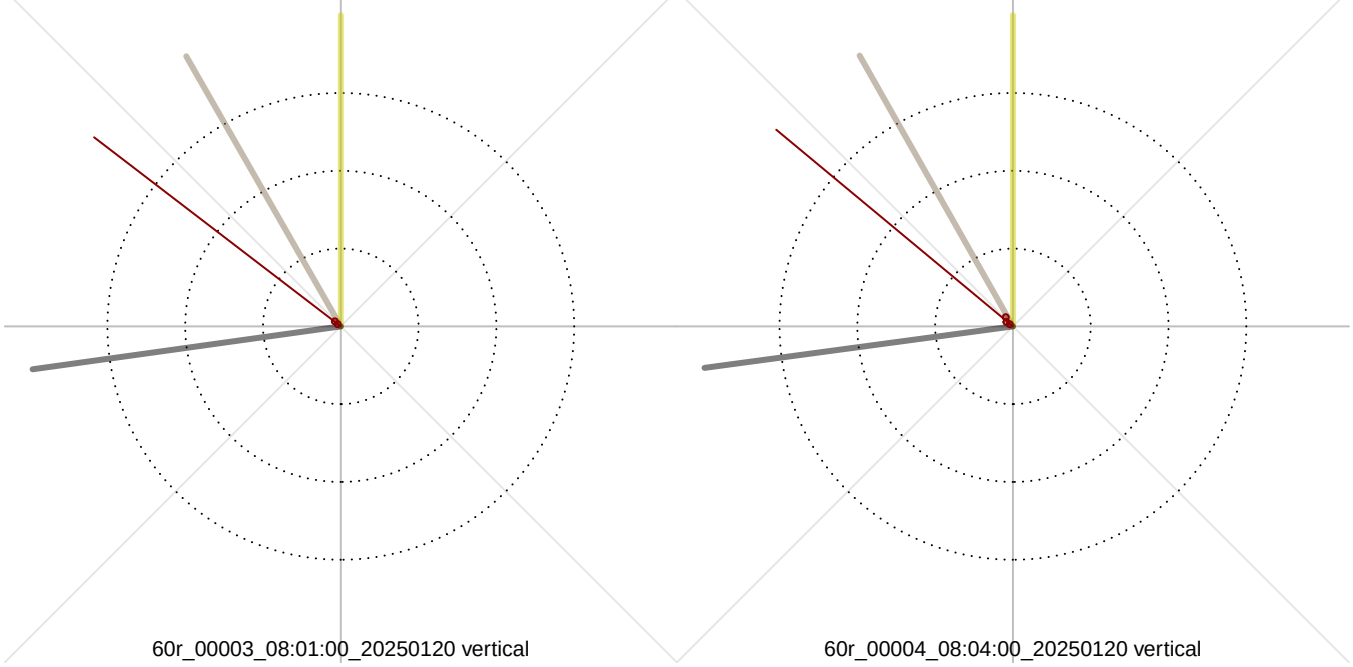


22b_00059_18:10:00_20250119 horizontal

62b_00063_18:18:00_20250119 horizontal



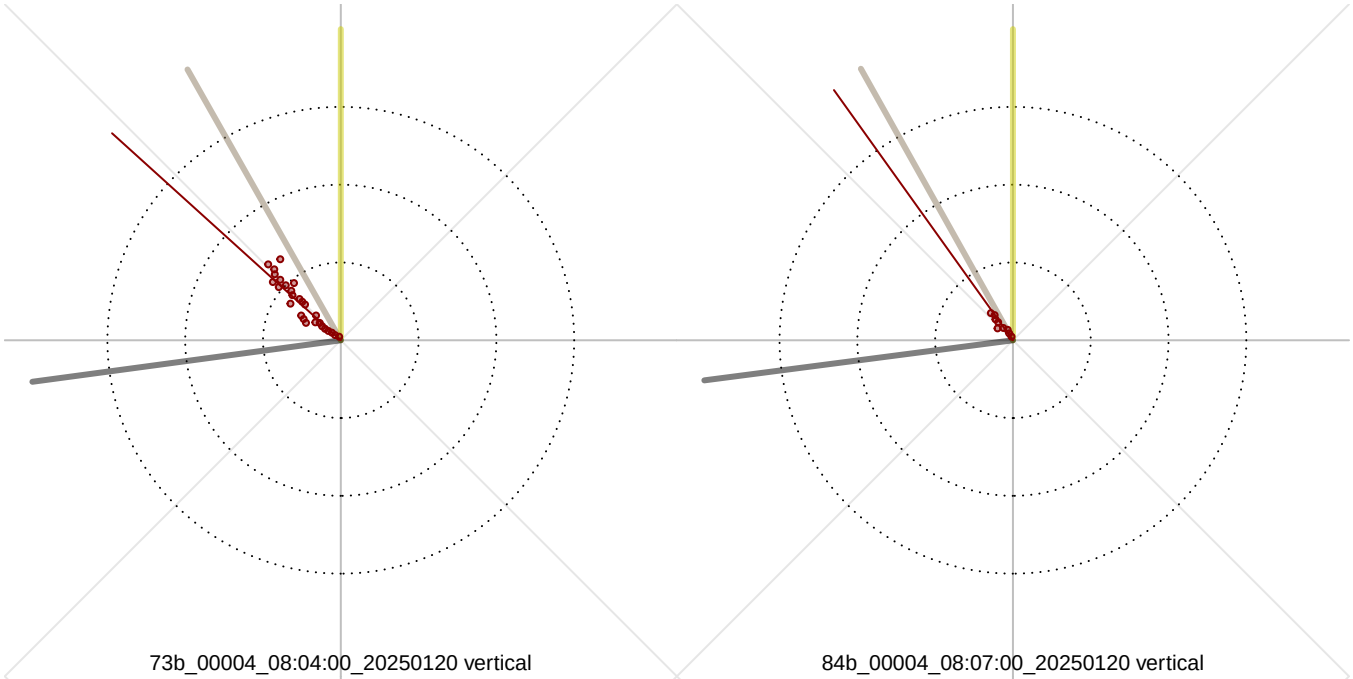
73b_00002_07:58:00_20250120 vertical



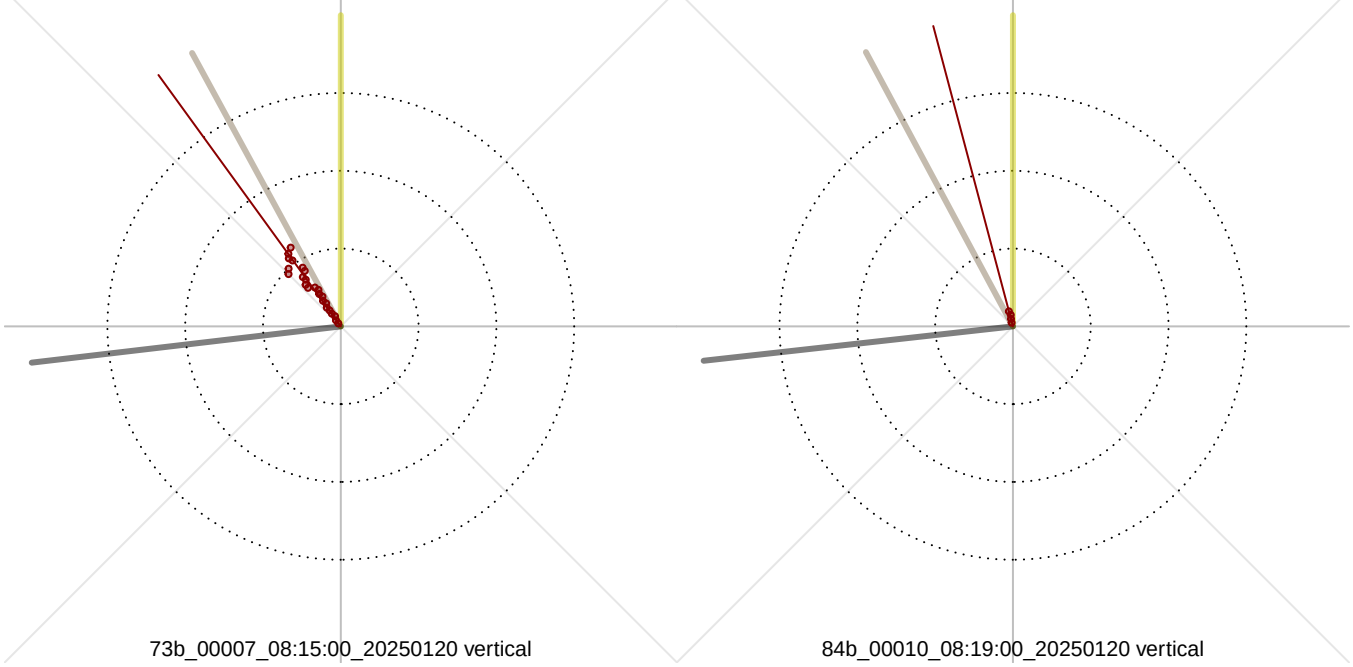
60r_00003_08:01:00_20250120 vertical

58r_00003_08:01:00_20250120 vertical

60r_00004_08:04:00_20250120 vertical



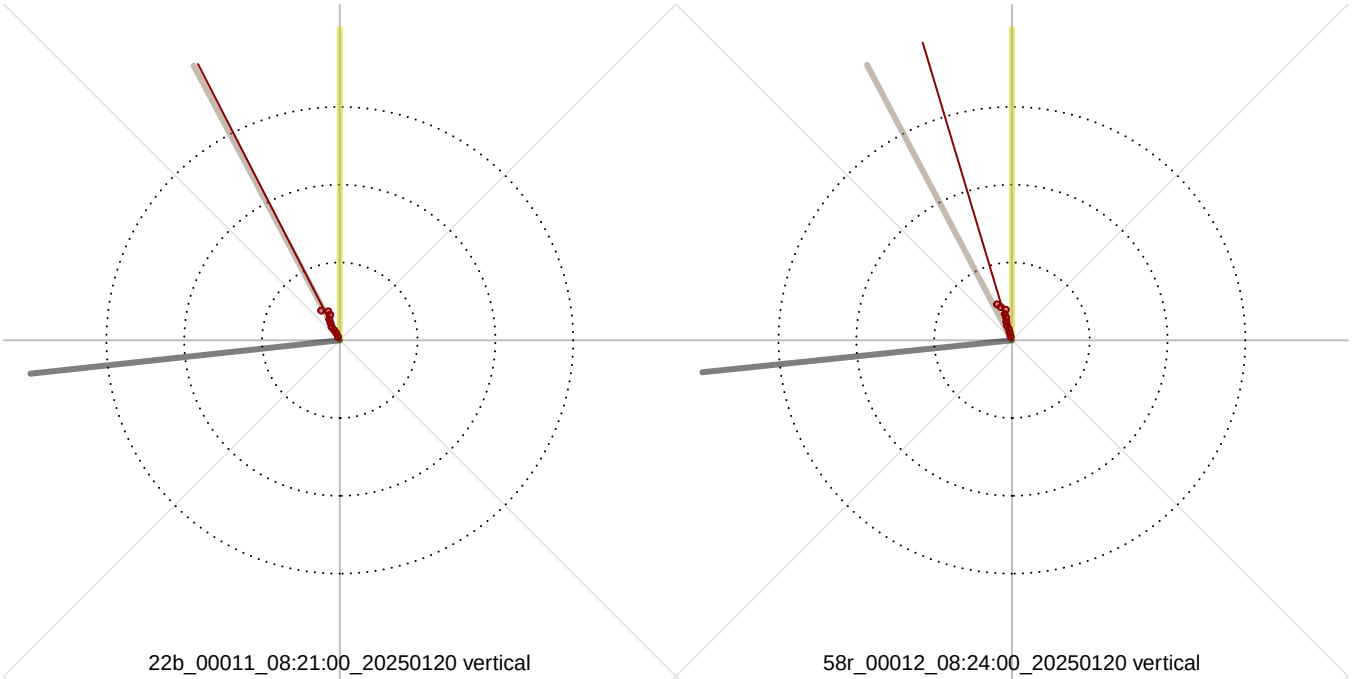
73b_00004_08:04:00_20250120 vertical



73b_00007_08:15:00_20250120 vertical

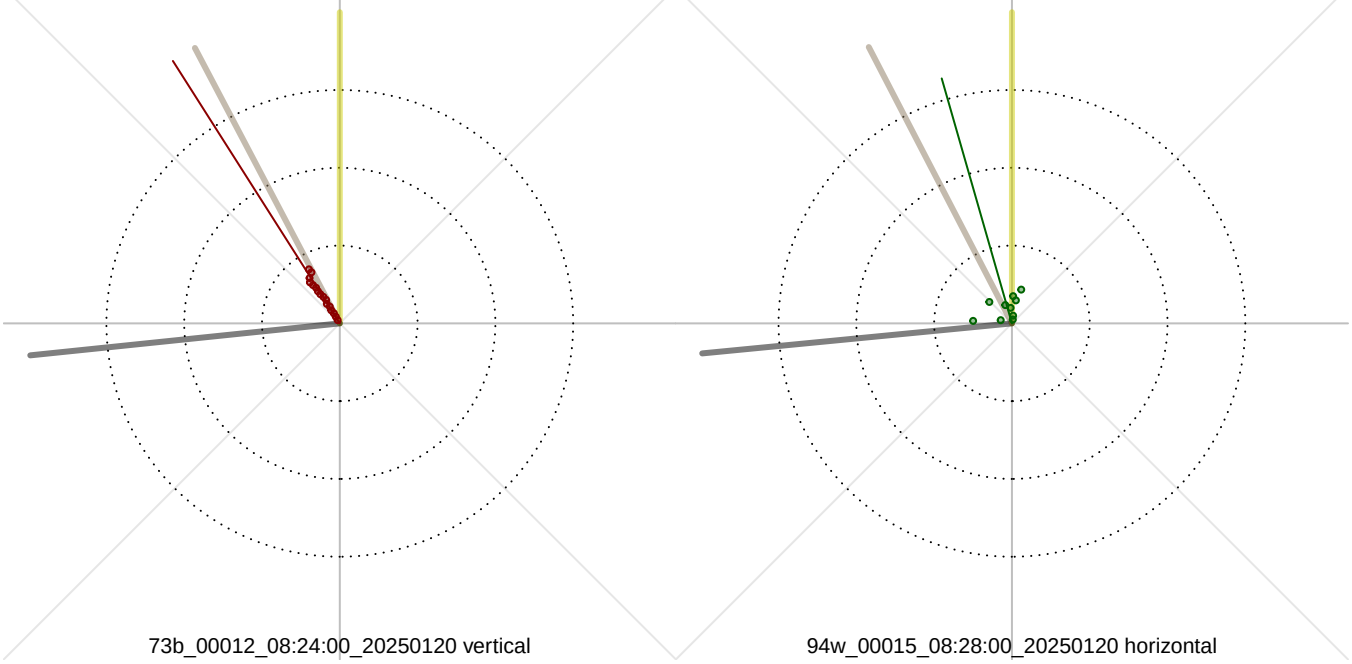
84b_00004_08:07:00_20250120 vertical

84b_00010_08:19:00_20250120 vertical



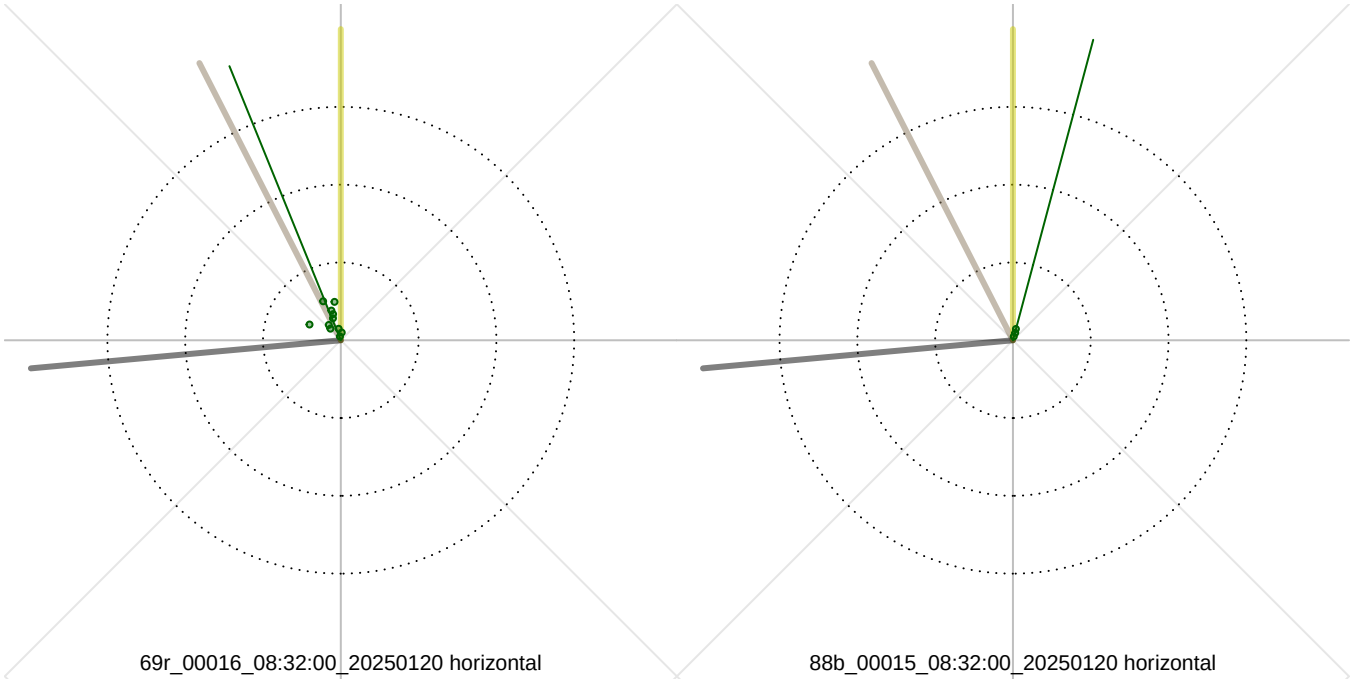
22b_00011_08:21:00_20250120 vertical

58r_00012_08:24:00_20250120 vertical



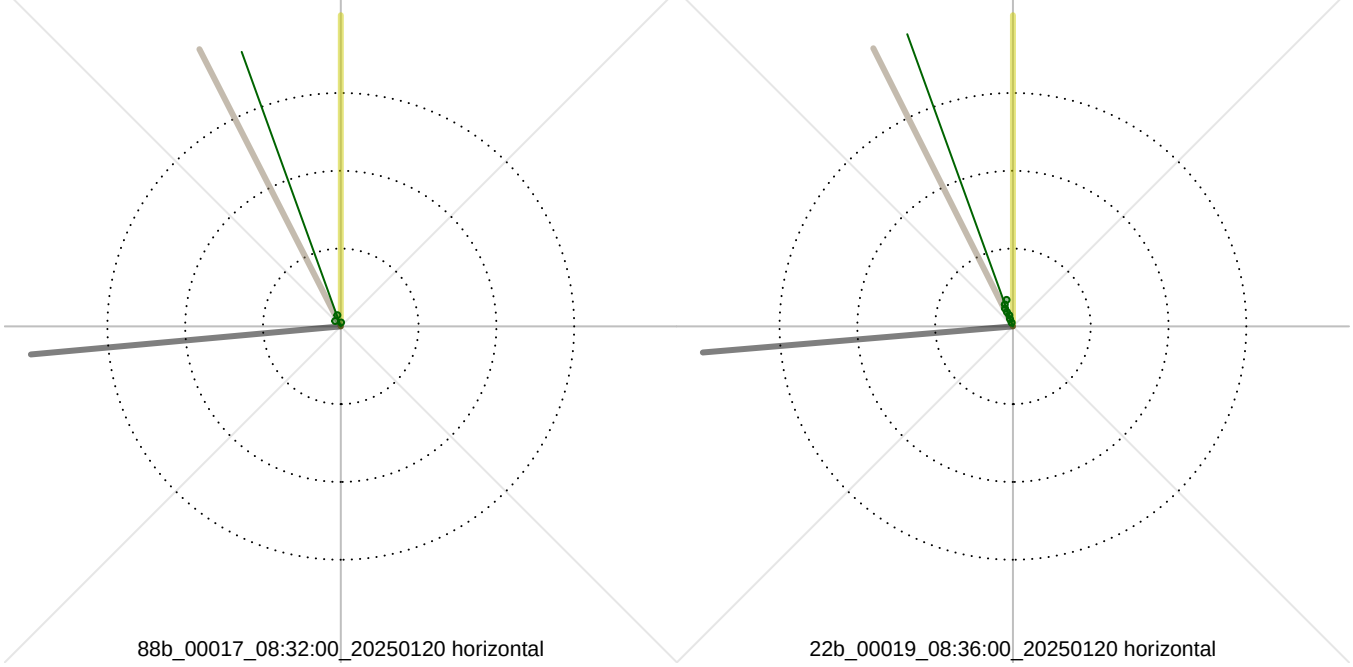
73b_00012_08:24:00_20250120 vertical

94w_00015_08:28:00_20250120 horizontal



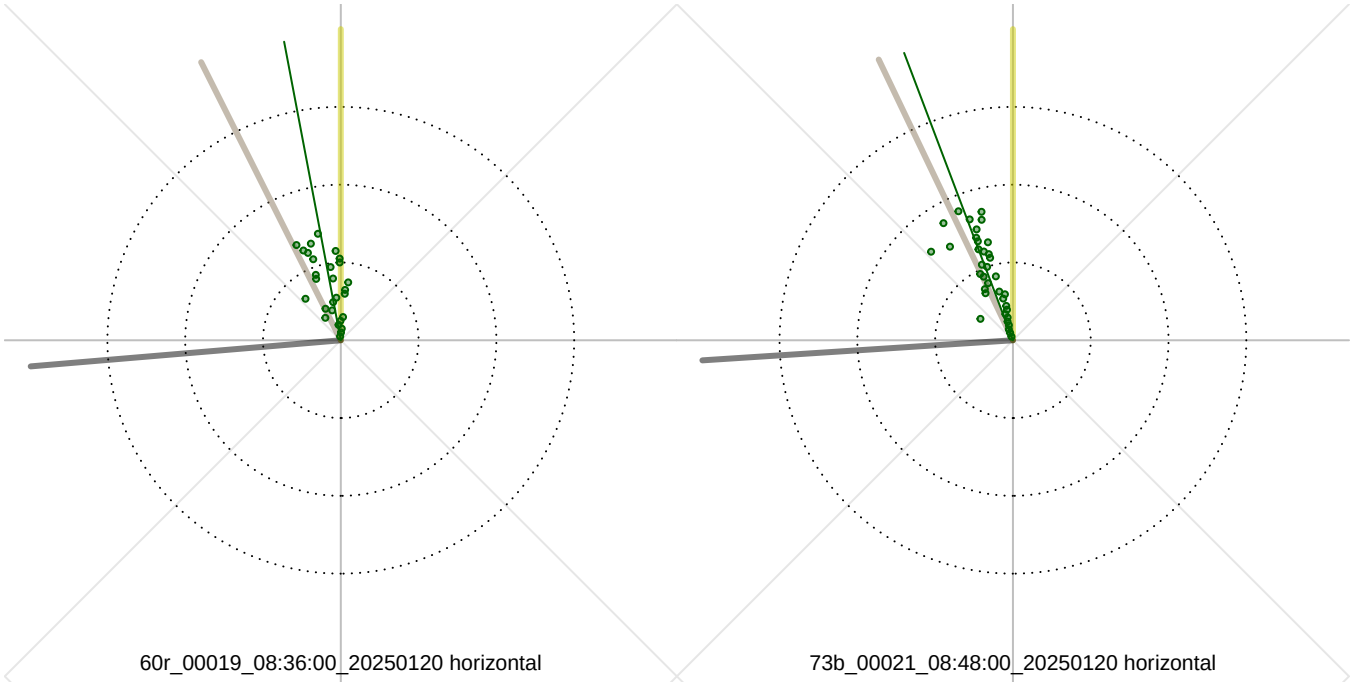
69r_00016_08:32:00_20250120 horizontal

88b_00015_08:32:00_20250120 horizontal

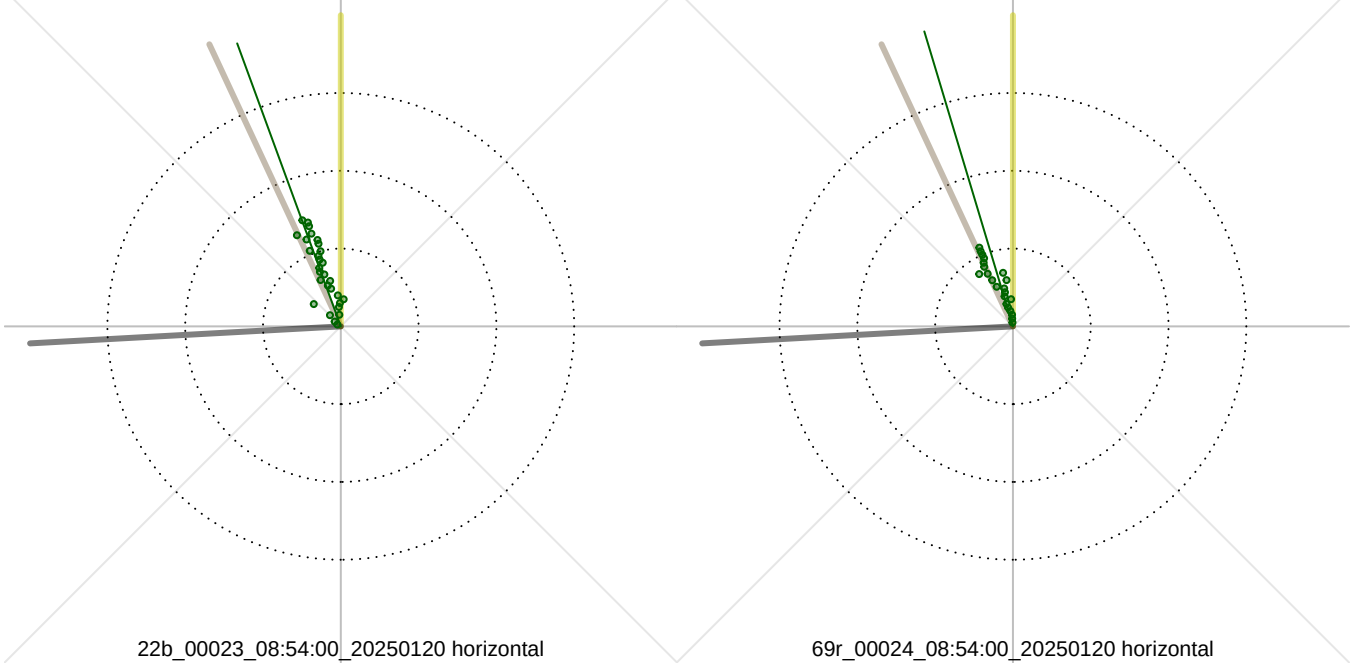


88b_00017_08:32:00_20250120 horizontal

22b_00019_08:36:00_20250120 horizontal



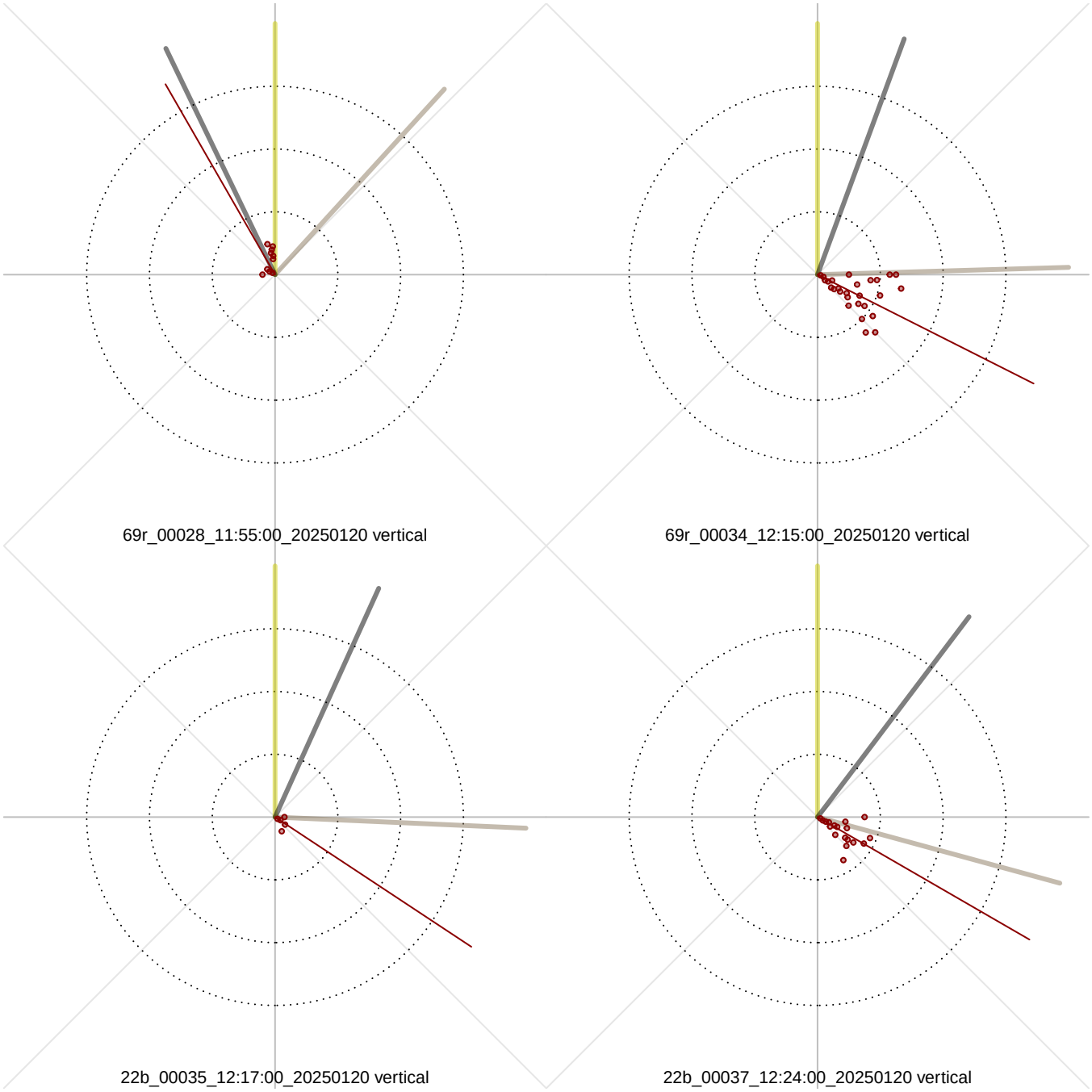
60r_00019_08:36:00_20250120 horizontal

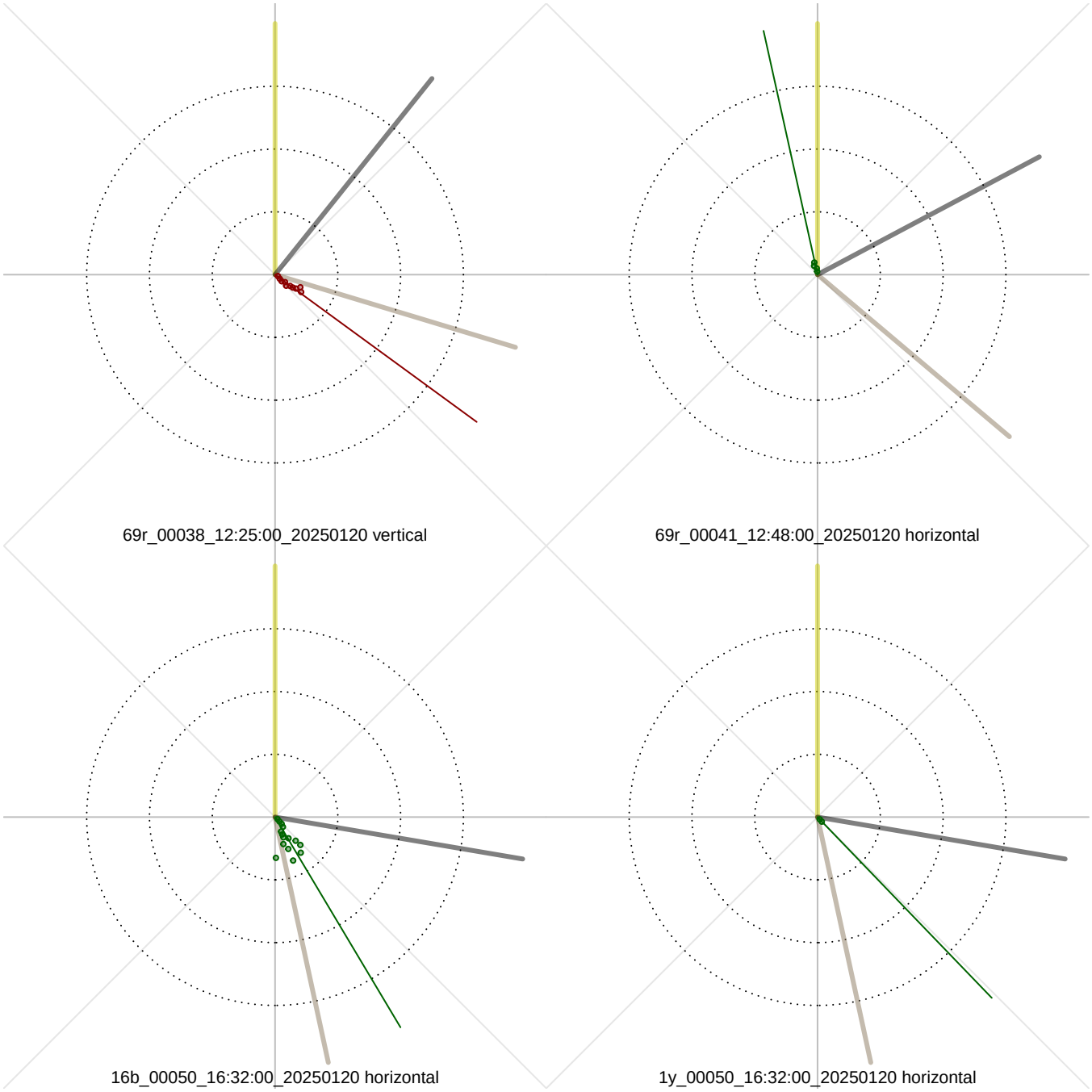


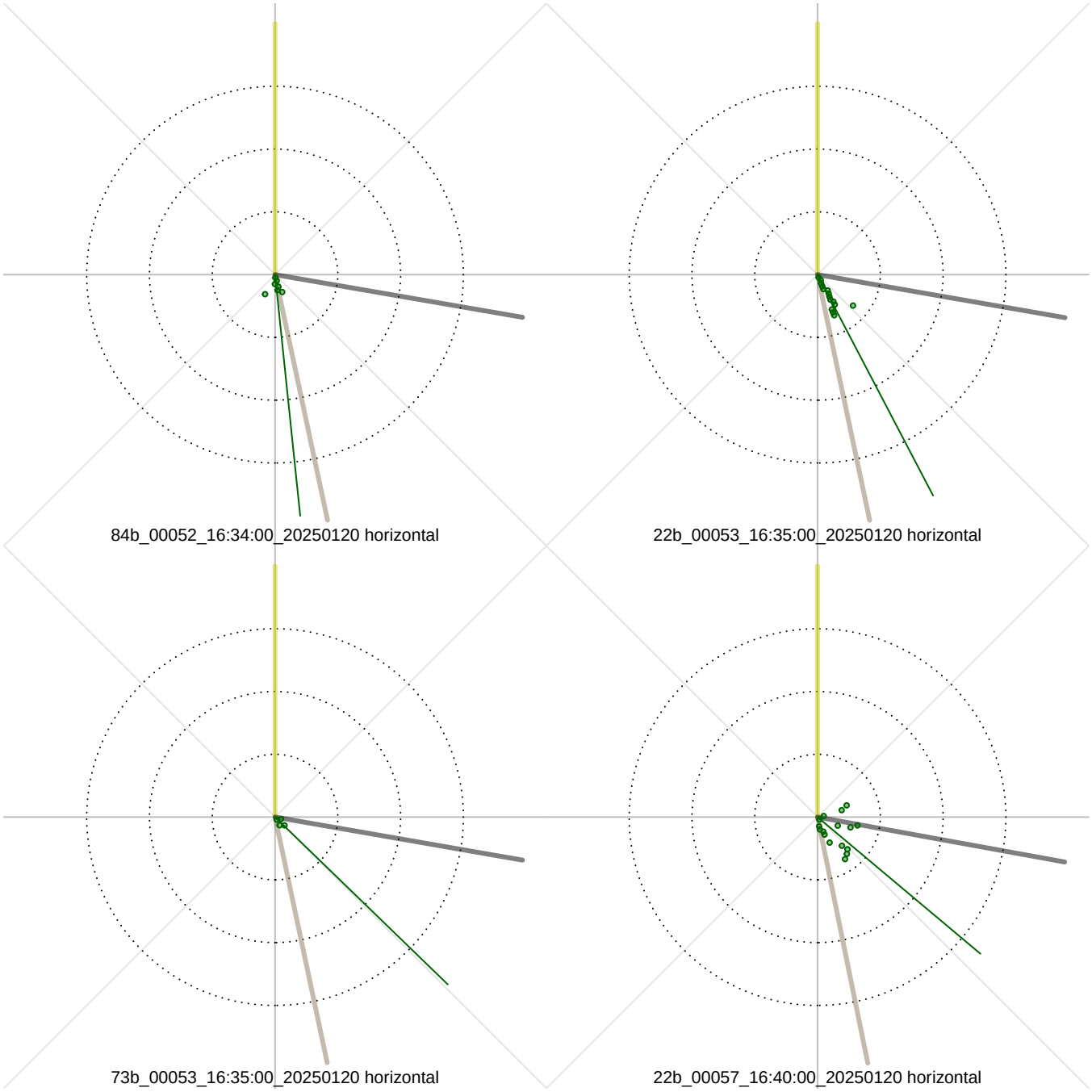
22b_00023_08:54:00_20250120 horizontal

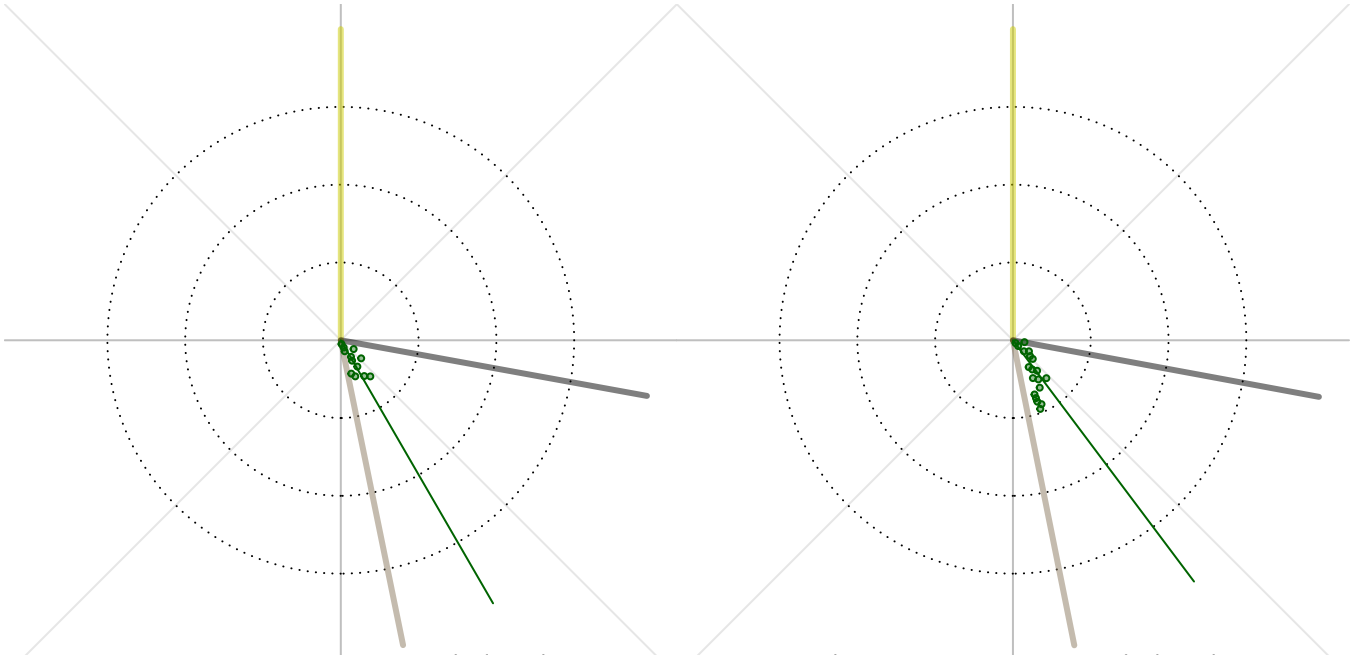
73b_00021_08:48:00_20250120 horizontal

69r_00024_08:54:00_20250120 horizontal


69r_00028_11:55:00_20250120 vertical
69r_00034_12:15:00_20250120 vertical
22b_00035_12:17:00_20250120 vertical
22b_00037_12:24:00_20250120 vertical

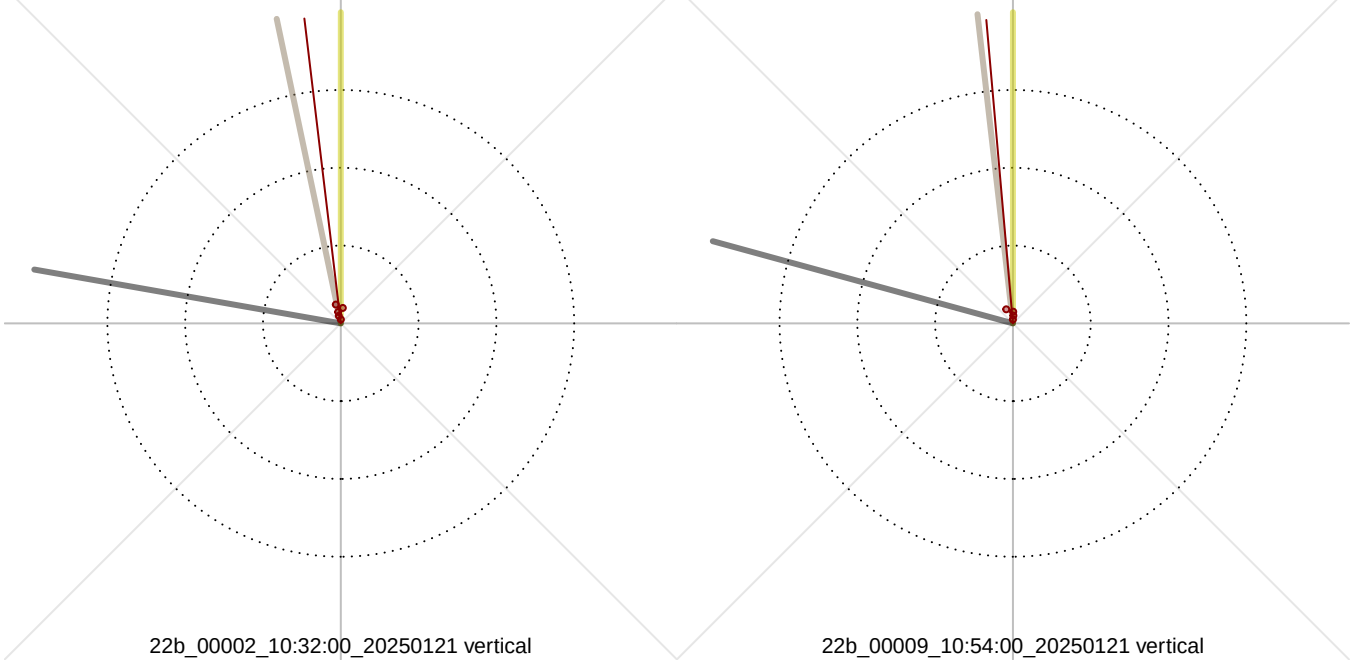

69r_00038_12:25:00_20250120 vertical
69r_00041_12:48:00_20250120 horizontal
16b_00050_16:32:00_20250120 horizontal
1y_00050_16:32:00_20250120 horizontal


84b_00052_16:34:00_20250120 horizontal
22b_00053_16:35:00_20250120 horizontal
73b_00053_16:35:00_20250120 horizontal
22b_00057_16:40:00_20250120 horizontal



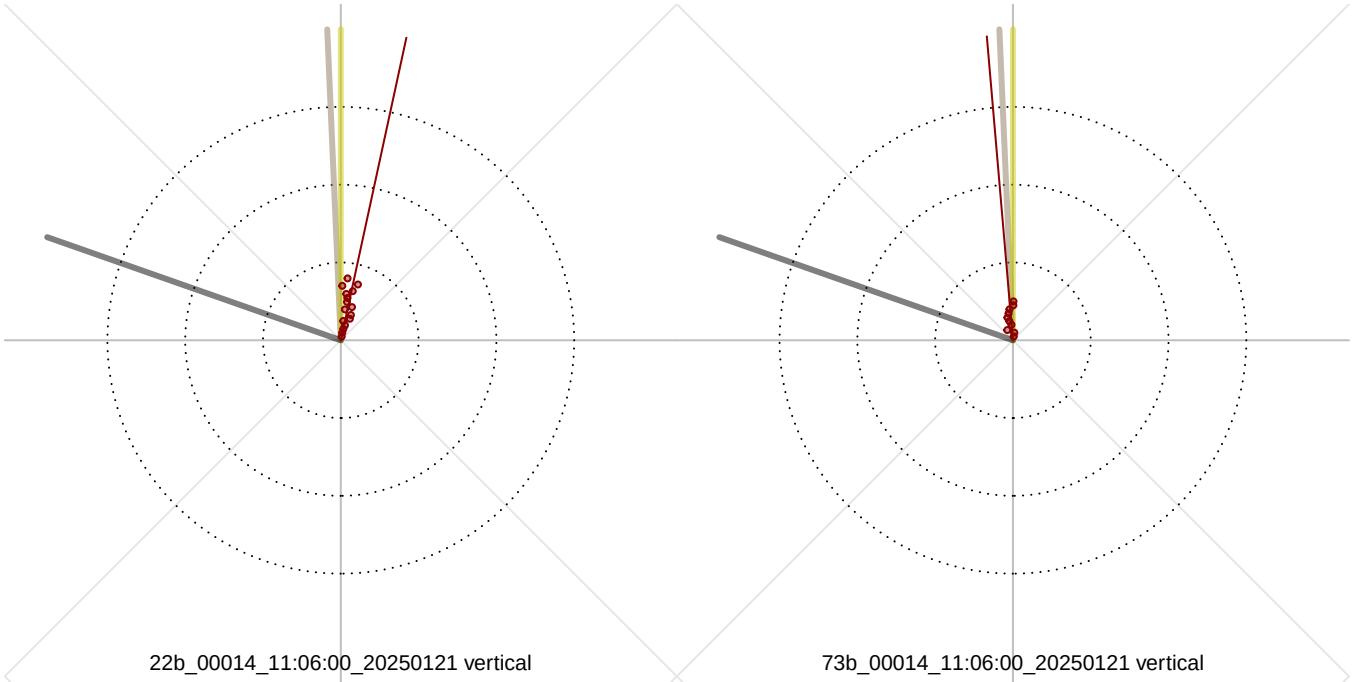
92w_00057_16:40:00_20250120 horizontal

73b_00057_16:42:00_20250120 horizontal



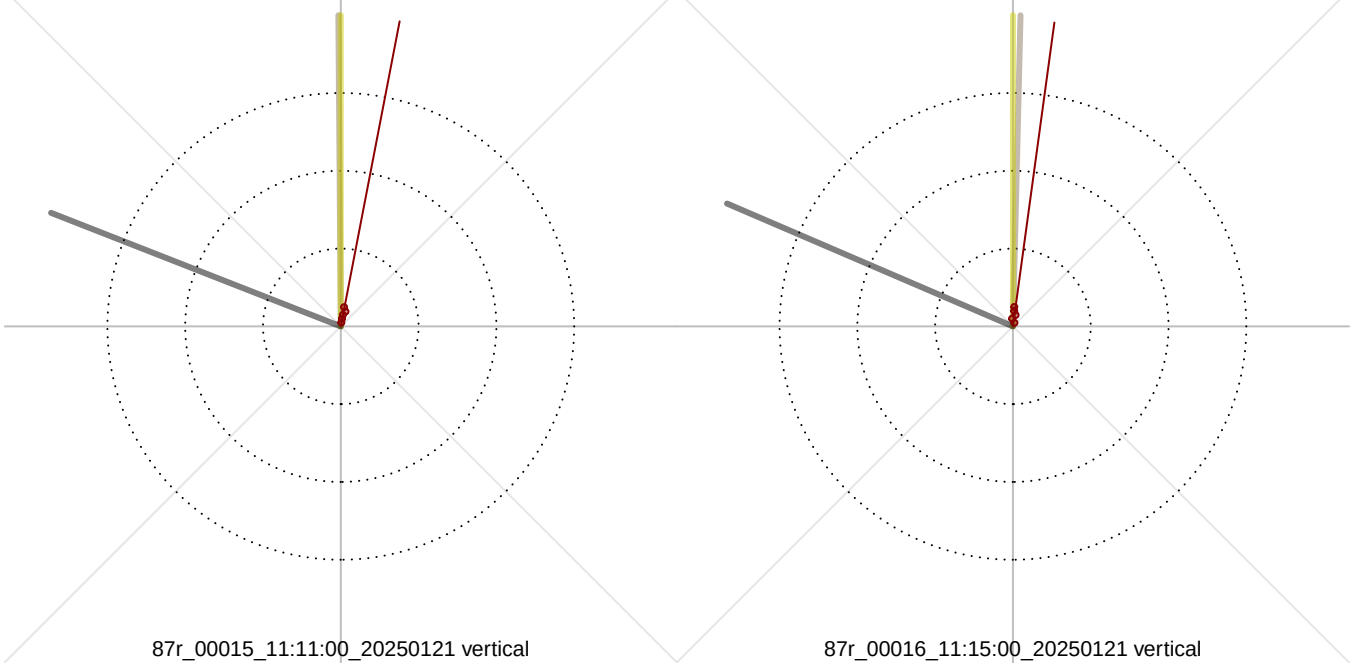
22b_00002_10:32:00_20250121 vertical

22b_00009_10:54:00_20250121 vertical



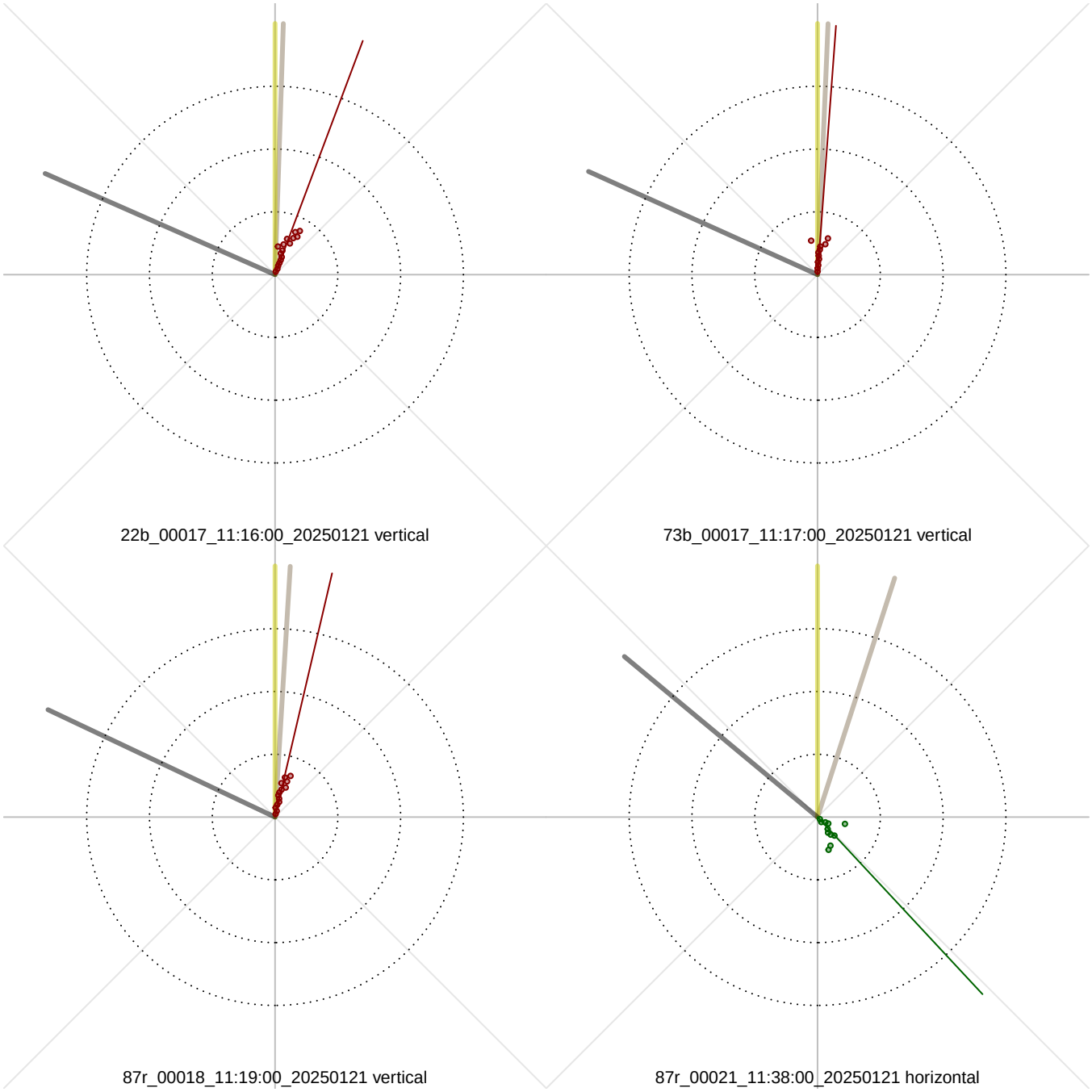
22b_00014_11:06:00_20250121 vertical

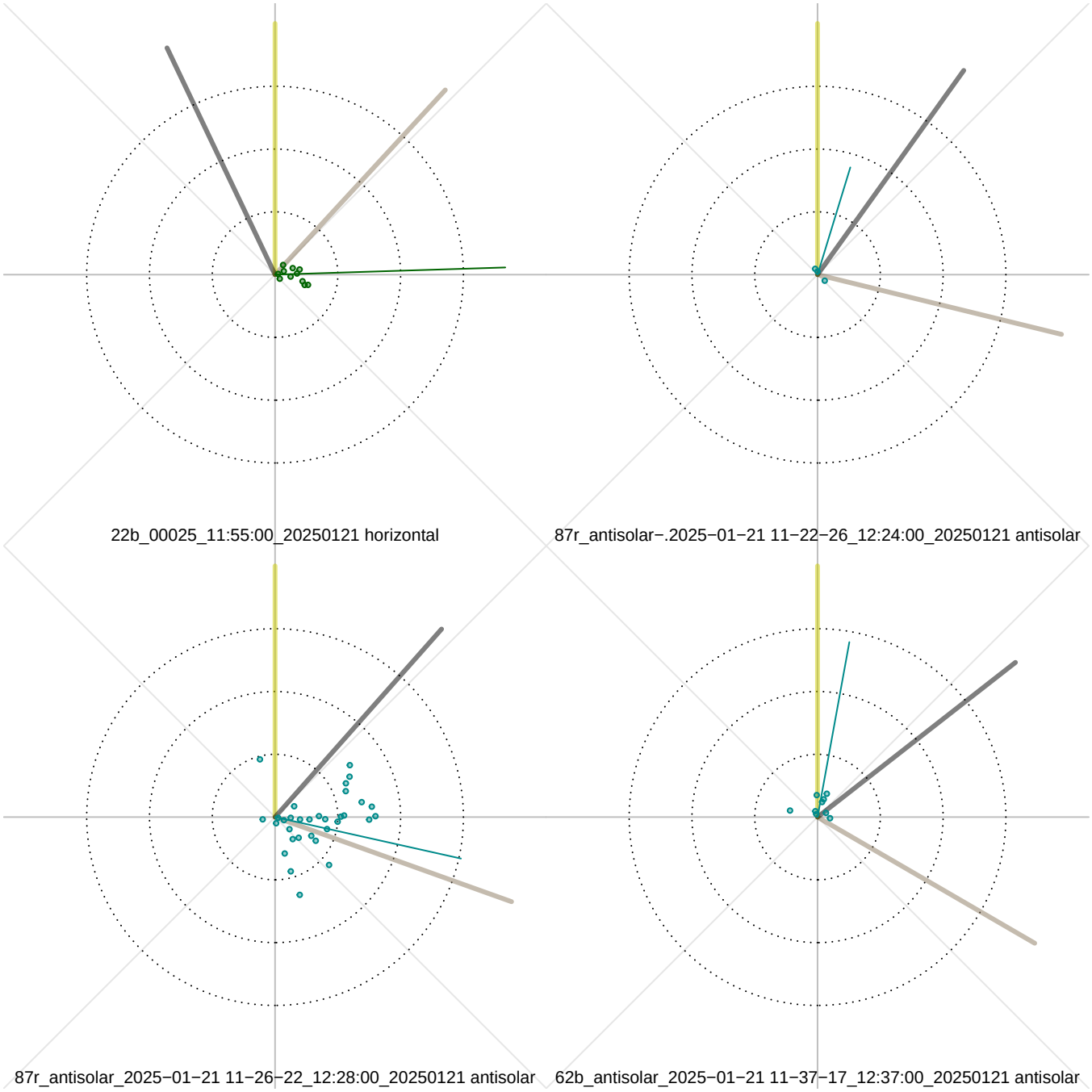
73b_00014_11:06:00_20250121 vertical

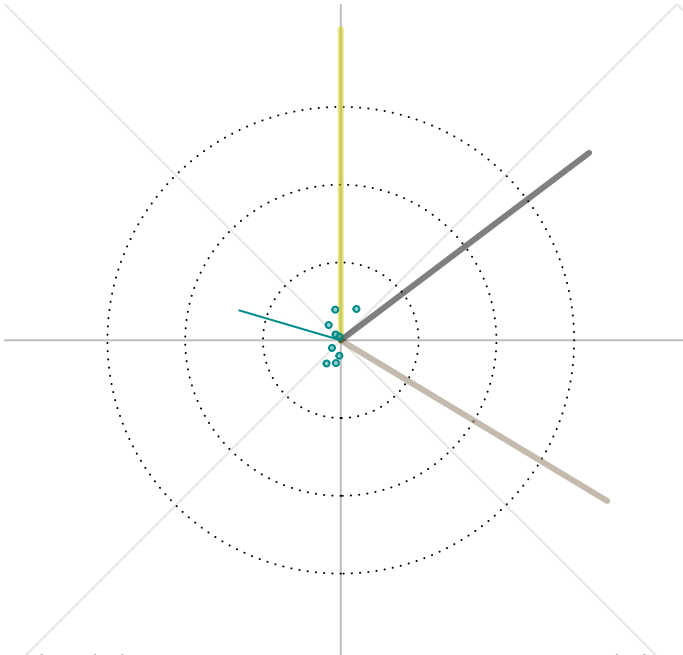


87r_00015_11:11:00_20250121 vertical

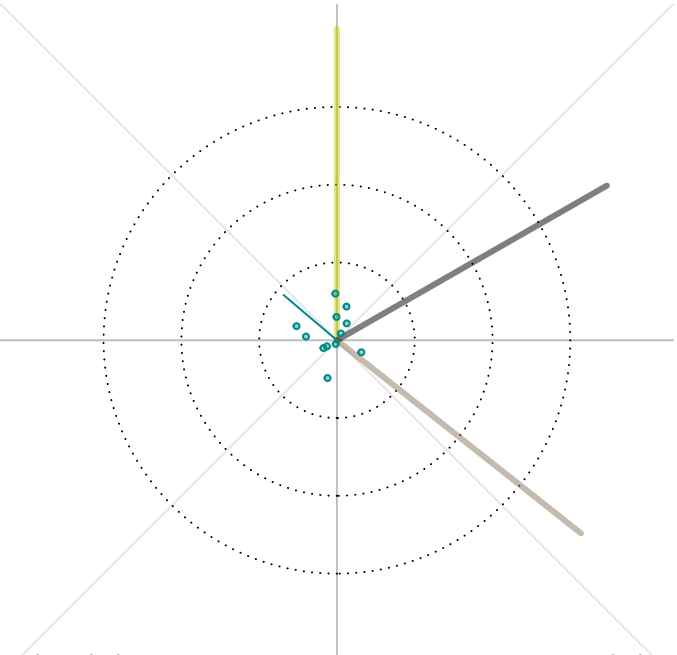
87r_00016_11:15:00_20250121 vertical


22b_00017_11:16:00_20250121 vertical
73b_00017_11:17:00_20250121 vertical
87r_00018_11:19:00_20250121 vertical
87r_00021_11:38:00_20250121 horizontal

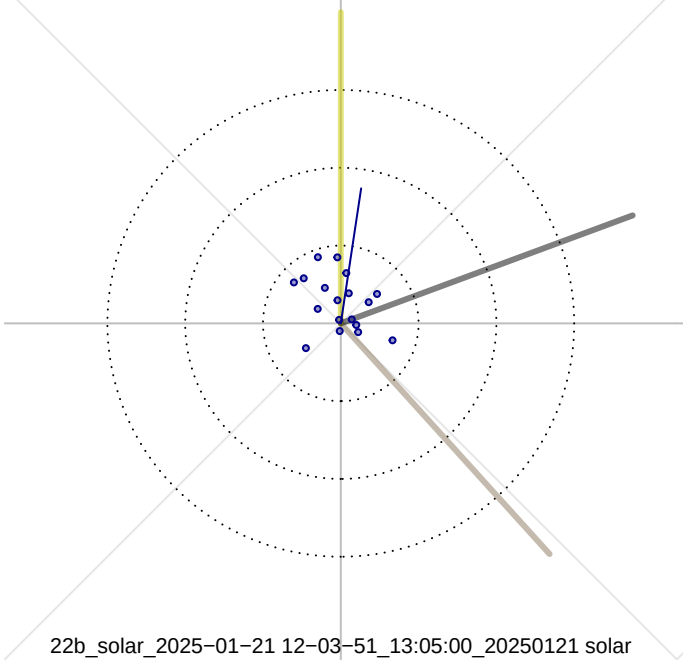

22b_00025_11:55:00_20250121 horizontal
87r_antisolar−.2025−01−21 11−22−26_12:24:00_20250121 antisolar
87r_antisolar_2025−01−21 11−26−22_12:28:00_20250121 antisolar
62b_antisolar_2025−01−21 11−37−17_12:37:00_20250121 antisolar



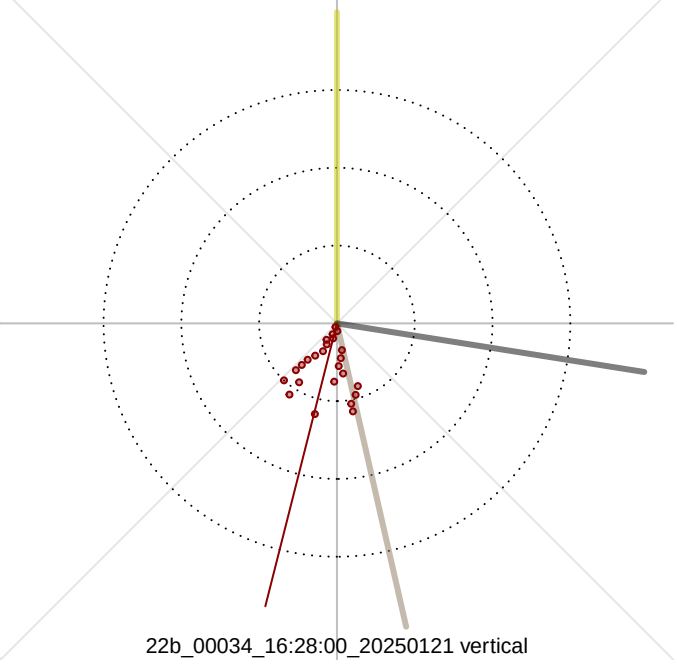
62b_antisolar_2025−01−21 11−38−00_12:38:00_20250121 antisolar



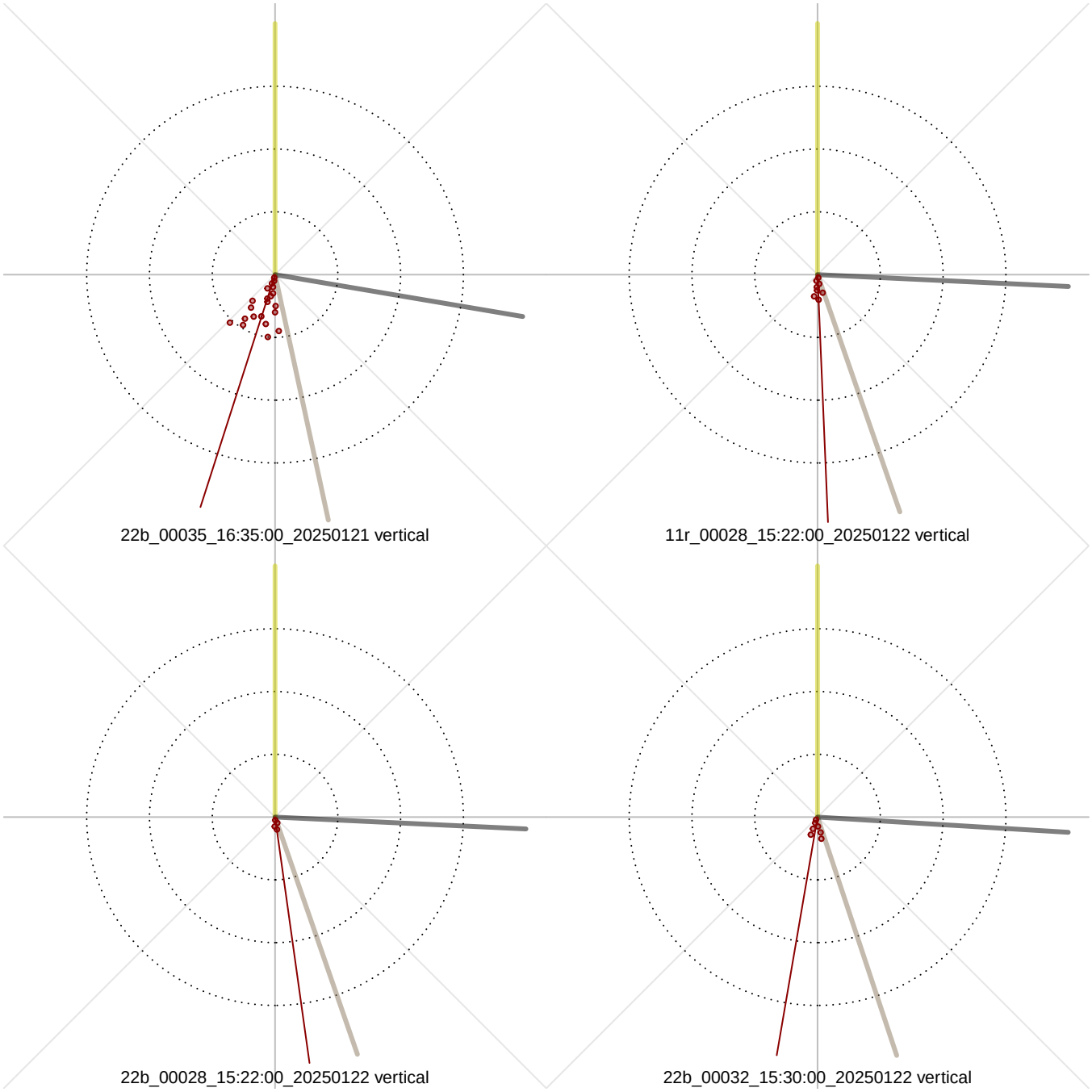
22b_antisolar_2025−01−21 11−47−07_12:47:00_20250121 antisolar

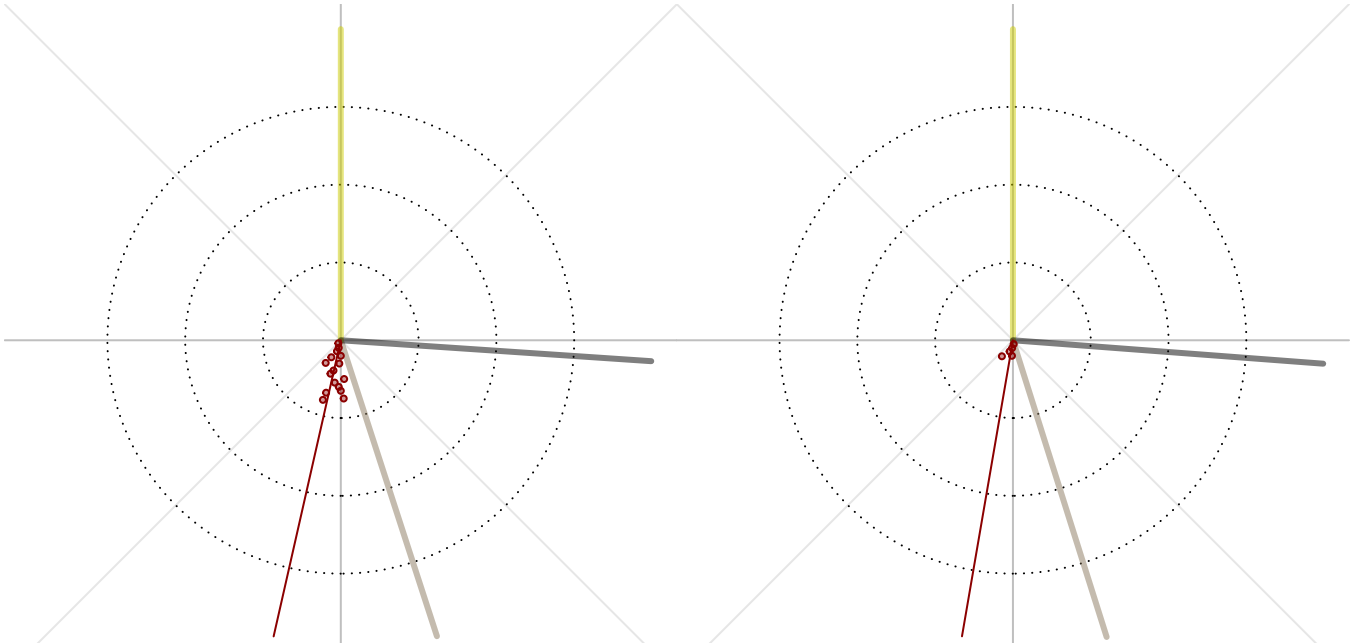


22b_solar_2025−01−21 12−03−51_13:05:00_20250121 solar



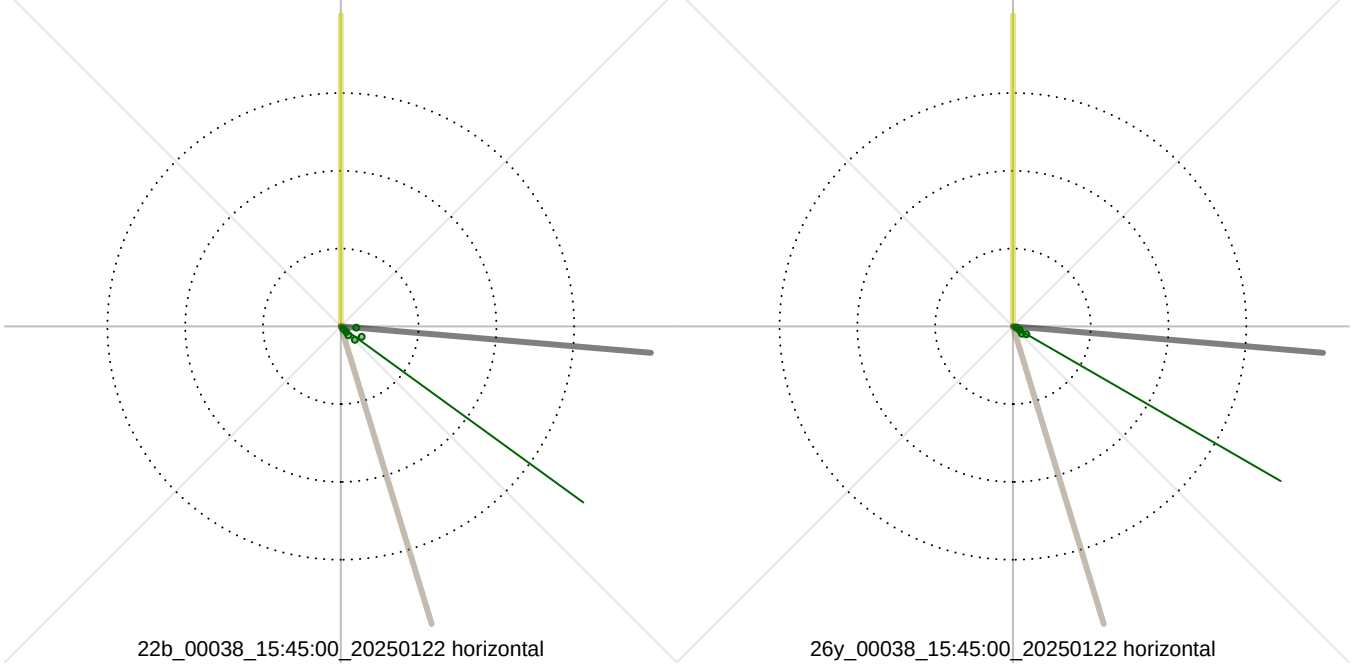
22b_00034_16:28:00_20250121 vertical


22b_00035_16:35:00_20250121 vertical
11r_00028_15:22:00_20250122 vertical
22b_00028_15:22:00_20250122 vertical
22b_00032_15:30:00_20250122 vertical



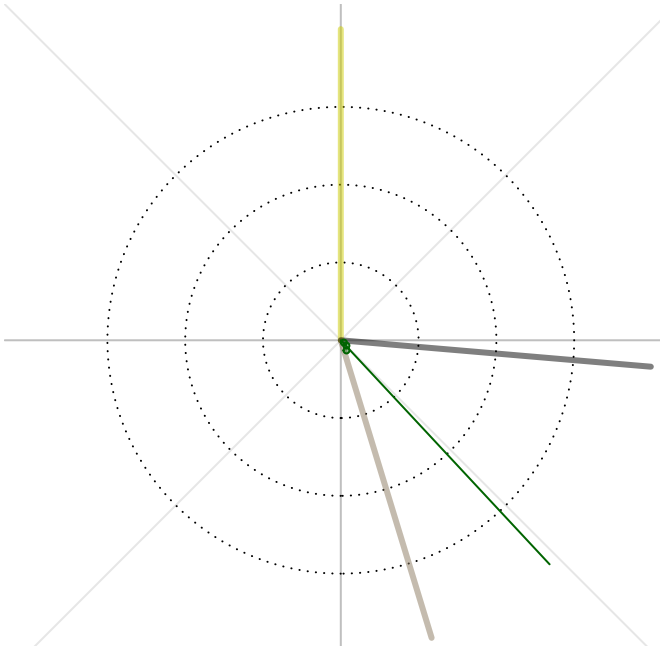
11r_00034_15:34:00_20250122 vertical

22b_00036_15:39:00_20250122 vertical

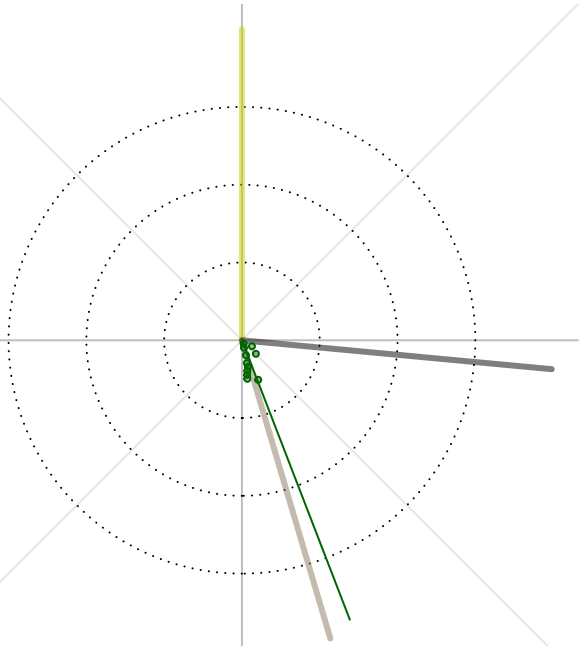


22b_00038_15:45:00_20250122 horizontal

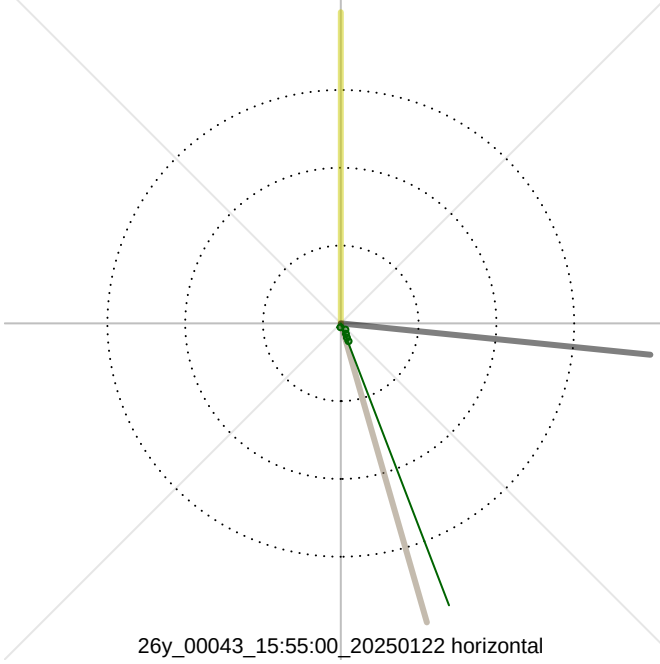
26y_00038_15:45:00_20250122 horizontal



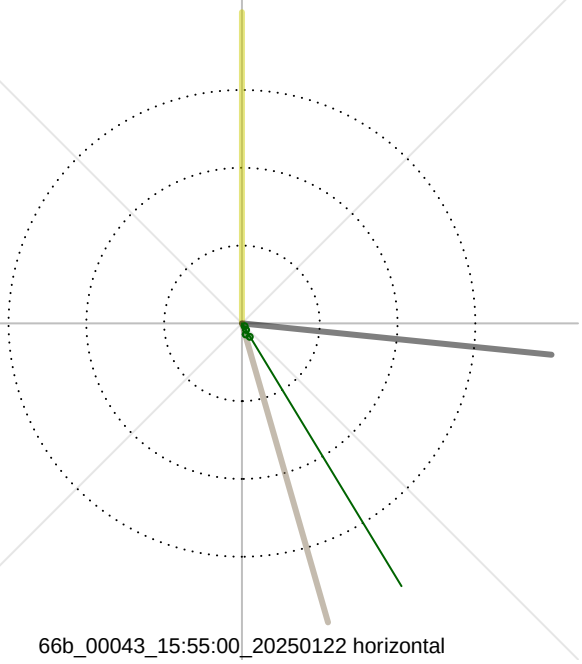
84b_00038_15:45:00_20250122 horizontal



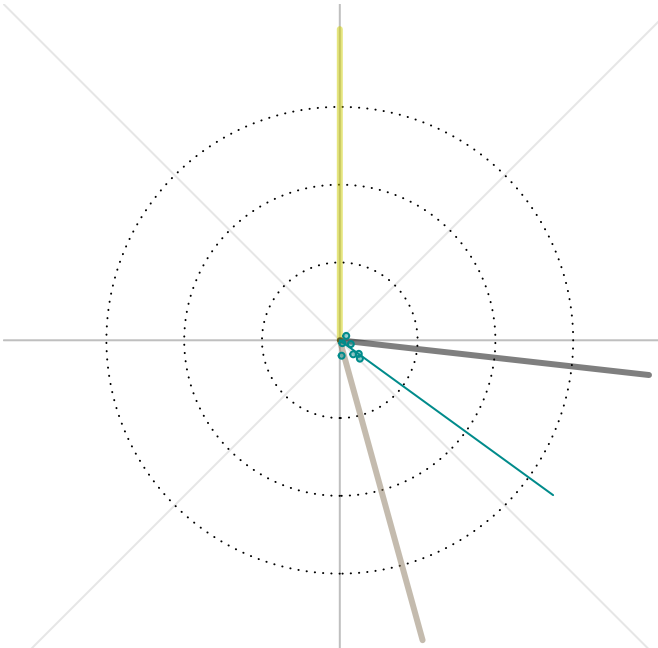
84b_00040_15:50:00_20250122 horizontal



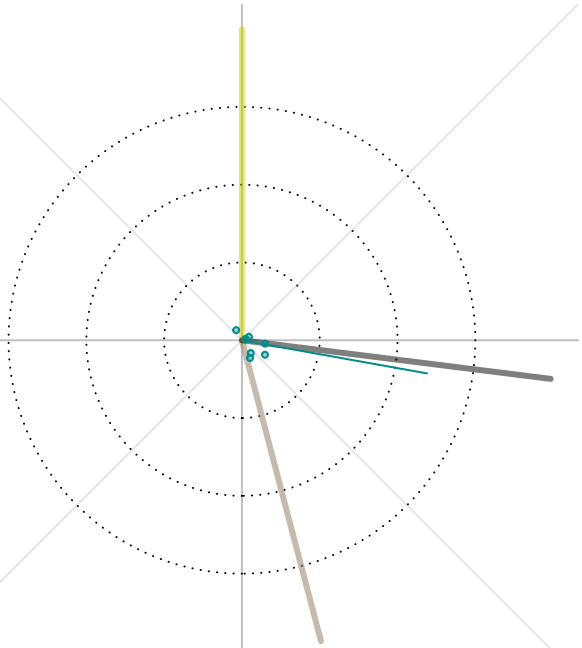
26y_00043_15:55:00_20250122 horizontal



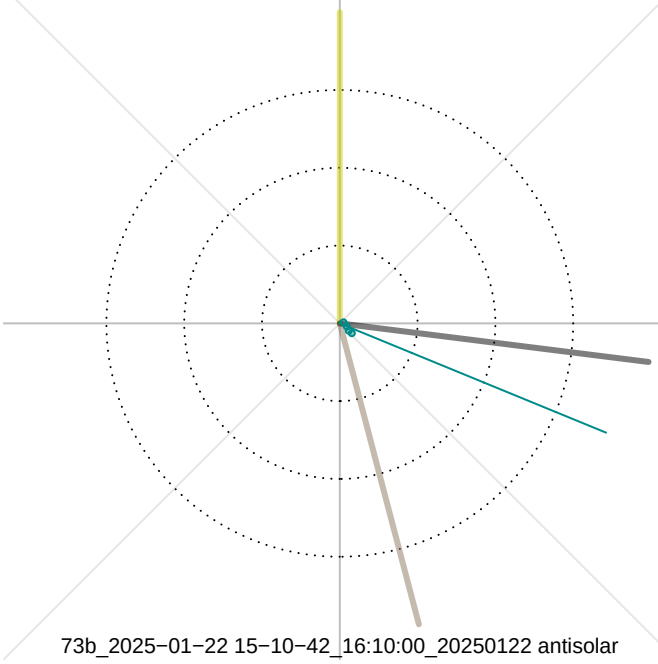
66b_00043_15:55:00_20250122 horizontal



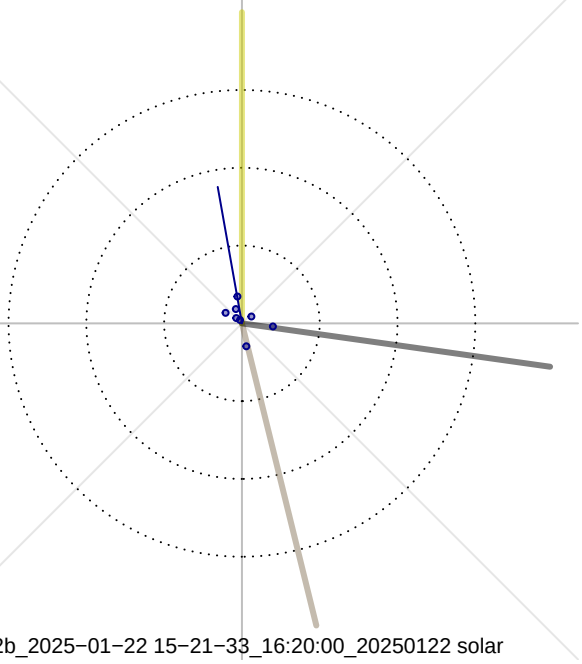
73b_2025-01-22 15-00-37_16:02:00_20250122 antisolar



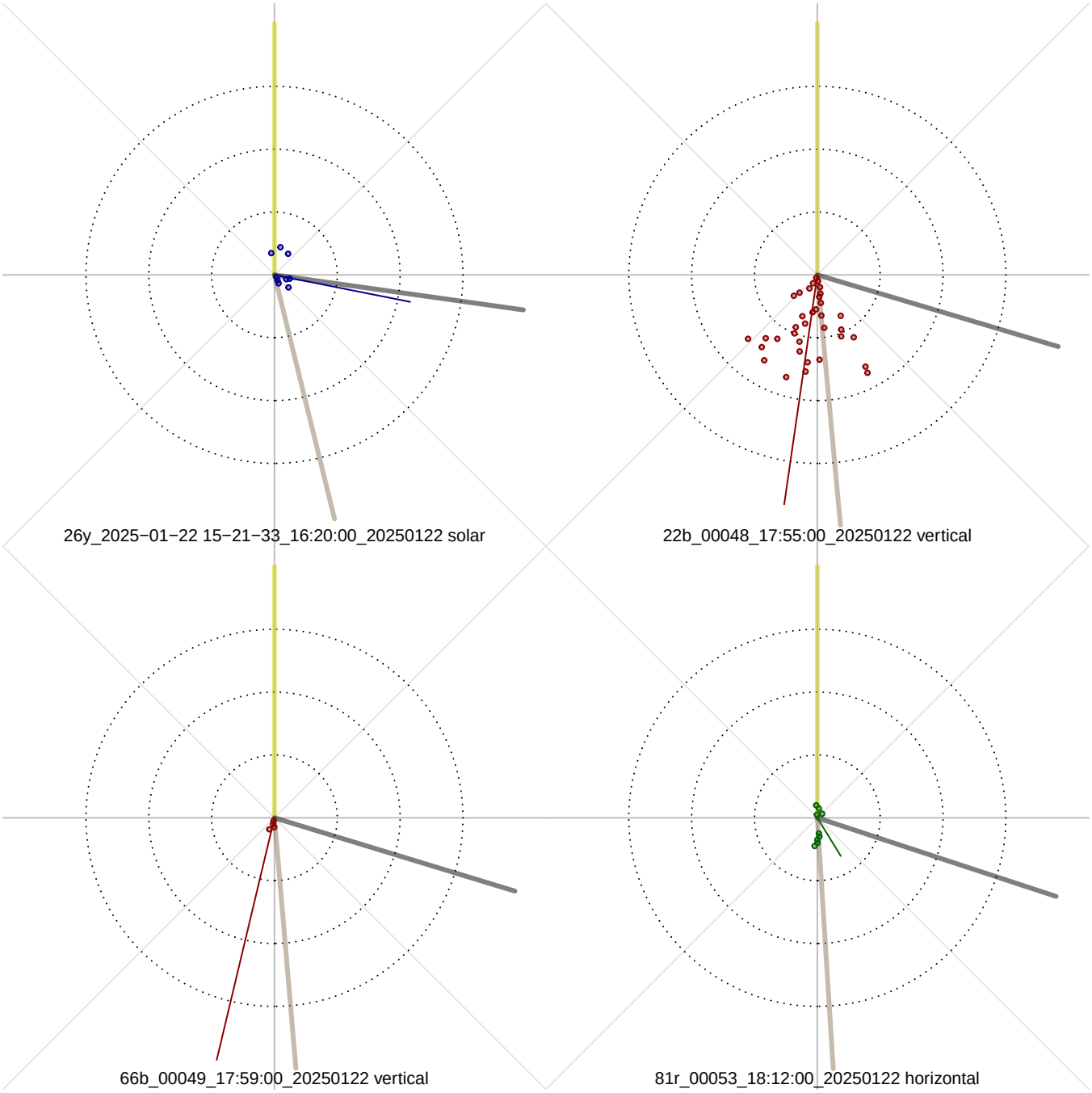
22b_2025-01-22 15-10-42_16:10:00_20250122 antisolar

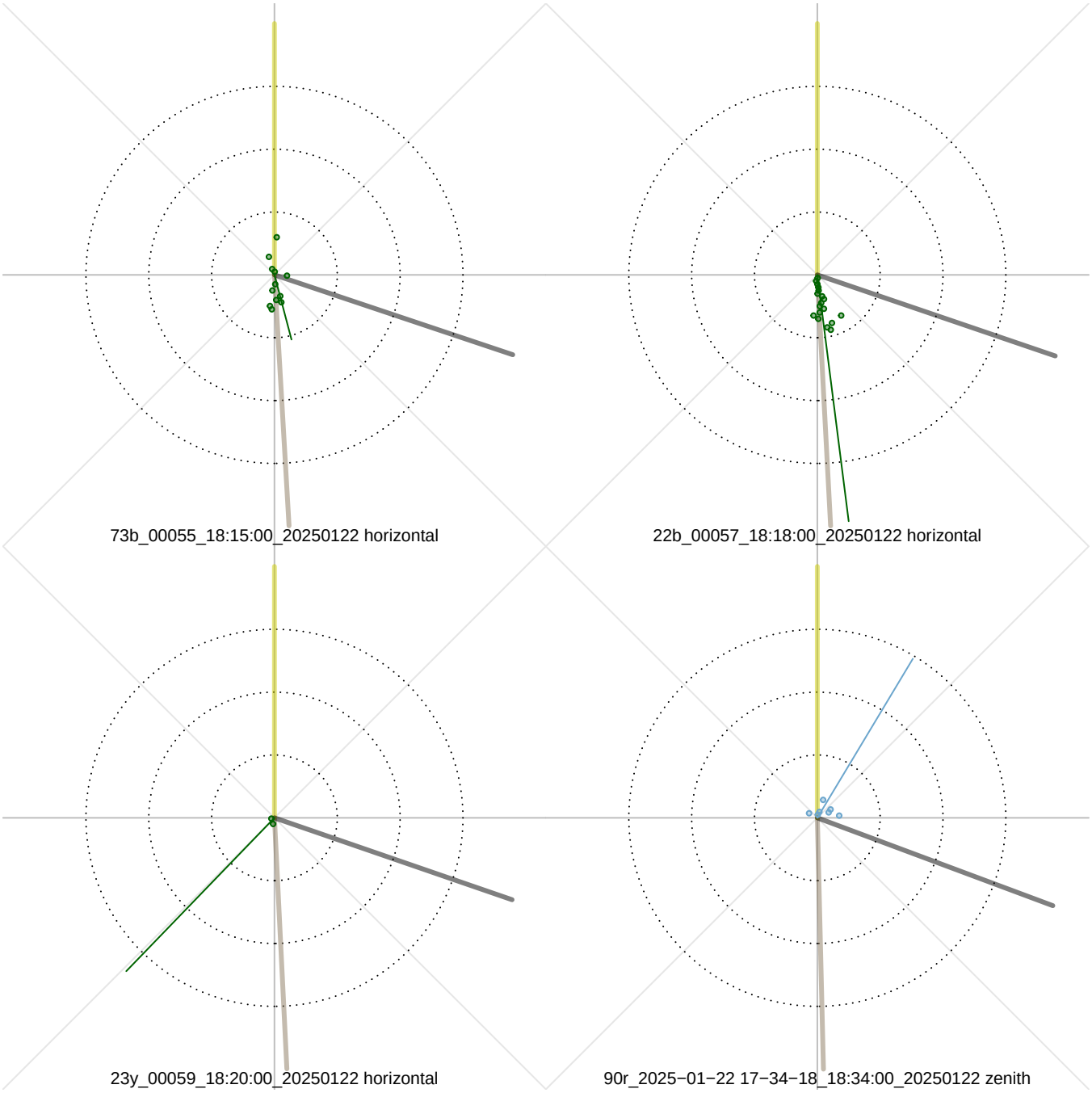


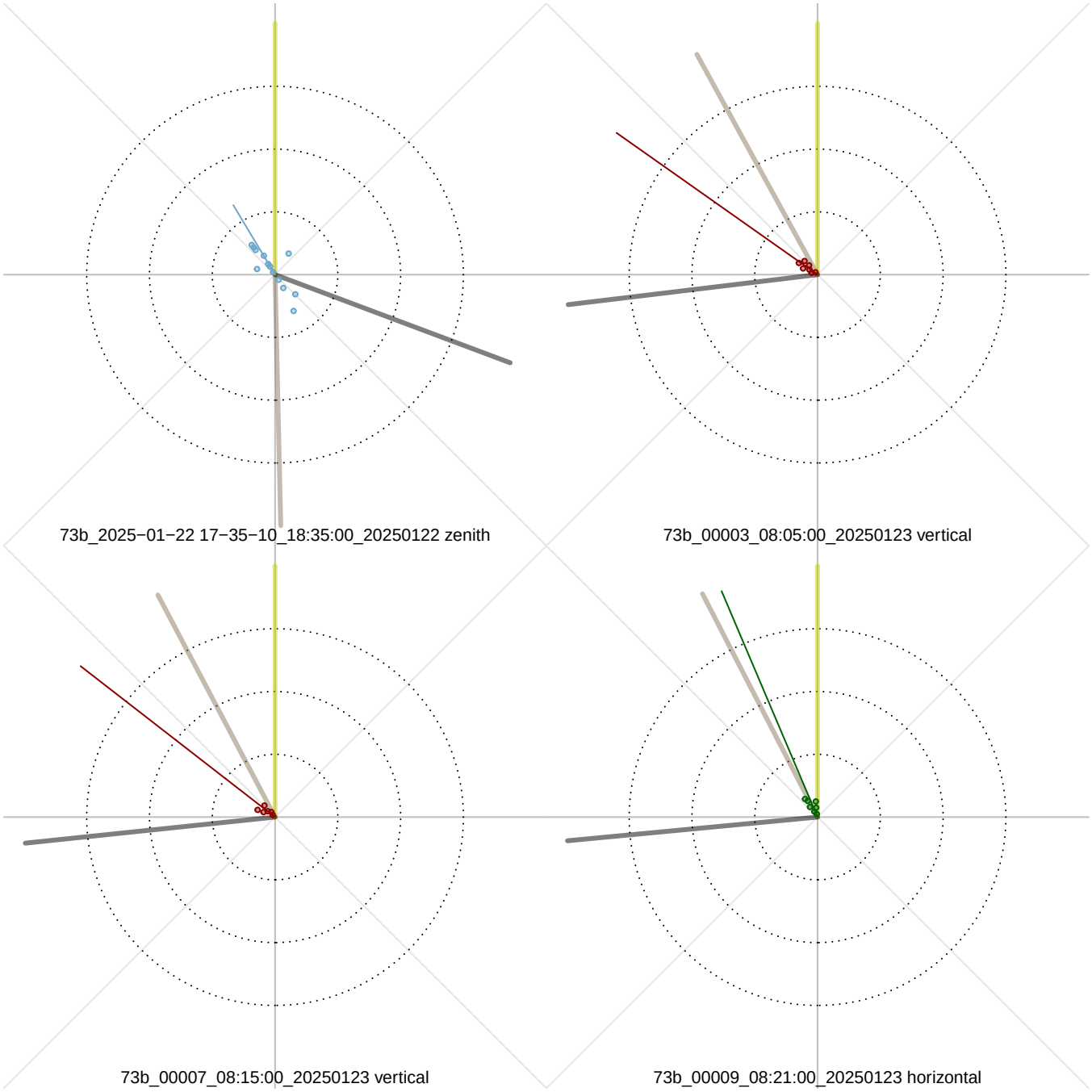
73b_2025-01-22 15-10-42_16:10:00_20250122 antisolar

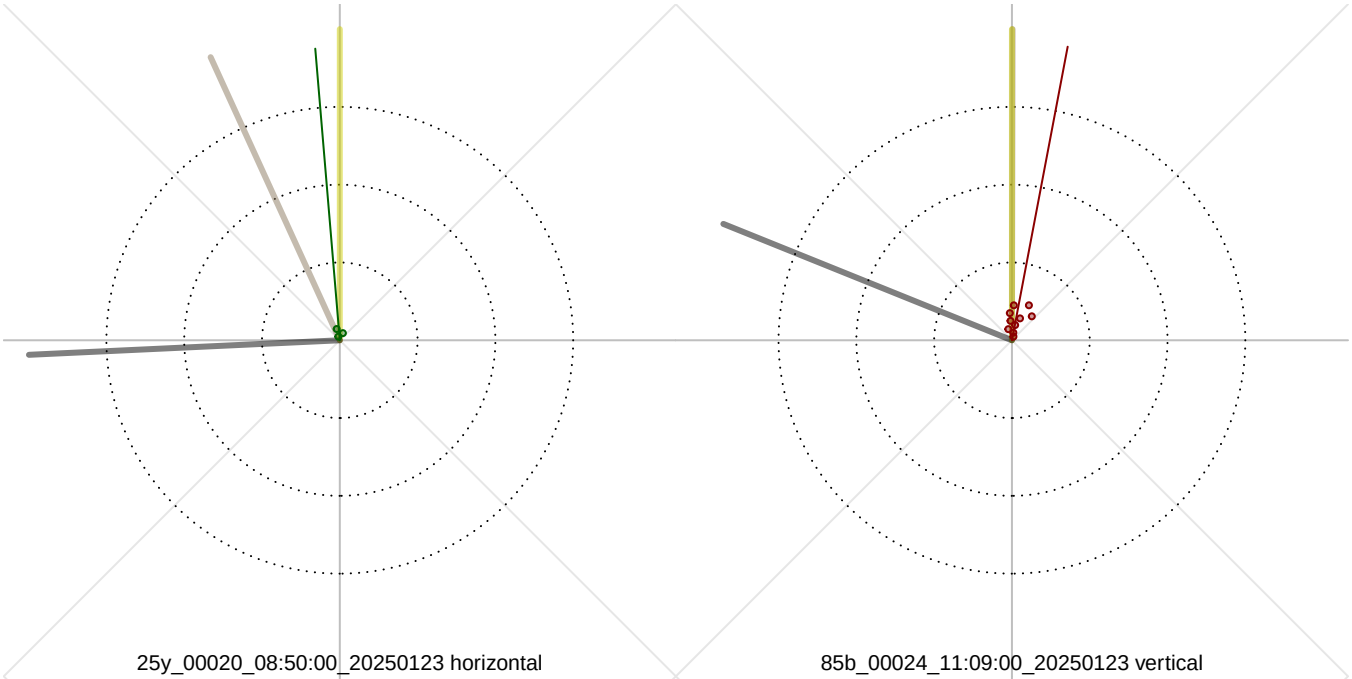


22b_2025-01-22 15-21-33_16:20:00_20250122 solar


26y_2025−01−22 15−21−33_16:20:00_20250122 solar
22b_00048_17:55:00_20250122 vertical
66b_00049_17:59:00_20250122 vertical
81r_00053_18:12:00_20250122 horizontal

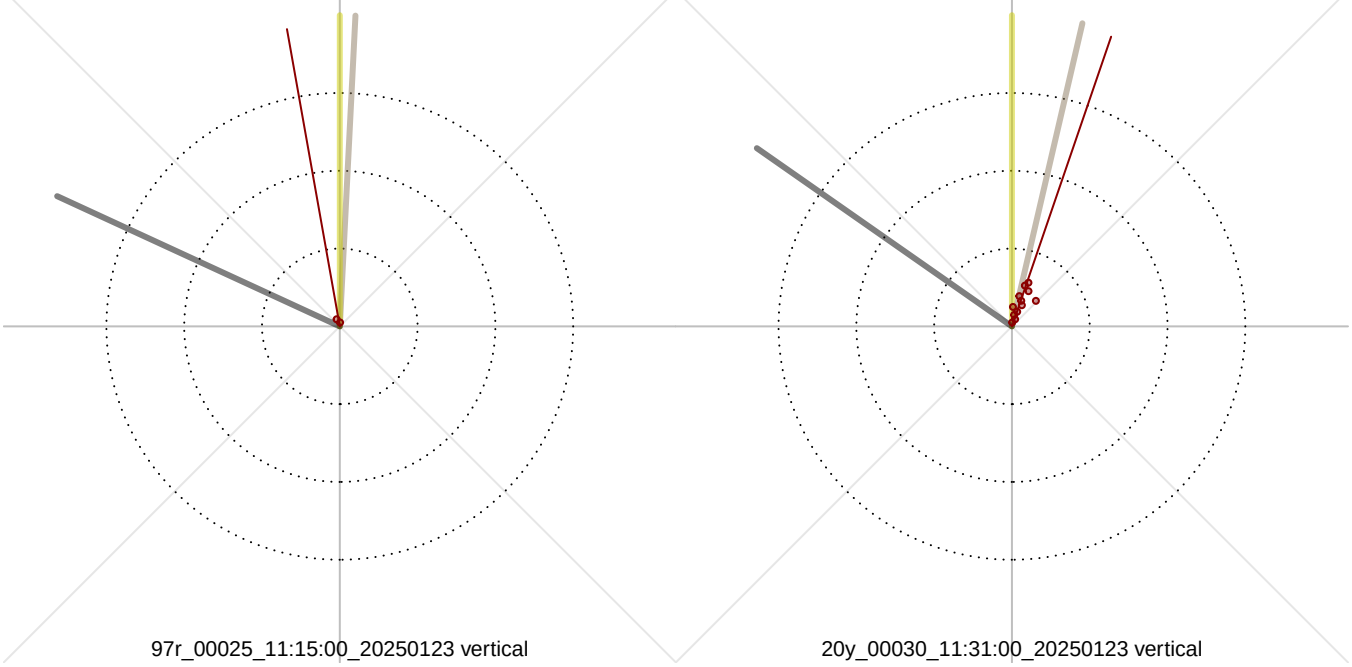

73b_00055_18:15:00_20250122 horizontal
22b_00057_18:18:00_20250122 horizontal
23y_00059_18:20:00_20250122 horizontal
90r_2025−01−22 17−34−18_18:34:00_20250122 zenith


73b_2025−01−22 17−35−10_18:35:00_20250122 zenith
73b_00003_08:05:00_20250123 vertical
73b_00007_08:15:00_20250123 vertical
73b_00009_08:21:00_20250123 horizontal



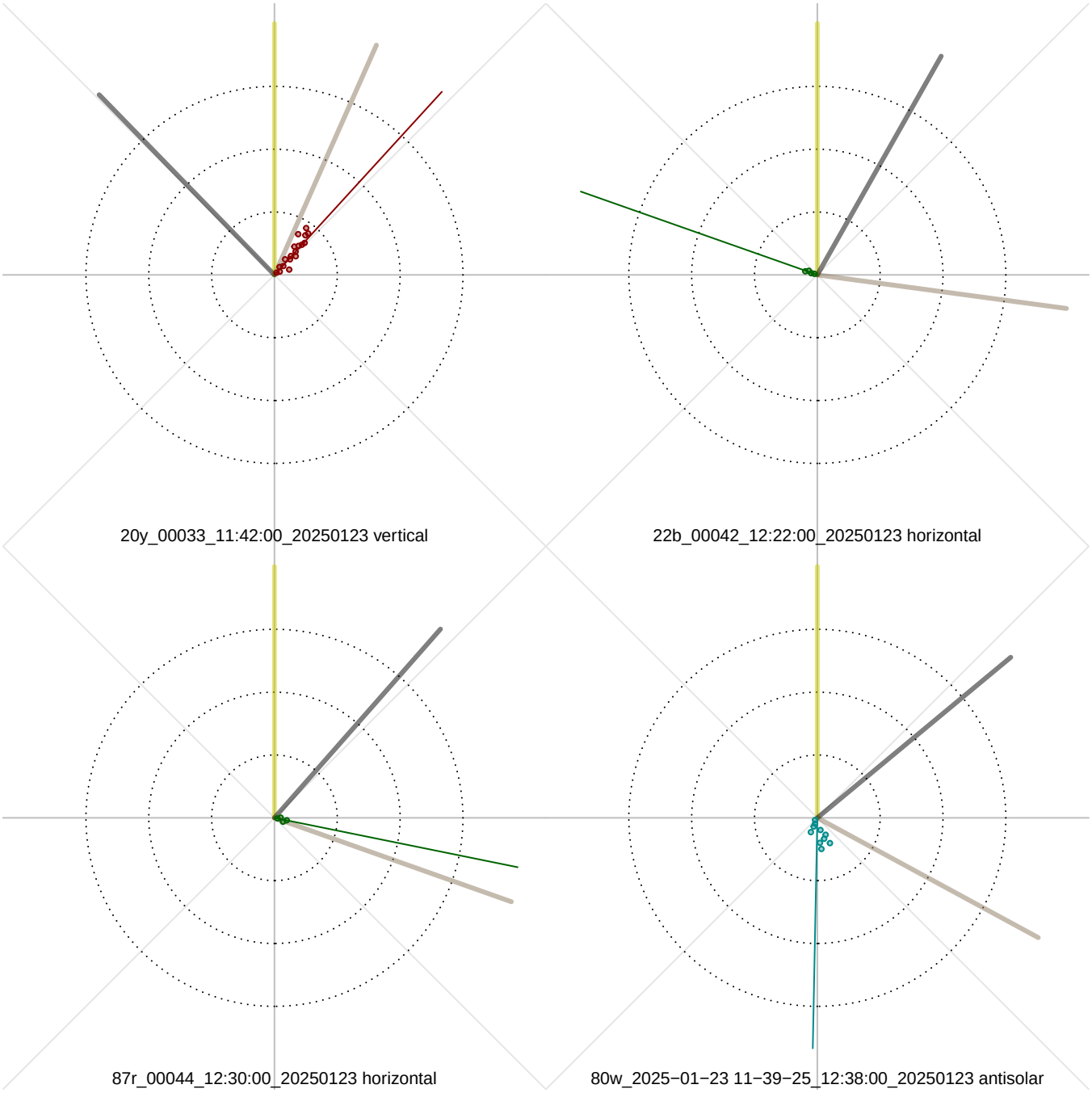
25y_00020_08:50:00_20250123 horizontal

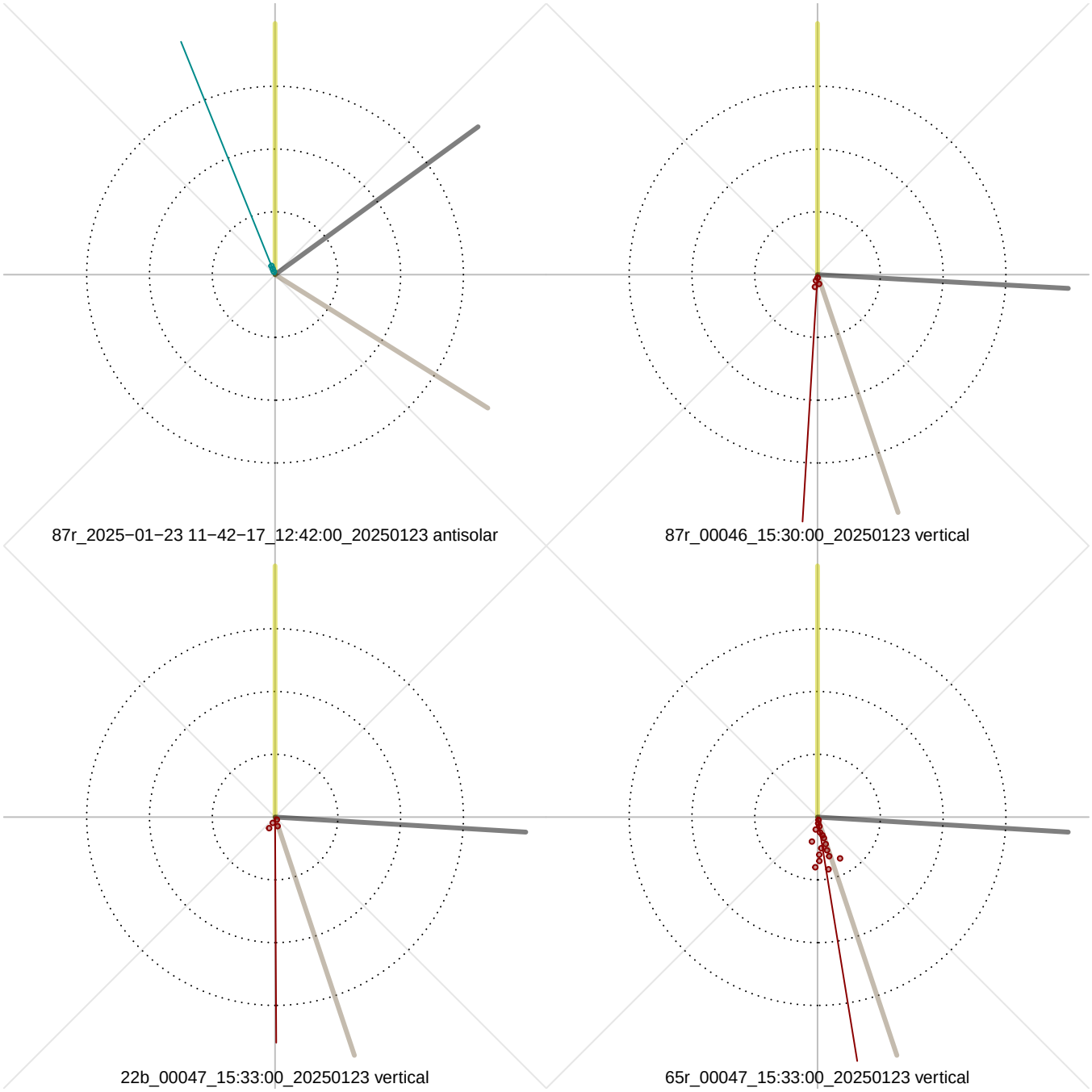
85b_00024_11:09:00_20250123 vertical

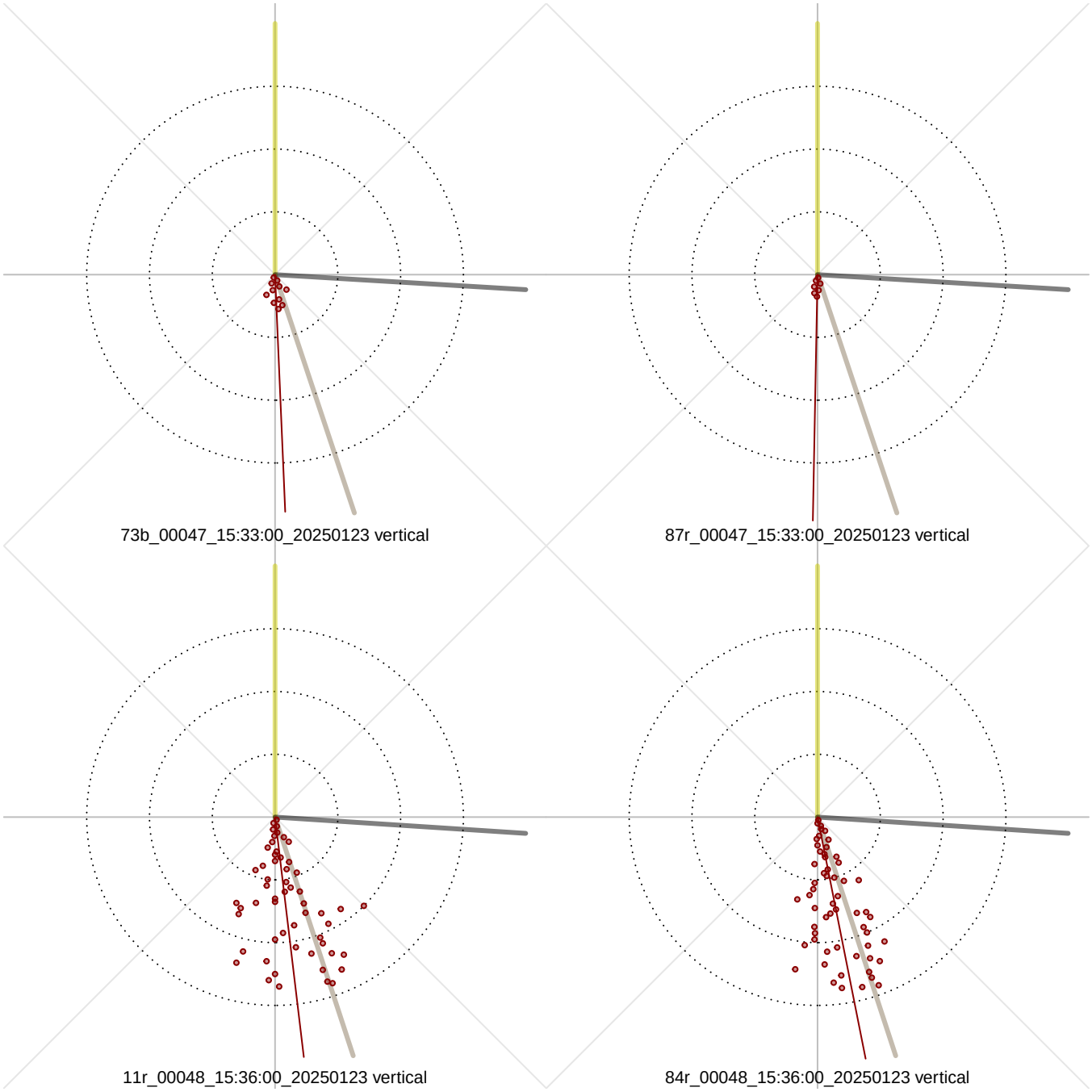


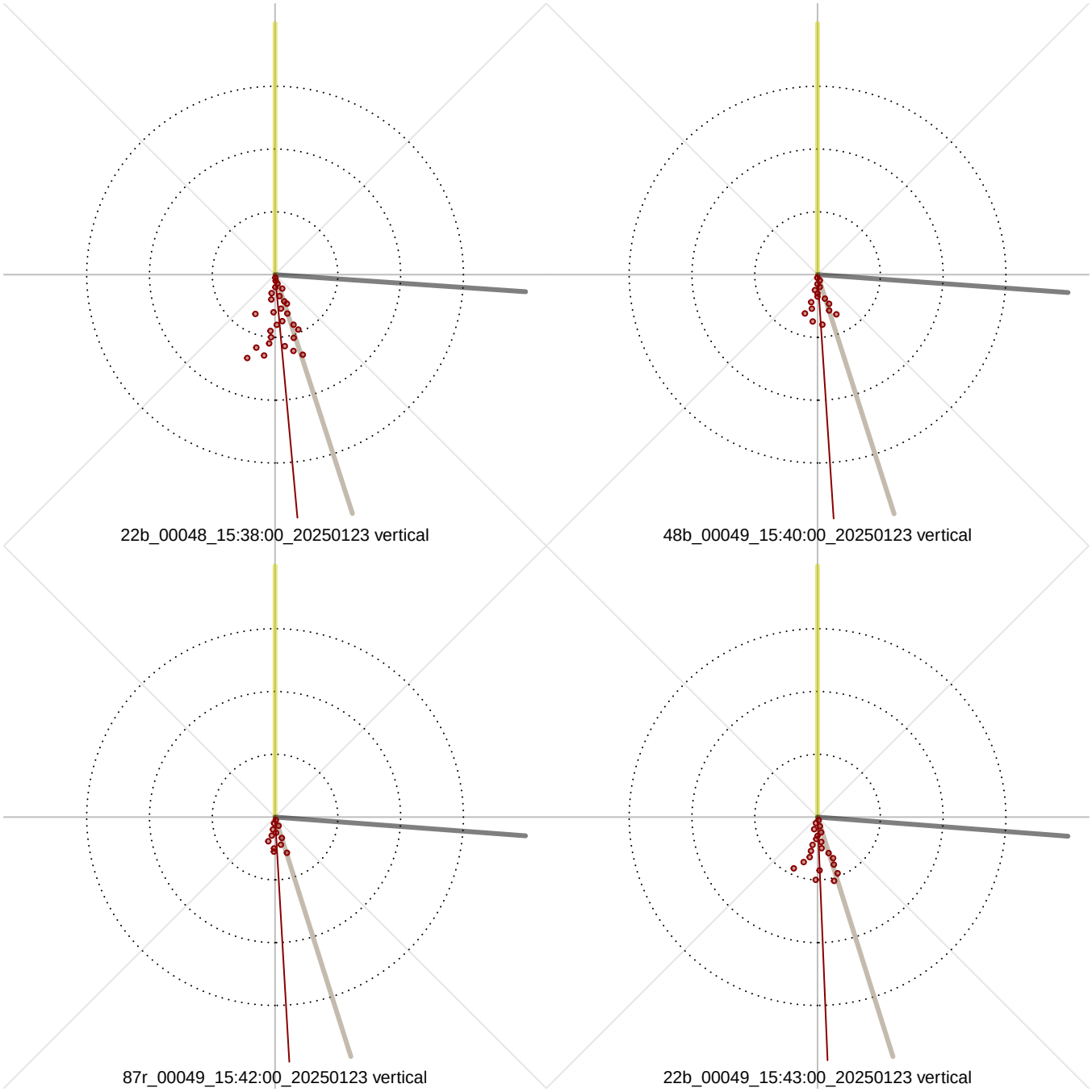
97r_00025_11:15:00_20250123 vertical

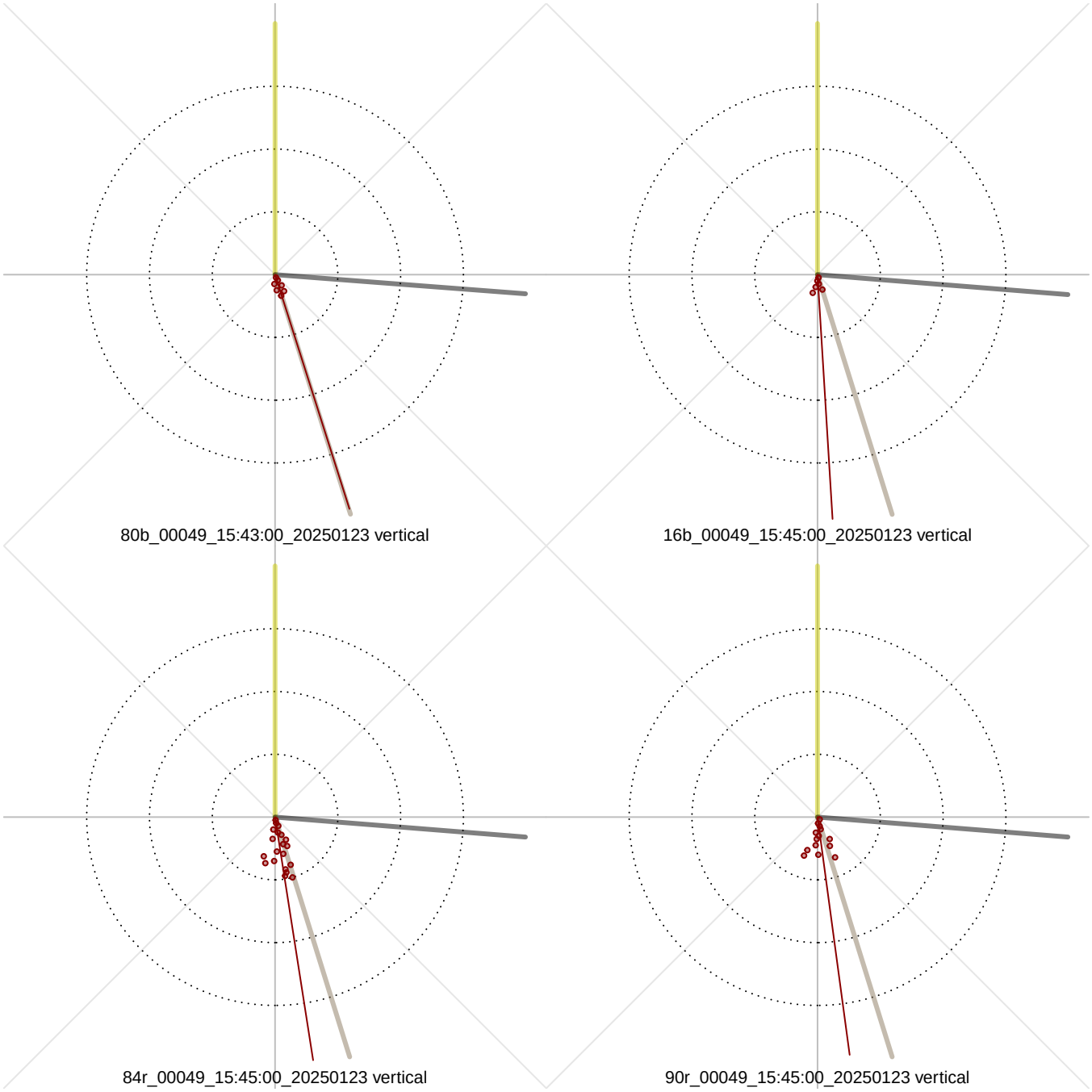
20y_00030_11:31:00_20250123 vertical

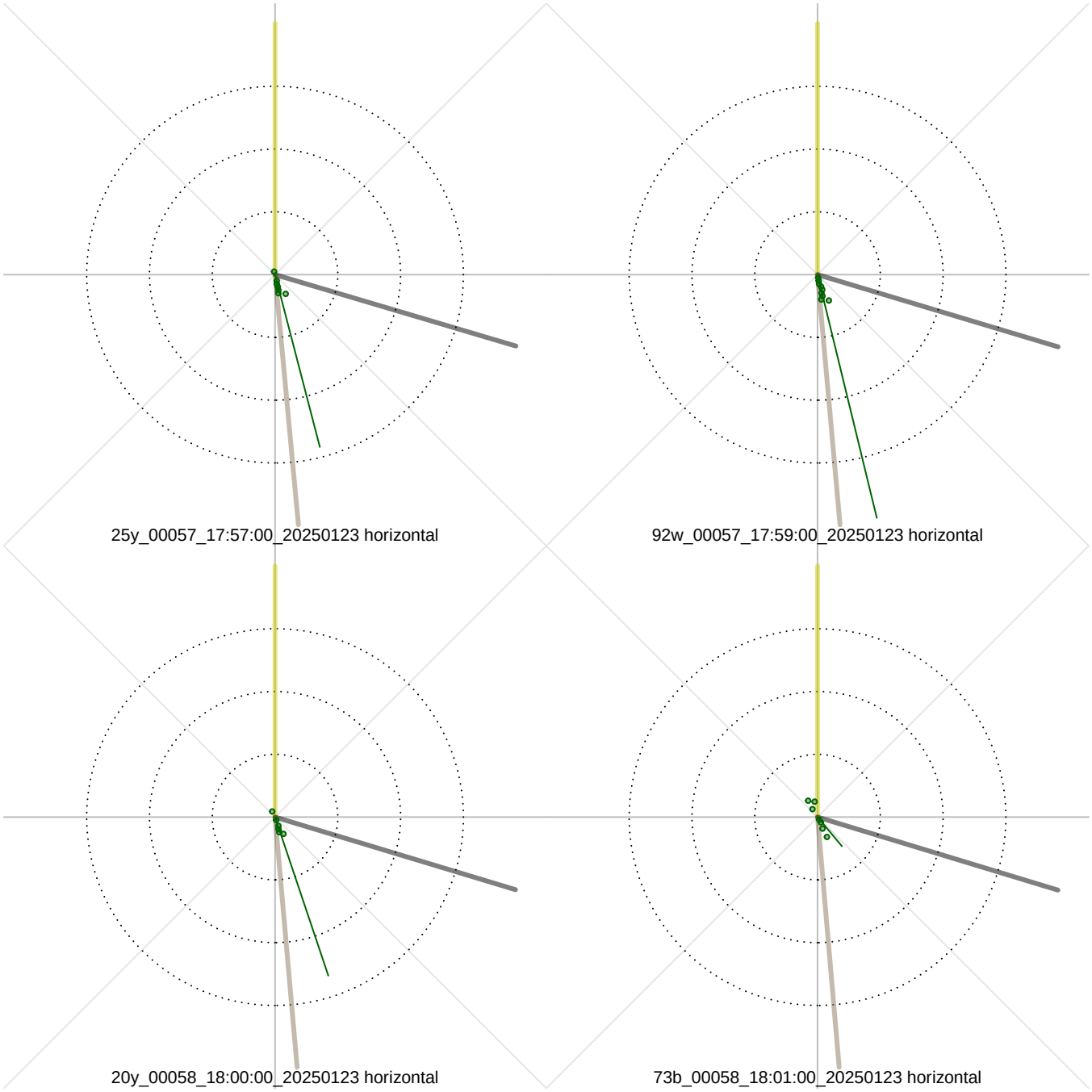

20y_00033_11:42:00_20250123 vertical
22b_00042_12:22:00_20250123 horizontal
87r_00044_12:30:00_20250123 horizontal
80w_2025−01−23 11−39−25_12:38:00_20250123 antisolar

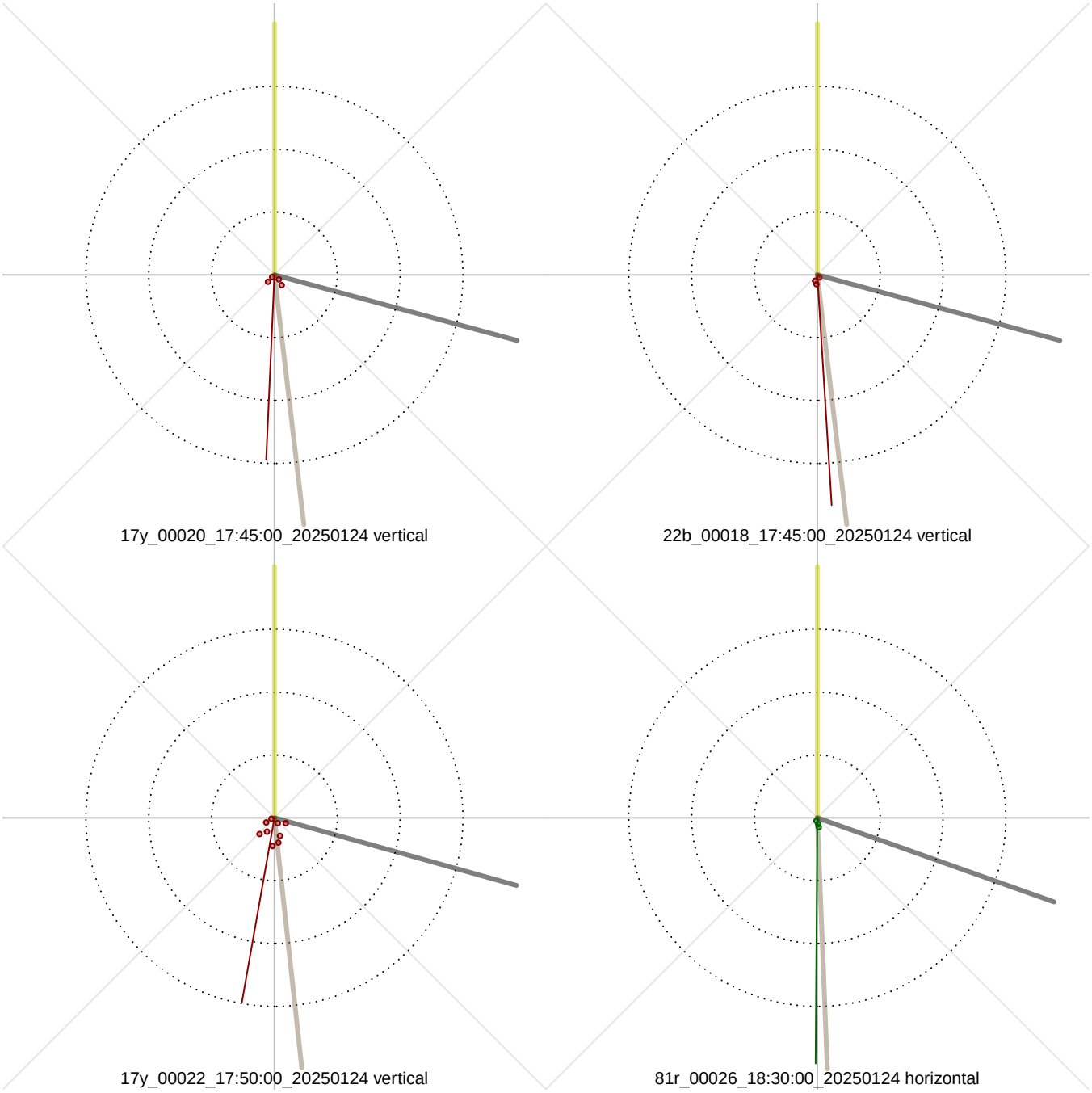

87r_2025−01−23 11−42−17_12:42:00_20250123 antisolar
87r_00046_15:30:00_20250123 vertical
22b_00047_15:33:00_20250123 vertical
65r_00047_15:33:00_20250123 vertical

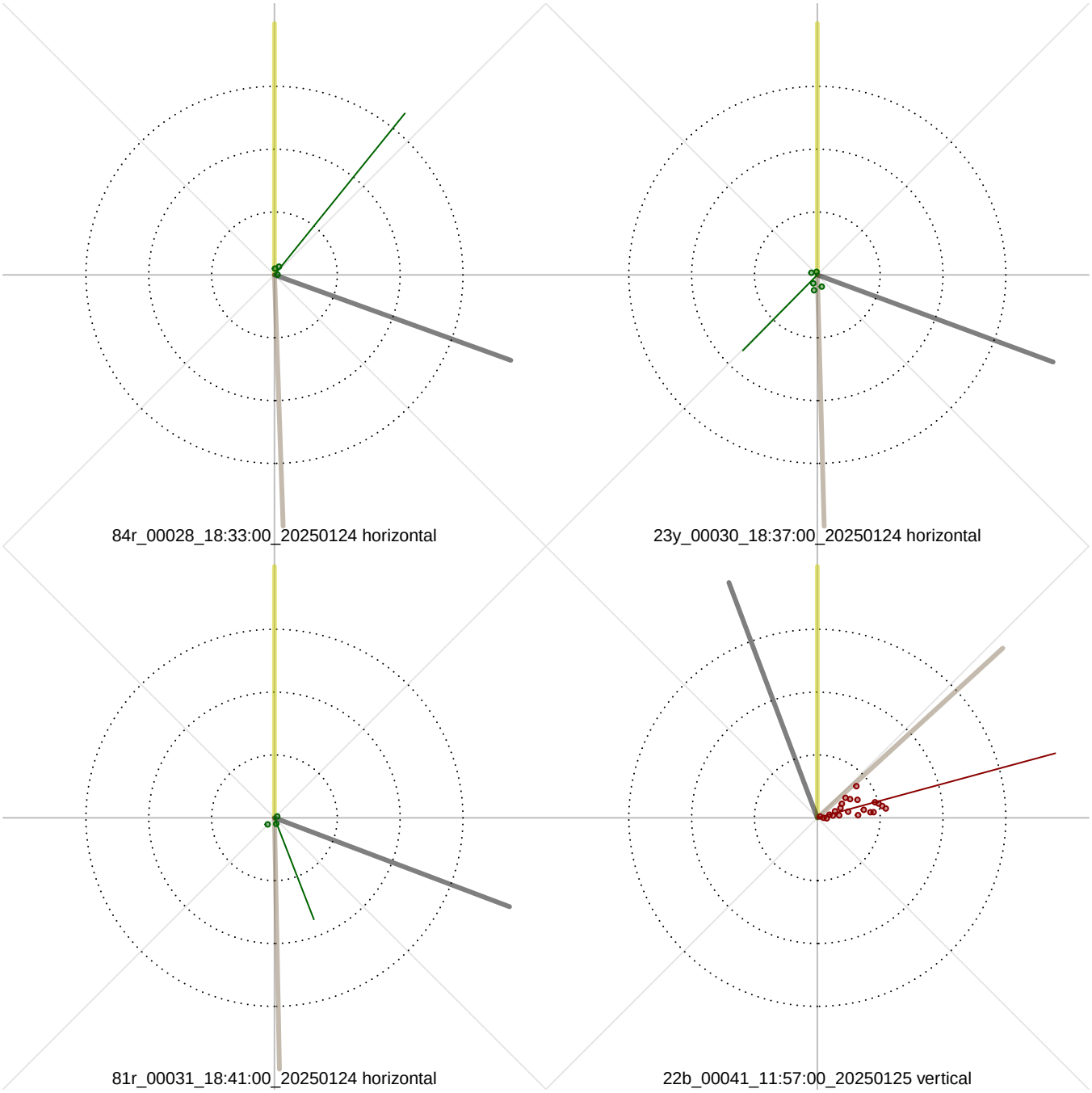

73b_00047_15:33:00_20250123 vertical
87r_00047_15:33:00_20250123 vertical
11r_00048_15:36:00_20250123 vertical
84r_00048_15:36:00_20250123 vertical

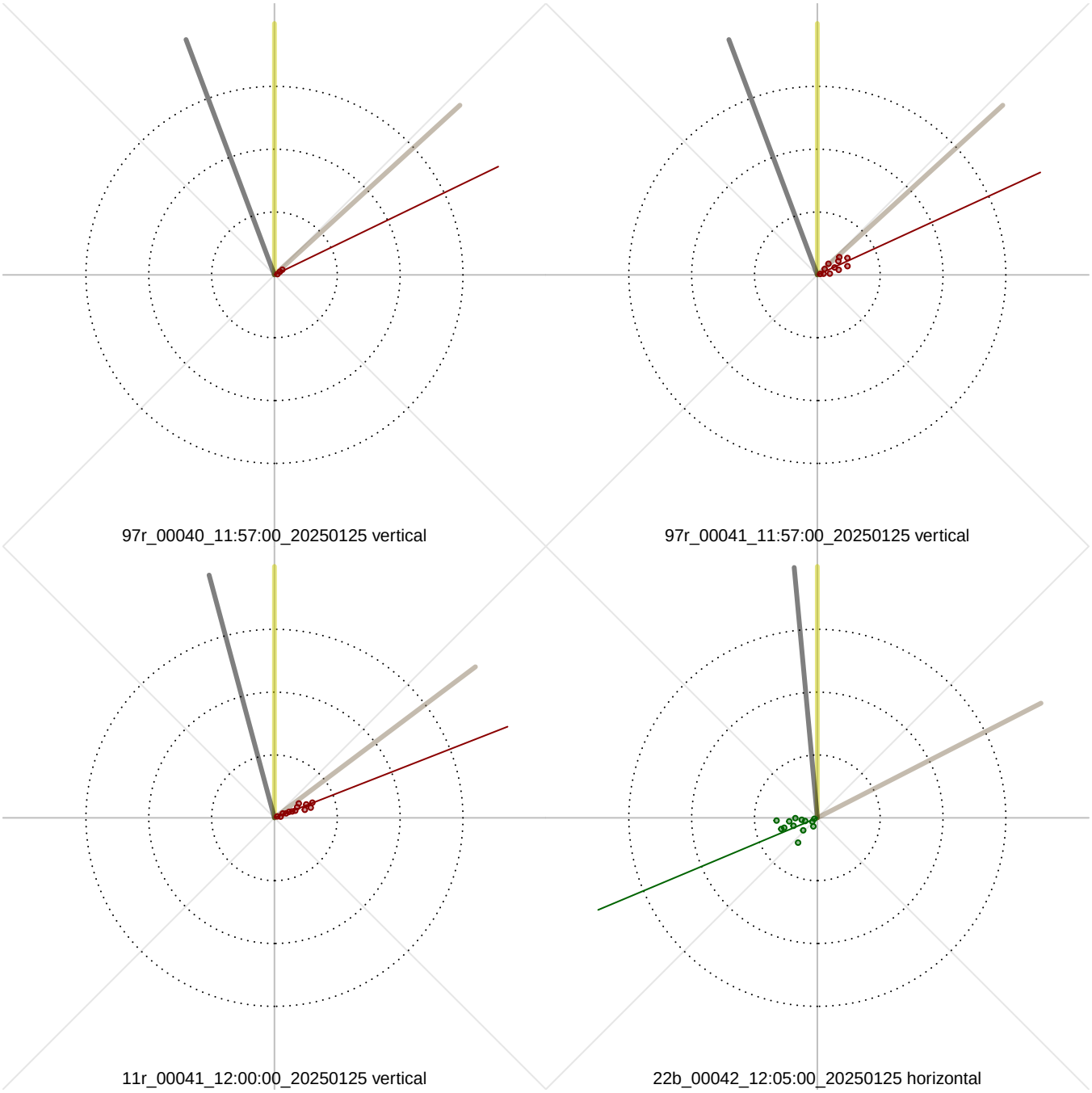

22b_00048_15:38:00_20250123 vertical
48b_00049_15:40:00_20250123 vertical
87r_00049_15:42:00_20250123 vertical
22b_00049_15:43:00_20250123 vertical

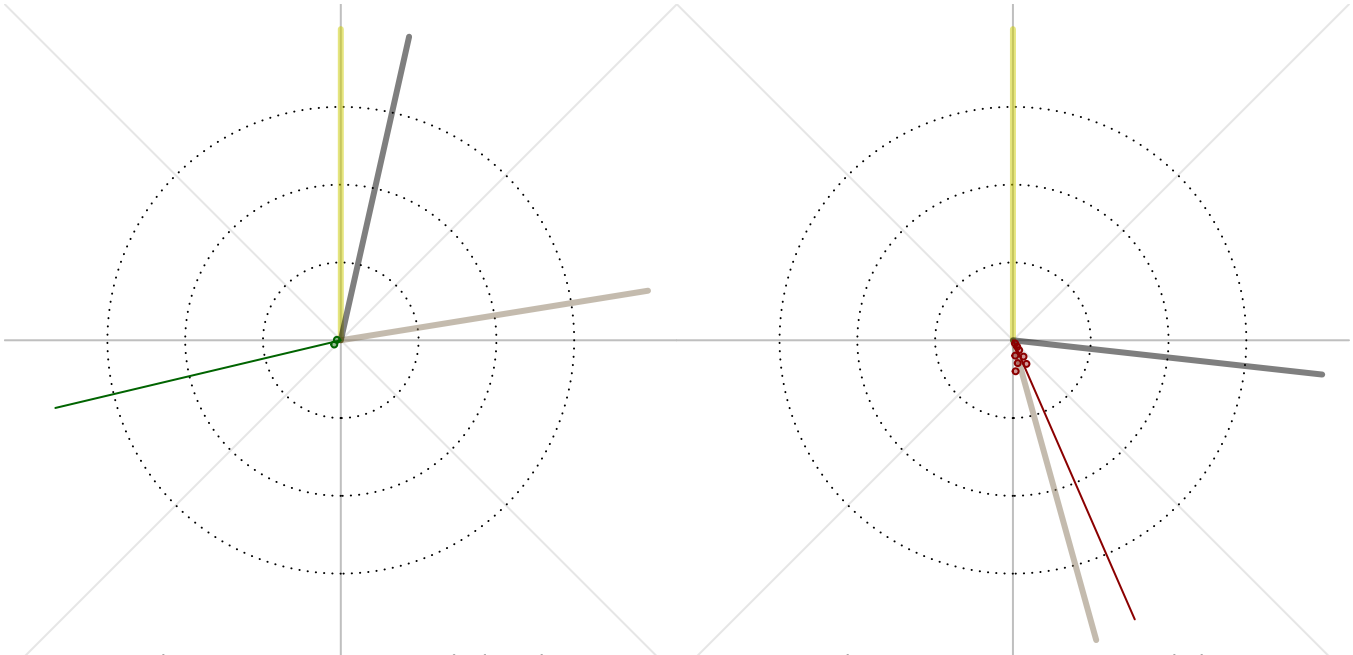

80b_00049_15:43:00_20250123 vertical
16b_00049_15:45:00_20250123 vertical
84r_00049_15:45:00_20250123 vertical
90r_00049_15:45:00_20250123 vertical


25y_00057_17:57:00_20250123 horizontal
92w_00057_17:59:00_20250123 horizontal
20y_00058_18:00:00_20250123 horizontal
73b_00058_18:01:00_20250123 horizontal


17y_00020_17:45:00_20250124 vertical
22b_00018_17:45:00_20250124 vertical
17y_00022_17:50:00_20250124 vertical
81r_00026_18:30:00_20250124 horizontal

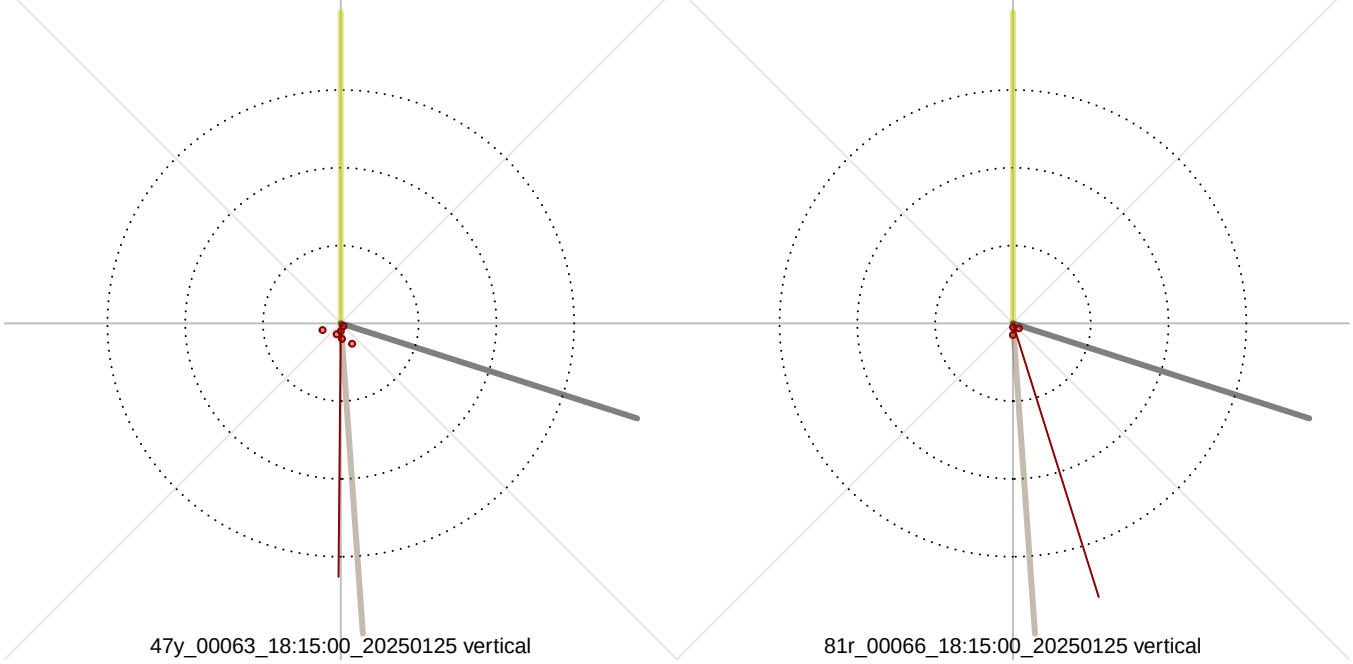

84r_00028_18:33:00_20250124 horizontal
23y_00030_18:37:00_20250124 horizontal
81r_00031_18:41:00_20250124 horizontal
22b_00041_11:57:00_20250125 vertical


97r_00040_11:57:00_20250125 vertical
97r_00041_11:57:00_20250125 vertical
11r_00041_12:00:00_20250125 vertical
22b_00042_12:05:00_20250125 horizontal



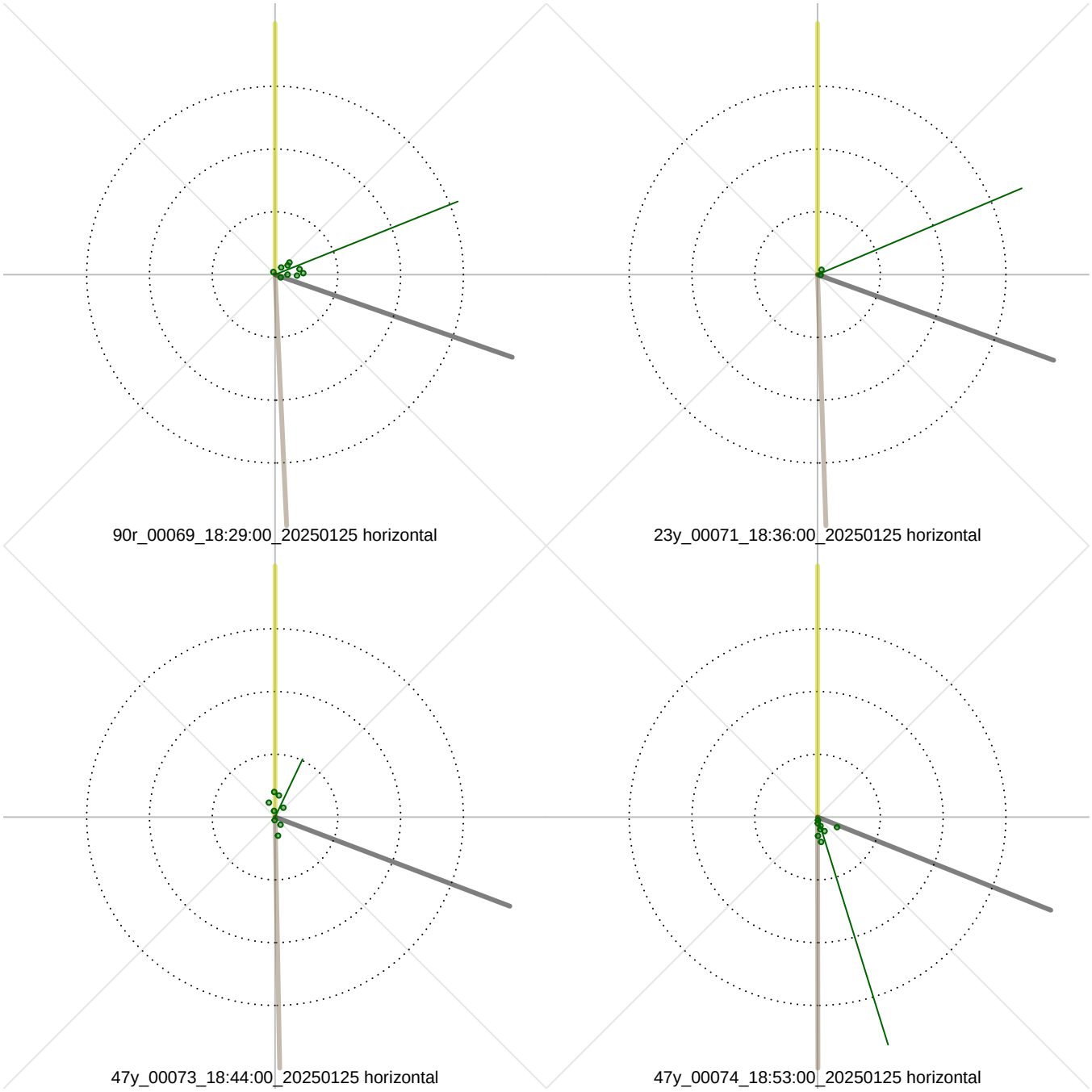
22b_00046_12:14:00_20250125 horizontal

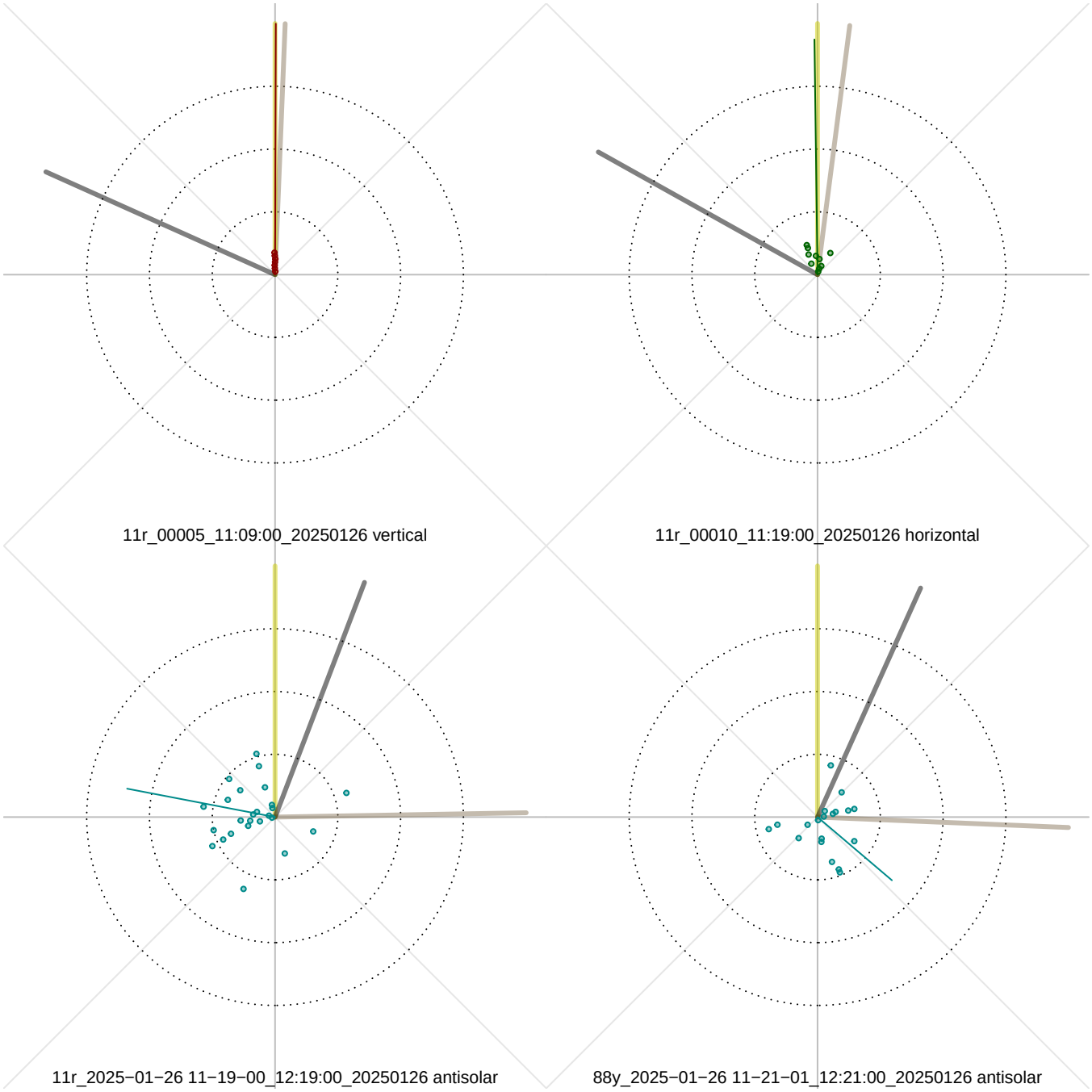
88b_00053_16:11:00_20250125 vertical

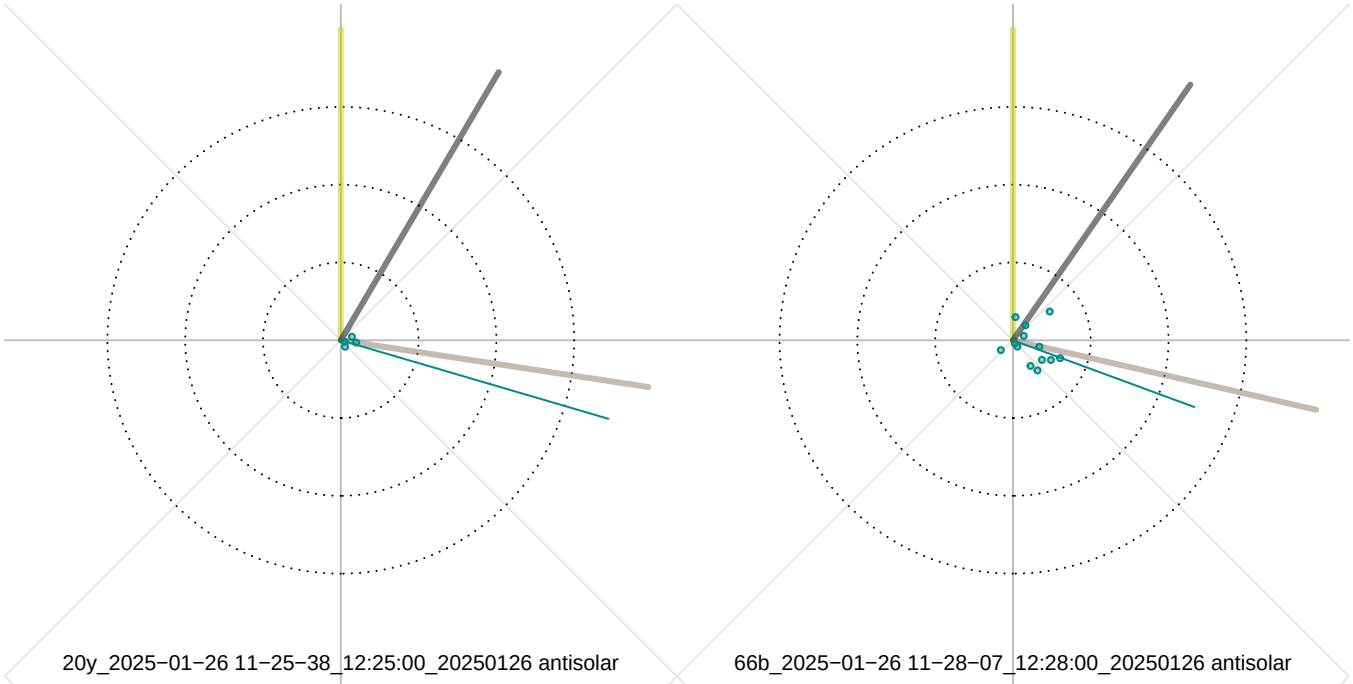


47y_00063_18:15:00_20250125 vertical

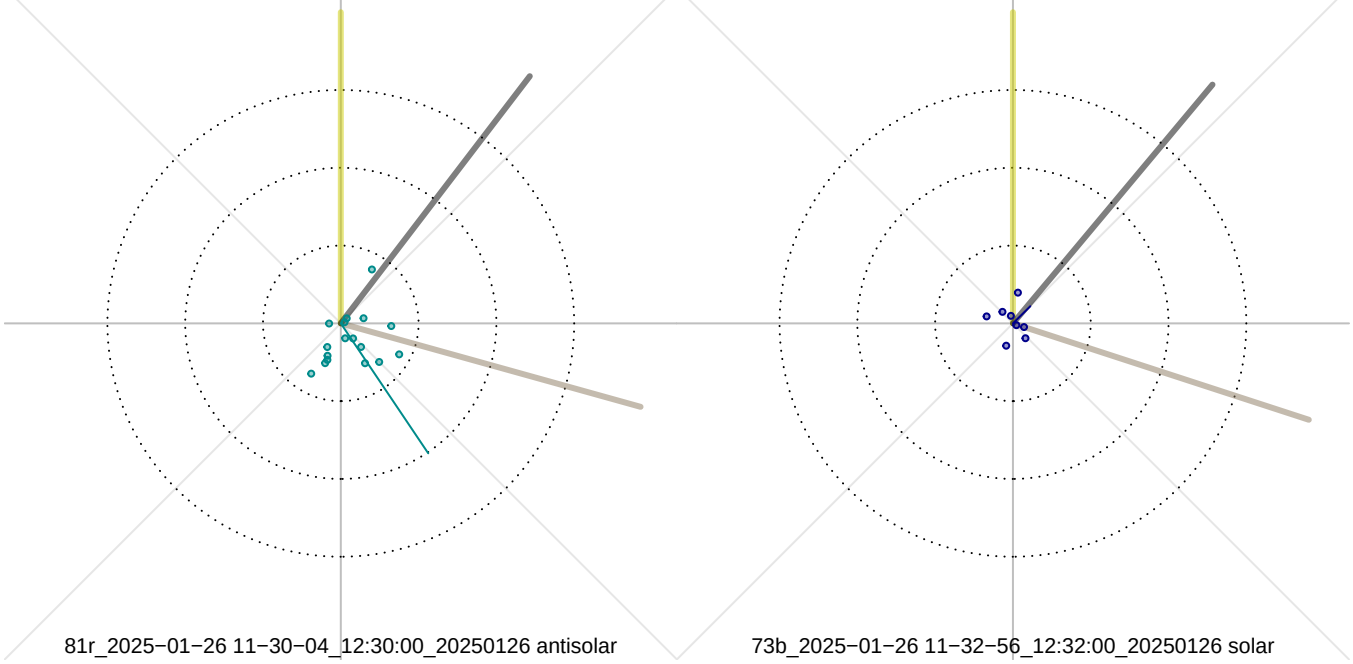
81r_00066_18:15:00_20250125 vertical


90r_00069_18:29:00_20250125 horizontal
23y_00071_18:36:00_20250125 horizontal
47y_00073_18:44:00_20250125 horizontal
47y_00074_18:53:00_20250125 horizontal


11r_00005_11:09:00_20250126 vertical
11r_00010_11:19:00_20250126 horizontal
11r_2025−01−26 11−19−00_12:19:00_20250126 antisolar
88y_2025−01−26 11−21−01_12:21:00_20250126 antisolar



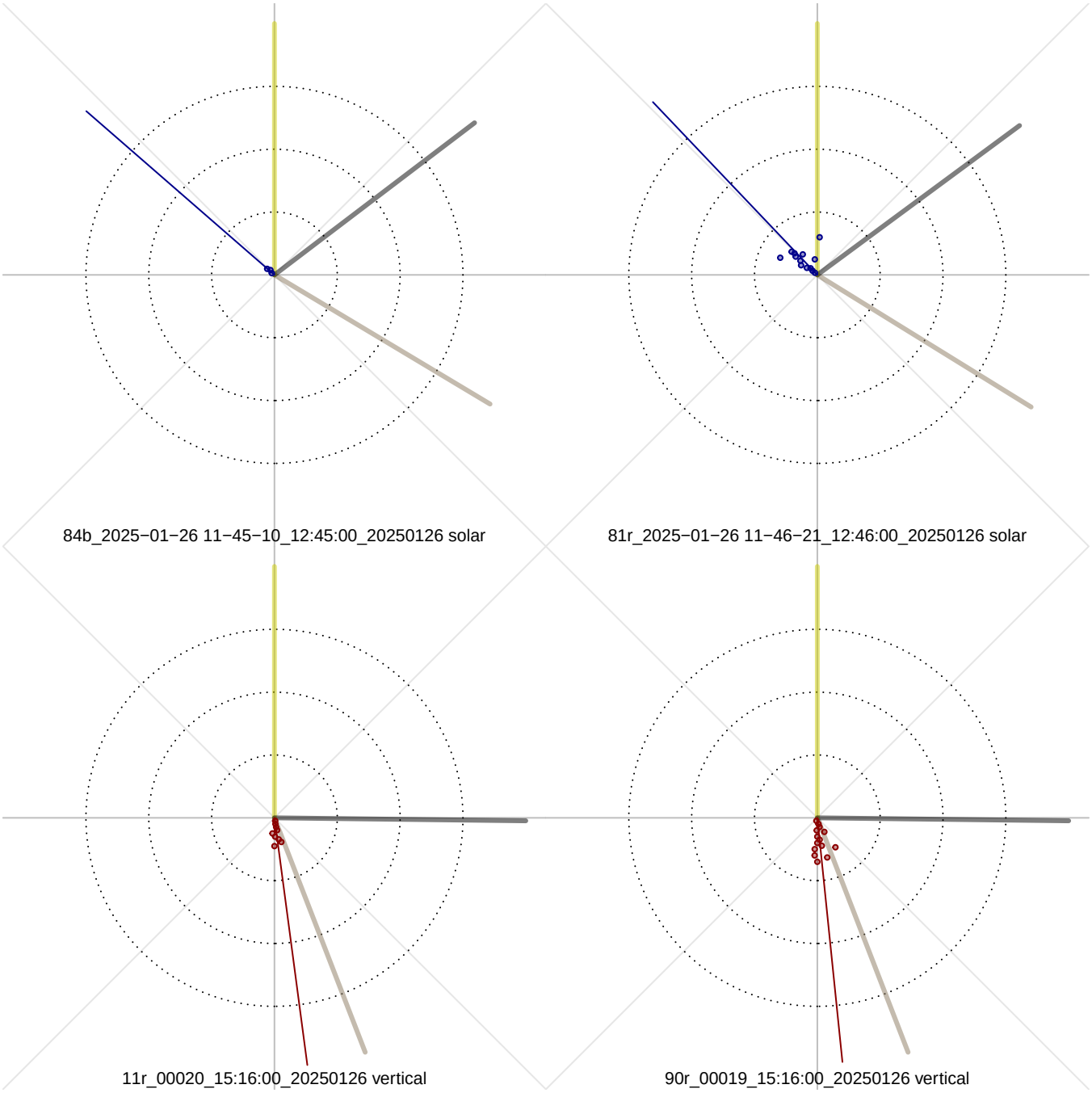
20y_2025−01−26 11−25−38_12:25:00_20250126 antisolar

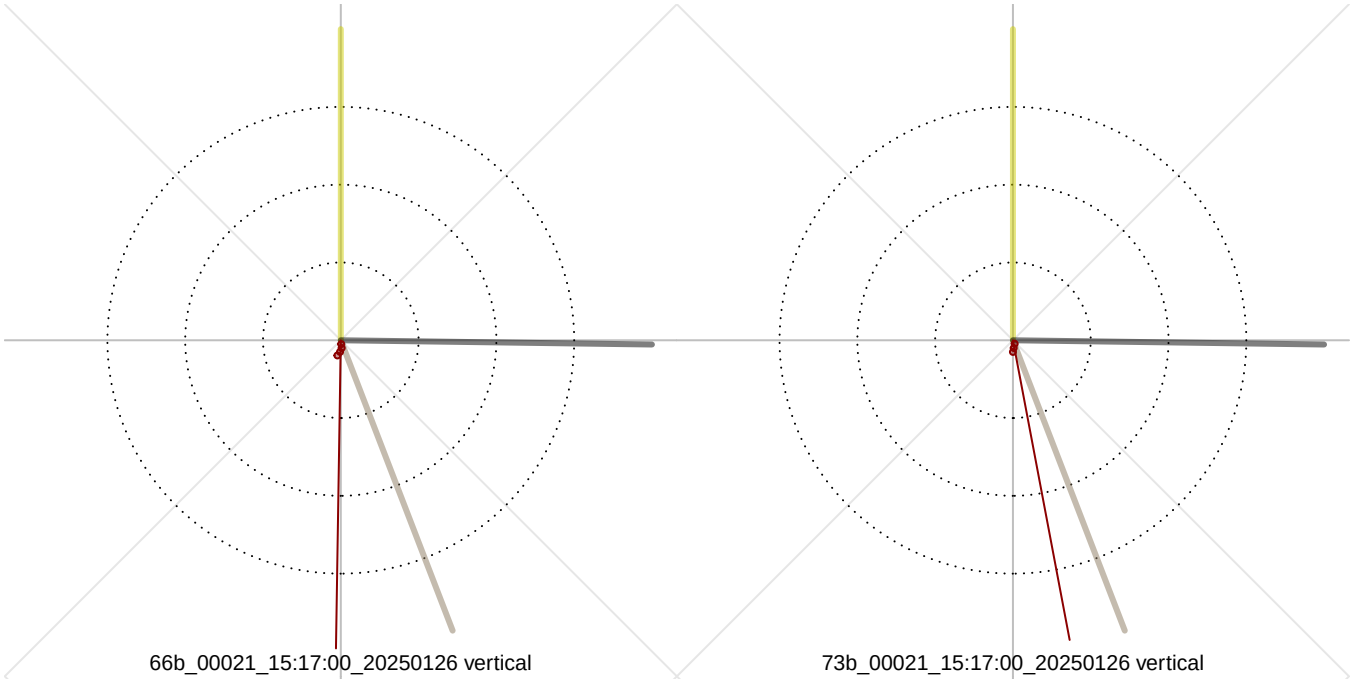


81r_2025−01−26 11−30−04_12:30:00_20250126 antisolar

66b_2025−01−26 11−28−07_12:28:00_20250126 antisolar

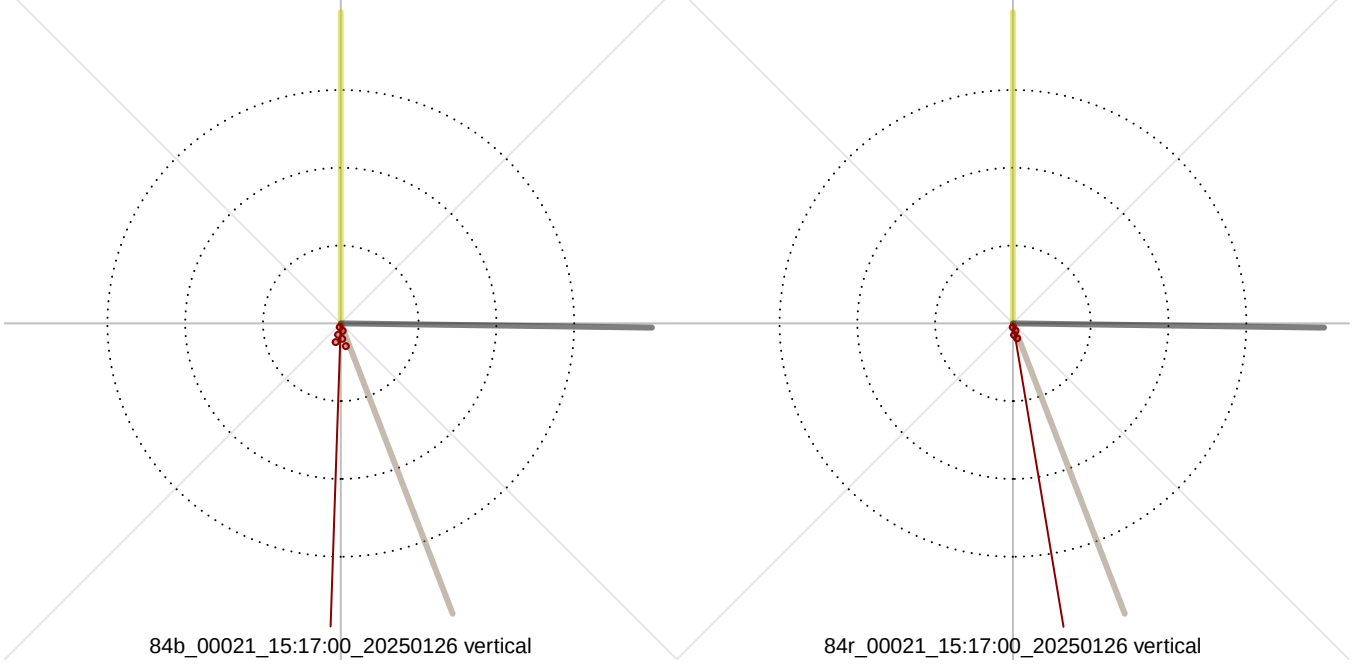
73b_2025−01−26 11−32−56_12:32:00_20250126 solar


84b_2025−01−26 11−45−10_12:45:00_20250126 solar
81r_2025−01−26 11−46−21_12:46:00_20250126 solar
11r_00020_15:16:00_20250126 vertical
90r_00019_15:16:00_20250126 vertical



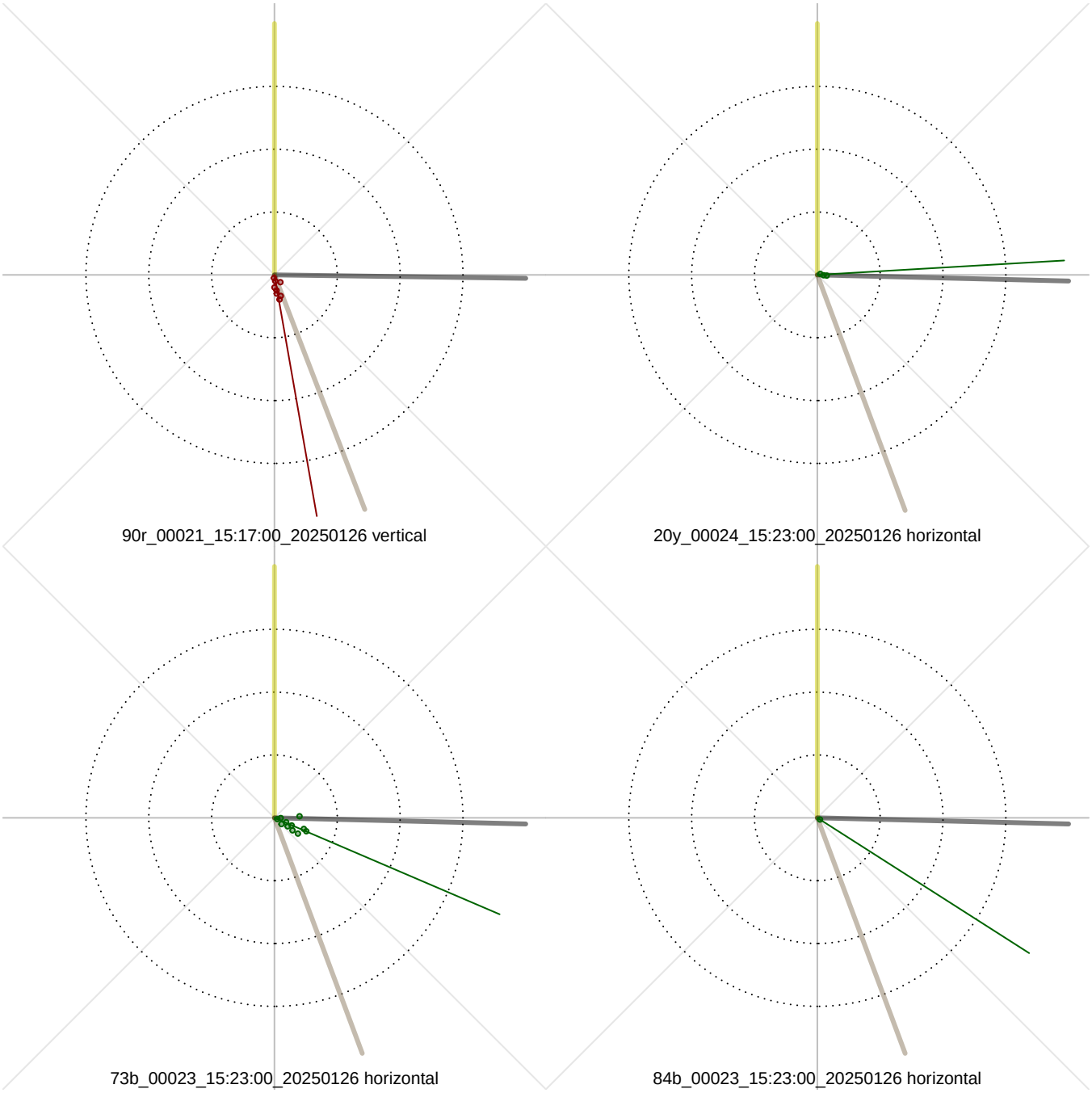
66b_00021_15:17:00_20250126 vertical

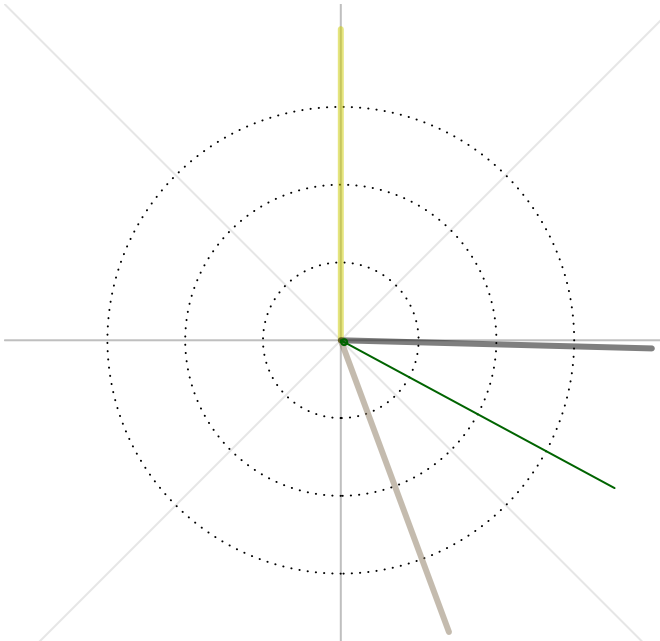
73b_00021_15:17:00_20250126 vertical



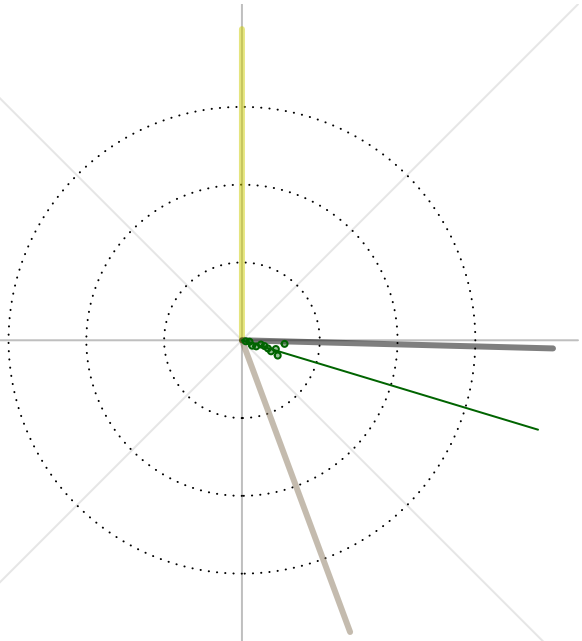
84b_00021_15:17:00_20250126 vertical

84r_00021_15:17:00_20250126 vertical

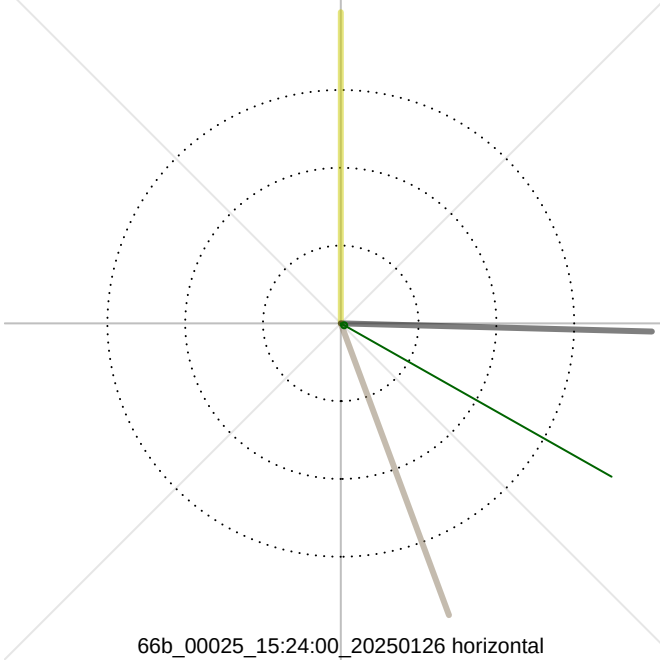

90r_00021_15:17:00_20250126 vertical
20y_00024_15:23:00_20250126 horizontal
73b_00023_15:23:00_20250126 horizontal
84b_00023_15:23:00_20250126 horizontal



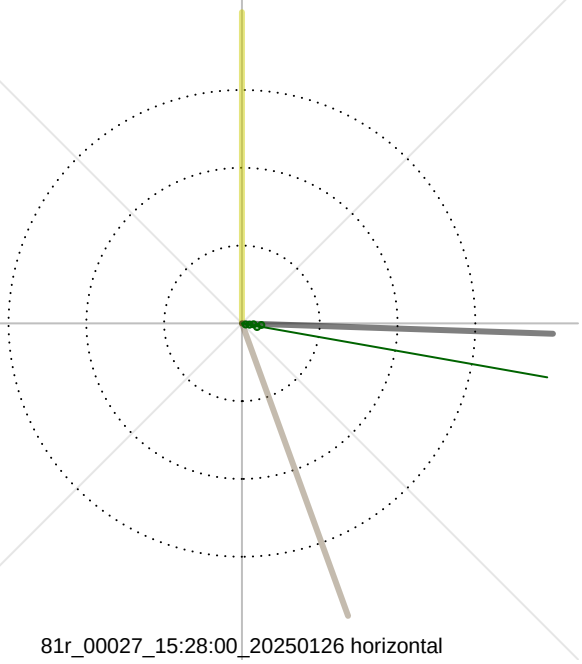
22b_00026_15:24:00_20250126 horizontal



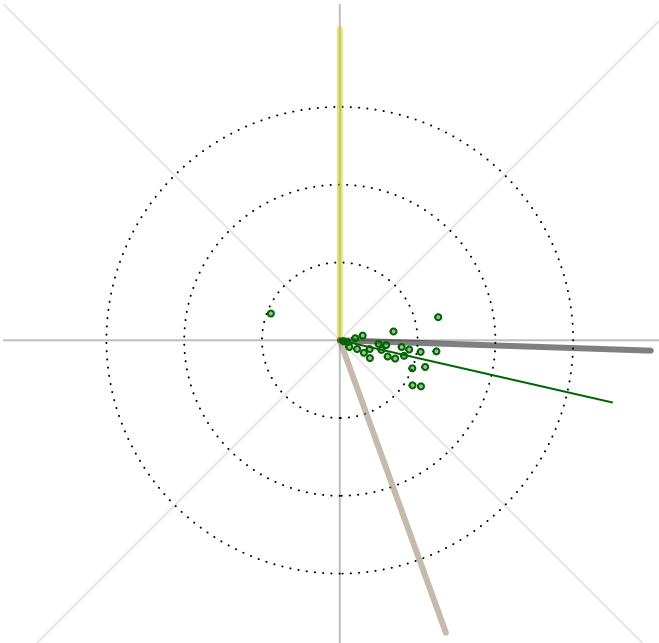
66b_00024_15:24:00_20250126 horizontal



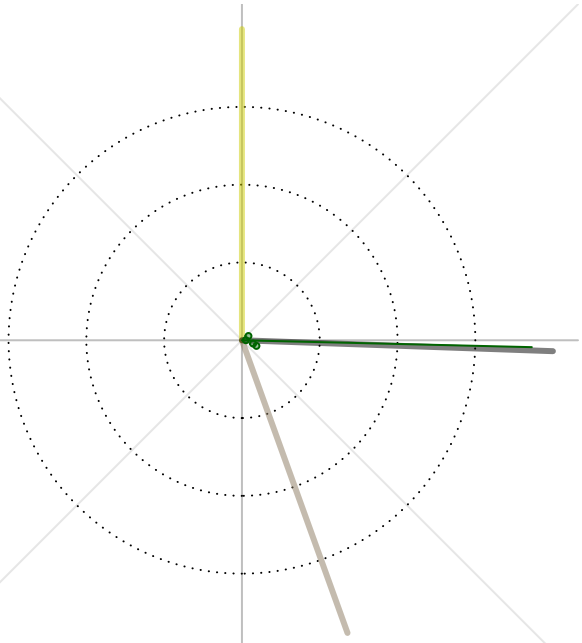
66b_00025_15:24:00_20250126 horizontal



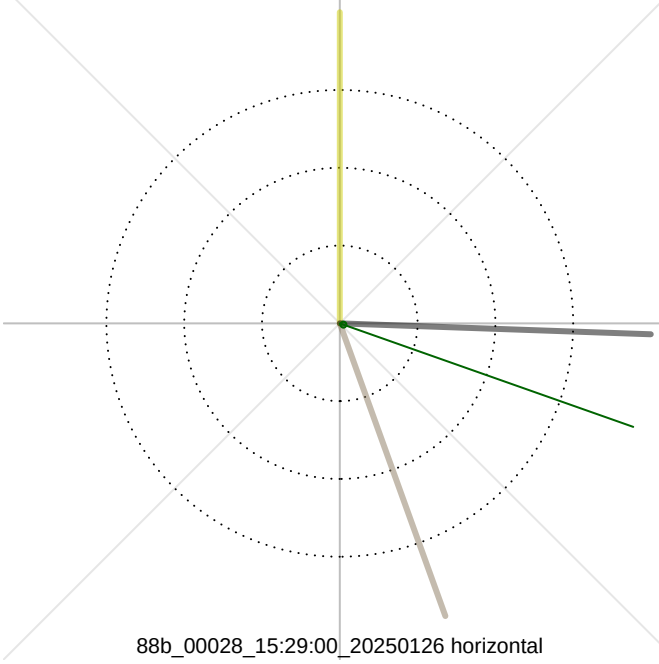
81r_00027_15:28:00_20250126 horizontal



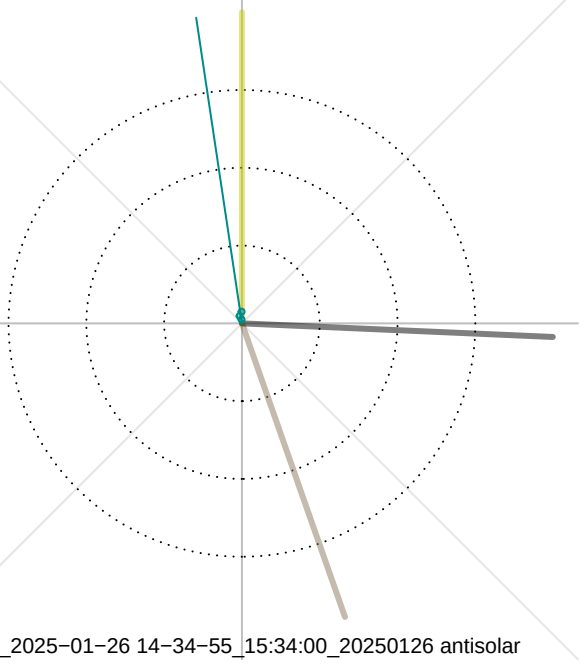
90r_00027_15:28:00_20250126 horizontal



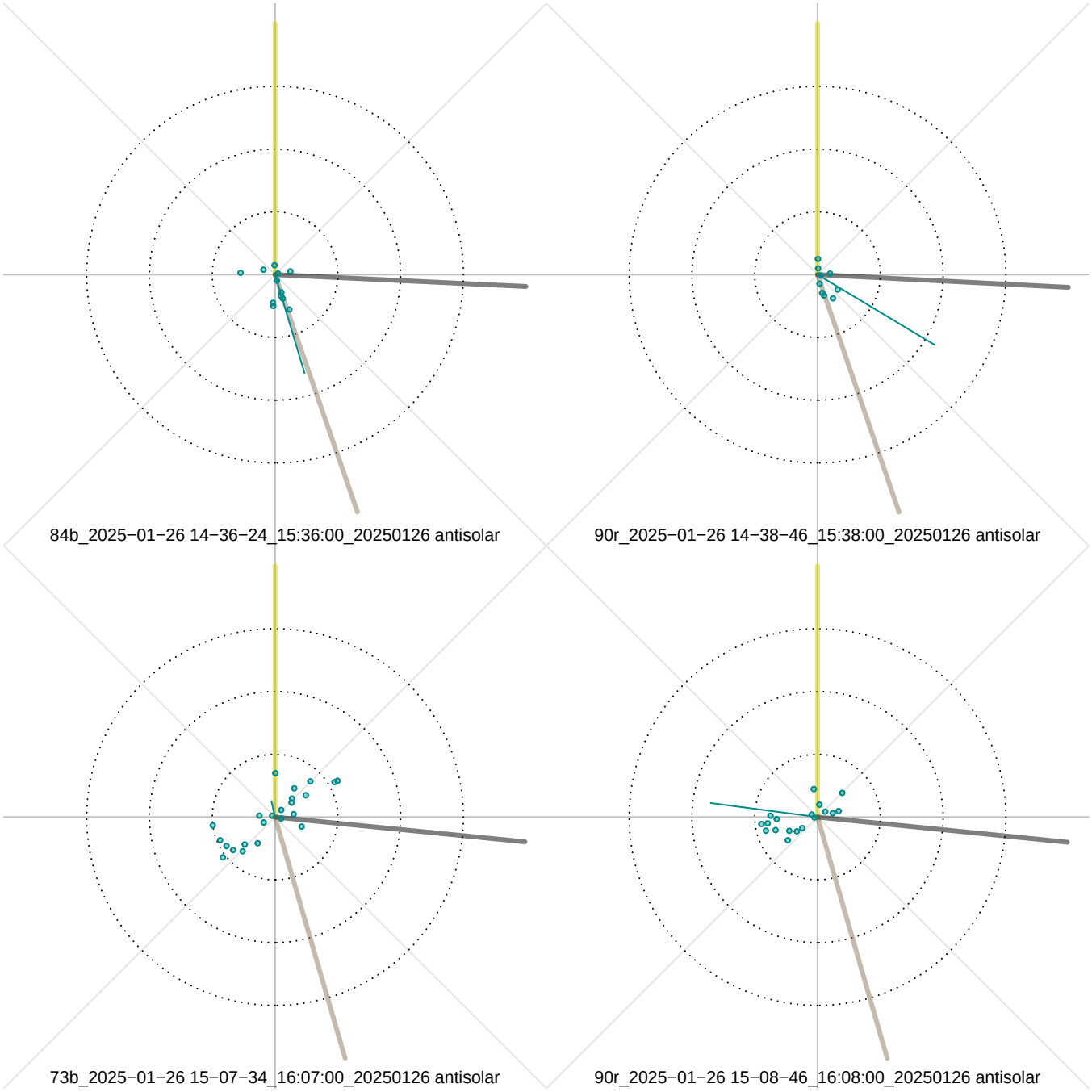
66b_00028_15:29:00_20250126 horizontal

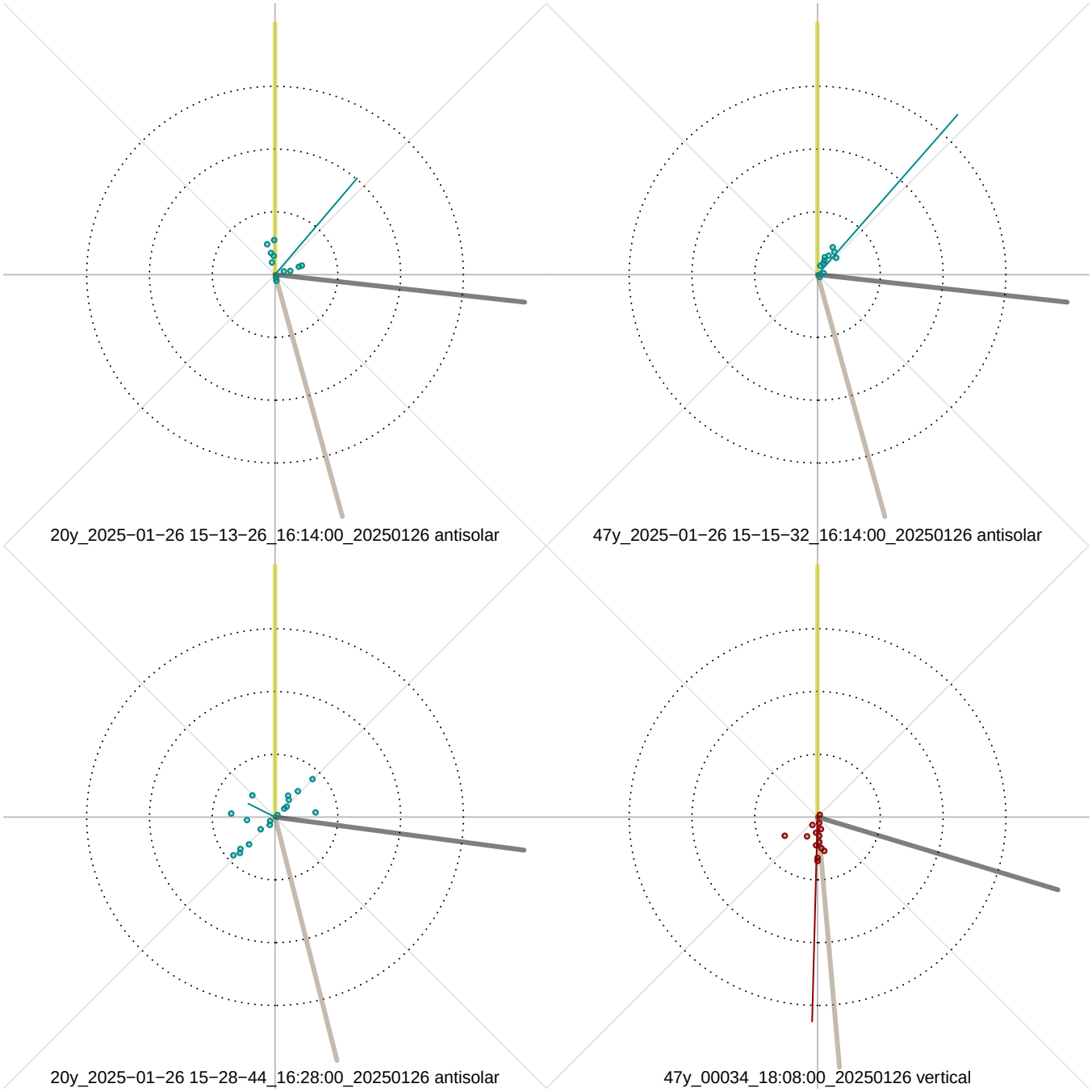


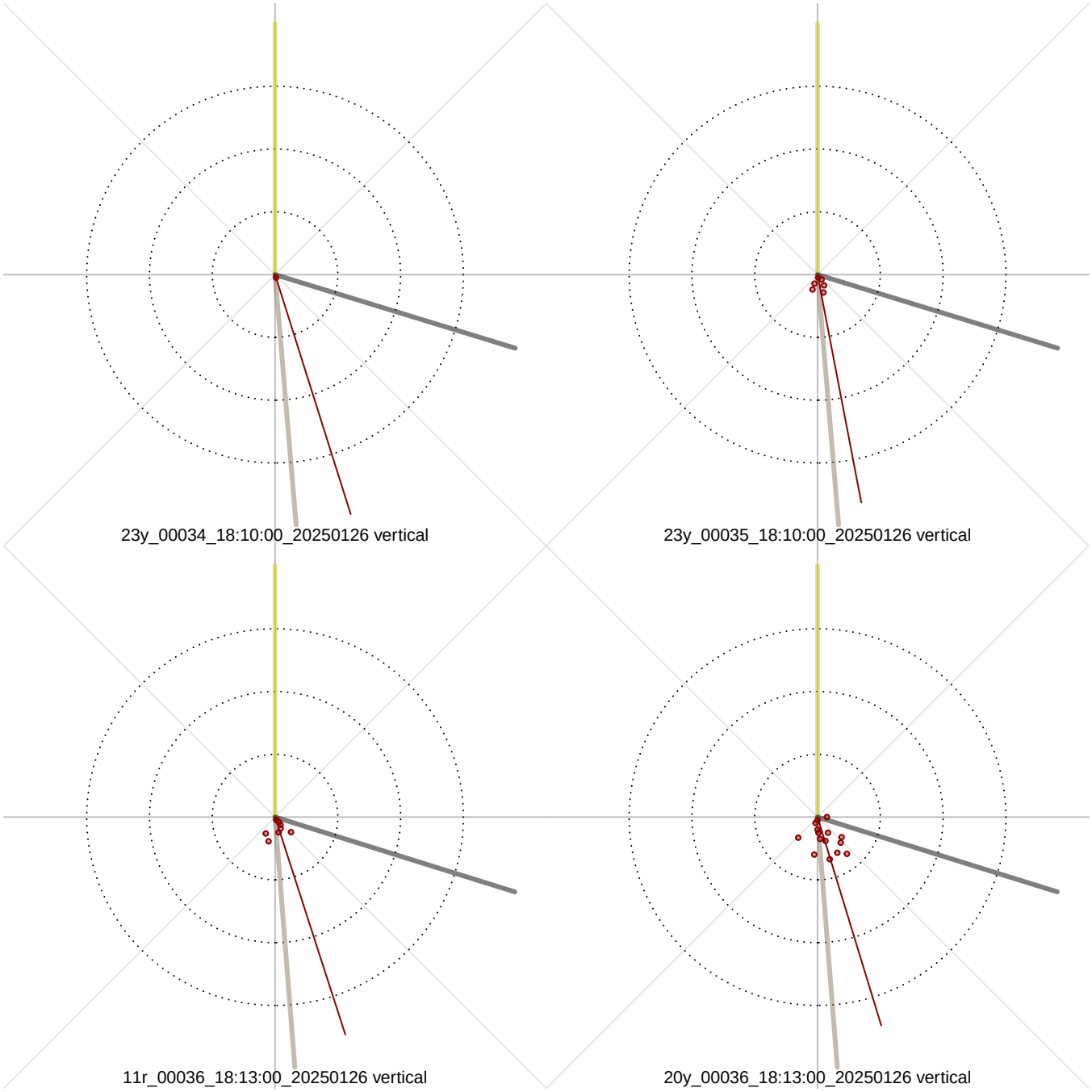
88b_00028_15:29:00_20250126 horizontal

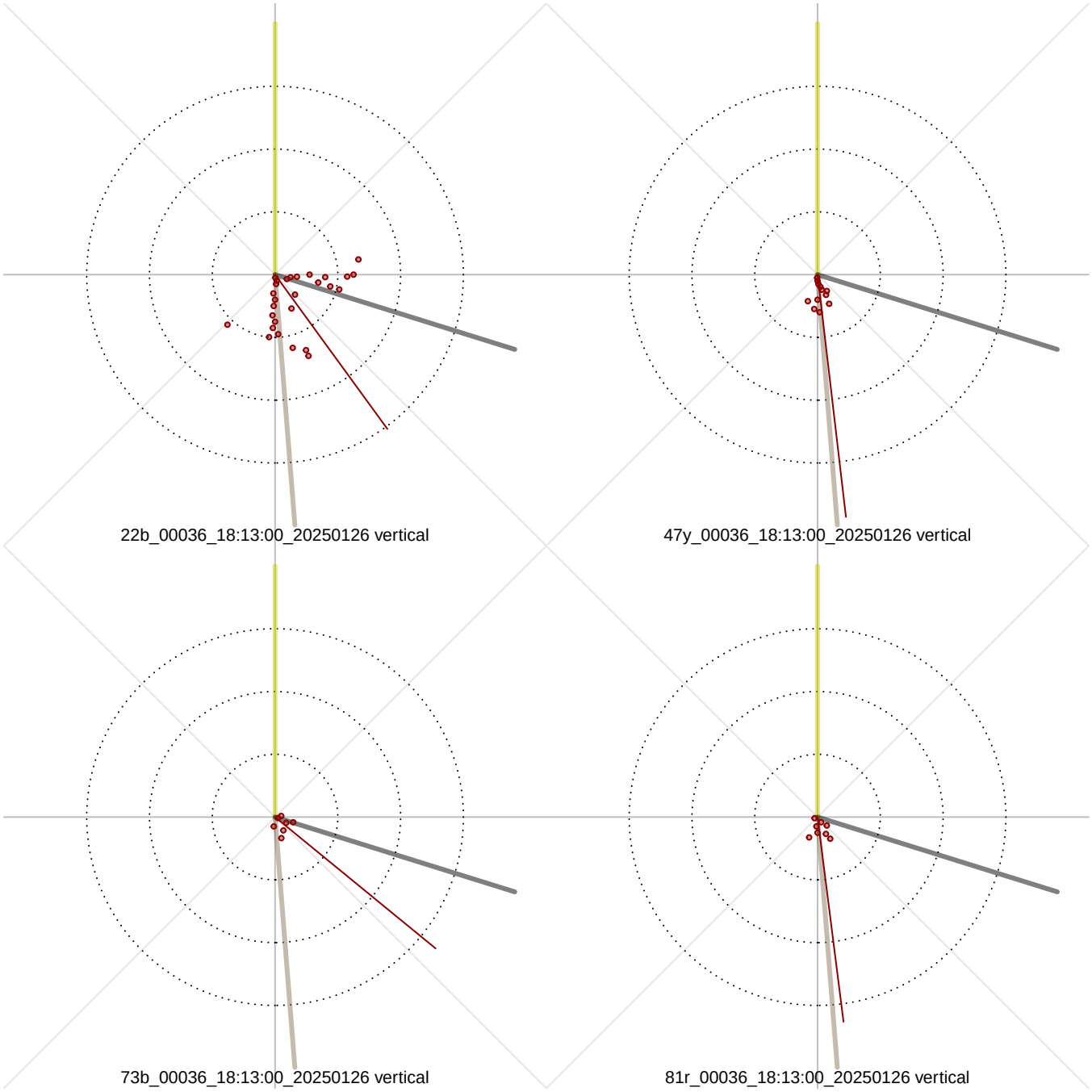


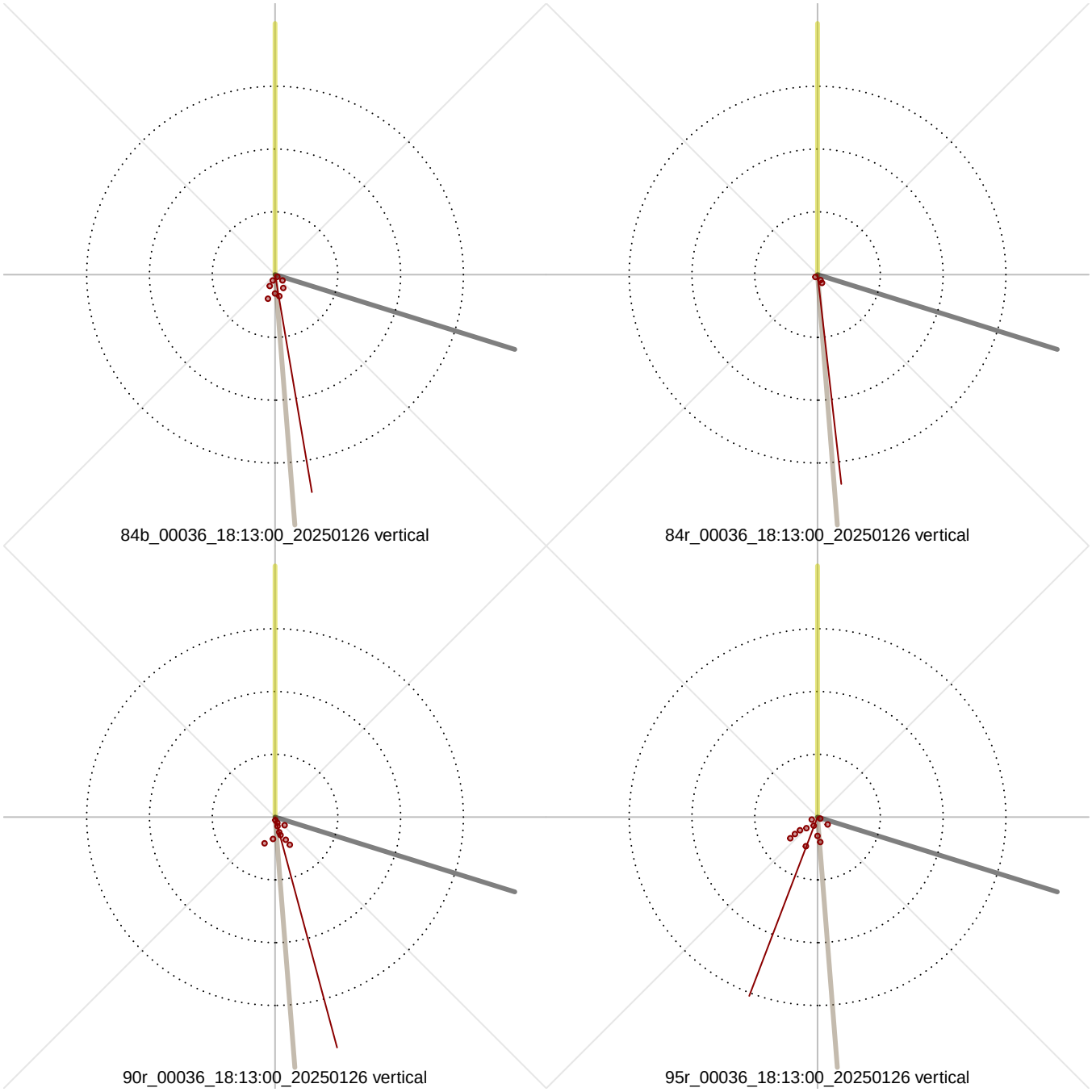
22b_2025−01−26 14−34−55_15:34:00_20250126 antisolar

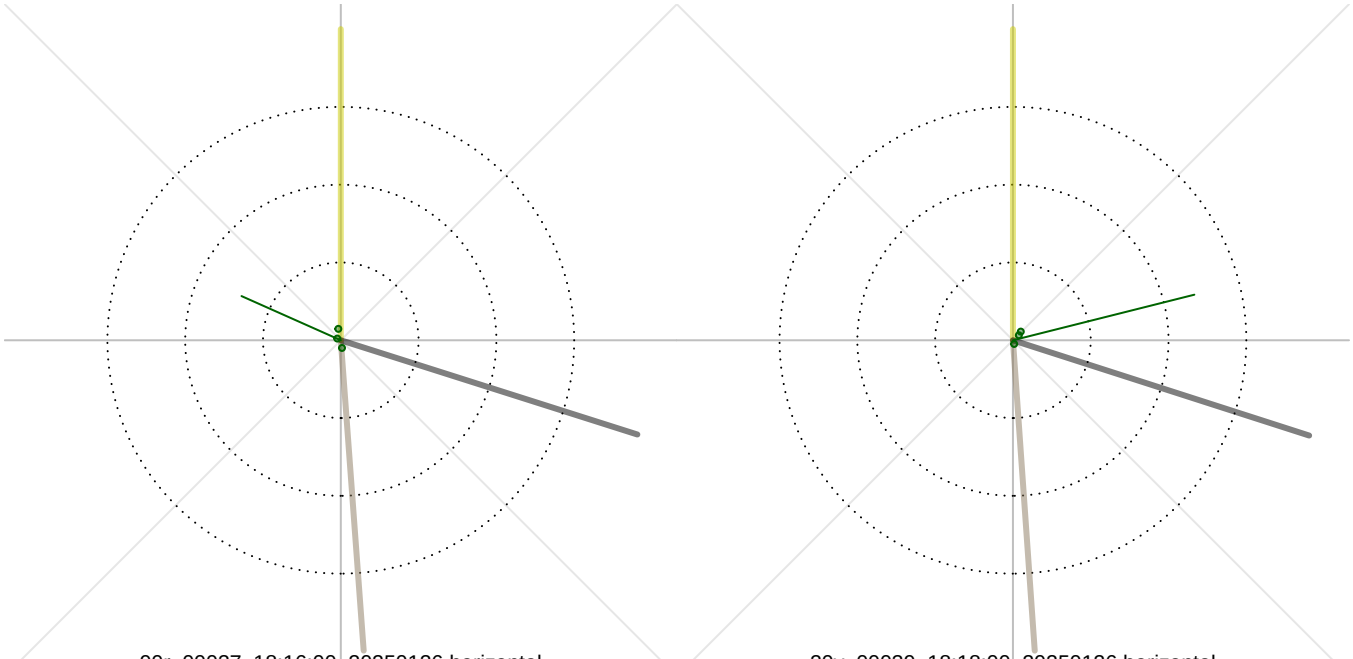

84b_2025−01−26 14−36−24_15:36:00_20250126 antisolar
90r_2025−01−26 14−38−46_15:38:00_20250126 antisolar
73b_2025−01−26 15−07−34_16:07:00_20250126 antisolar
90r_2025−01−26 15−08−46_16:08:00_20250126 antisolar


20y_2025−01−26 15−13−26_16:14:00_20250126 antisolar
47y_2025−01−26 15−15−32_16:14:00_20250126 antisolar
20y_2025−01−26 15−28−44_16:28:00_20250126 antisolar
47y_00034_18:08:00_20250126 vertical


23y_00034_18:10:00_20250126 vertical
23y_00035_18:10:00_20250126 vertical
11r_00036_18:13:00_20250126 vertical
20y_00036_18:13:00_20250126 vertical

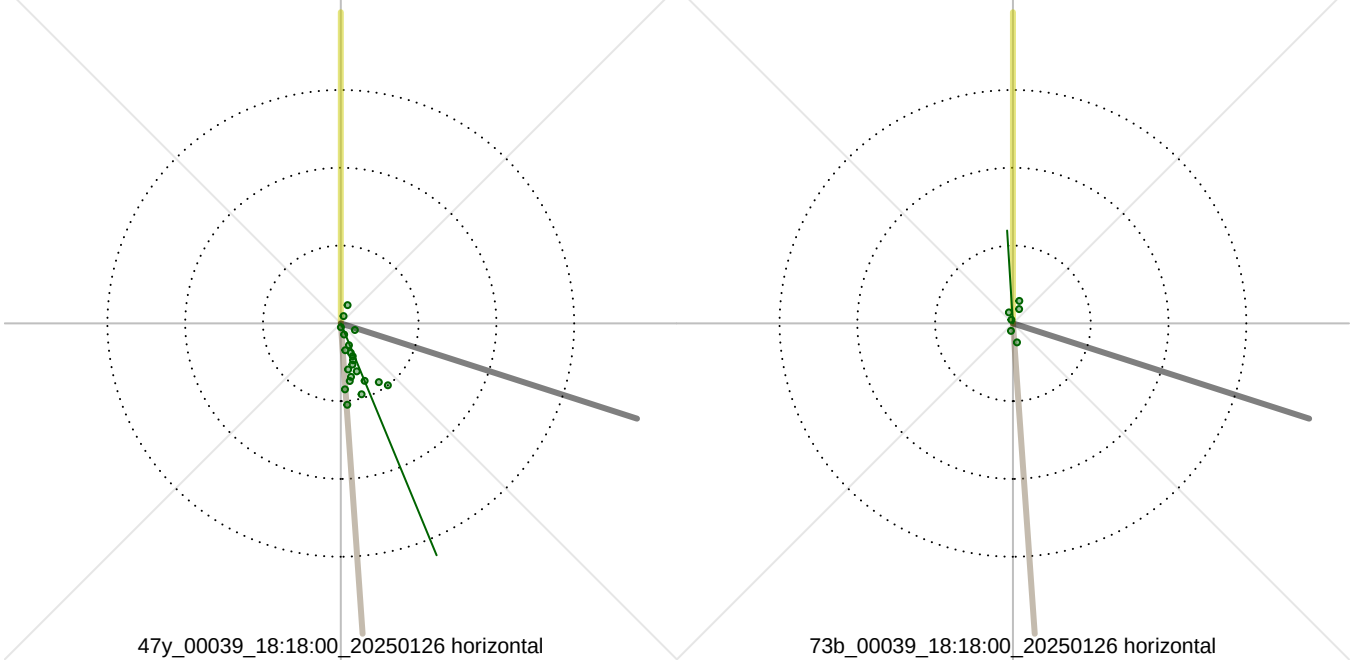

22b_00036_18:13:00_20250126 vertical
47y_00036_18:13:00_20250126 vertical
73b_00036_18:13:00_20250126 vertical
81r_00036_18:13:00_20250126 vertical


84b_00036_18:13:00_20250126 vertical
84r_00036_18:13:00_20250126 vertical
90r_00036_18:13:00_20250126 vertical
95r_00036_18:13:00_20250126 vertical



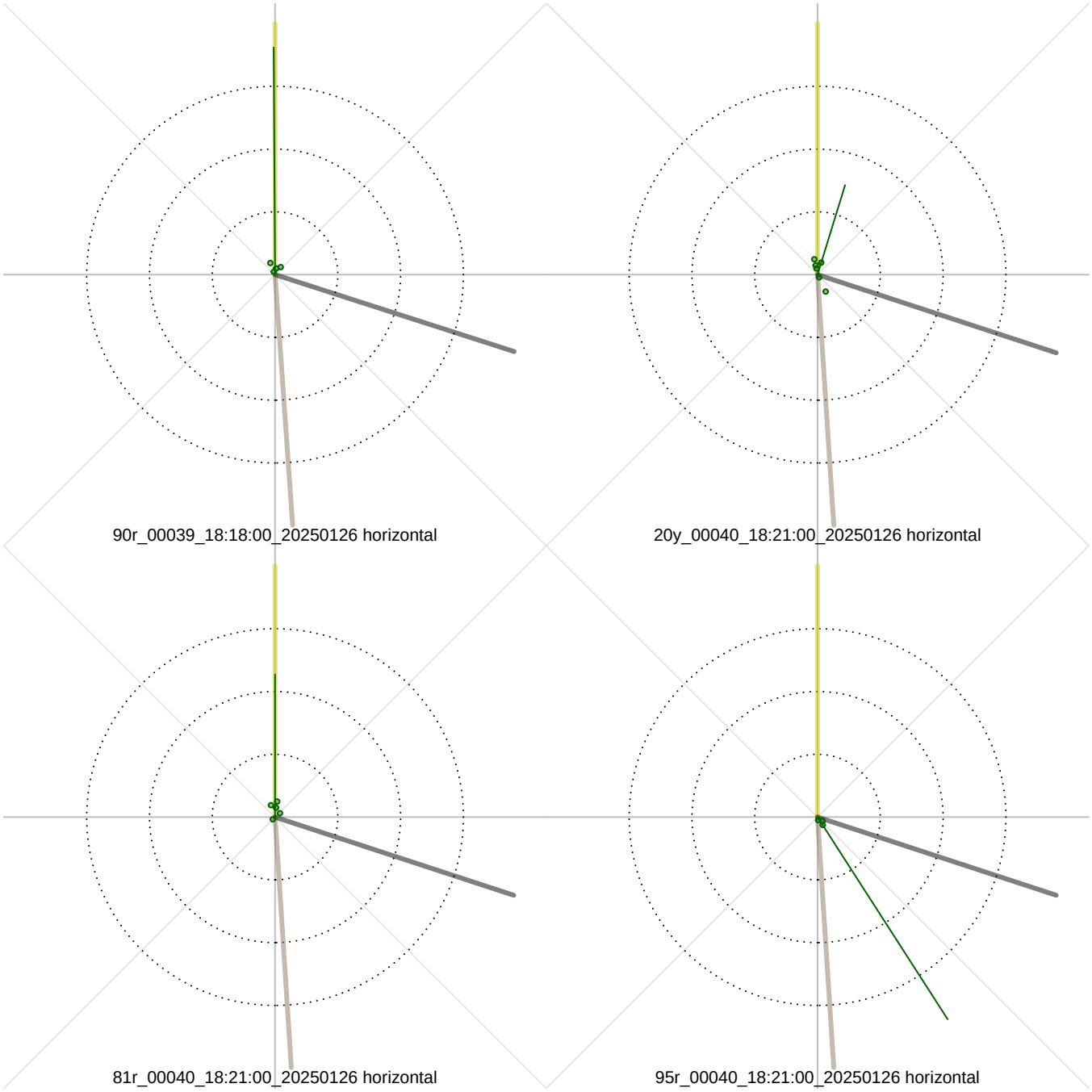
90r_00037_18:16:00_20250126 horizontal

20y_00039_18:18:00_20250126 horizontal



47y_00039_18:18:00_20250126 horizontal

73b_00039_18:18:00_20250126 horizontal

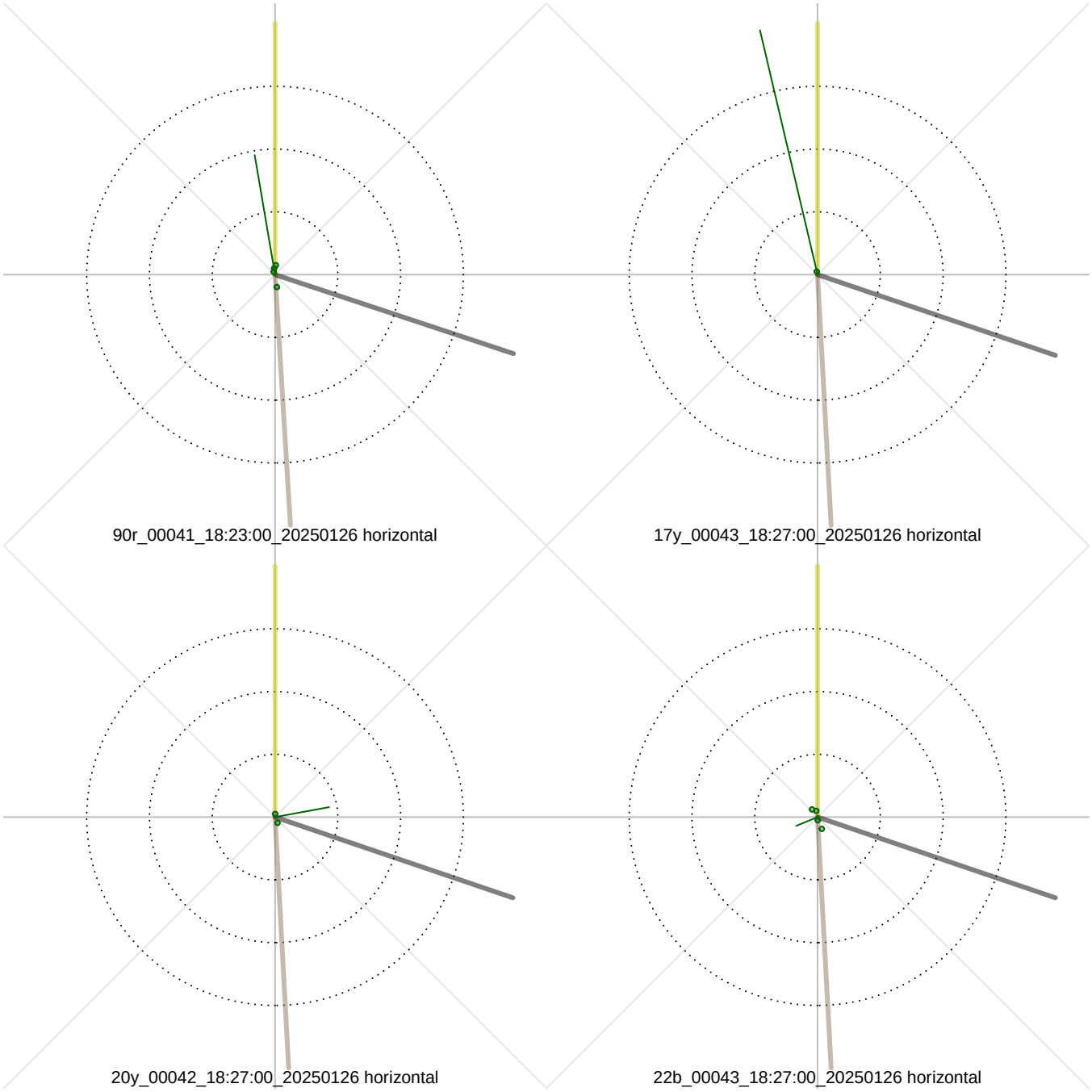


90r_00039_18:18:00_20250126 horizontal

20y_00040_18:21:00_20250126 horizontal

81r_00040_18:21:00_20250126 horizontal

95r_00040_18:21:00_20250126 horizontal

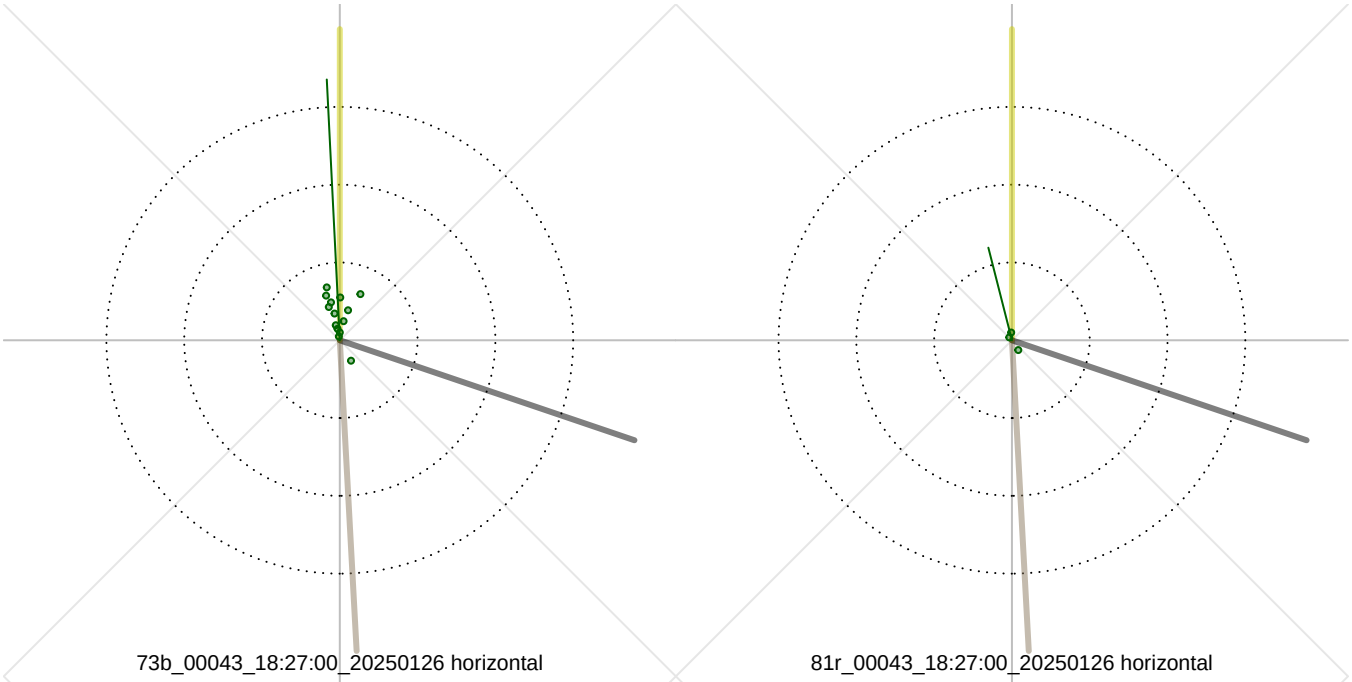


90r_00041_18:23:00_20250126 horizontal

17y_00043_18:27:00_20250126 horizontal

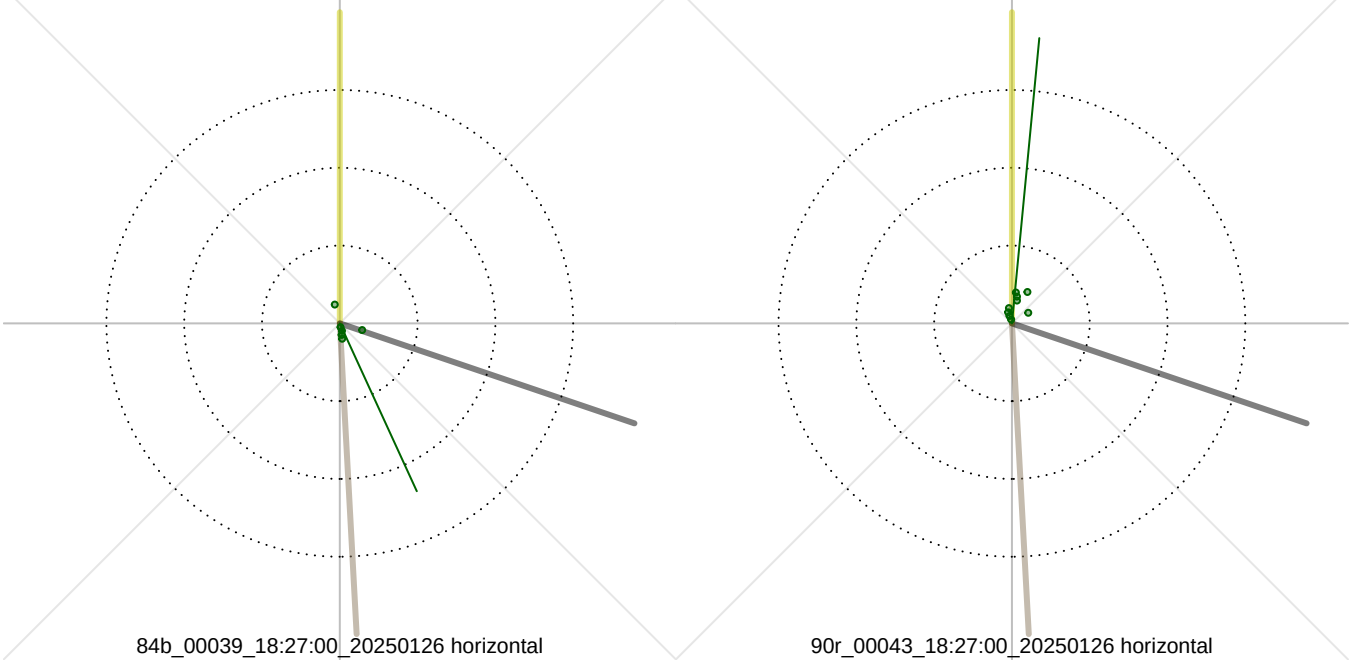
20y_00042_18:27:00_20250126 horizontal

22b_00043_18:27:00_20250126 horizontal



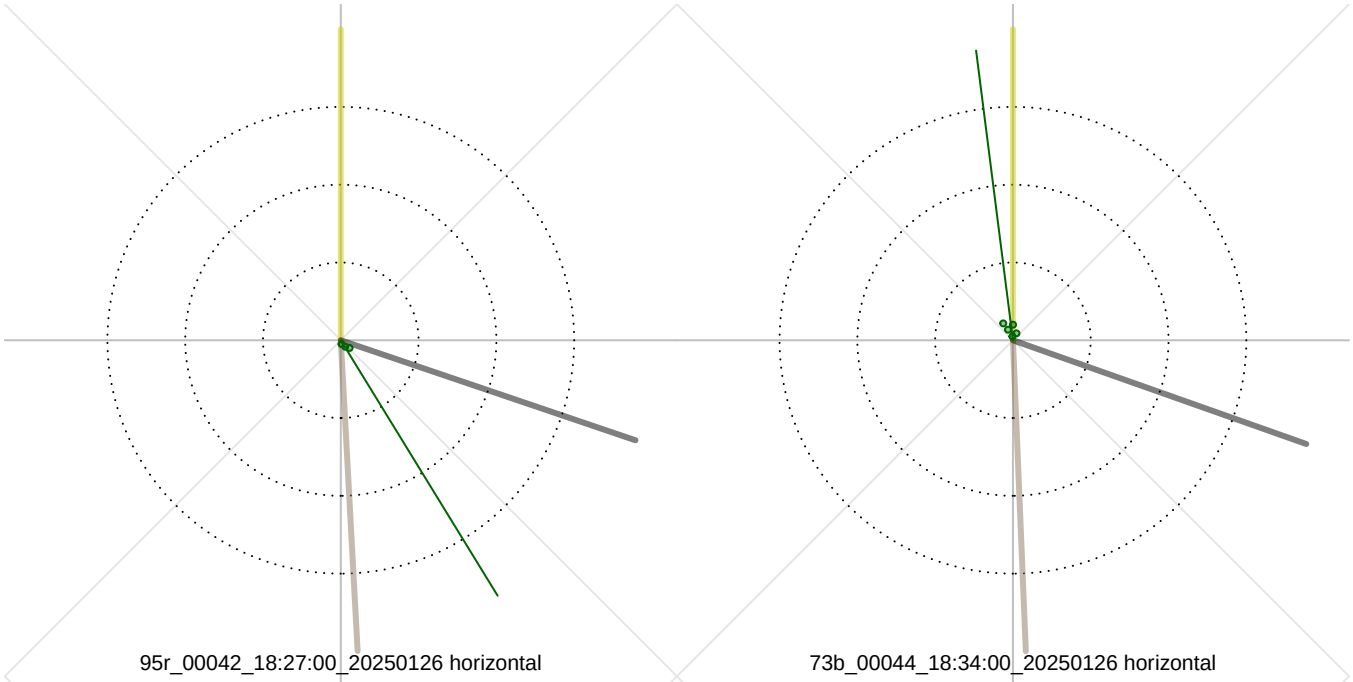
73b_00043_18:27:00_20250126 horizontal

81r_00043_18:27:00_20250126 horizontal



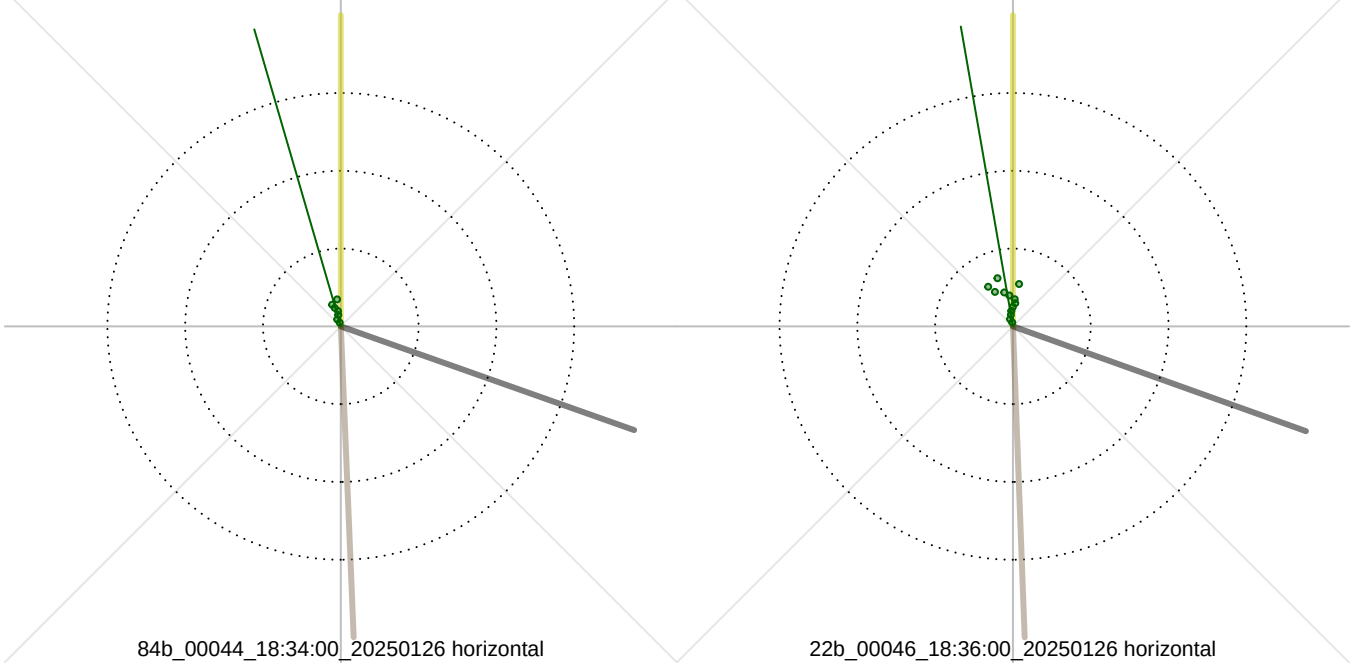
84b_00039_18:27:00_20250126 horizontal

90r_00043_18:27:00_20250126 horizontal



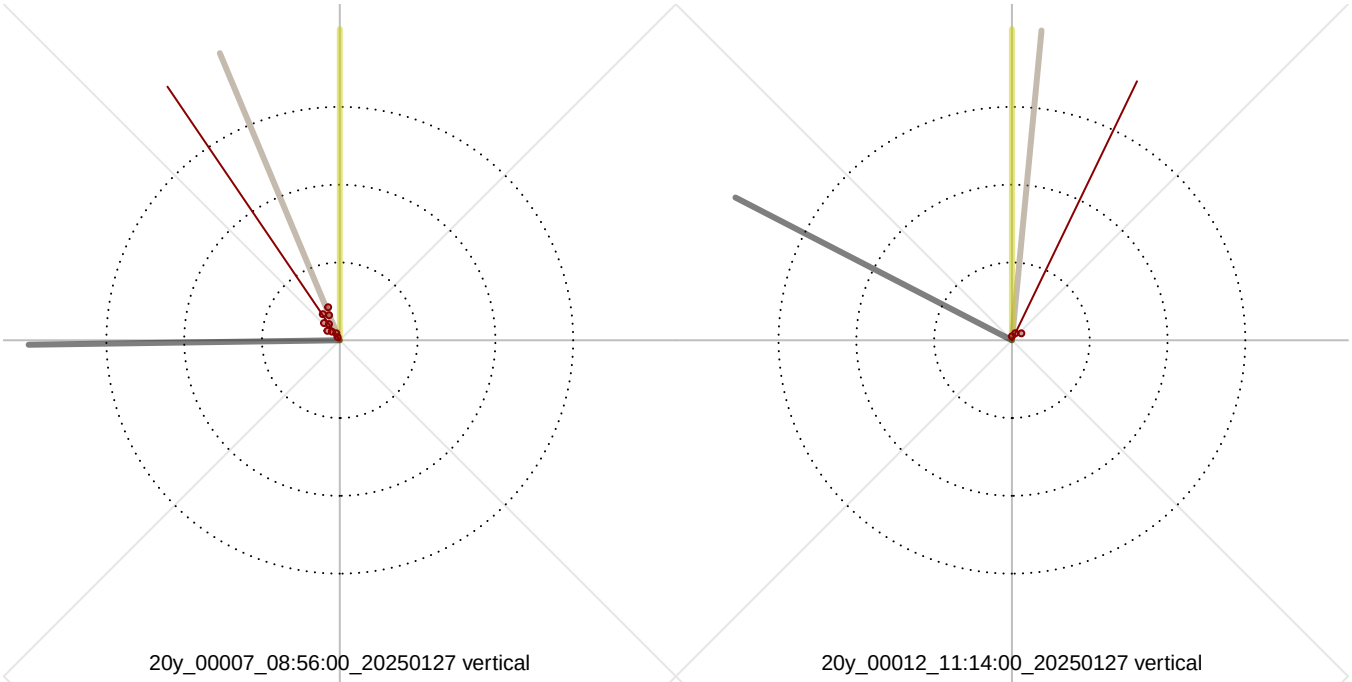
95r_00042_18:27:00_20250126 horizontal

73b_00044_18:34:00_20250126 horizontal



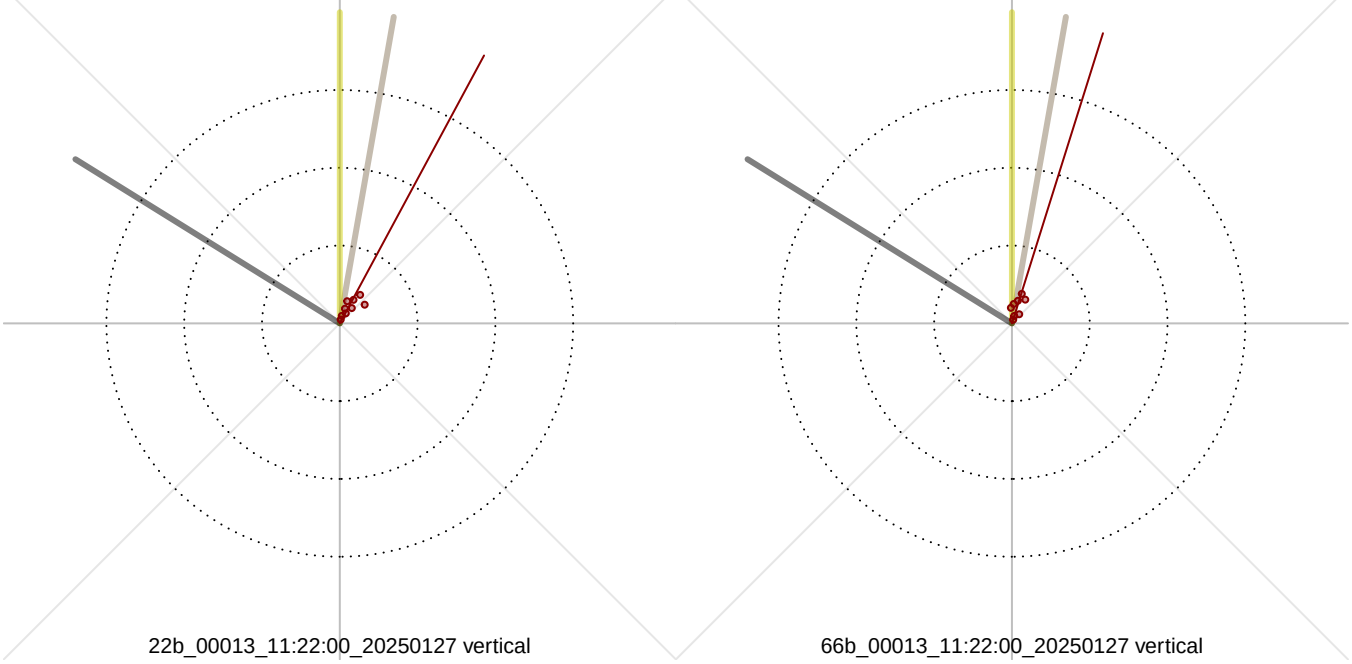
84b_00044_18:34:00_20250126 horizontal

22b_00046_18:36:00_20250126 horizontal



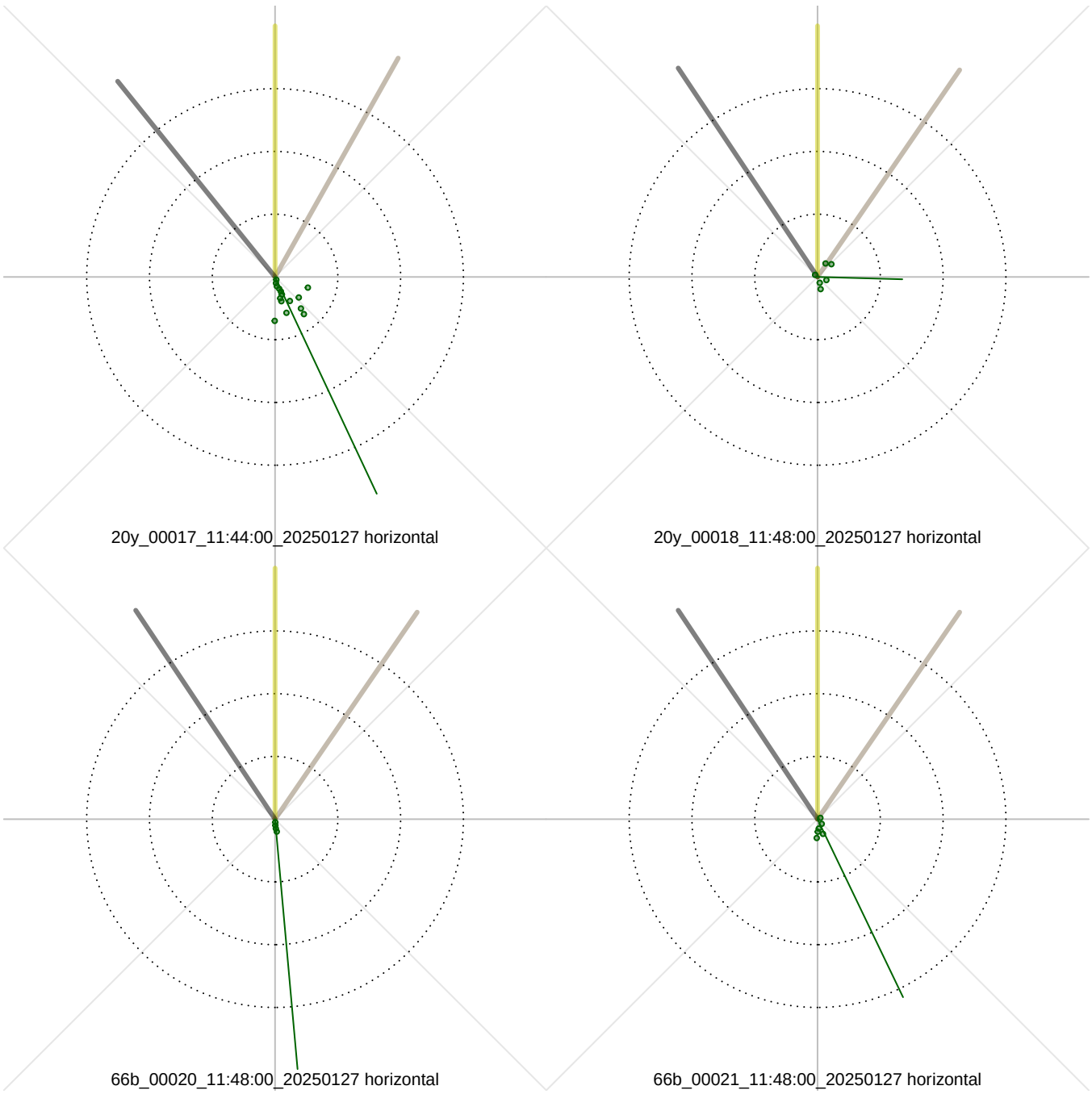
20y_00007_08:56:00_20250127 vertical

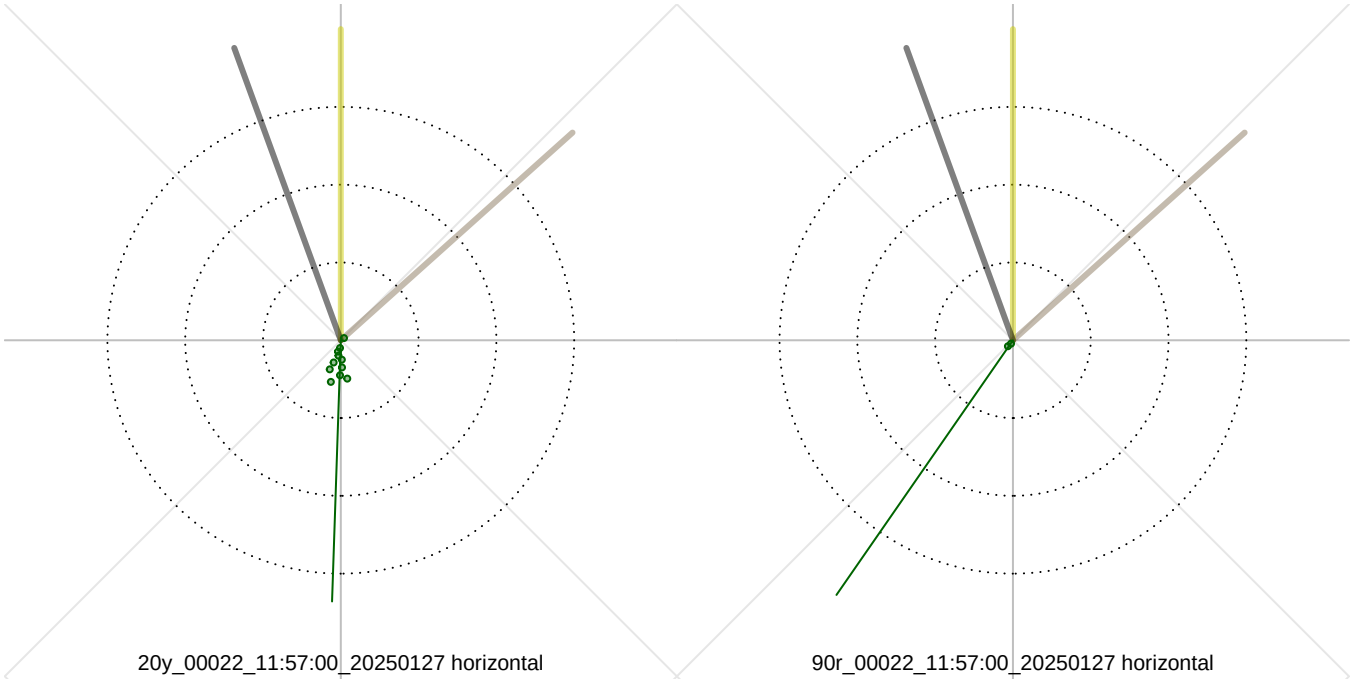
20y_00012_11:14:00_20250127 vertical



22b_00013_11:22:00_20250127 vertical

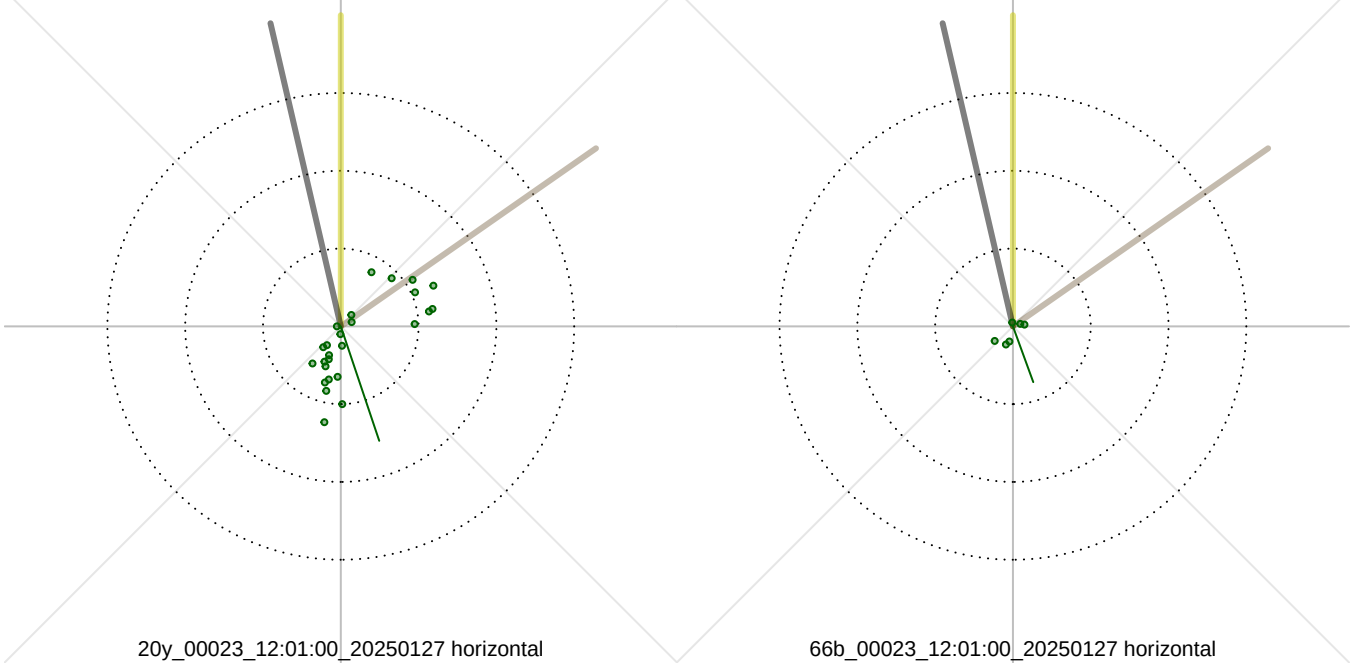
66b_00013_11:22:00_20250127 vertical


20y_00017_11:44:00_20250127 horizontal
20y_00018_11:48:00_20250127 horizontal
66b_00020_11:48:00_20250127 horizontal
66b_00021_11:48:00_20250127 horizontal



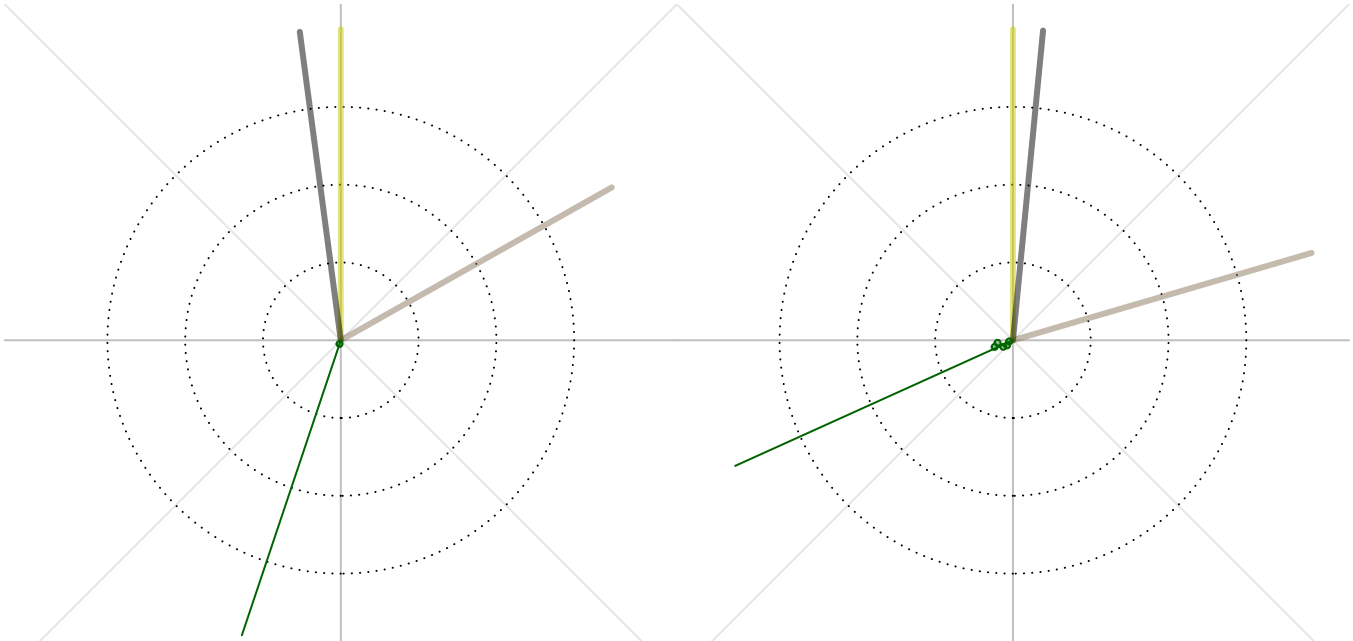
20y_00022_11:57:00_20250127 horizontal

90r_00022_11:57:00_20250127 horizontal



20y_00023_12:01:00_20250127 horizontal

66b_00023_12:01:00_20250127 horizontal



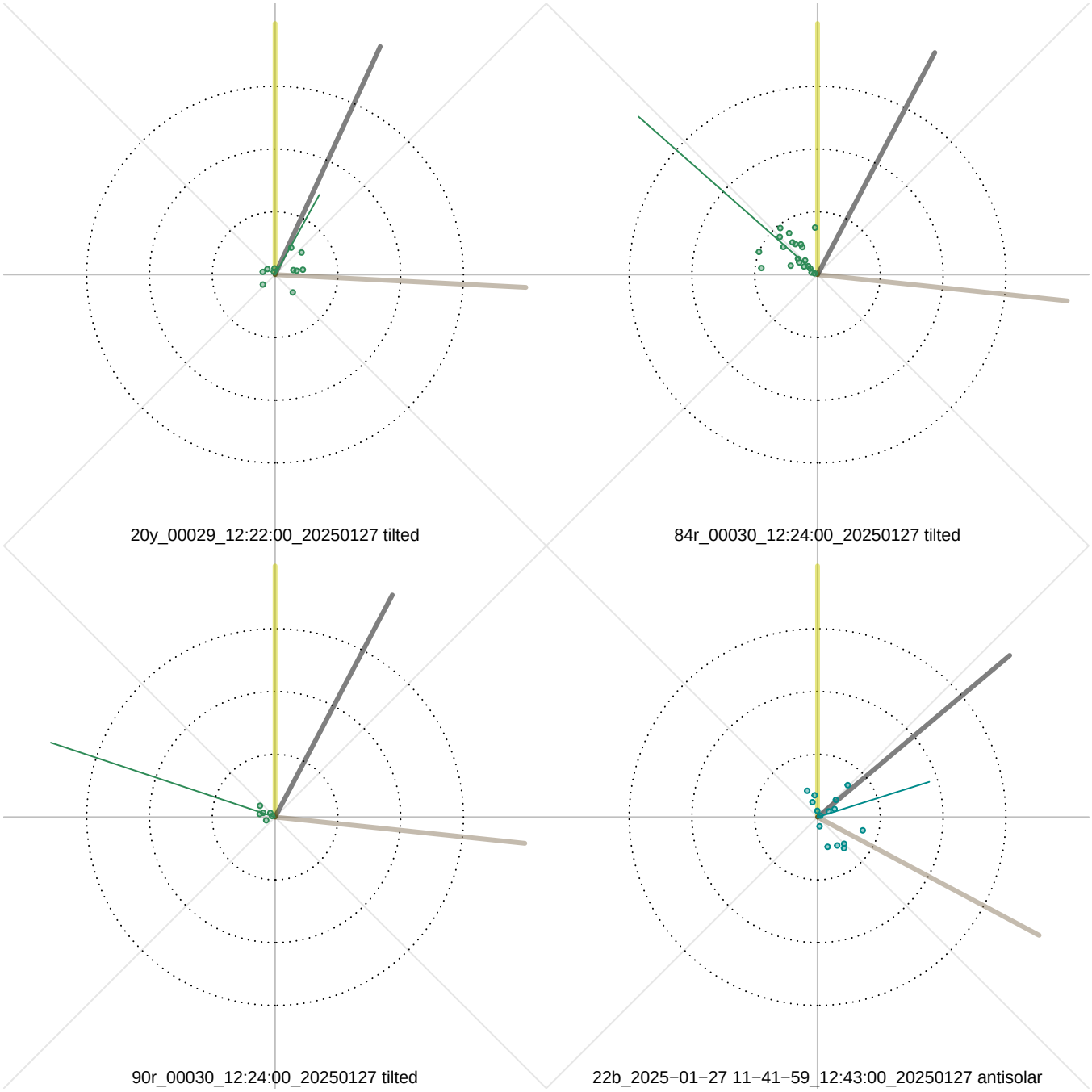
84b_00024_12:04:00_20250127 horizontal

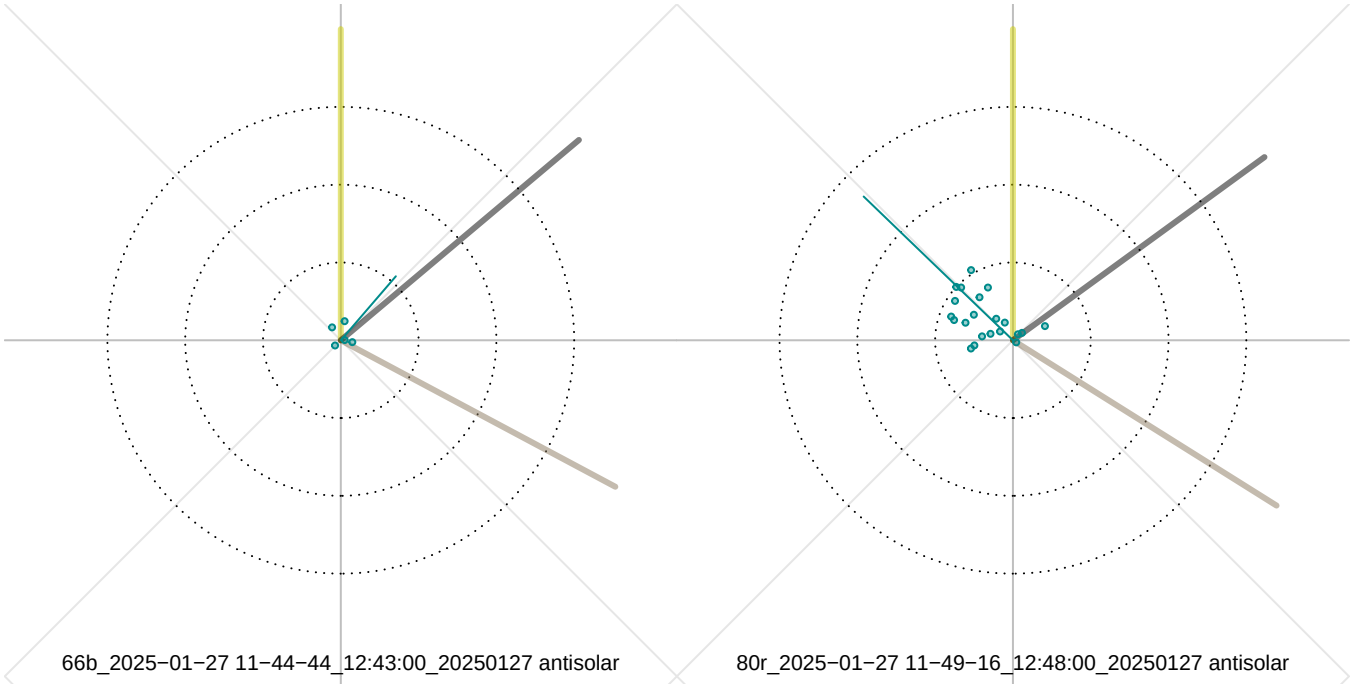
20y_00025_12:11:00_20250127 horizontal



66b_00026_12:11:00_20250127 horizontal

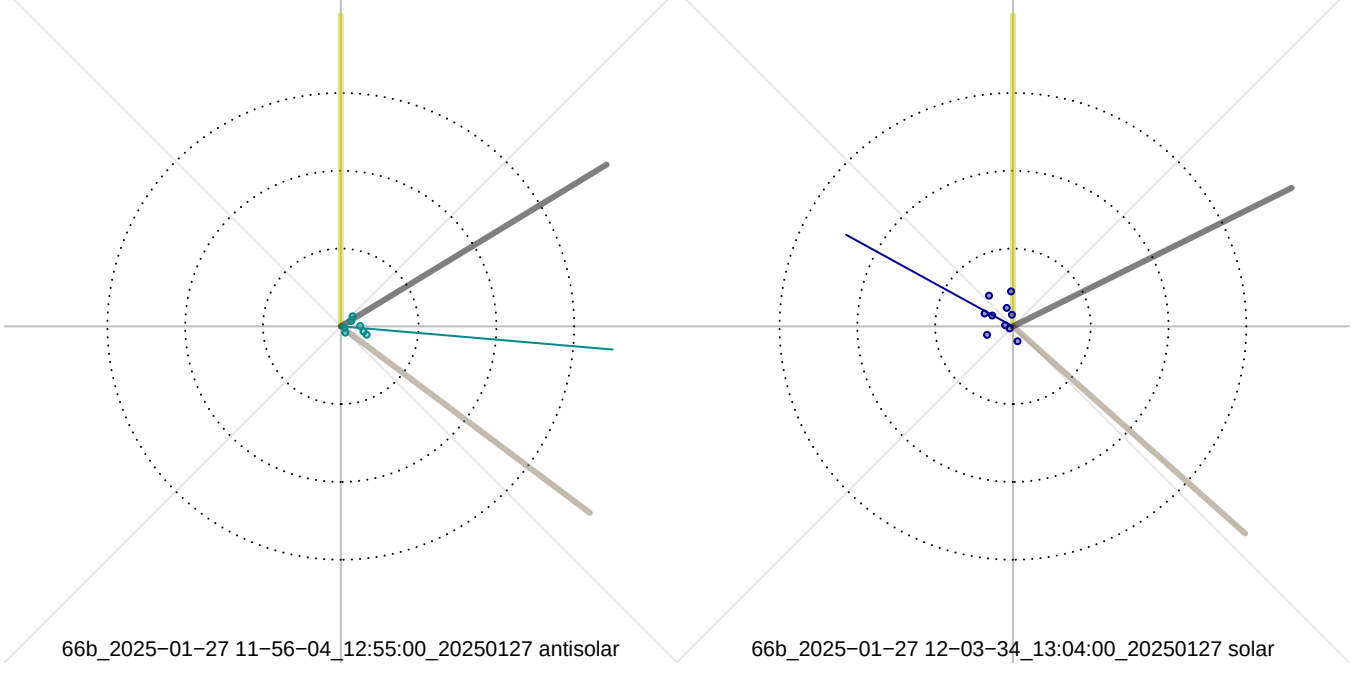
20y_00028_12:20:00_20250127 tilted


20y_00029_12:22:00_20250127 tilted
84r_00030_12:24:00_20250127 tilted
90r_00030_12:24:00_20250127 tilted
22b_2025−01−27 11−41−59_12:43:00_20250127 antisolar



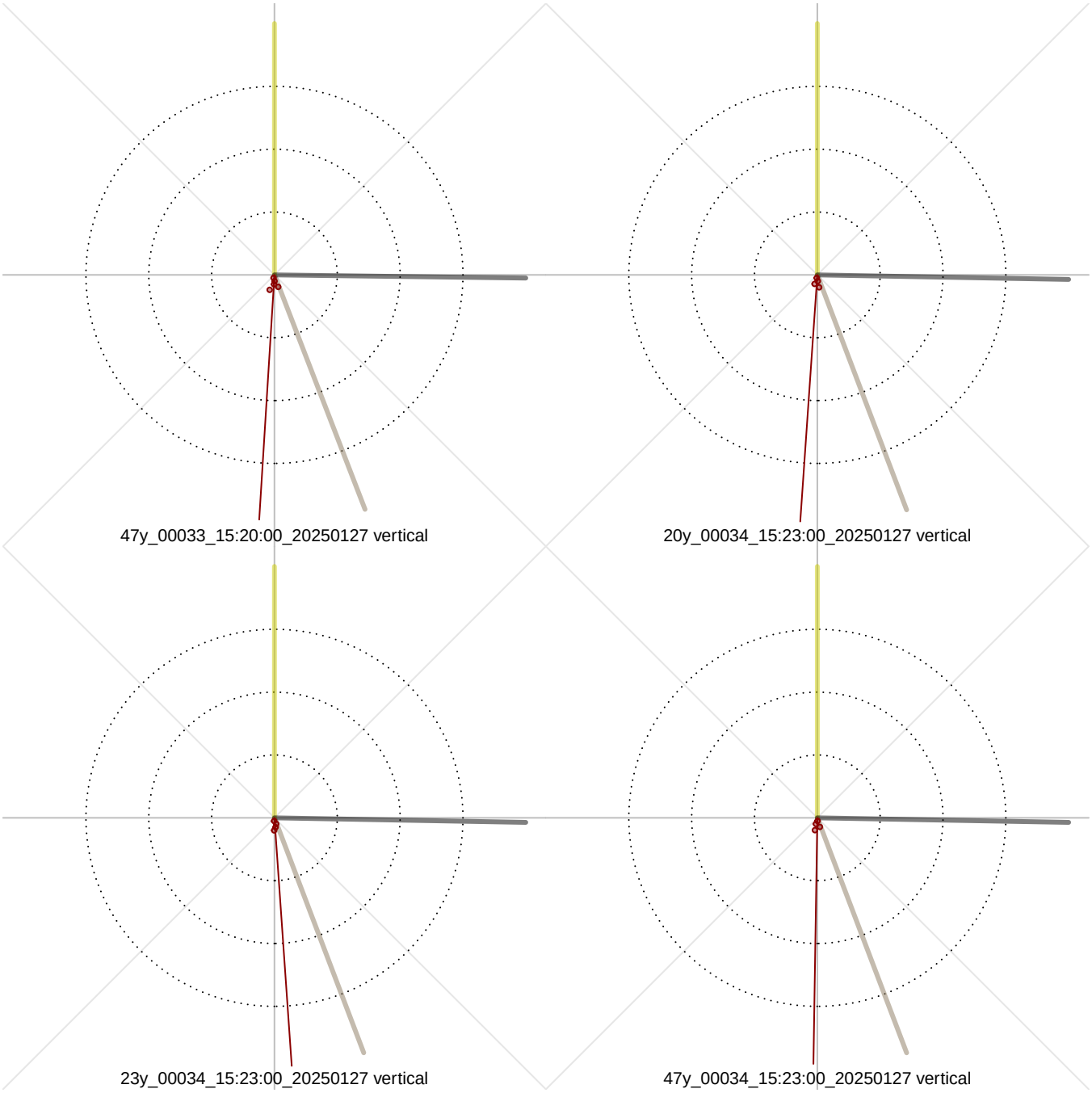
66b_2025-01-27 11-44-44_12:43:00_20250127 antisolar

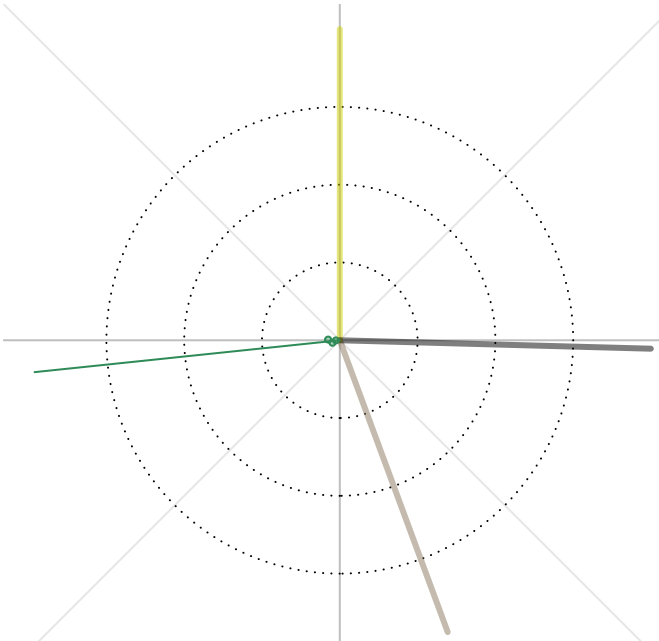
80r_2025-01-27 11-49-16_12:48:00_20250127 antisolar



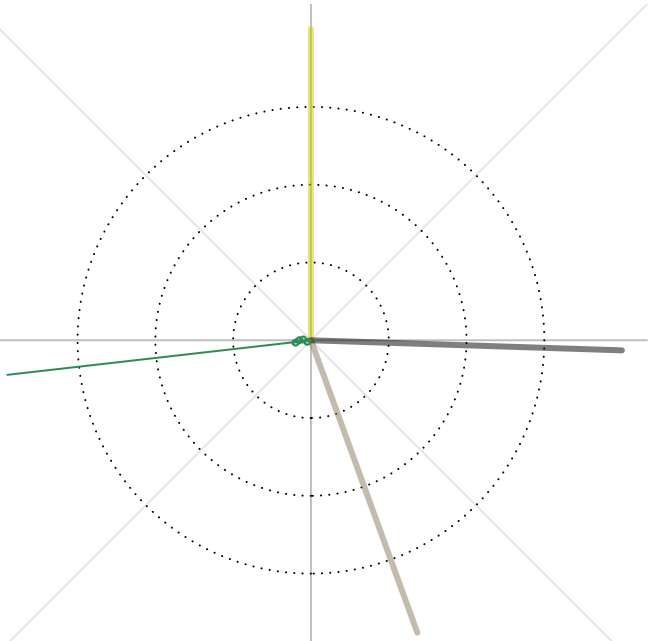
66b_2025-01-27 11-56-04_12:55:00_20250127 antisolar

66b_2025-01-27 12-03-34_13:04:00_20250127 solar

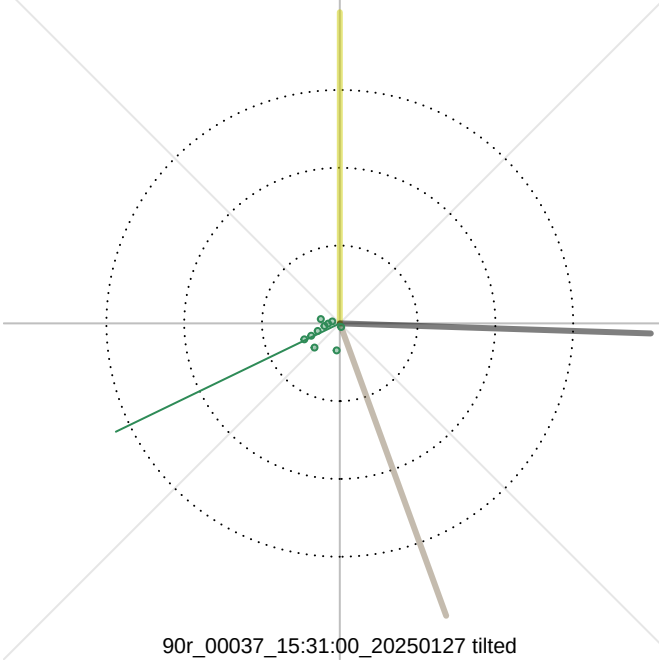

47y_00033_15:20:00_20250127 vertical
20y_00034_15:23:00_20250127 vertical
23y_00034_15:23:00_20250127 vertical
47y_00034_15:23:00_20250127 vertical



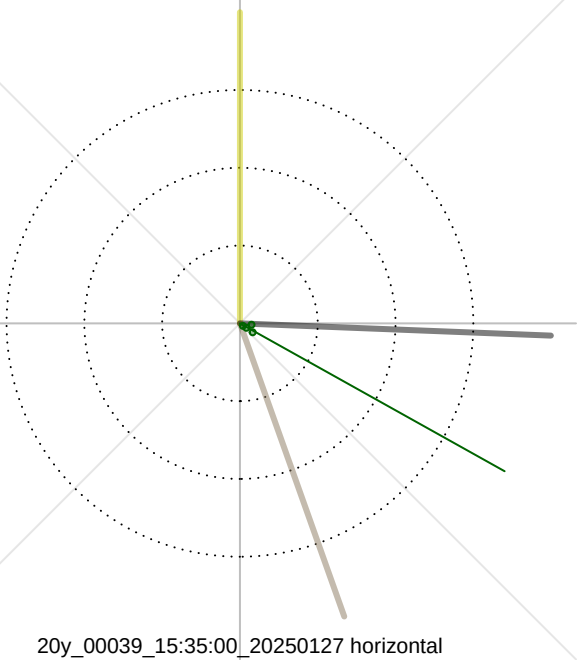
20y_00035_15:28:00_20250127 tilted



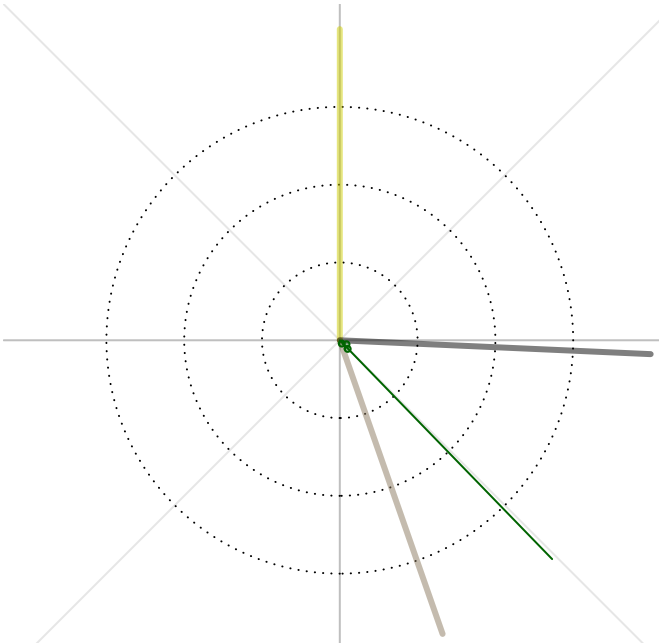
20y_00037_15:31:00_20250127 tilted



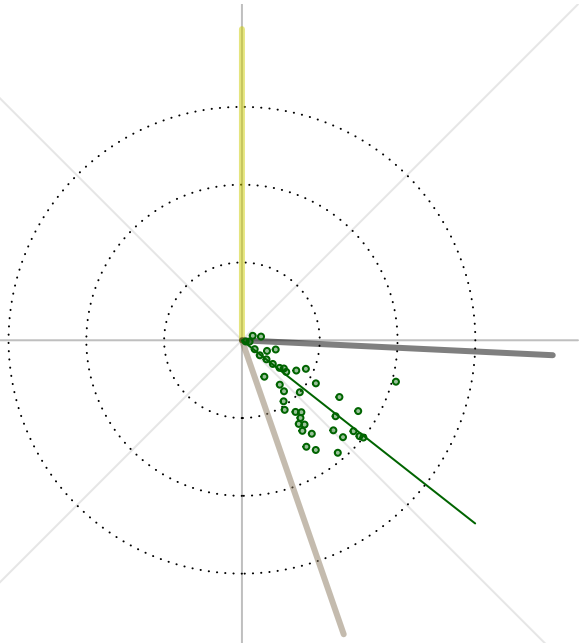
90r_00037_15:31:00_20250127 tilted



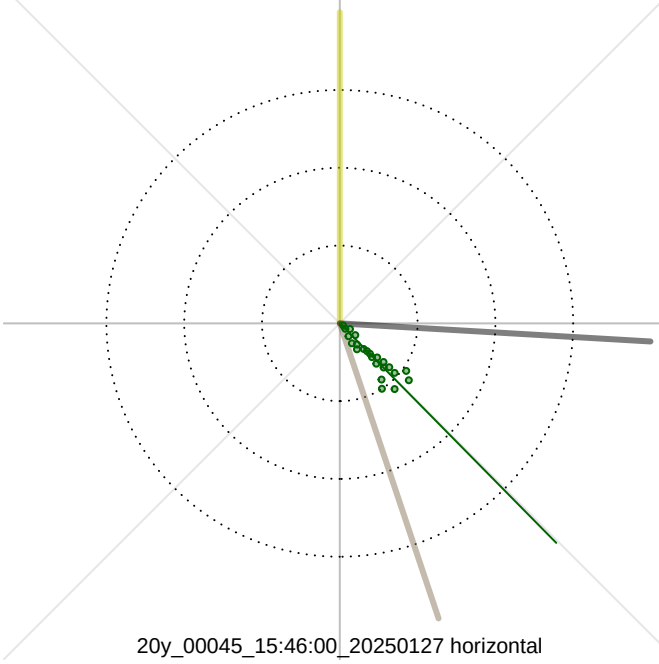
20y_00039_15:35:00_20250127 horizontal



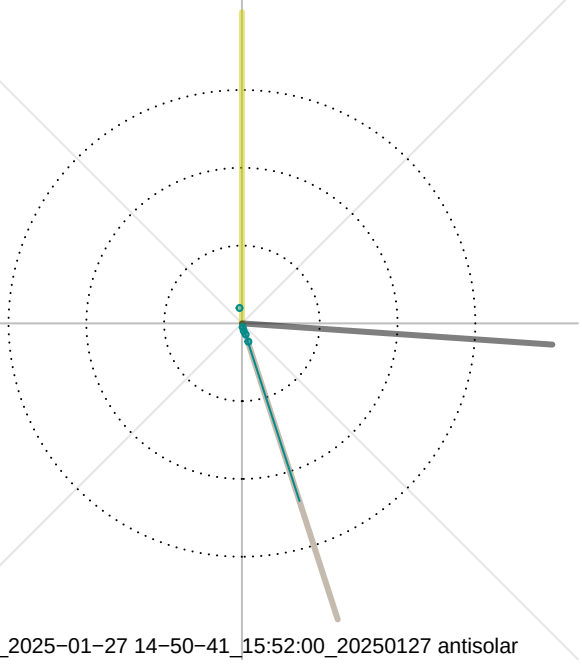
90r_00041_15:38:00_20250127 horizontal



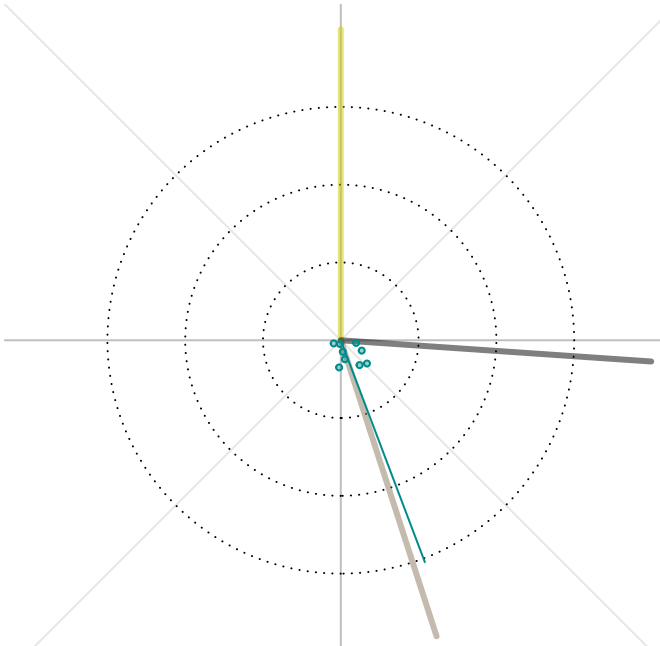
20y_00041_15:40:00_20250127 horizontal



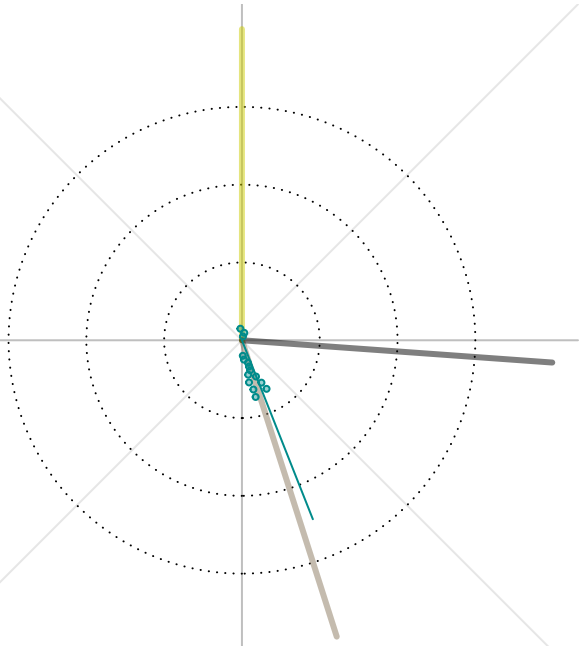
20y_00045_15:46:00_20250127 horizontal



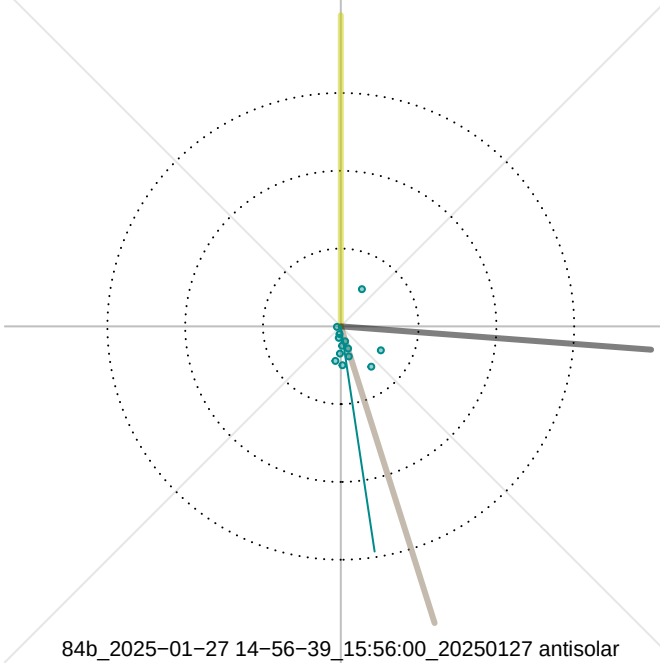
90r_2025−01−27 14−50−41_15:52:00_20250127 antisolar



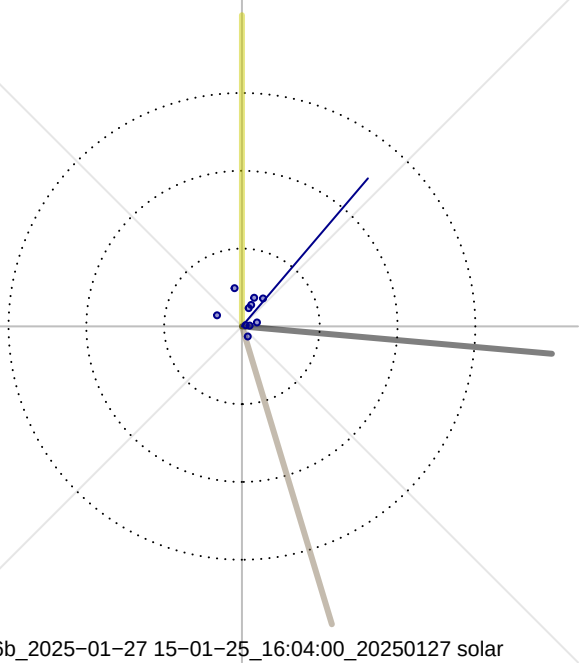
95r_2025−01−27 14−50−41_15:52:00_20250127 antisolar



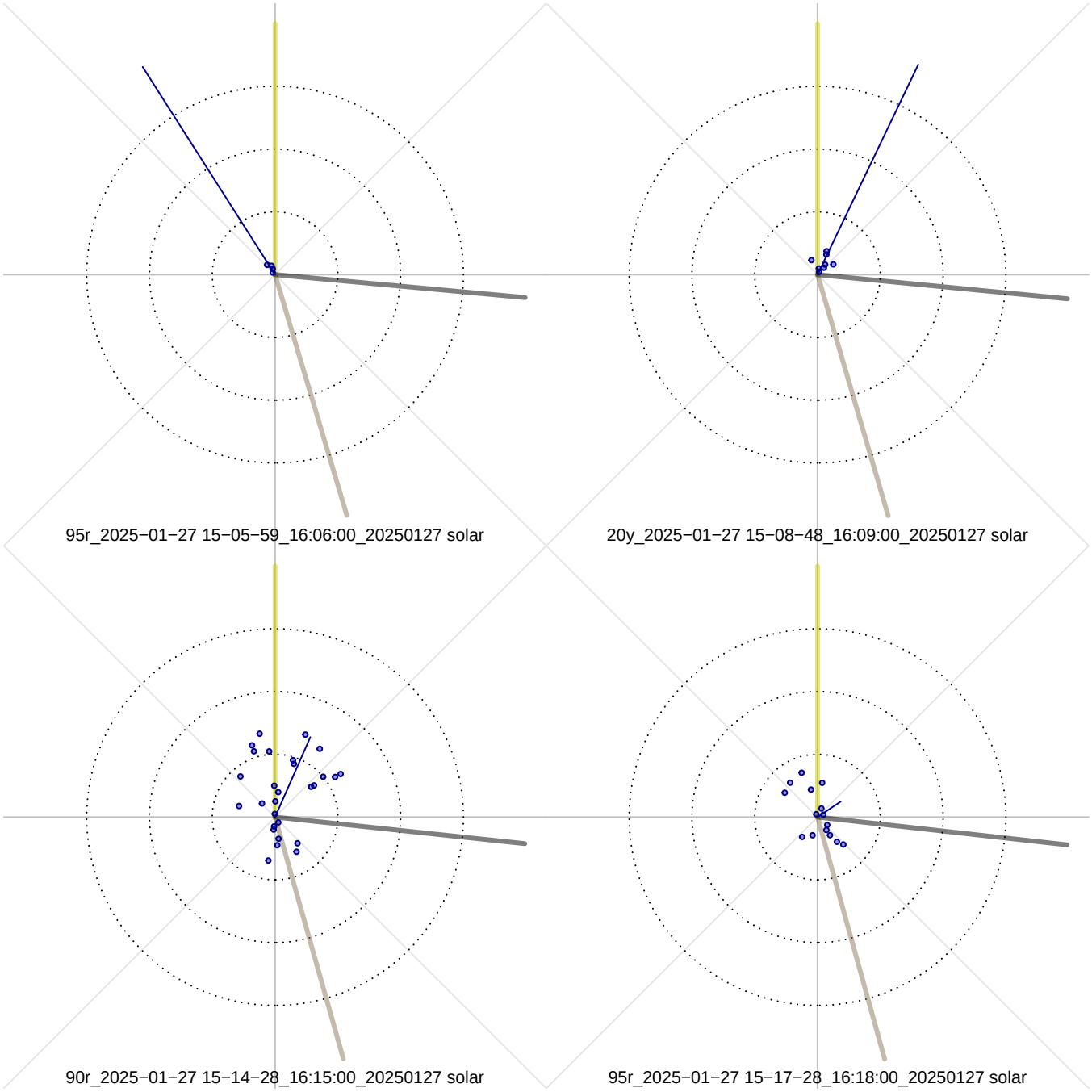
84b_2025−01−27 14−52−42_15:54:00_20250127 antisolar

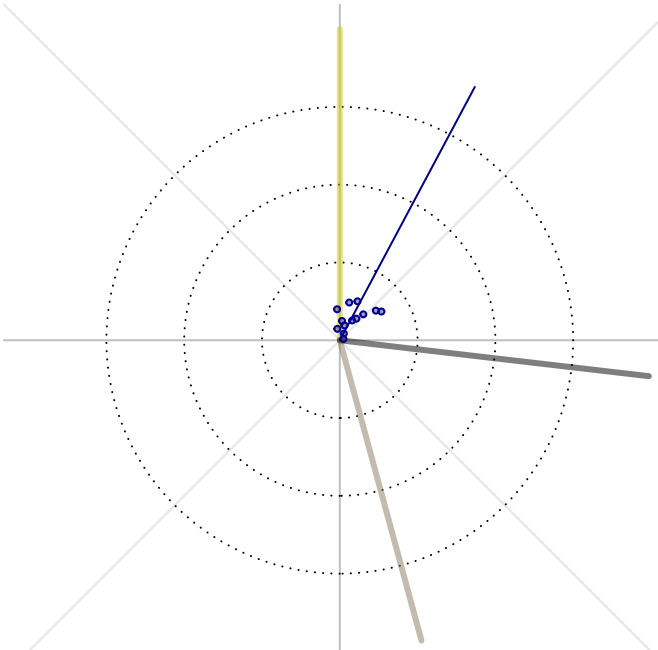


84b_2025−01−27 14−56−39_15:56:00_20250127 antisolar

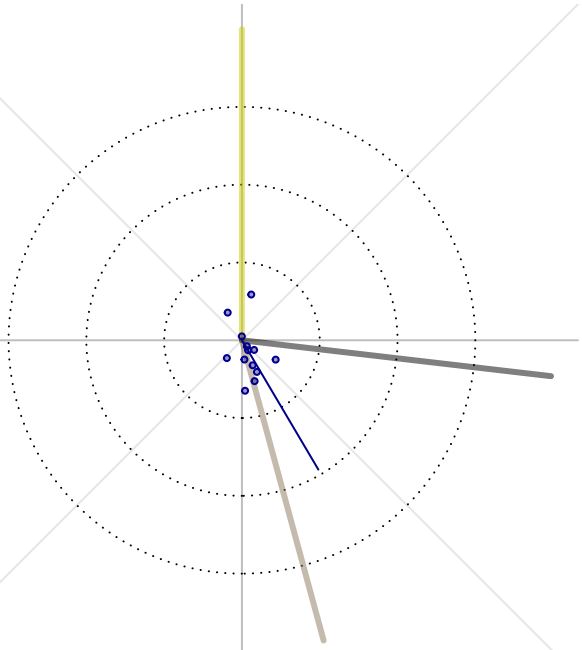


66b_2025−01−27 15−01−25_16:04:00_20250127 solar

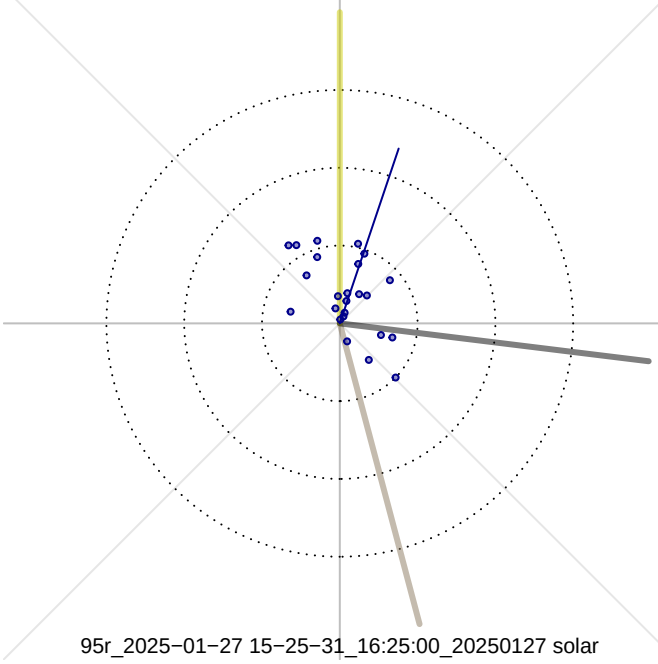

95r_2025-01-27 15-05-59_16:06:00_20250127 solar
20y_2025-01-27 15-08-48_16:09:00_20250127 solar
90r_2025-01-27 15-14-28_16:15:00_20250127 solar
95r_2025-01-27 15-17-28_16:18:00_20250127 solar



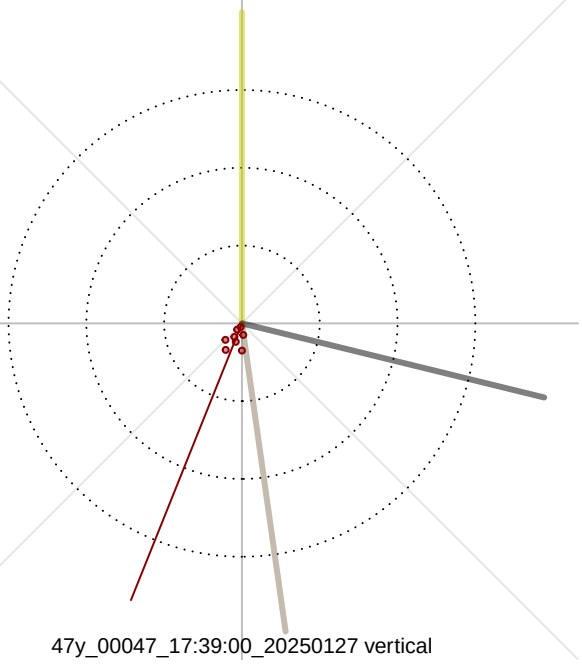
20y_2025−01−27 15−21−13_16:21:00_20250127 solar



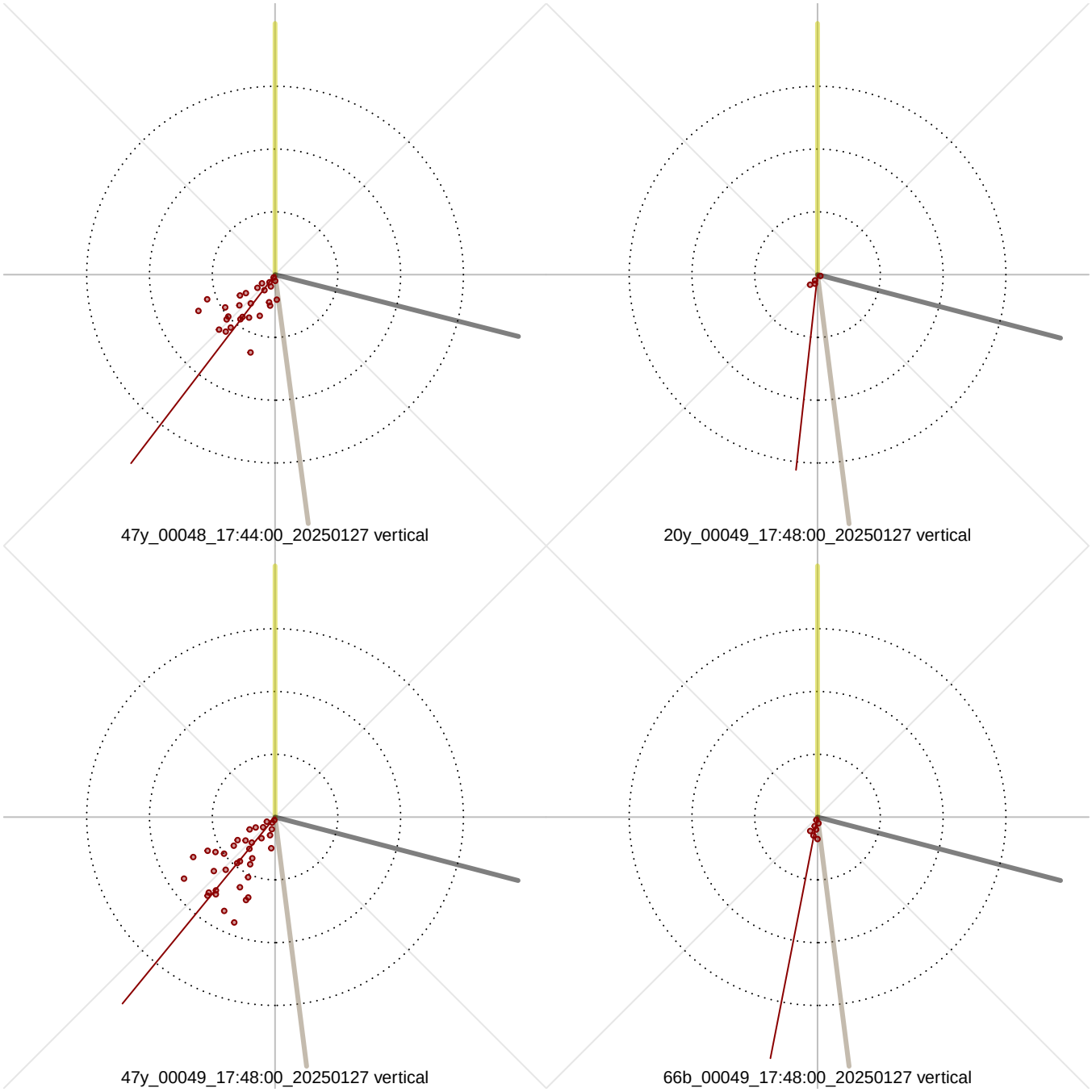
95r_2025−01−27 15−21−13_16:21:00_20250127 solar

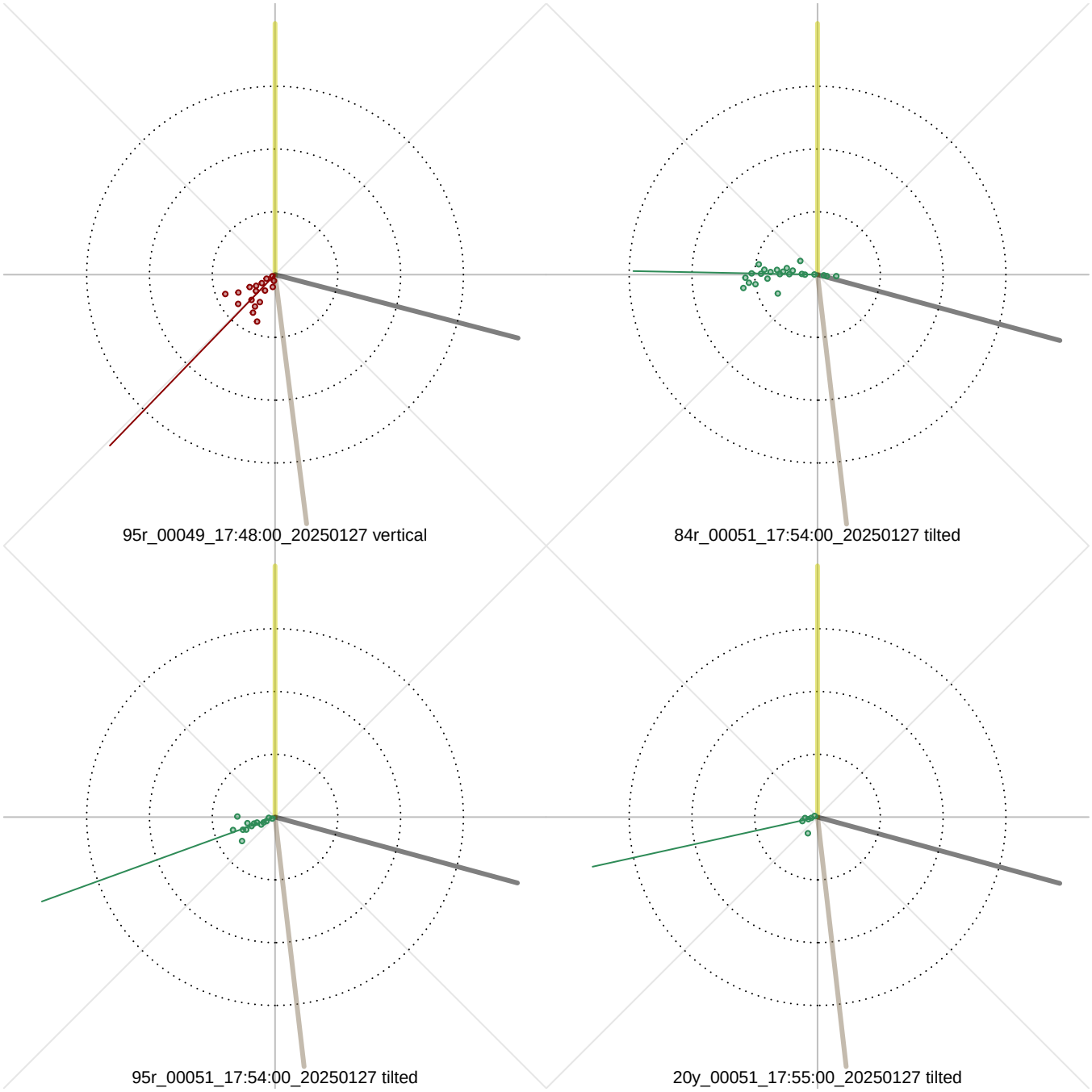


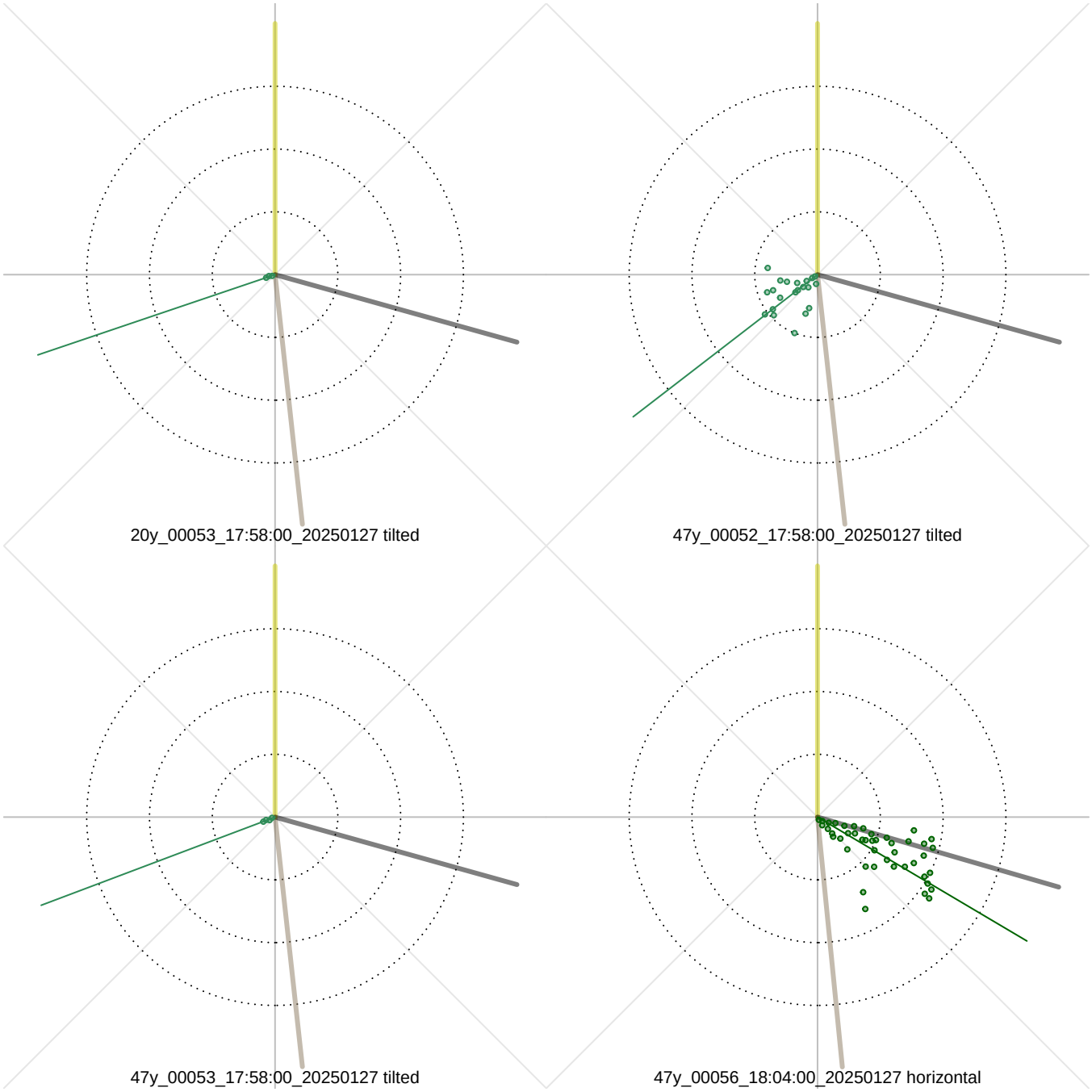
95r_2025−01−27 15−25−31_16:25:00_20250127 solar

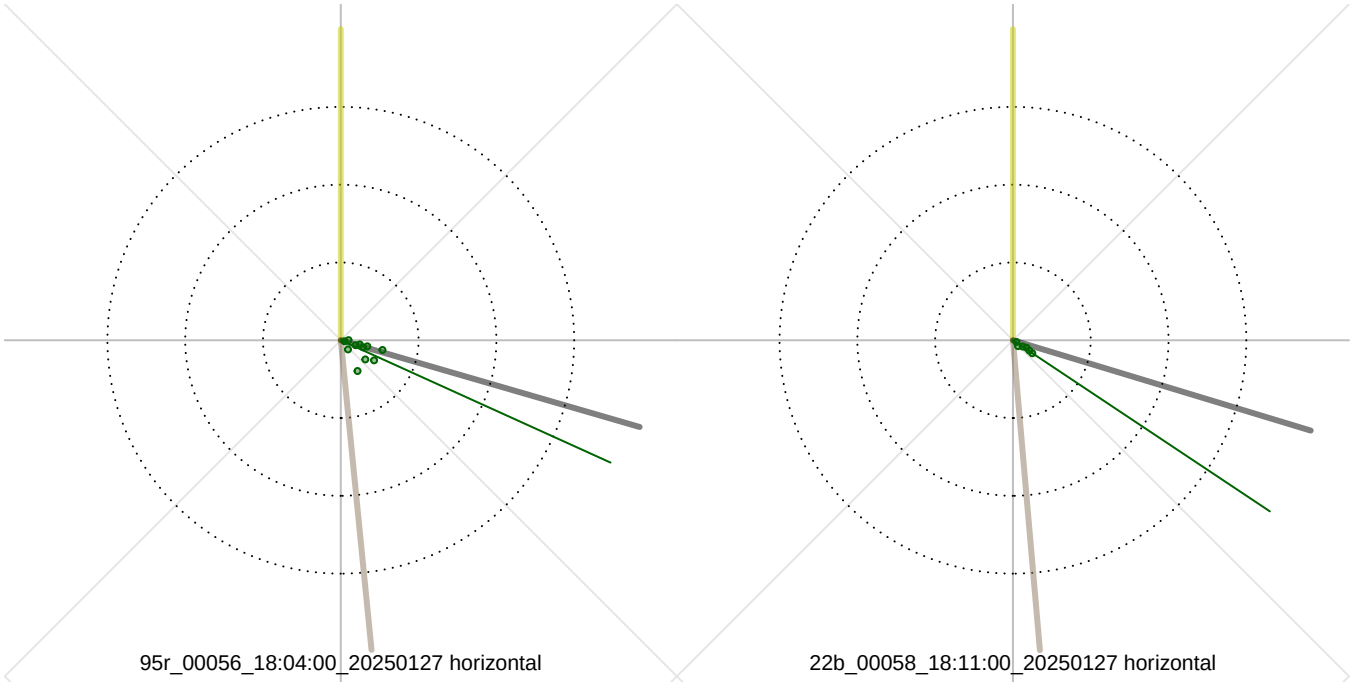


47y_00047_17:39:00_20250127 vertical


47y_00048_17:44:00_20250127 vertical
20y_00049_17:48:00_20250127 vertical
47y_00049_17:48:00_20250127 vertical
66b_00049_17:48:00_20250127 vertical

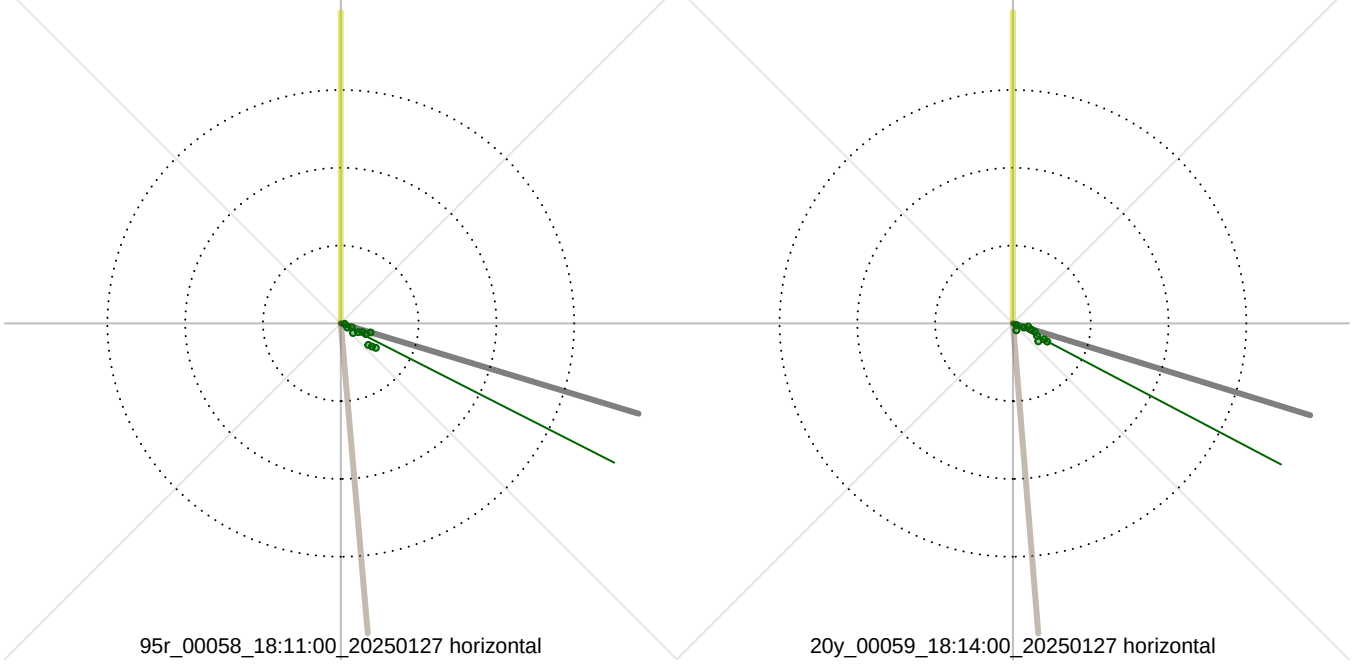

95r_00049_17:48:00_20250127 vertical
84r_00051_17:54:00_20250127 tilted
95r_00051_17:54:00_20250127 tilted
20y_00051_17:55:00_20250127 tilted


20y_00053_17:58:00_20250127 tilted
47y_00052_17:58:00_20250127 tilted
47y_00053_17:58:00_20250127 tilted
47y_00056_18:04:00_20250127 horizontal



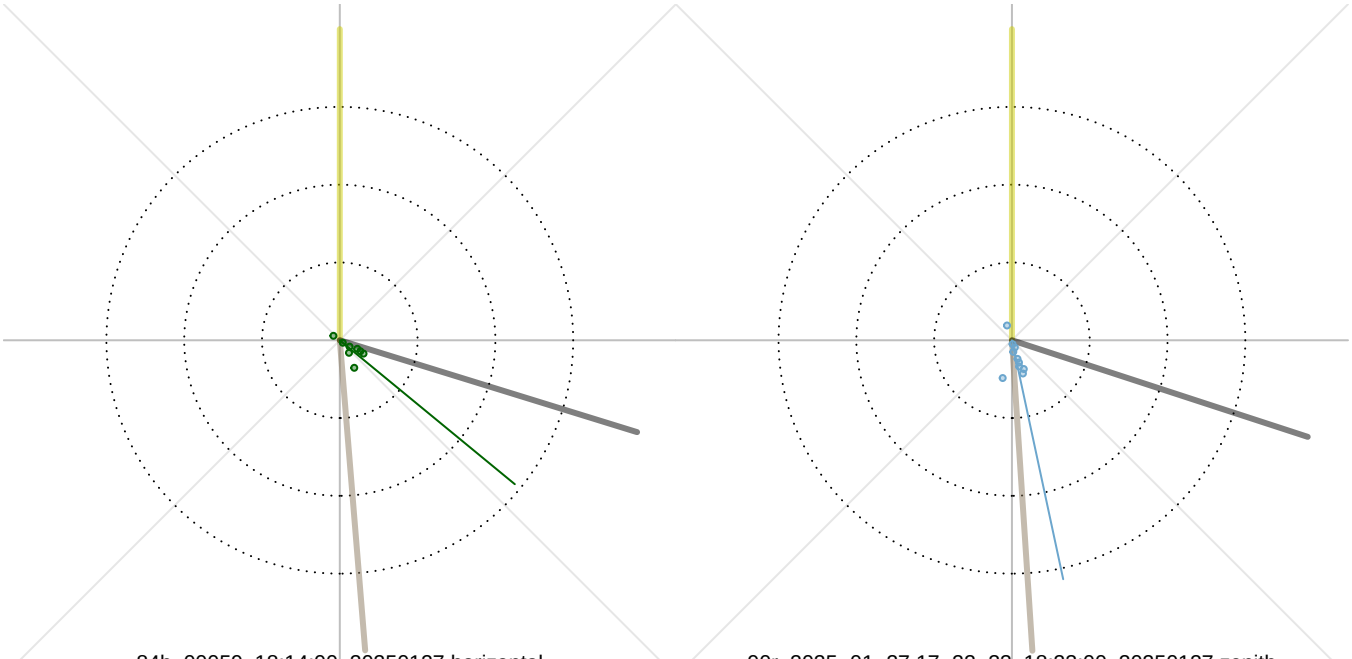
95r_00056_18:04:00_20250127 horizontal

22b_00058_18:11:00_20250127 horizontal



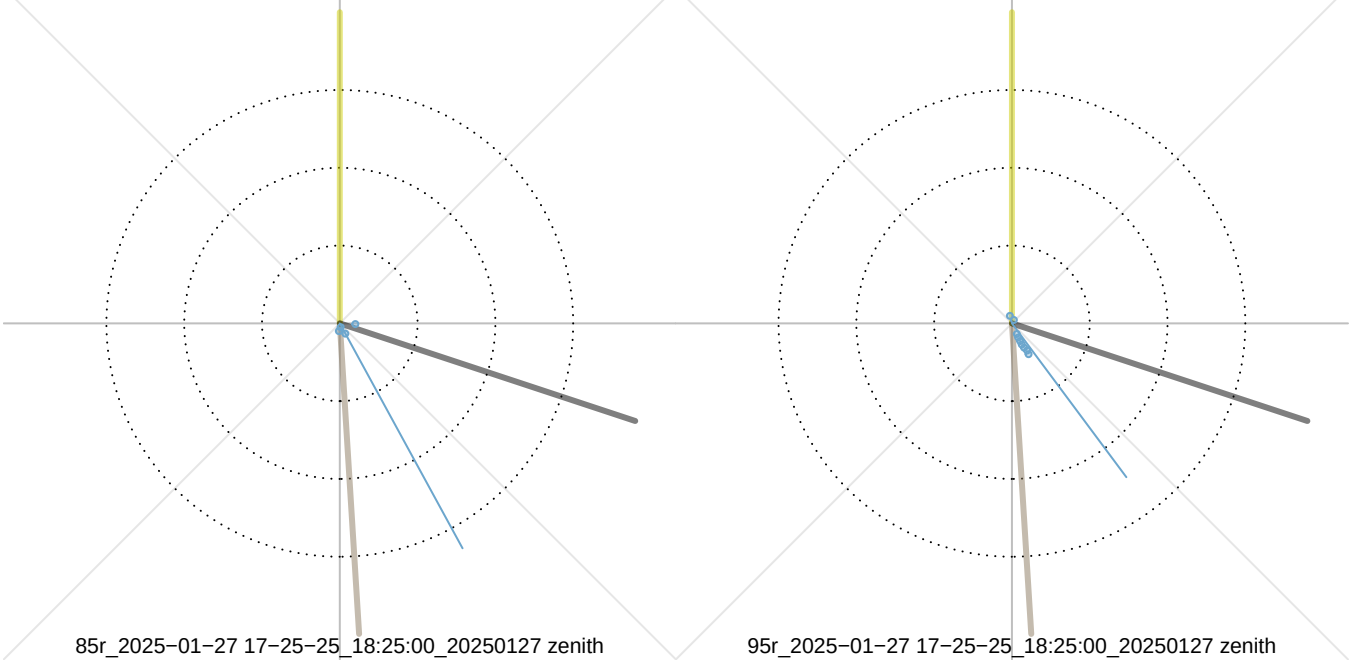
95r_00058_18:11:00_20250127 horizontal

20y_00059_18:14:00_20250127 horizontal



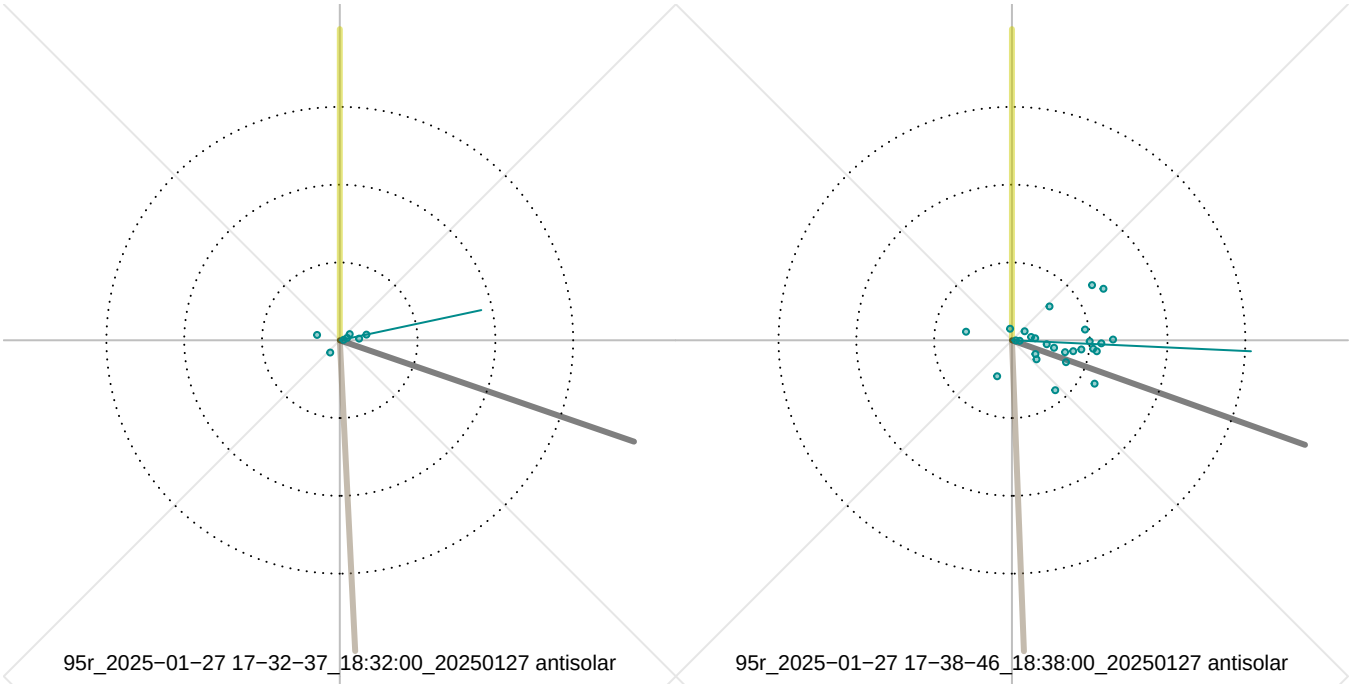
84b_00059_18:14:00_20250127 horizontal

90r_2025−01−27 17−22−33_18:23:00_20250127 zenith



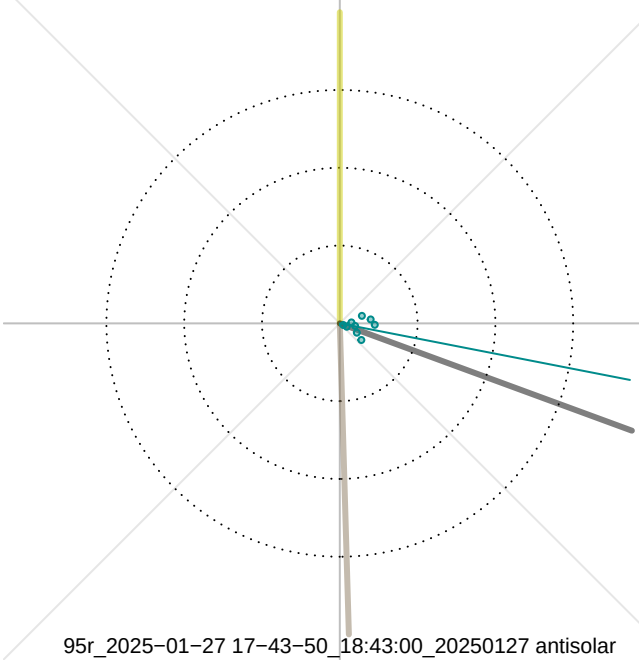
85r_2025−01−27 17−25−25_18:25:00_20250127 zenith

95r_2025−01−27 17−25−25_18:25:00_20250127 zenith

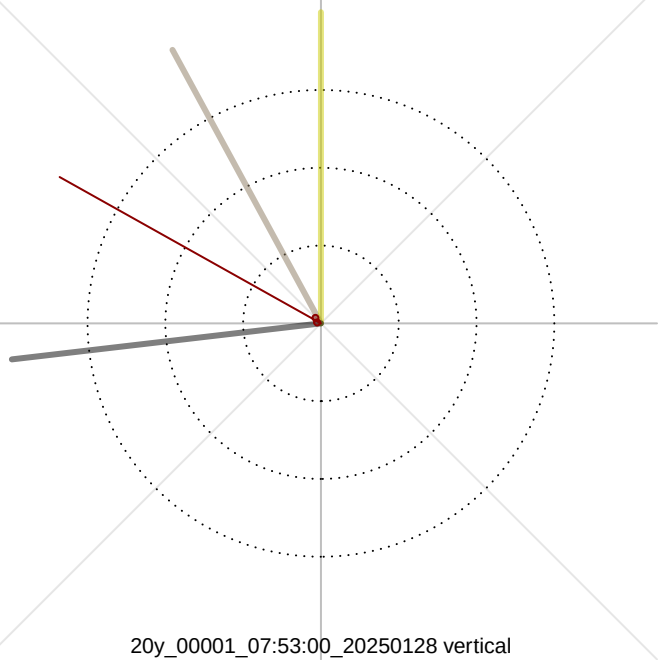


95r_2025−01−27 17−32−37_18:32:00_20250127 antisolar

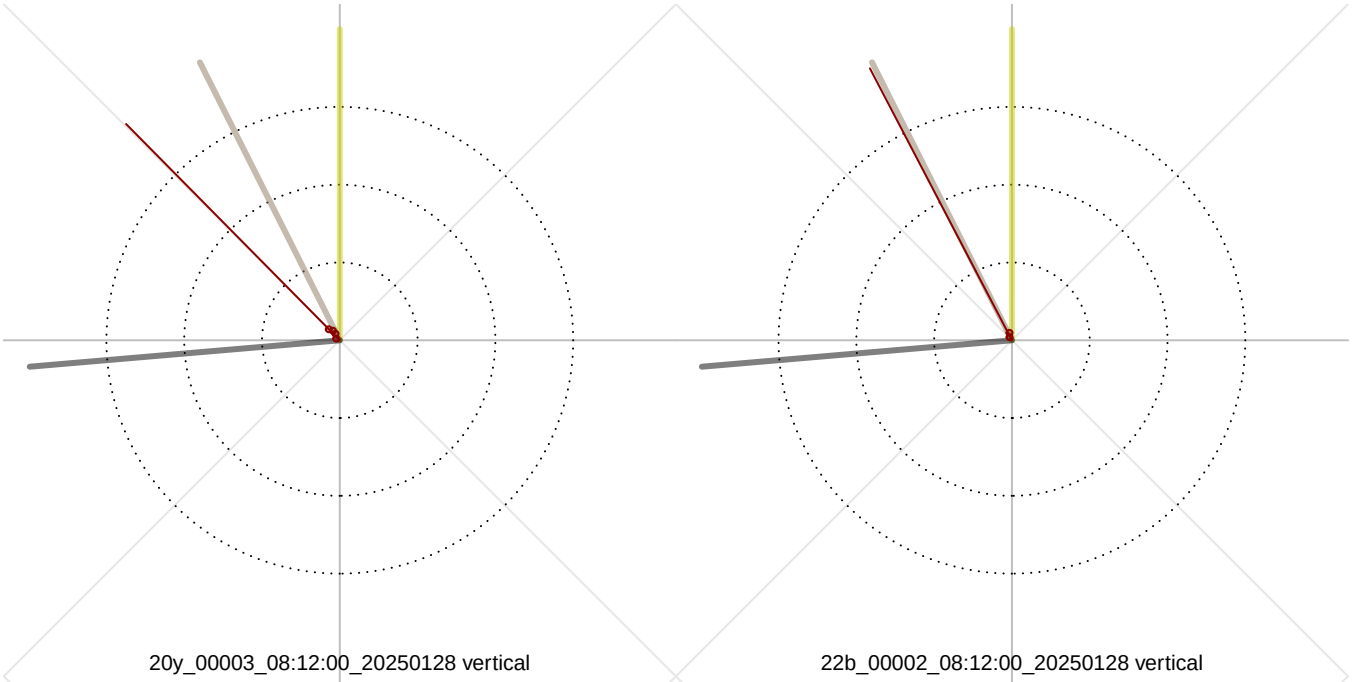
95r_2025−01−27 17−38−46_18:38:00_20250127 antisolar



95r_2025−01−27 17−43−50_18:43:00_20250127 antisolar

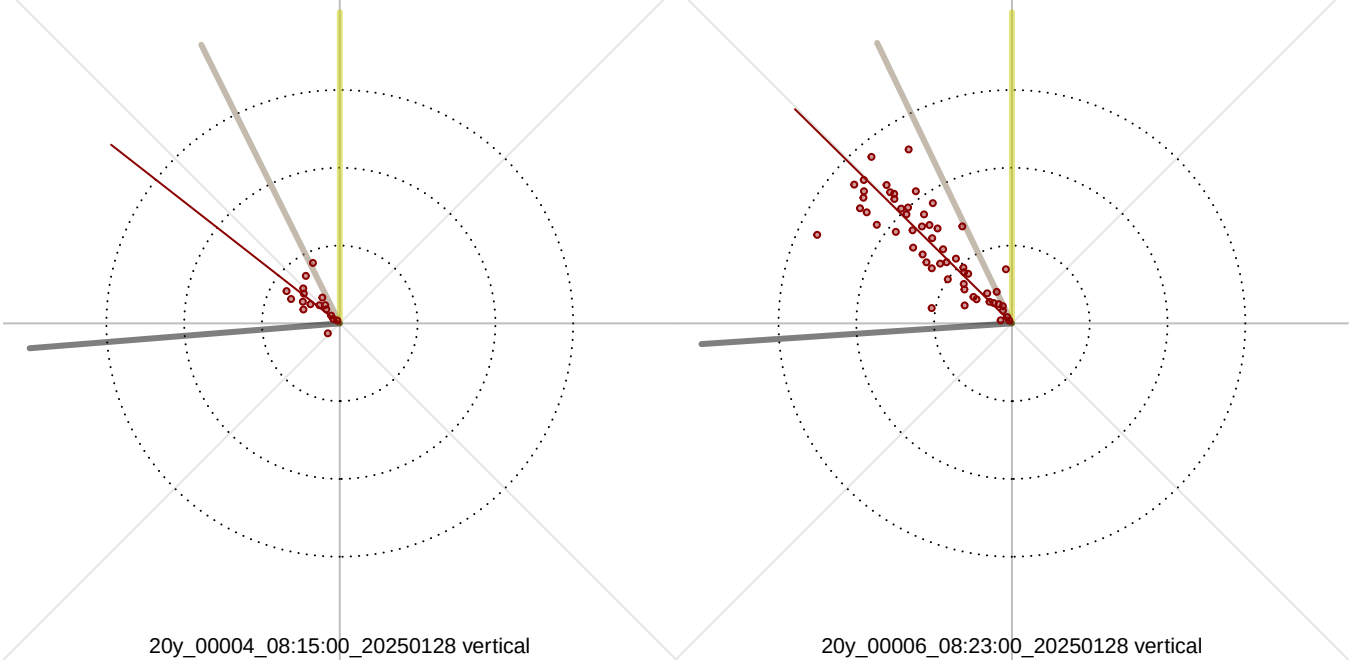


20y_00001_07:53:00_20250128 vertical



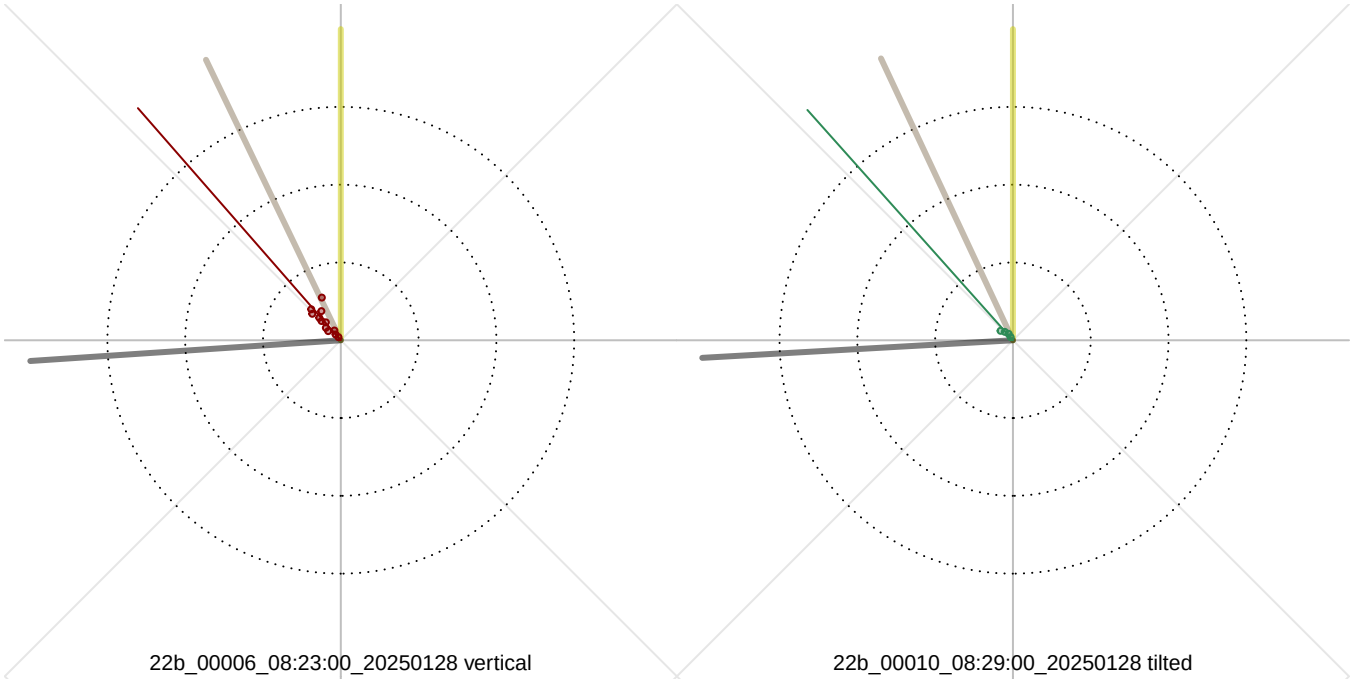
20y_00003_08:12:00_20250128 vertical

22b_00002_08:12:00_20250128 vertical



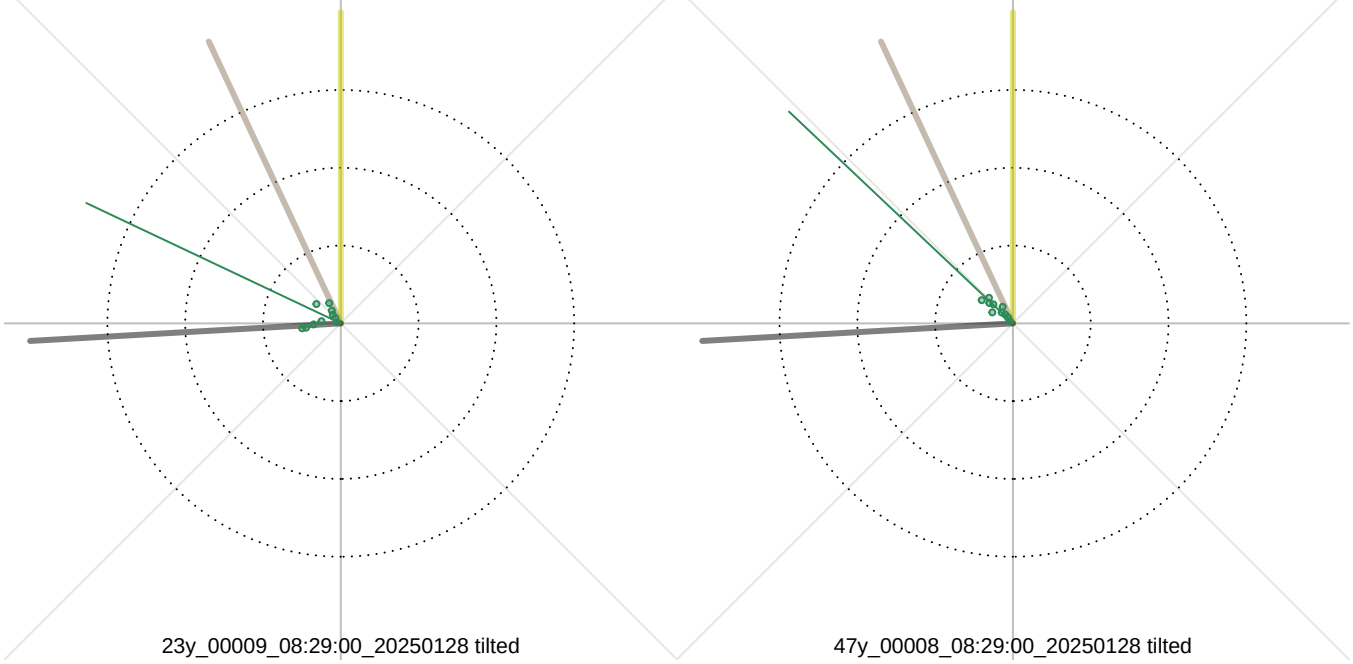
20y_00004_08:15:00_20250128 vertical

20y_00006_08:23:00_20250128 vertical



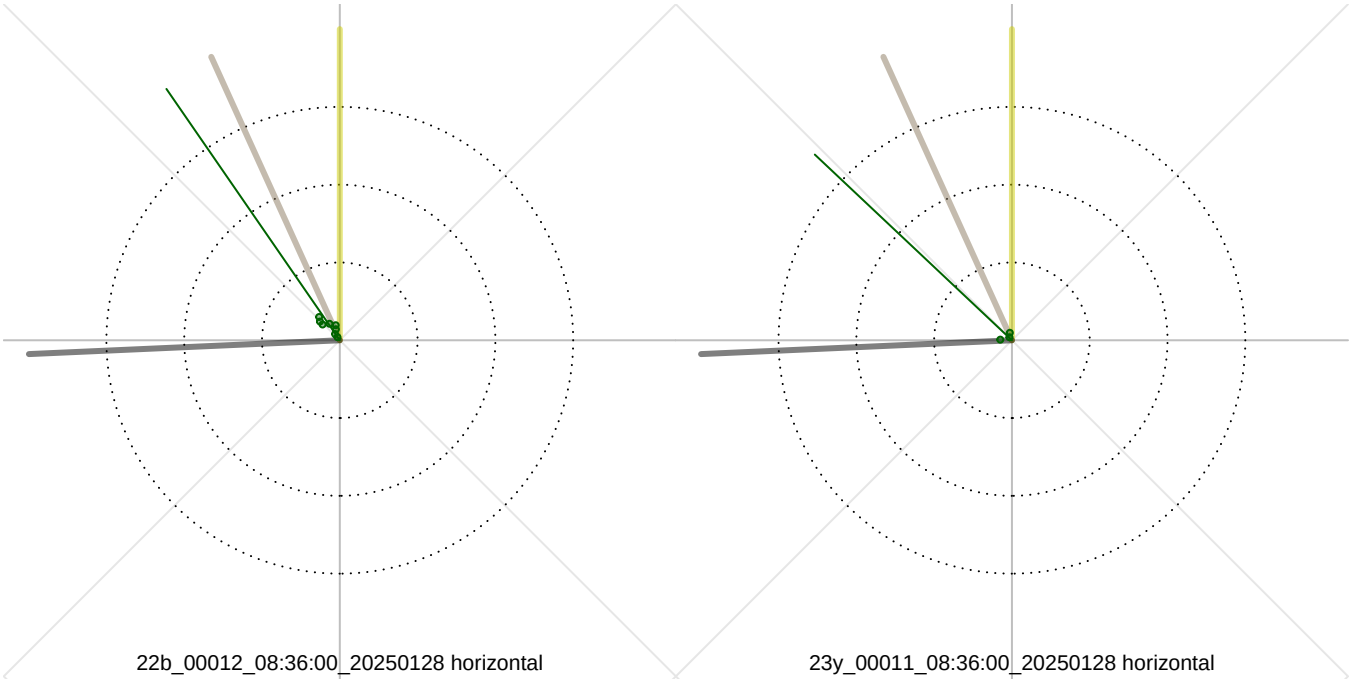
22b_00006_08:23:00_20250128 vertical

22b_00010_08:29:00_20250128 tilted



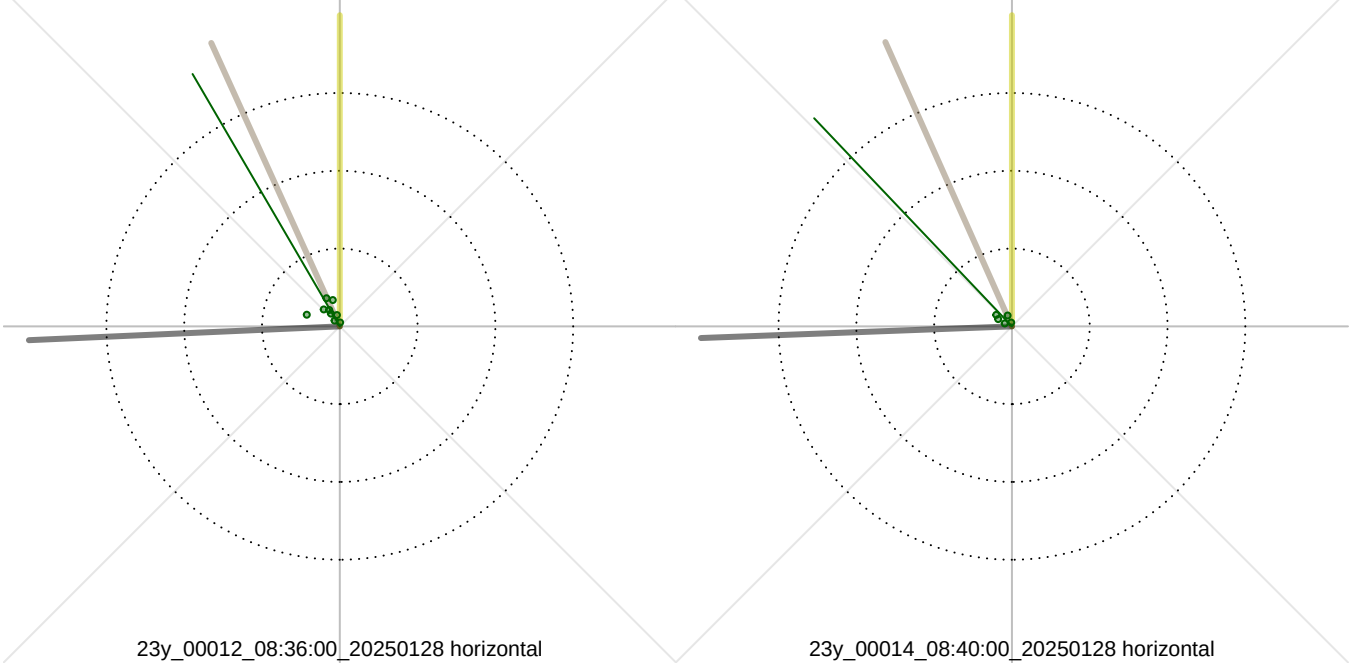
23y_00009_08:29:00_20250128 tilted

47y_00008_08:29:00_20250128 tilted



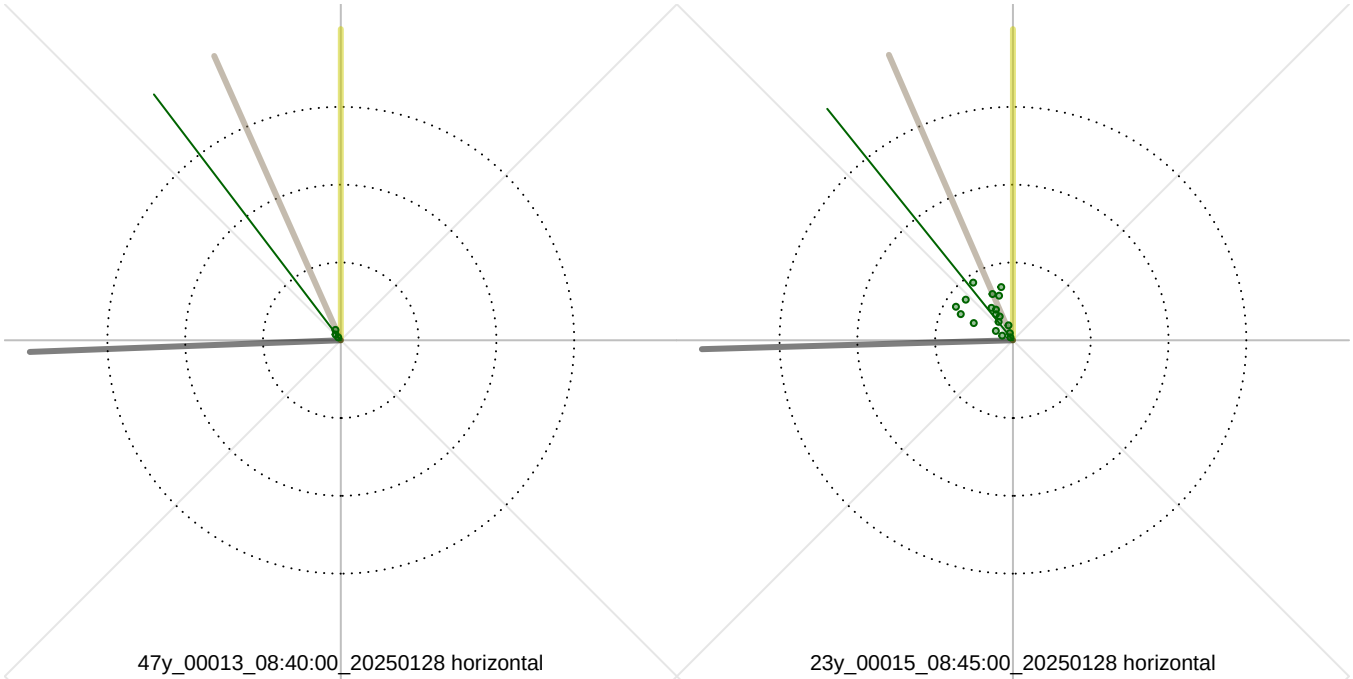
22b_00012_08:36:00_20250128 horizontal

23y_00011_08:36:00_20250128 horizontal



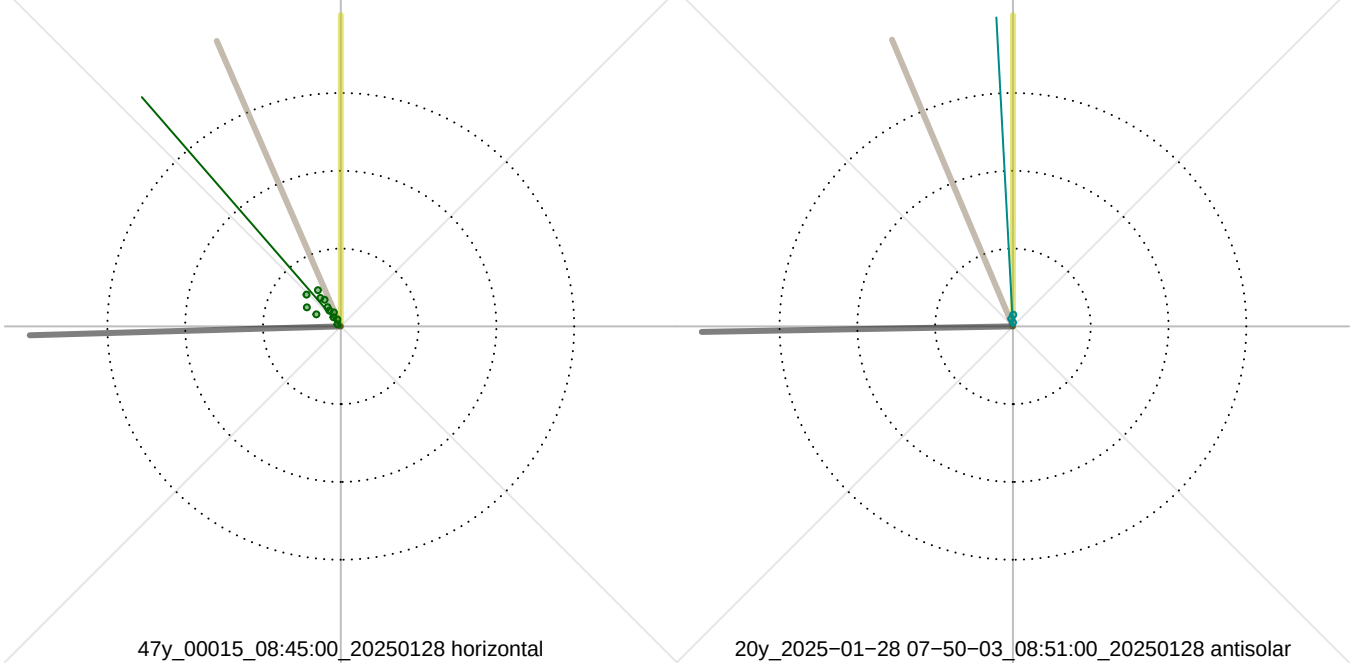
23y_00012_08:36:00_20250128 horizontal

23y_00014_08:40:00_20250128 horizontal



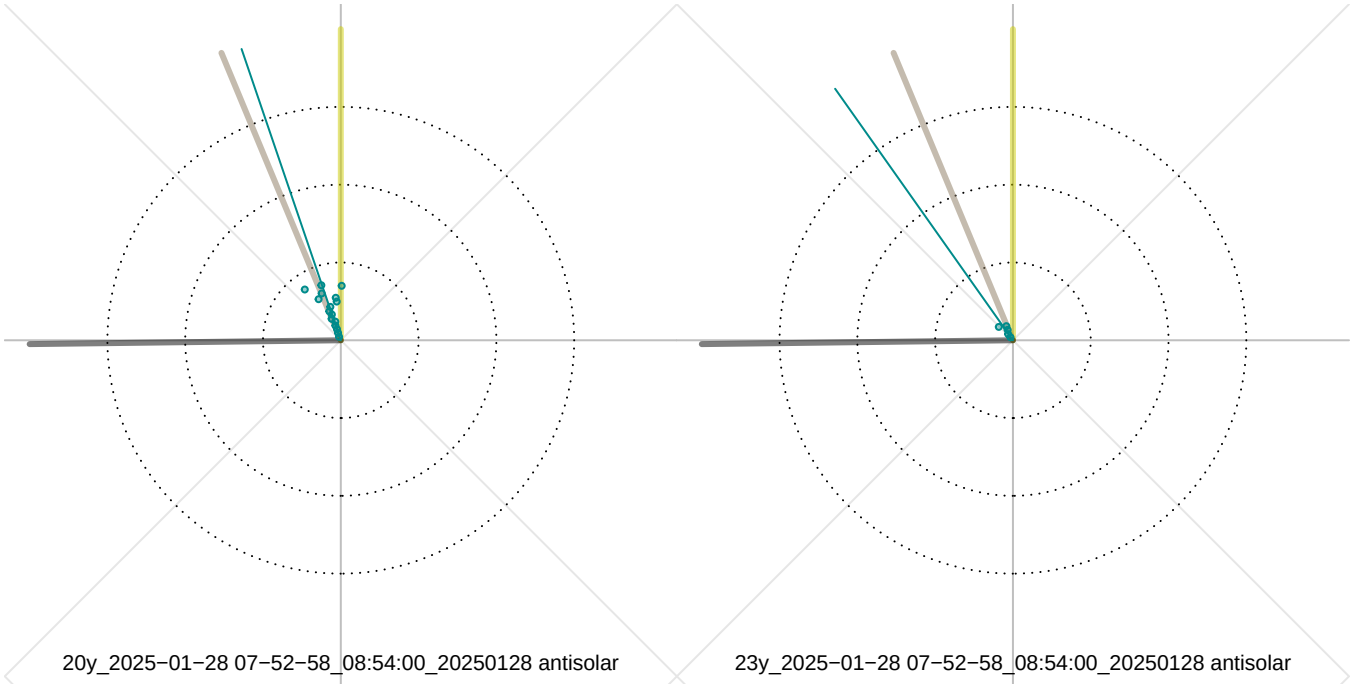
47y_00013_08:40:00_20250128 horizontal

23y_00015_08:45:00_20250128 horizontal



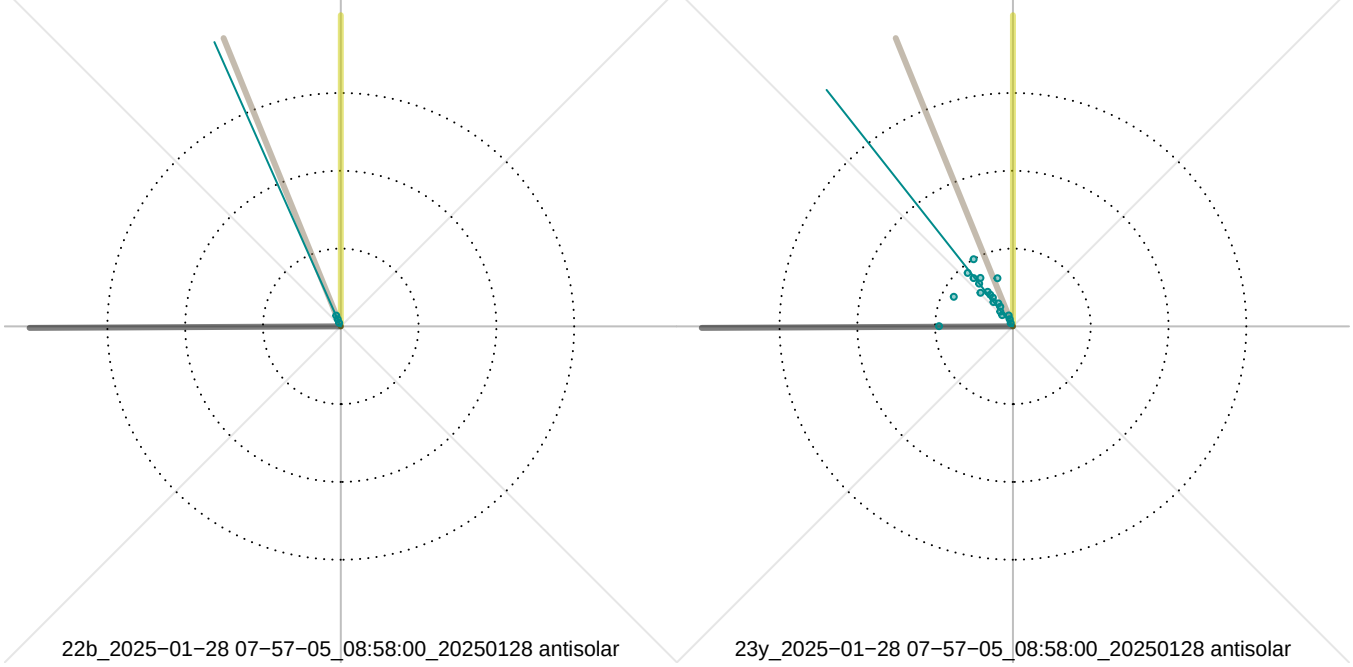
47y_00015_08:45:00_20250128 horizontal

20y_2025−01−28 07−50−03_08:51:00_20250128 antisolar



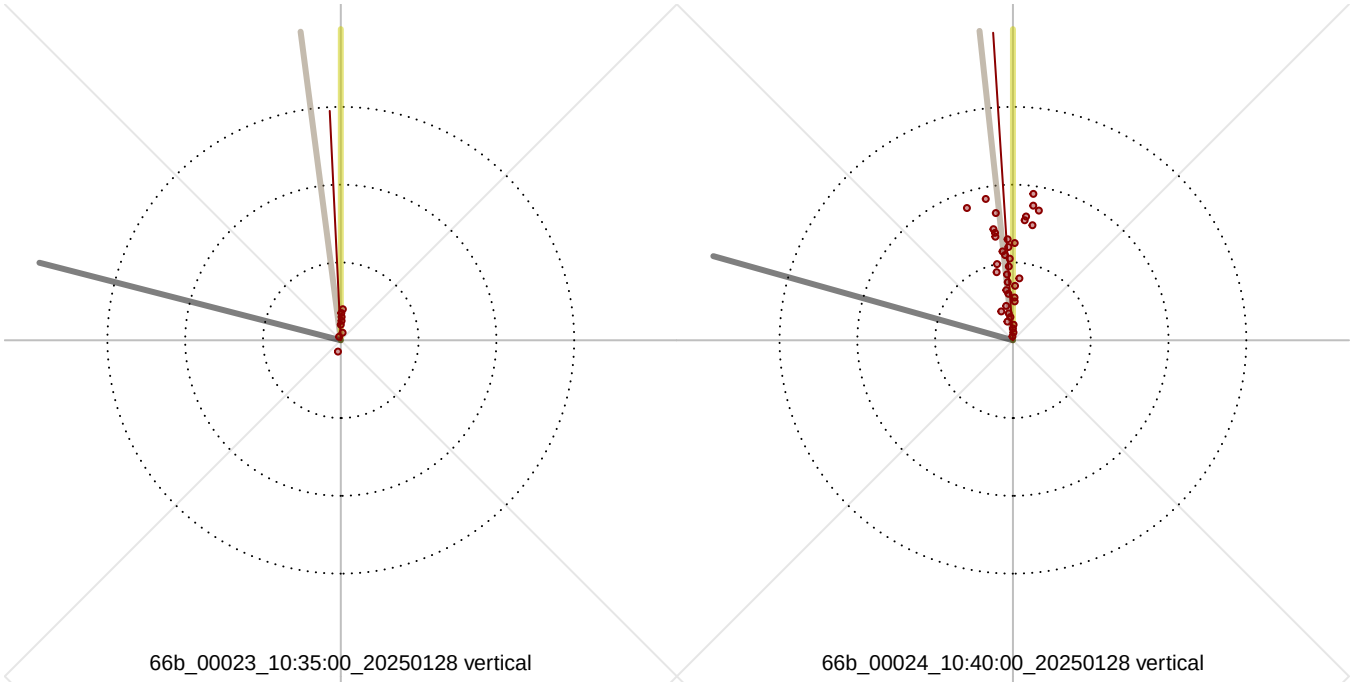
20y_2025−01−28 07−52−58_08:54:00_20250128 antisolar

23y_2025−01−28 07−52−58_08:54:00_20250128 antisolar



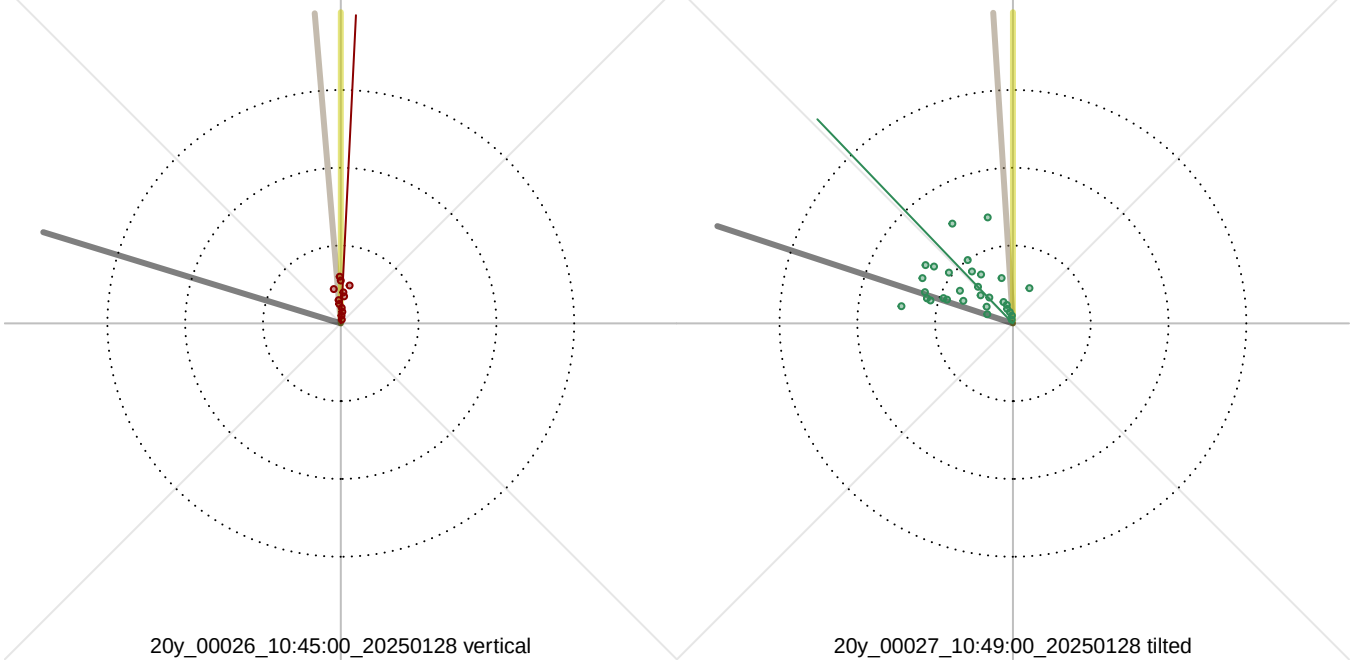
22b_2025−01−28 07−57−05_08:58:00_20250128 antisolar

23y_2025−01−28 07−57−05_08:58:00_20250128 antisolar



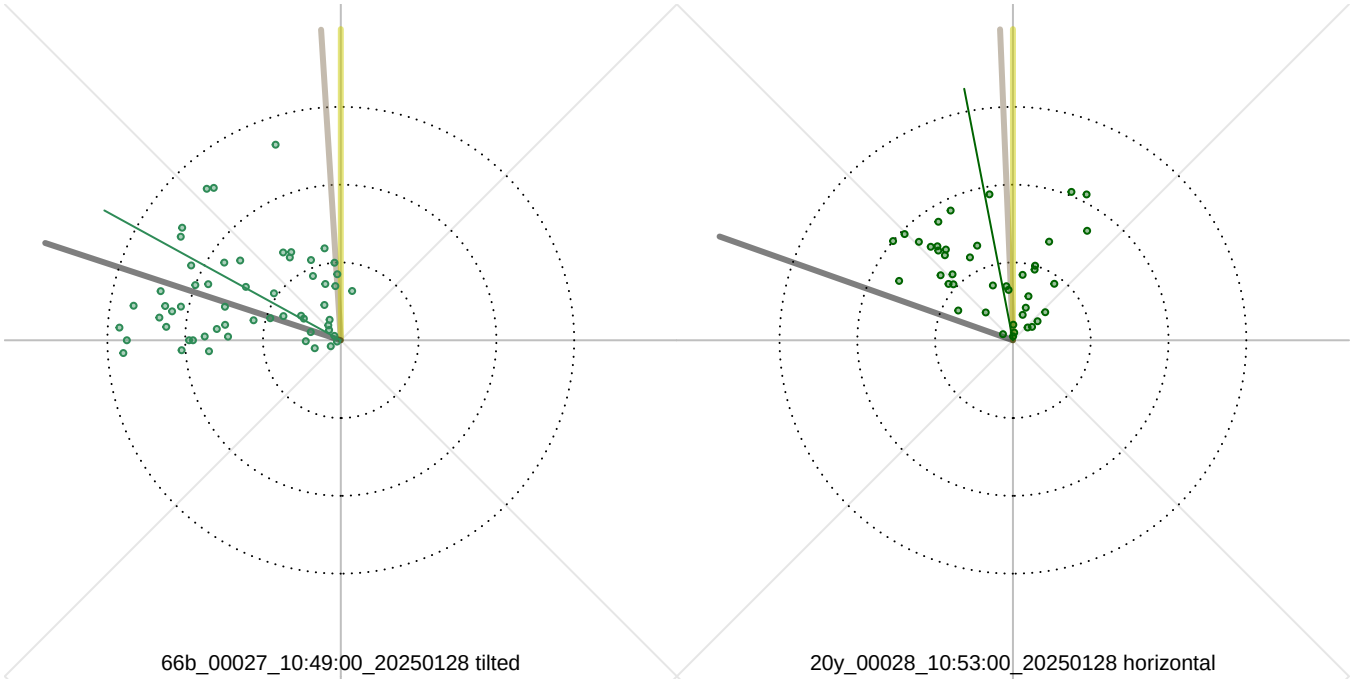
66b_00023_10:35:00_20250128 vertical

66b_00024_10:40:00_20250128 vertical



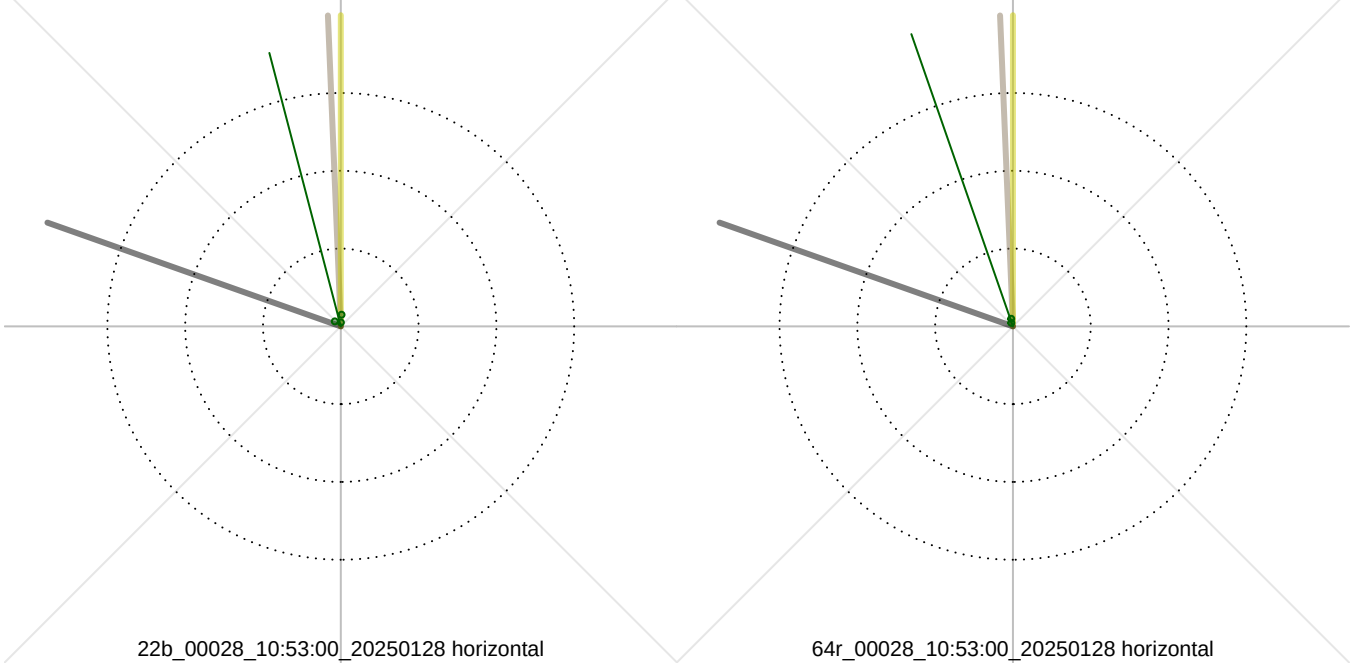
20y_00026_10:45:00_20250128 vertical

20y_00027_10:49:00_20250128 tilted



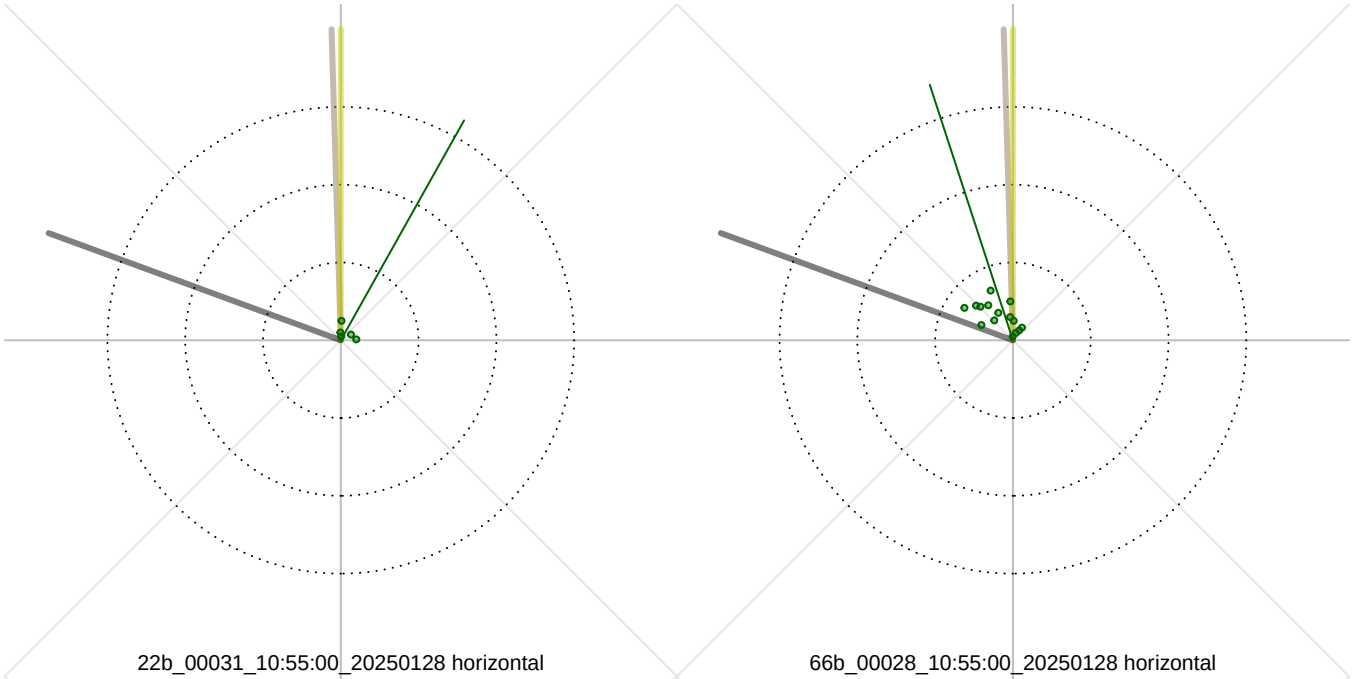
66b_00027_10:49:00_20250128 tilted

20y_00028_10:53:00_20250128 horizontal



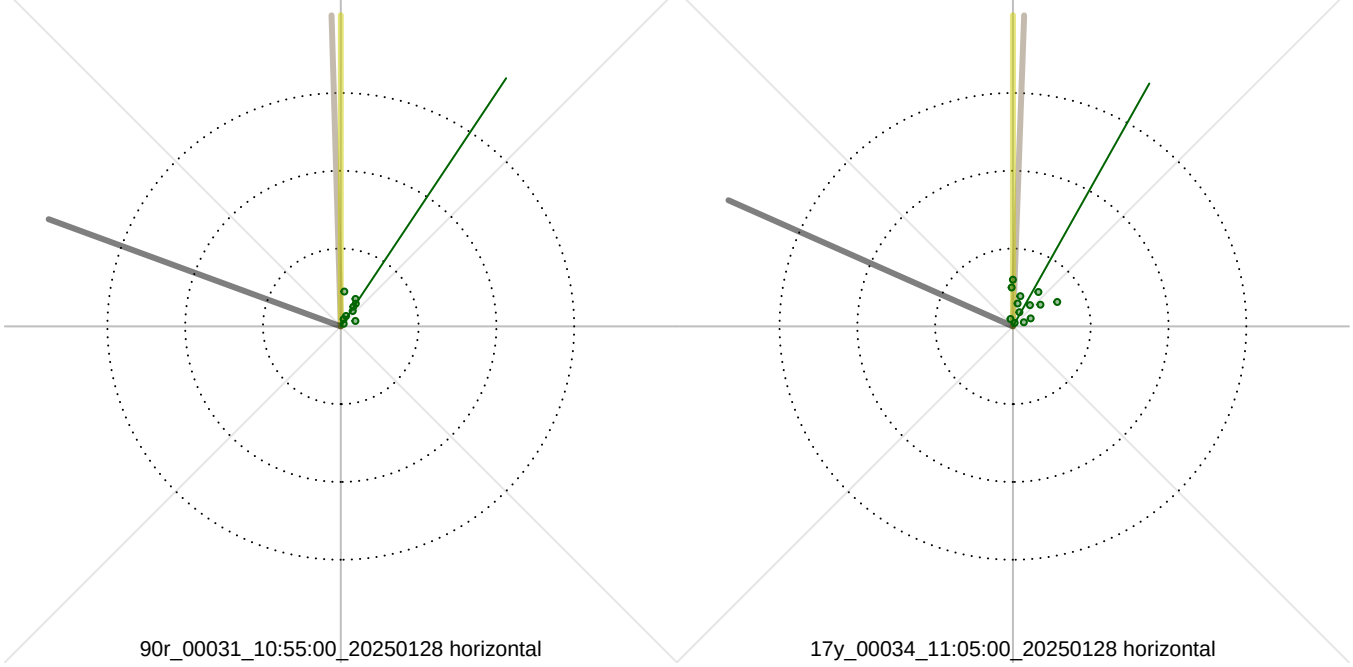
22b_00028_10:53:00_20250128 horizontal

64r_00028_10:53:00_20250128 horizontal



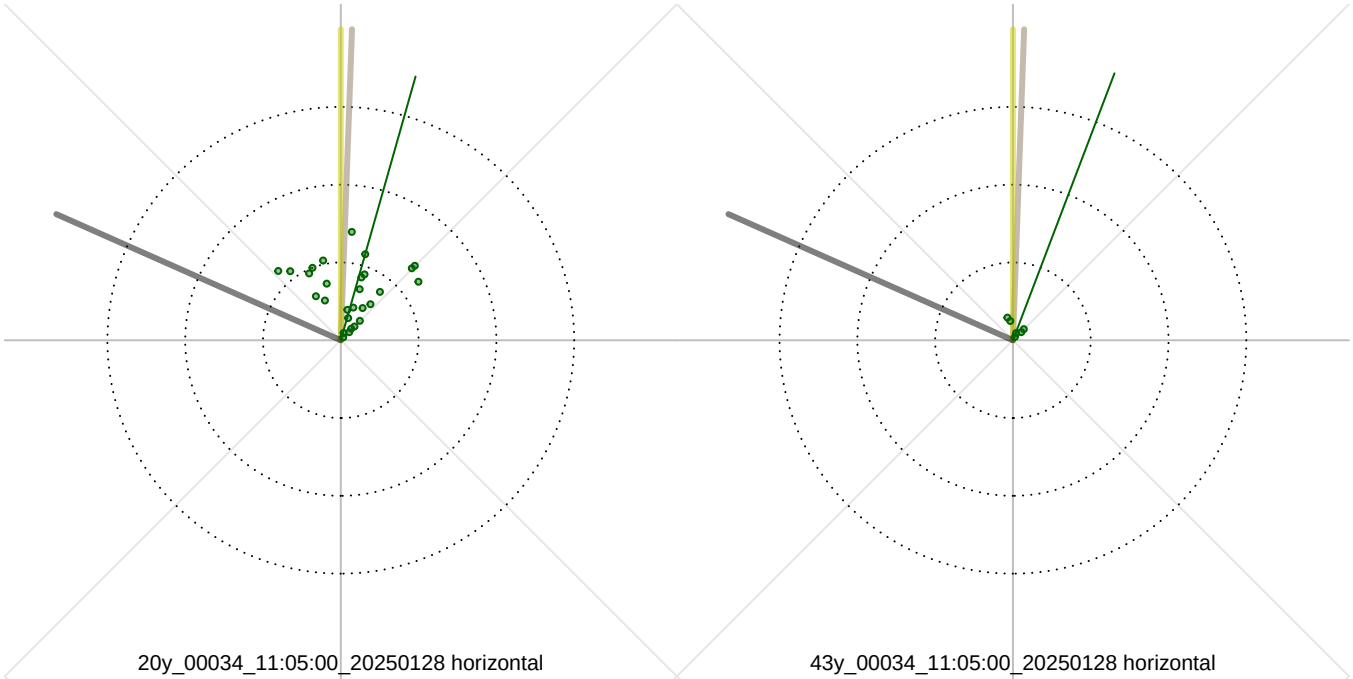
22b_00031_10:55:00_20250128 horizontal

66b_00028_10:55:00_20250128 horizontal



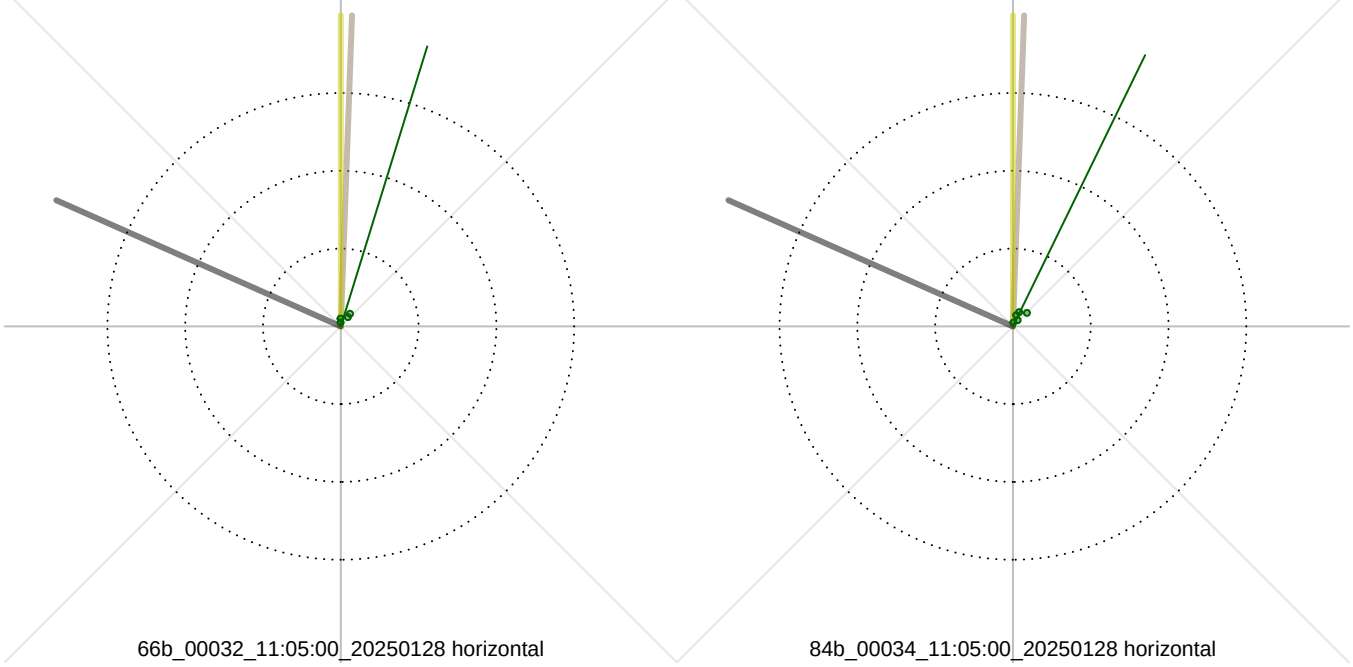
90r_00031_10:55:00_20250128 horizontal

17y_00034_11:05:00_20250128 horizontal



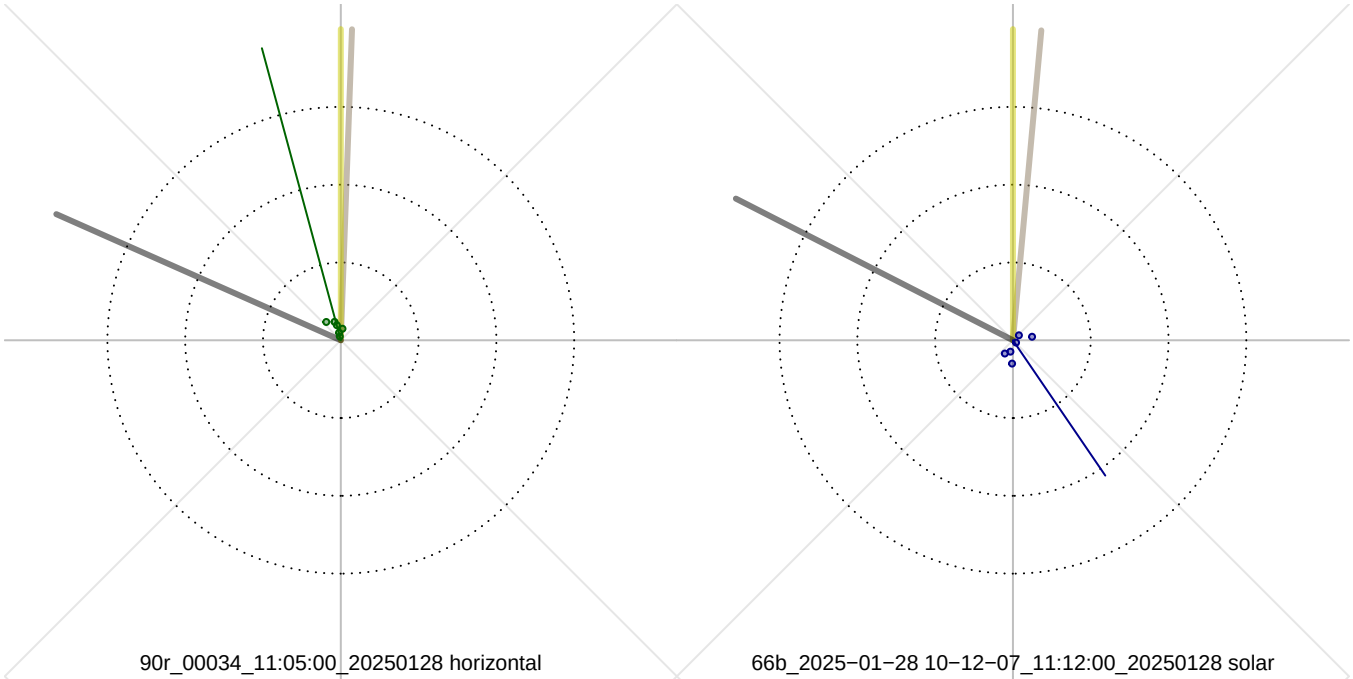
20y_00034_11:05:00_20250128 horizontal

43y_00034_11:05:00_20250128 horizontal



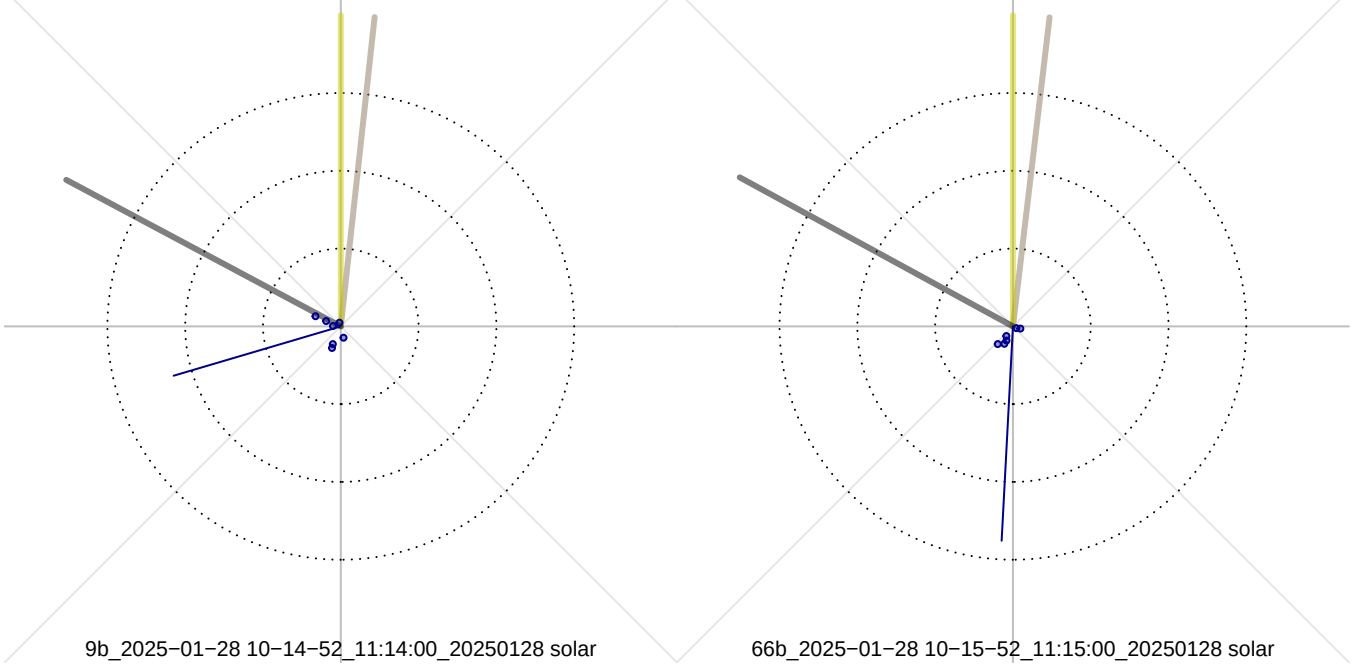
66b_00032_11:05:00_20250128 horizontal

84b_00034_11:05:00_20250128 horizontal



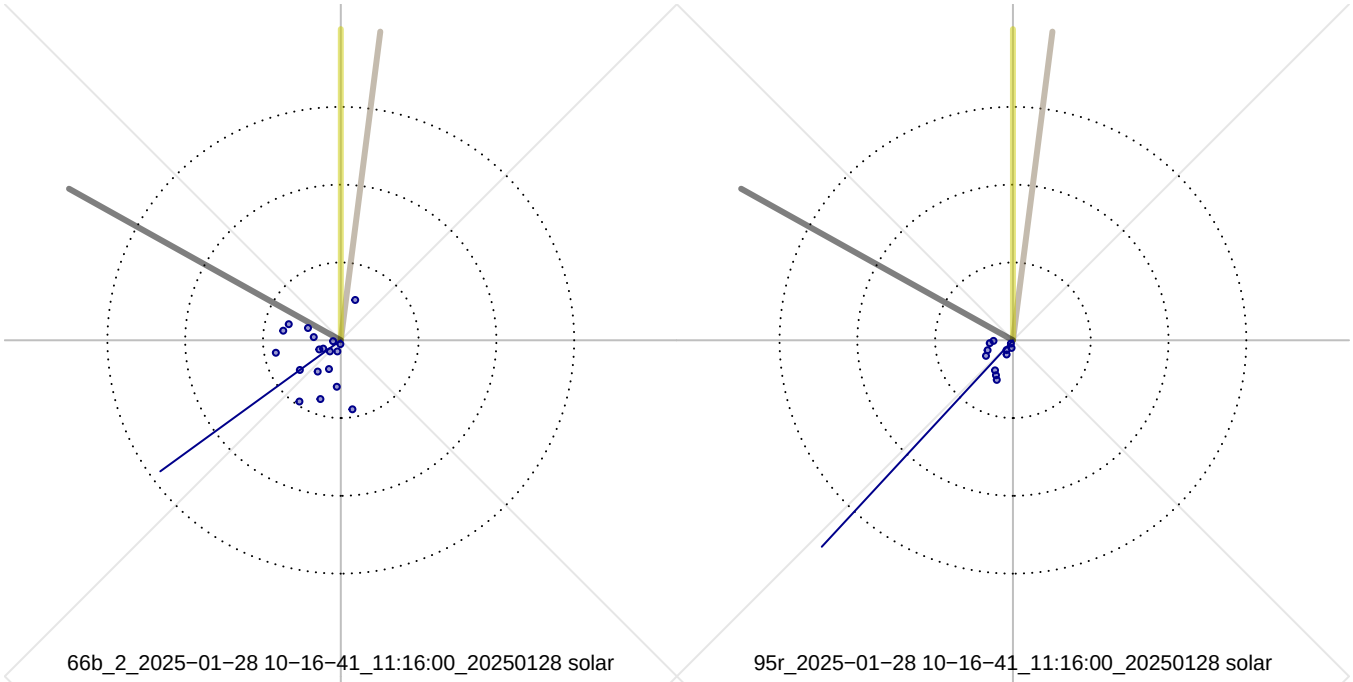
90r_00034_11:05:00_20250128 horizontal

66b_2025−01−28 10−12−07_11:12:00_20250128 solar

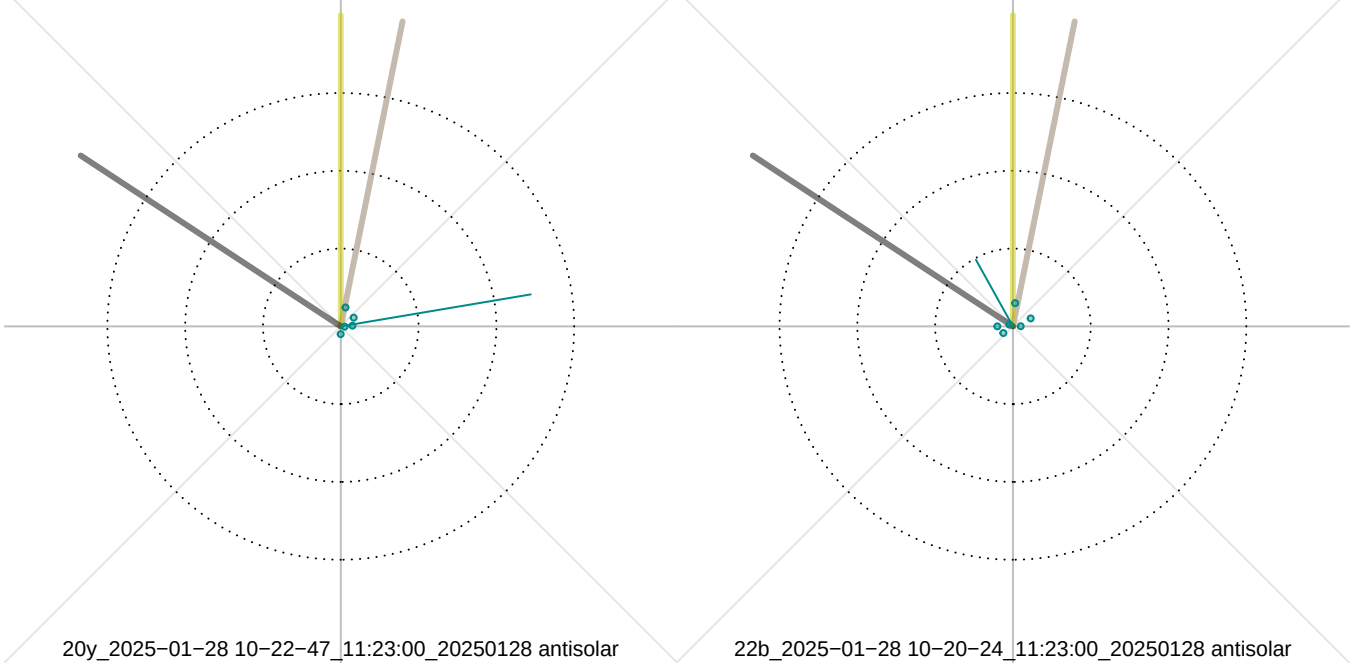


9b_2025−01−28 10−14−52_11:14:00_20250128 solar

66b_2025−01−28 10−15−52_11:15:00_20250128 solar



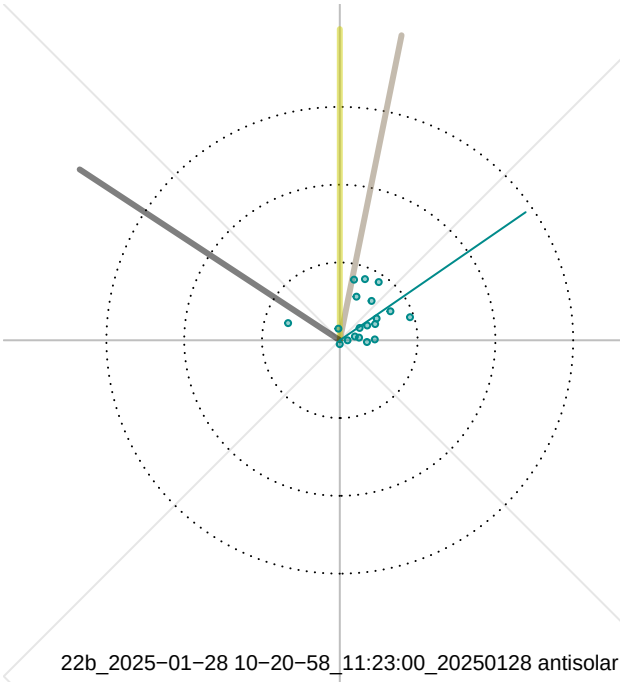
66b_2_2025−01−28 10−16−41_11:16:00_20250128 solar



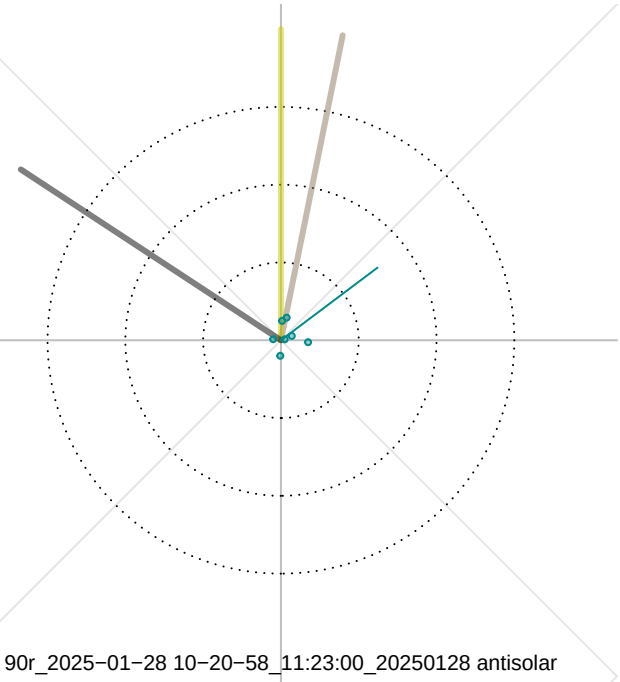
20y_2025−01−28 10−22−47_11:23:00_20250128 antisolar

95r_2025−01−28 10−16−41_11:16:00_20250128 solar

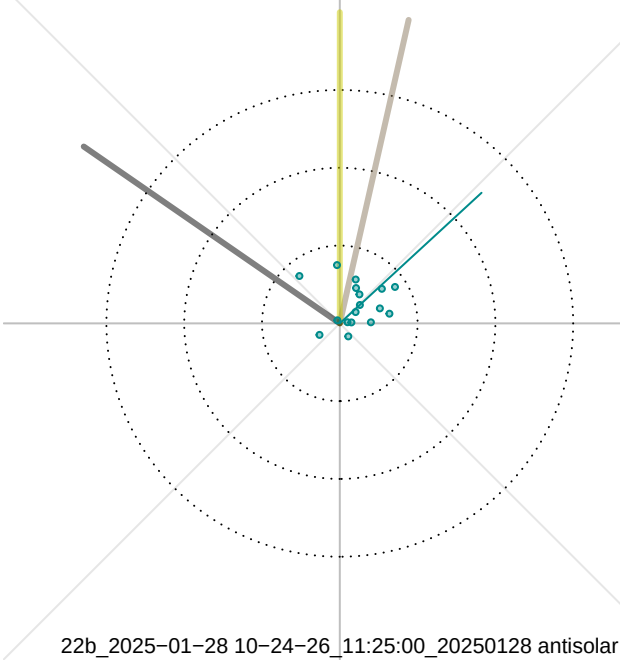
22b_2025−01−28 10−20−24_11:23:00_20250128 antisolar



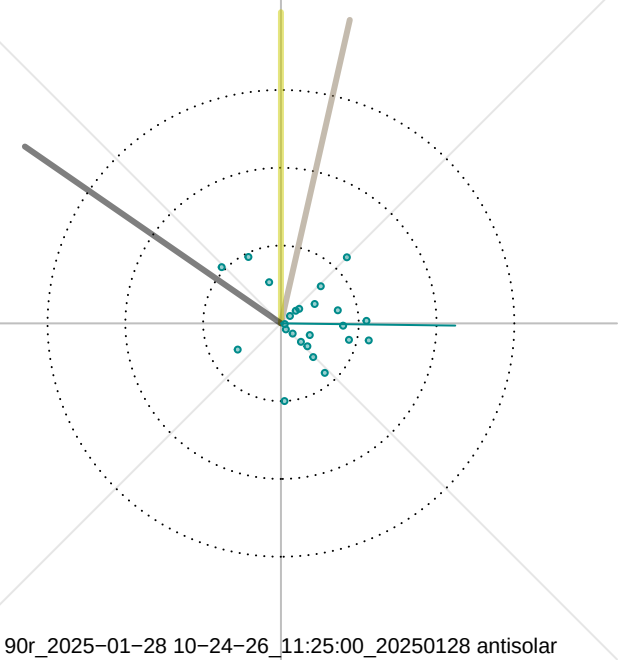
22b_2025−01−28 10−20−58_11:23:00_20250128 antisolar



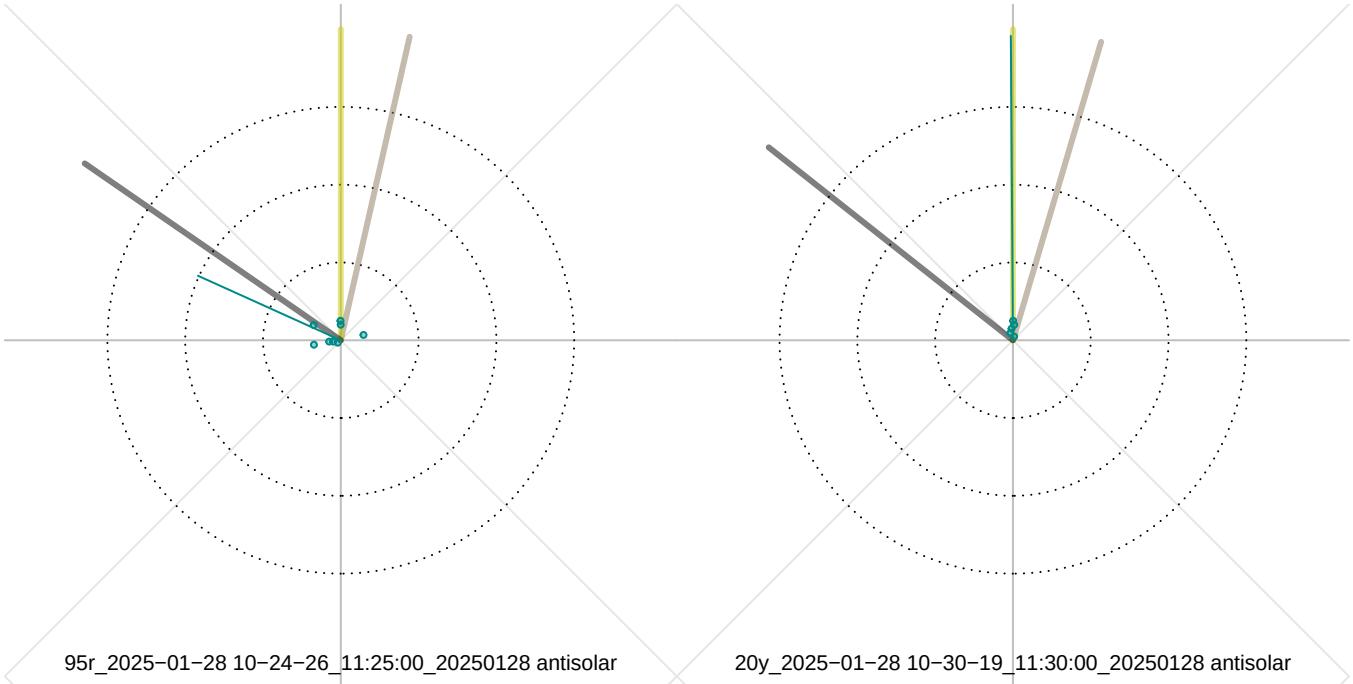
90r_2025−01−28 10−20−58_11:23:00_20250128 antisolar



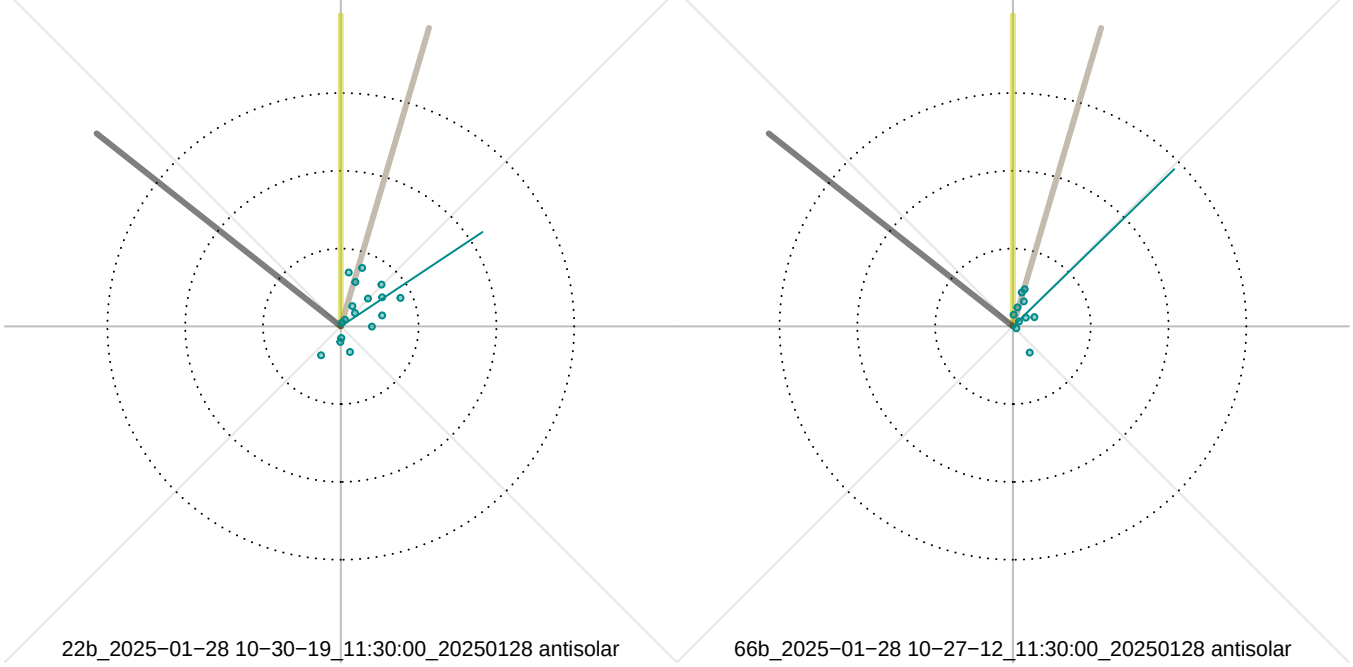
22b_2025−01−28 10−24−26_11:25:00_20250128 antisolar



90r_2025−01−28 10−24−26_11:25:00_20250128 antisolar



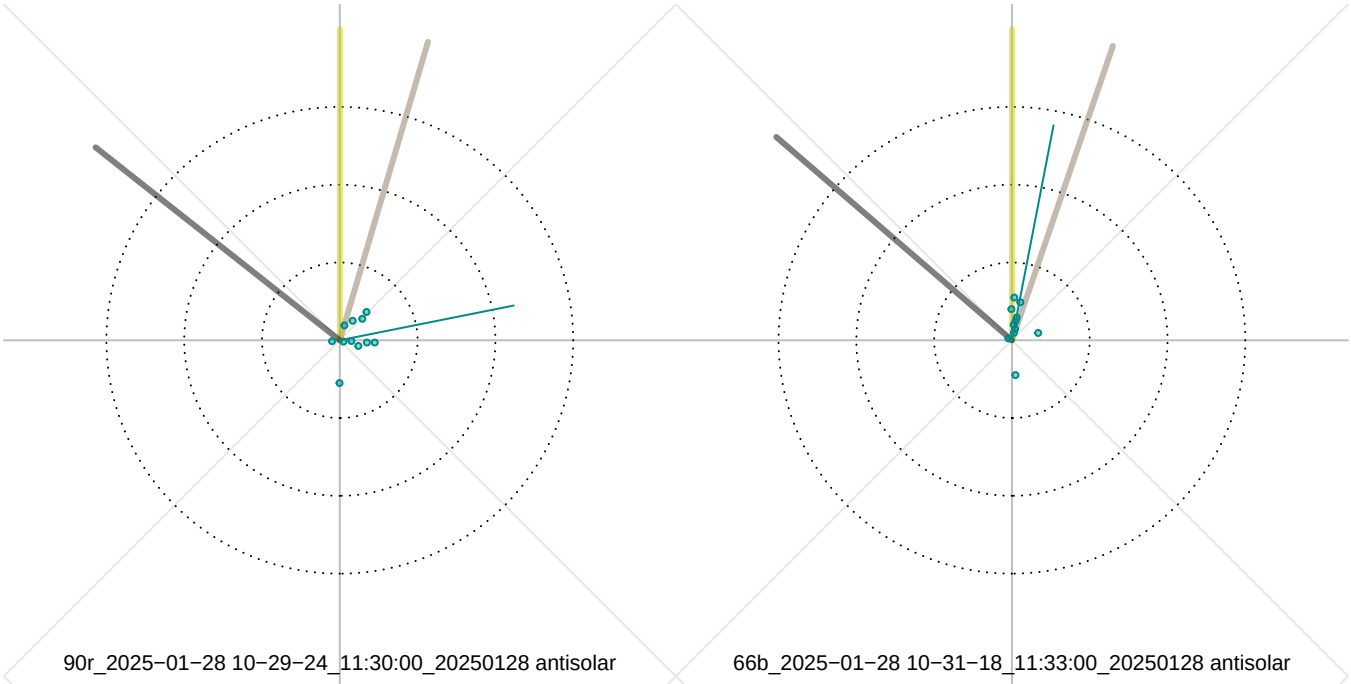
95r_2025−01−28 10−24−26_11:25:00_20250128 antisolar



22b_2025−01−28 10−30−19_11:30:00_20250128 antisolar

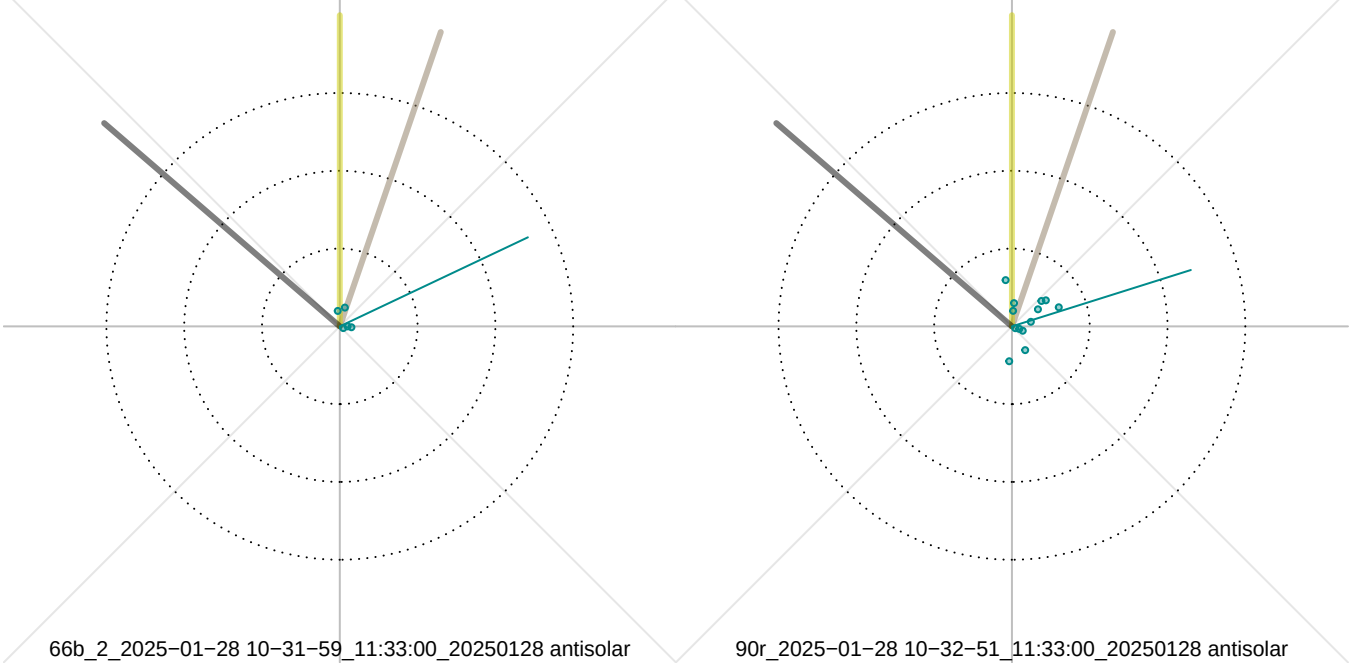
20y_2025−01−28 10−30−19_11:30:00_20250128 antisolar

66b_2025−01−28 10−27−12_11:30:00_20250128 antisolar



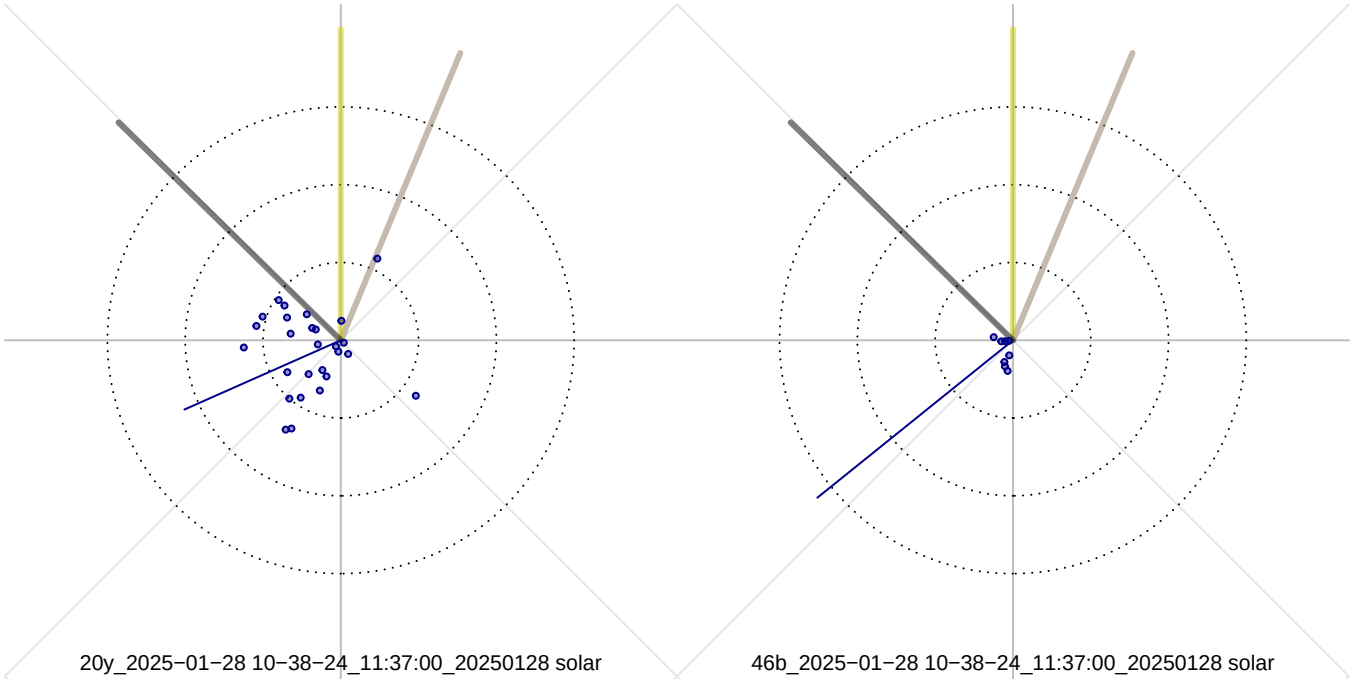
90r_2025−01−28 10−29−24_11:30:00_20250128 antisolar

66b_2025−01−28 10−31−18_11:33:00_20250128 antisolar



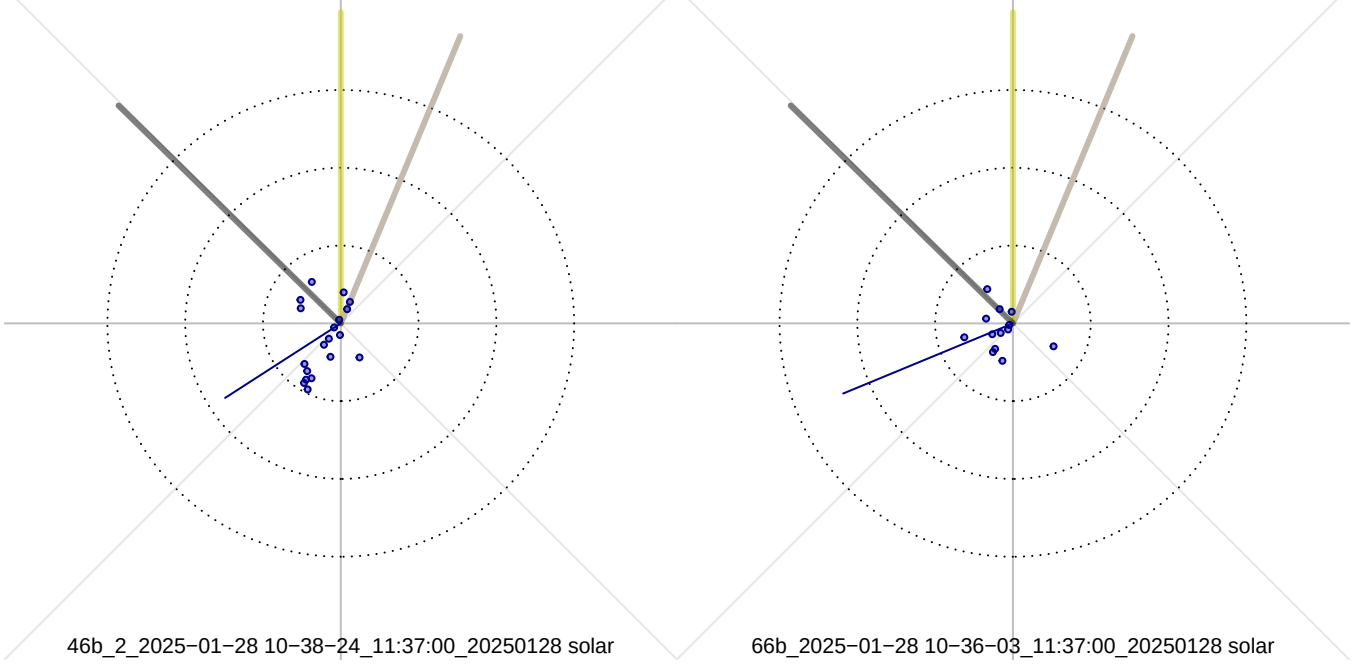
66b_2_2025−01−28 10−31−59_11:33:00_20250128 antisolar

90r_2025−01−28 10−32−51_11:33:00_20250128 antisolar



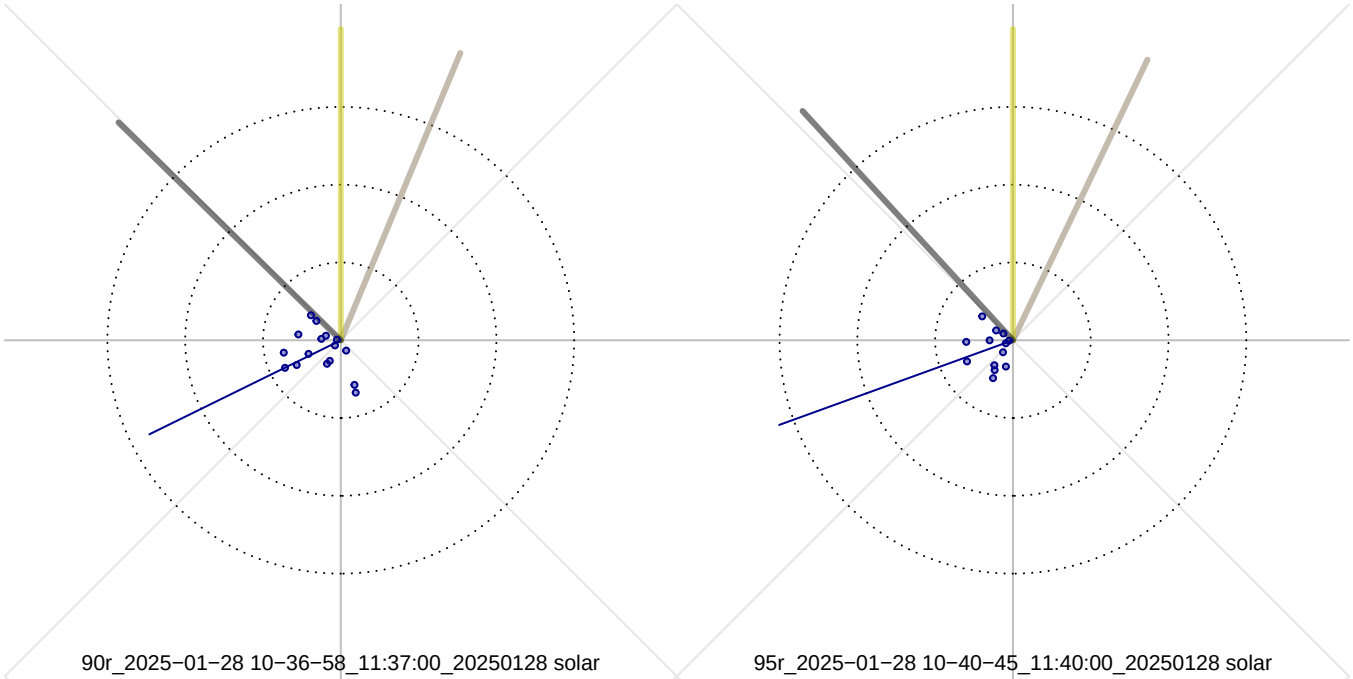
20y_2025-01-28 10-38-24_11:37:00_20250128 solar

46b_2025-01-28 10-38-24_11:37:00_20250128 solar

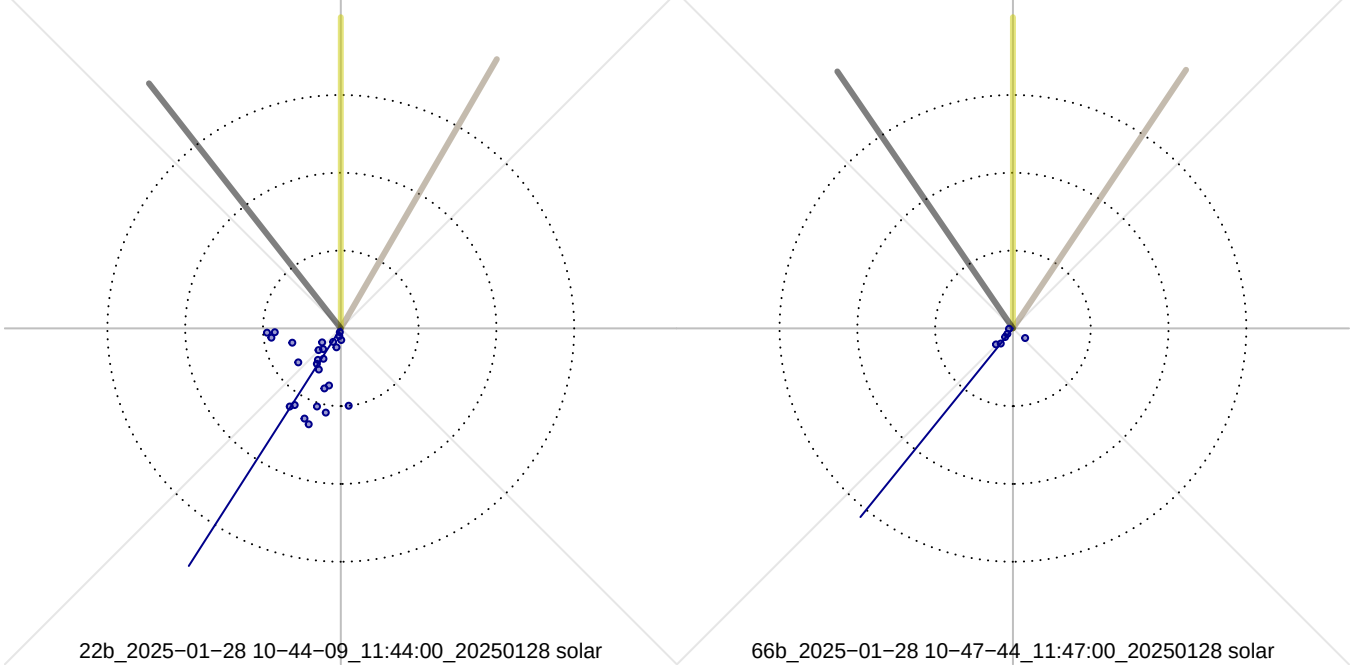


46b_2_2025-01-28 10-38-24_11:37:00_20250128 solar

66b_2025-01-28 10-36-03_11:37:00_20250128 solar



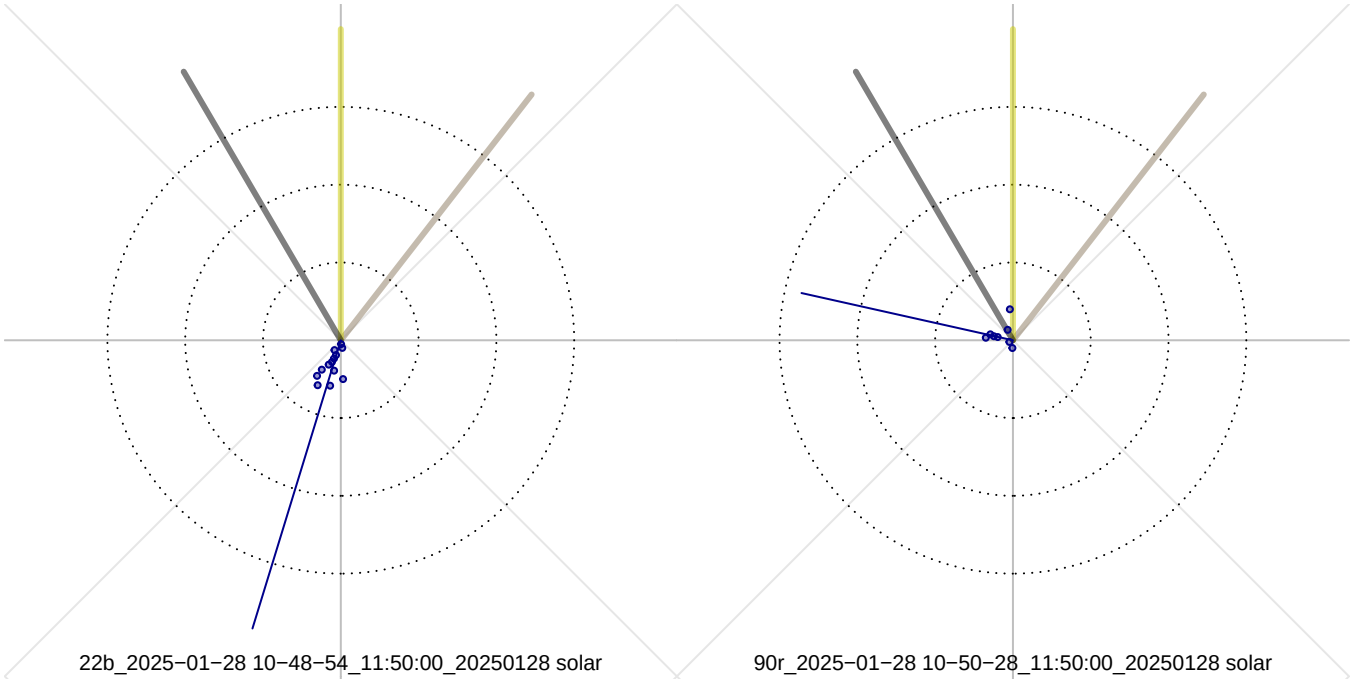
90r_2025−01−28 10−36−58_11:37:00_20250128 solar



22b_2025−01−28 10−44−09_11:44:00_20250128 solar

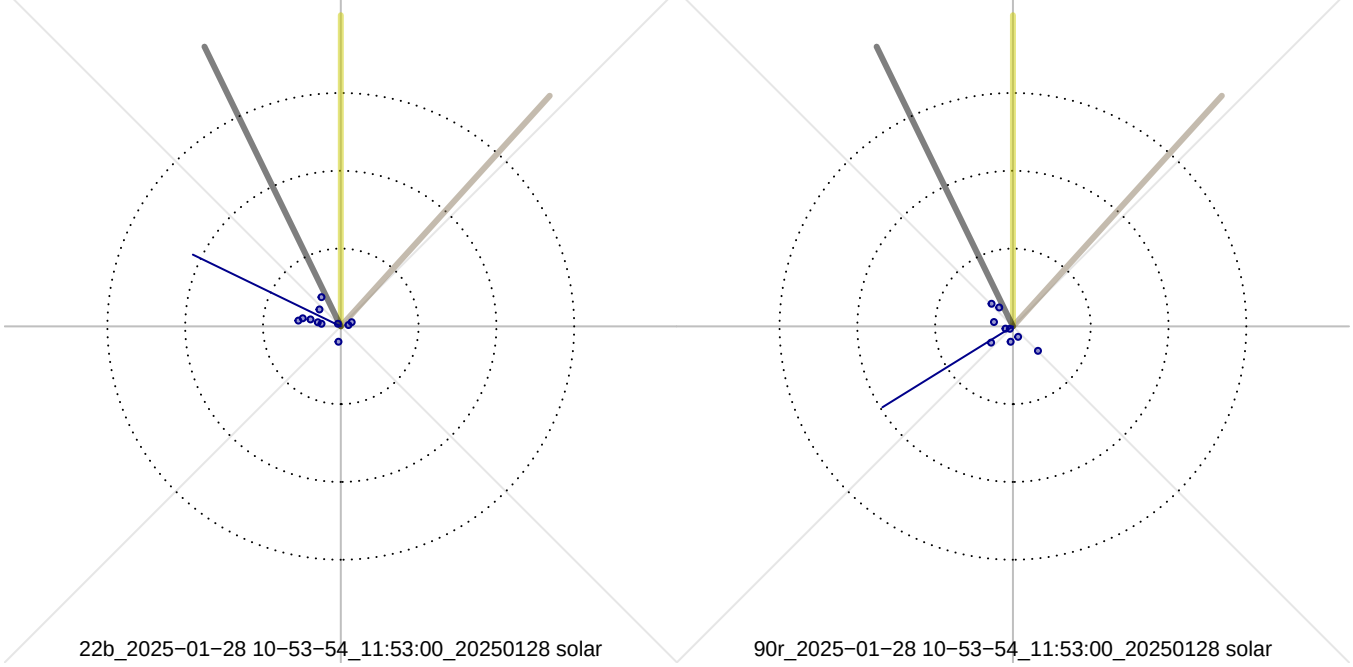
95r_2025−01−28 10−40−45_11:40:00_20250128 solar

66b_2025−01−28 10−47−44_11:47:00_20250128 solar



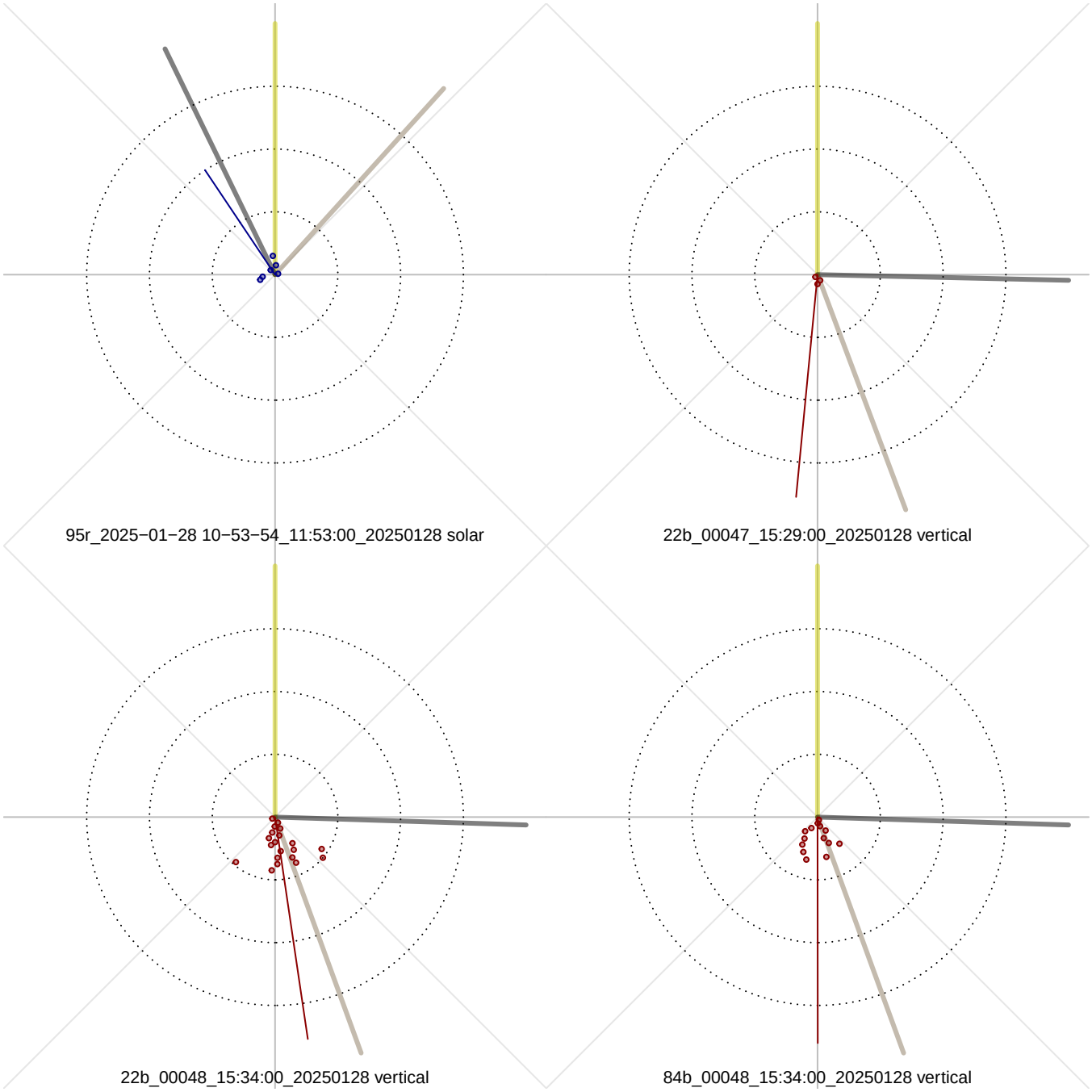
22b_2025-01-28 10-48-54_11:50:00_20250128 solar

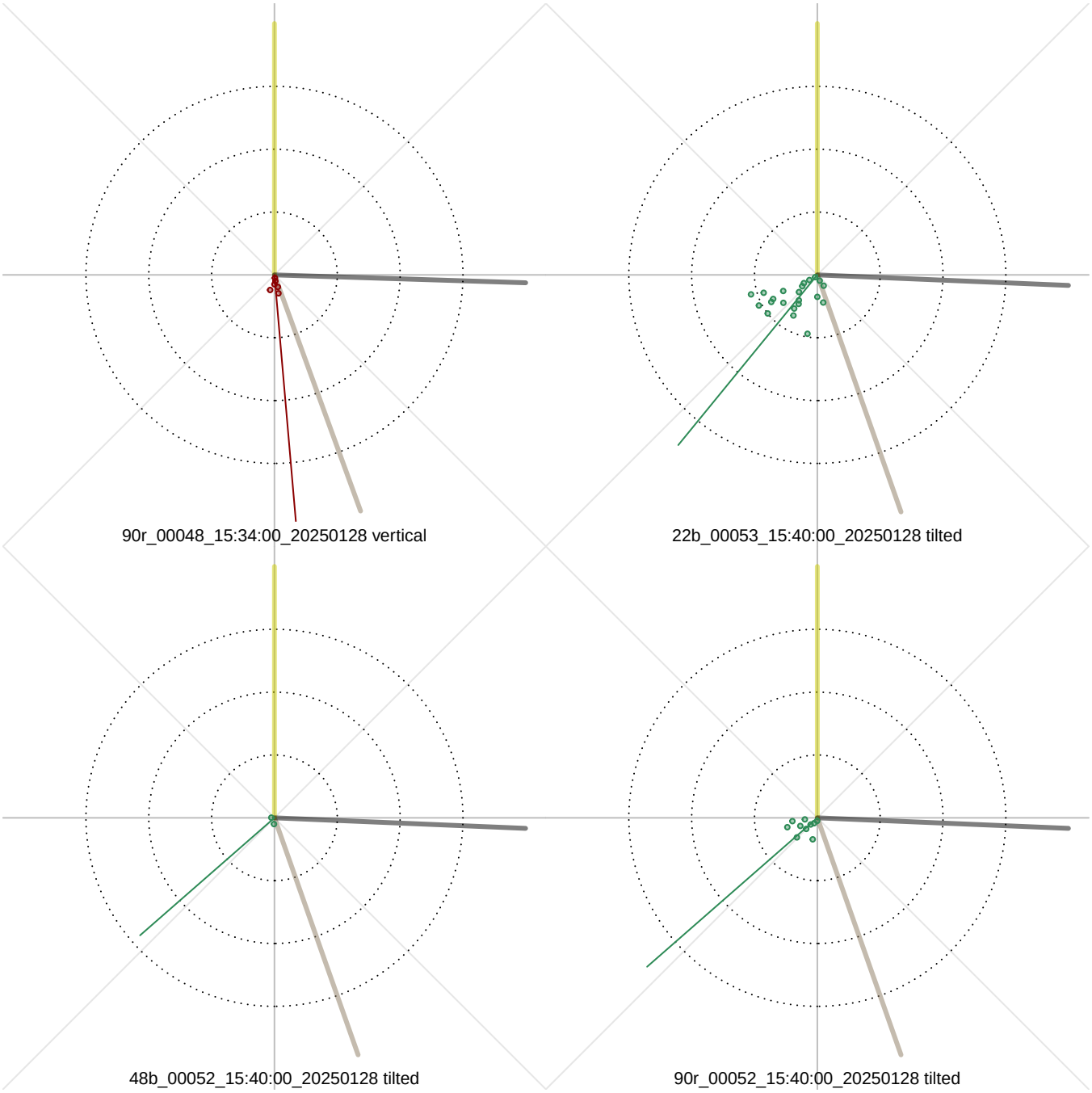
90r_2025-01-28 10-50-28_11:50:00_20250128 solar

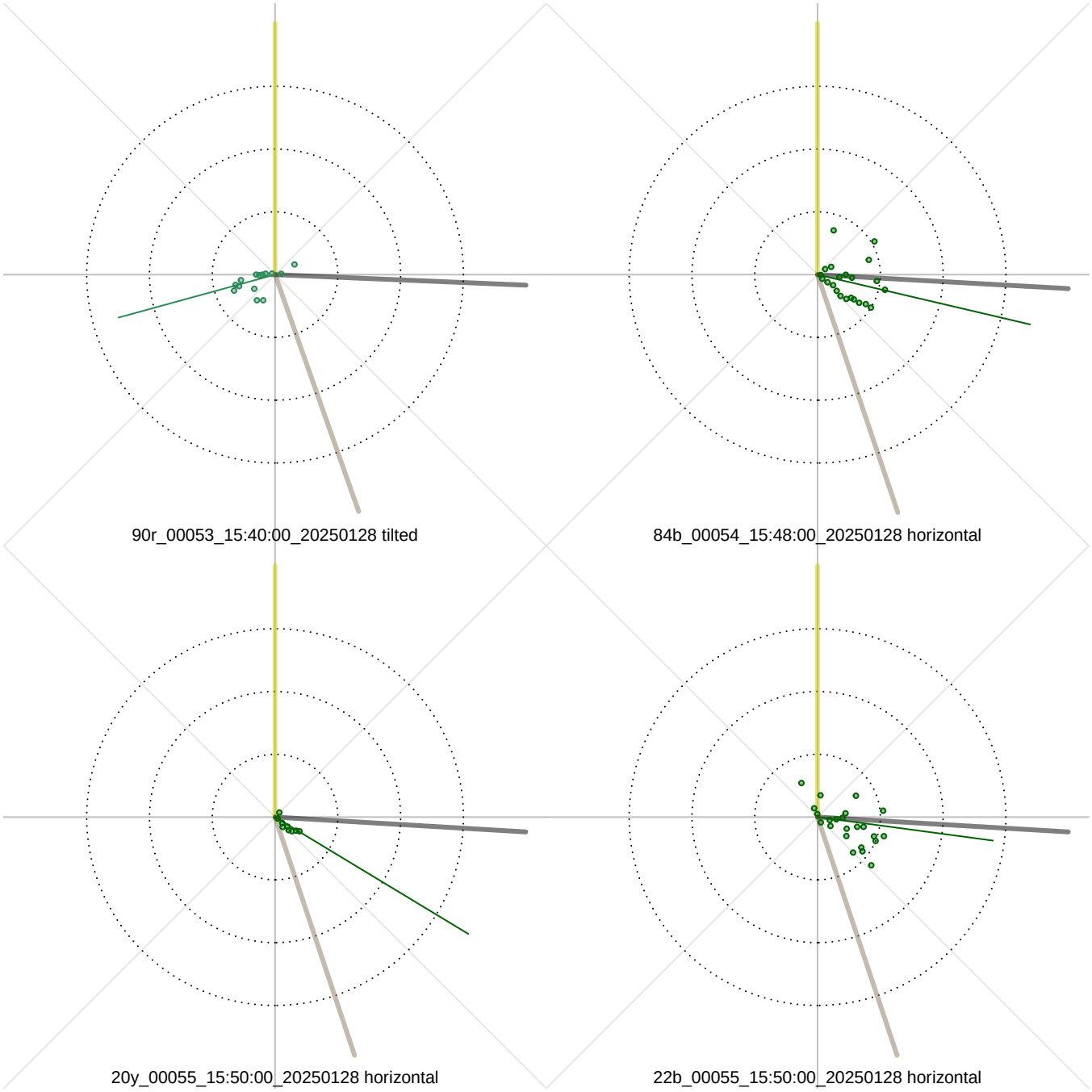


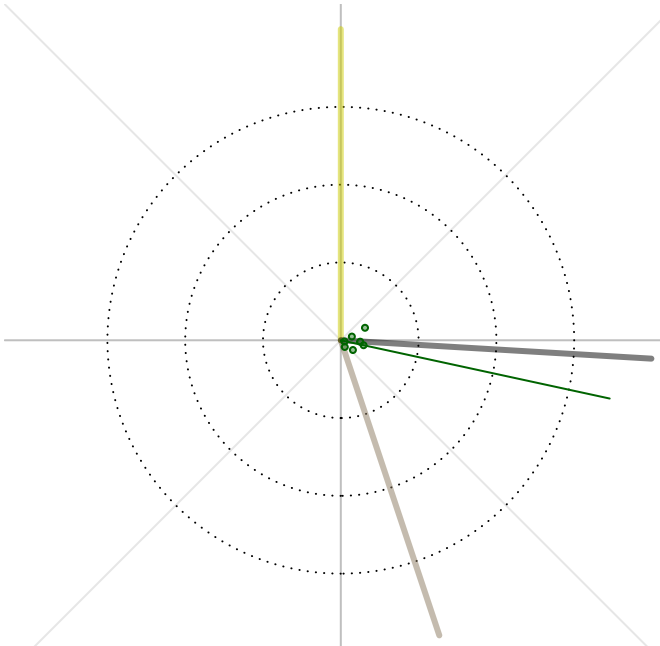
22b_2025-01-28 10-53-54_11:53:00_20250128 solar

90r_2025-01-28 10-53-54_11:53:00_20250128 solar

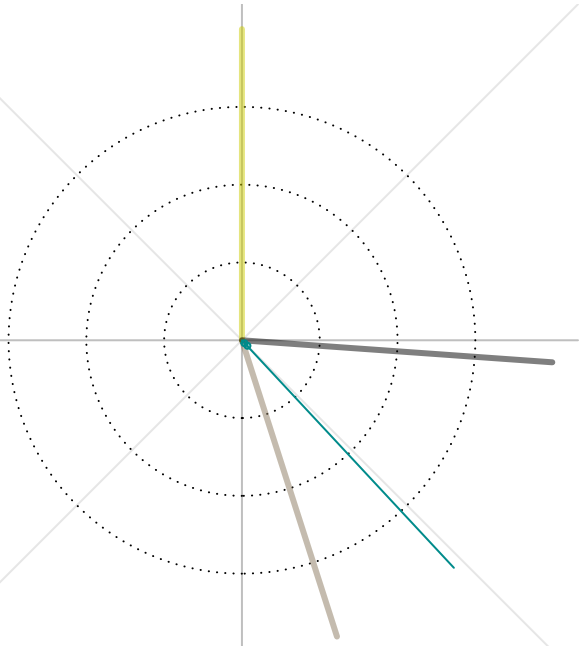

95r_2025−01−28 10−53−54_11:53:00_20250128 solar
22b_00047_15:29:00_20250128 vertical
22b_00048_15:34:00_20250128 vertical
84b_00048_15:34:00_20250128 vertical


90r_00048_15:34:00_20250128 vertical
22b_00053_15:40:00_20250128 tilted
48b_00052_15:40:00_20250128 tilted
90r_00052_15:40:00_20250128 tilted

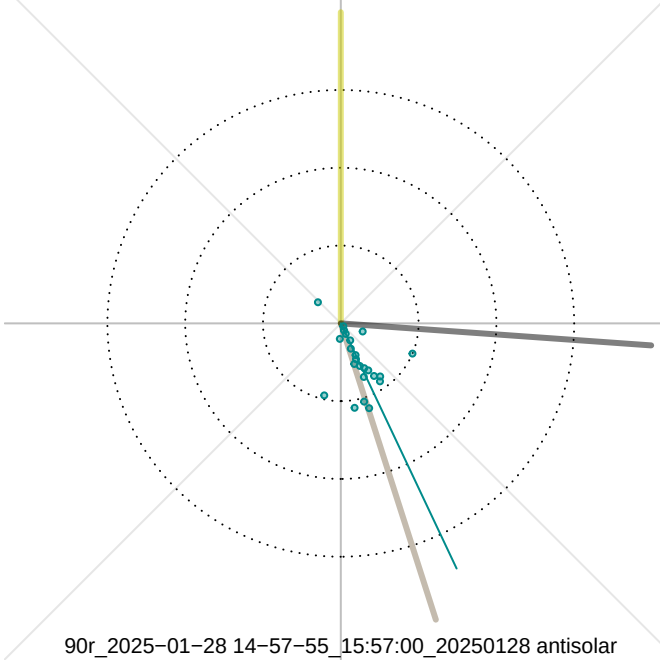

90r_00053_15:40:00_20250128 tilted
84b_00054_15:48:00_20250128 horizontal
20y_00055_15:50:00_20250128 horizontal
22b_00055_15:50:00_20250128 horizontal



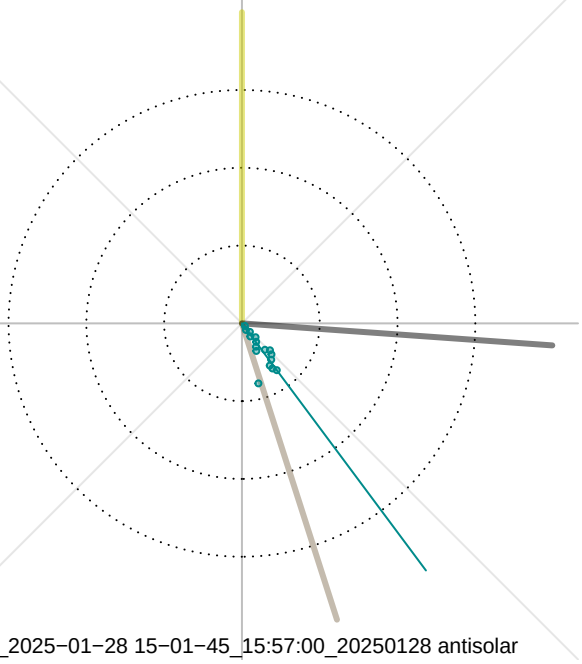
90r_00055_15:50:00_20250128 horizontal



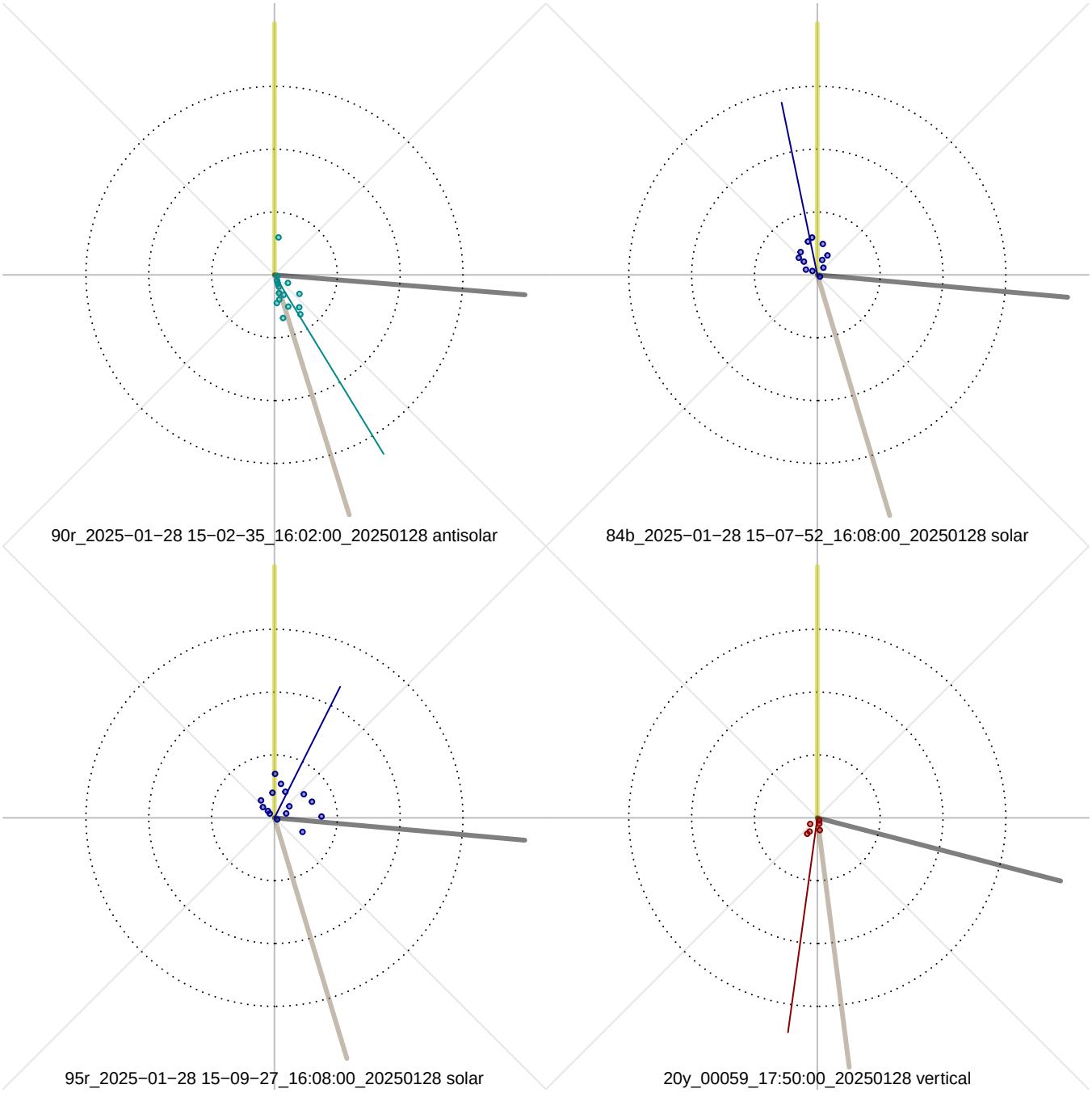
22b_2025−01−28 14−56−08_15:57:00_20250128 antisolar

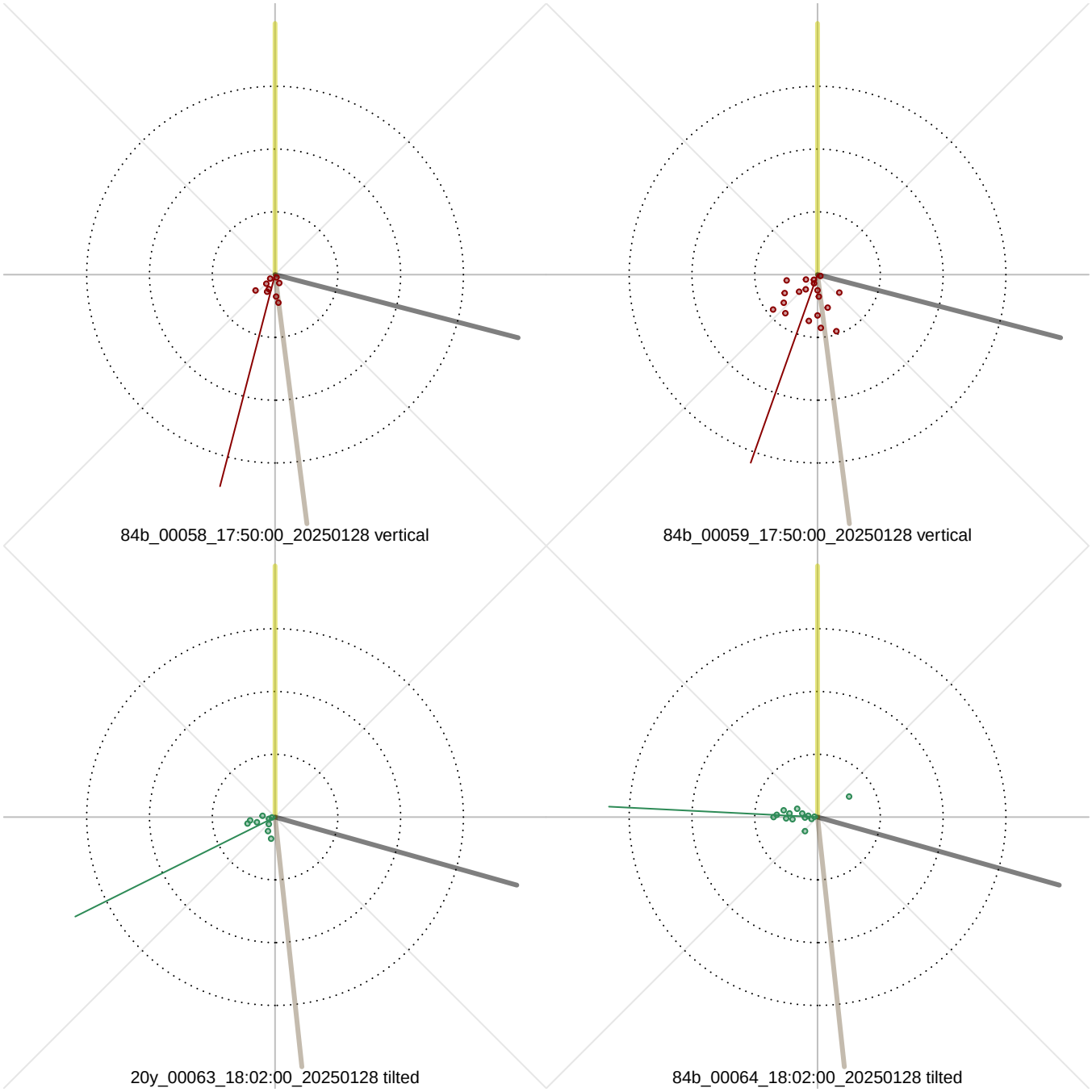


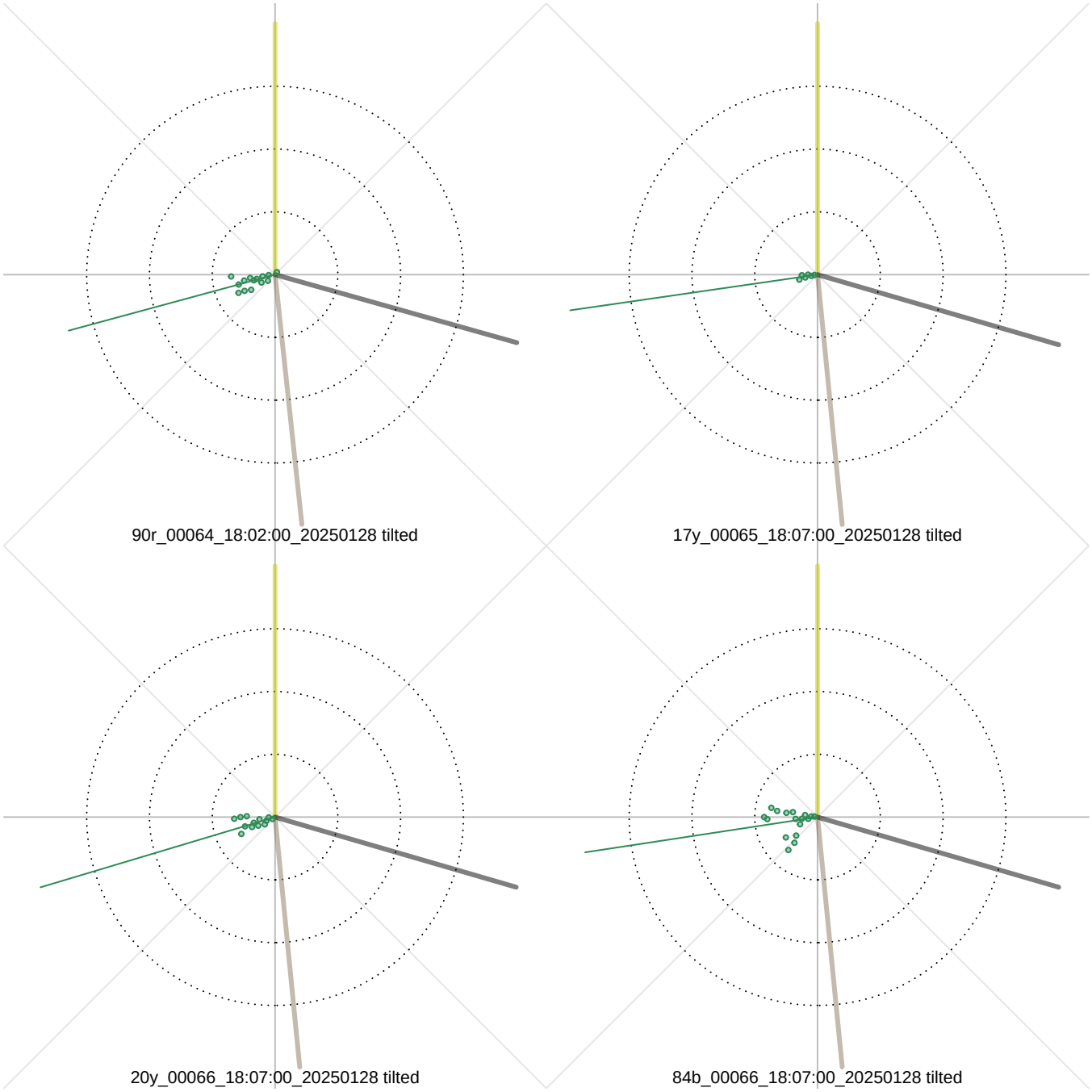
90r_2025−01−28 14−57−55_15:57:00_20250128 antisolar

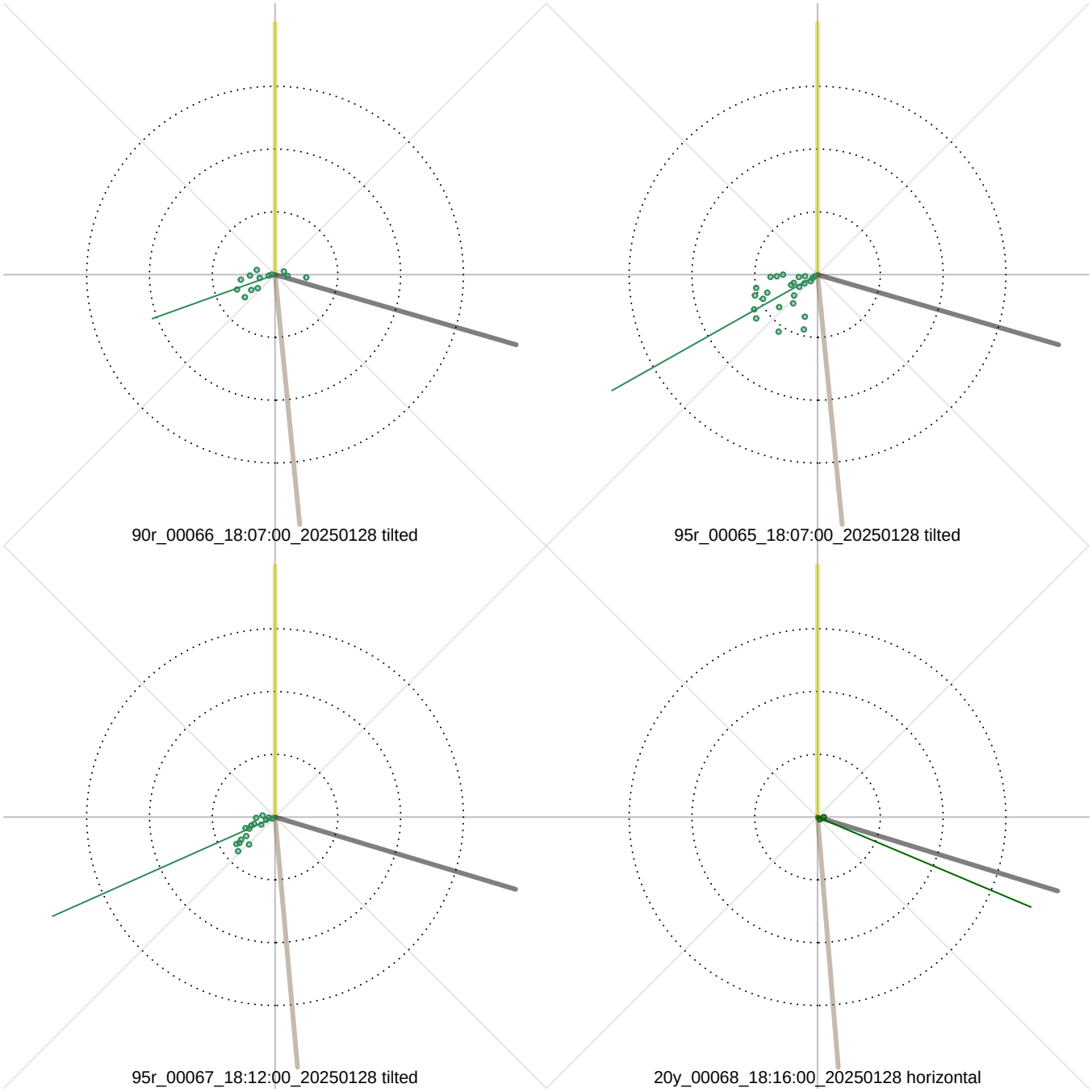


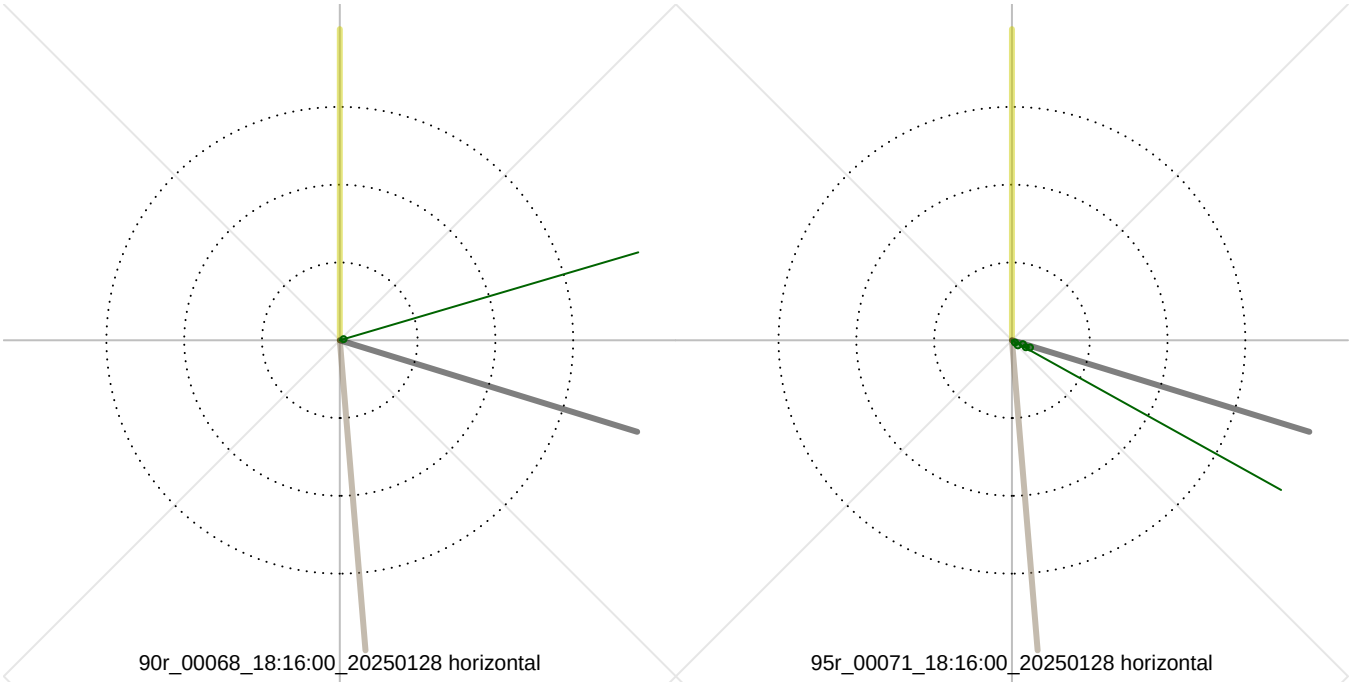
95r_2025−01−28 15−01−45_15:57:00_20250128 antisolar


90r_2025−01−28 15−02−35_16:02:00_20250128 antisolar
84b_2025−01−28 15−07−52_16:08:00_20250128 solar
95r_2025−01−28 15−09−27_16:08:00_20250128 solar
20y_00059_17:50:00_20250128 vertical


84b_00058_17:50:00_20250128 vertical
84b_00059_17:50:00_20250128 vertical
20y_00063_18:02:00_20250128 tilted
84b_00064_18:02:00_20250128 tilted

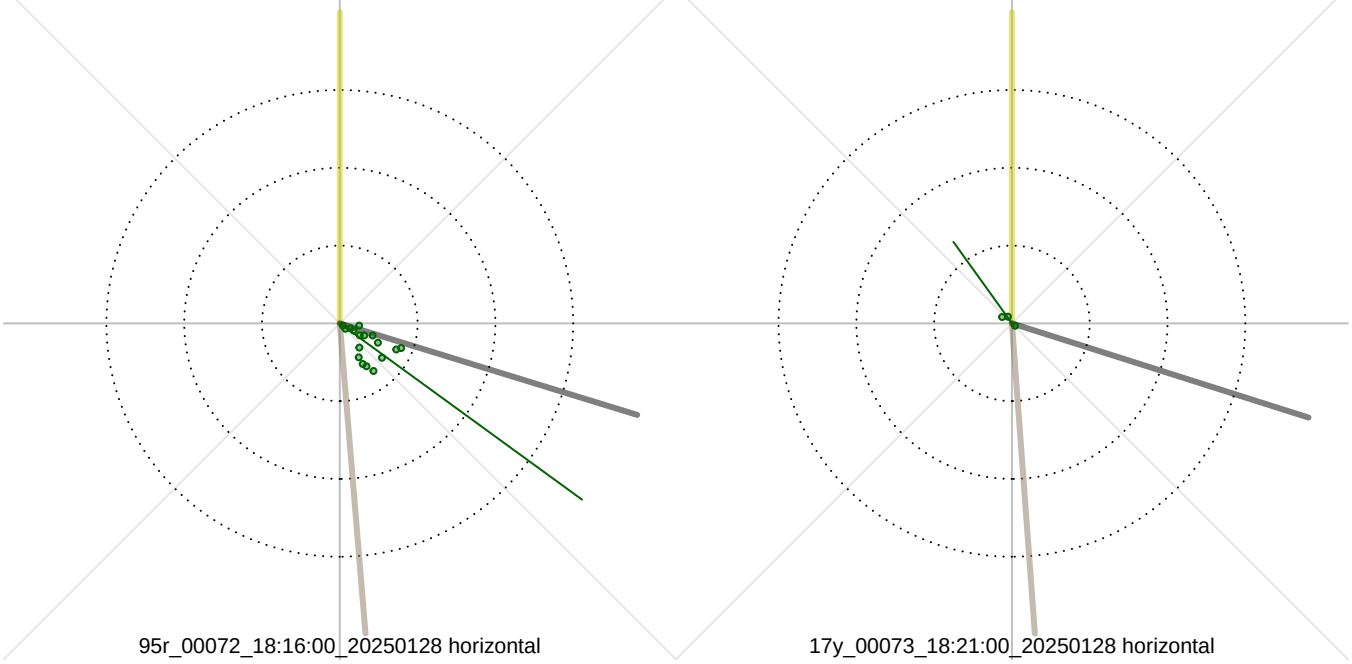

90r_00064_18:02:00_20250128 tilted
17y_00065_18:07:00_20250128 tilted
20y_00066_18:07:00_20250128 tilted
84b_00066_18:07:00_20250128 tilted


90r_00066_18:07:00_20250128 tilted
95r_00065_18:07:00_20250128 tilted
95r_00067_18:12:00_20250128 tilted
20y_00068_18:16:00_20250128 horizontal



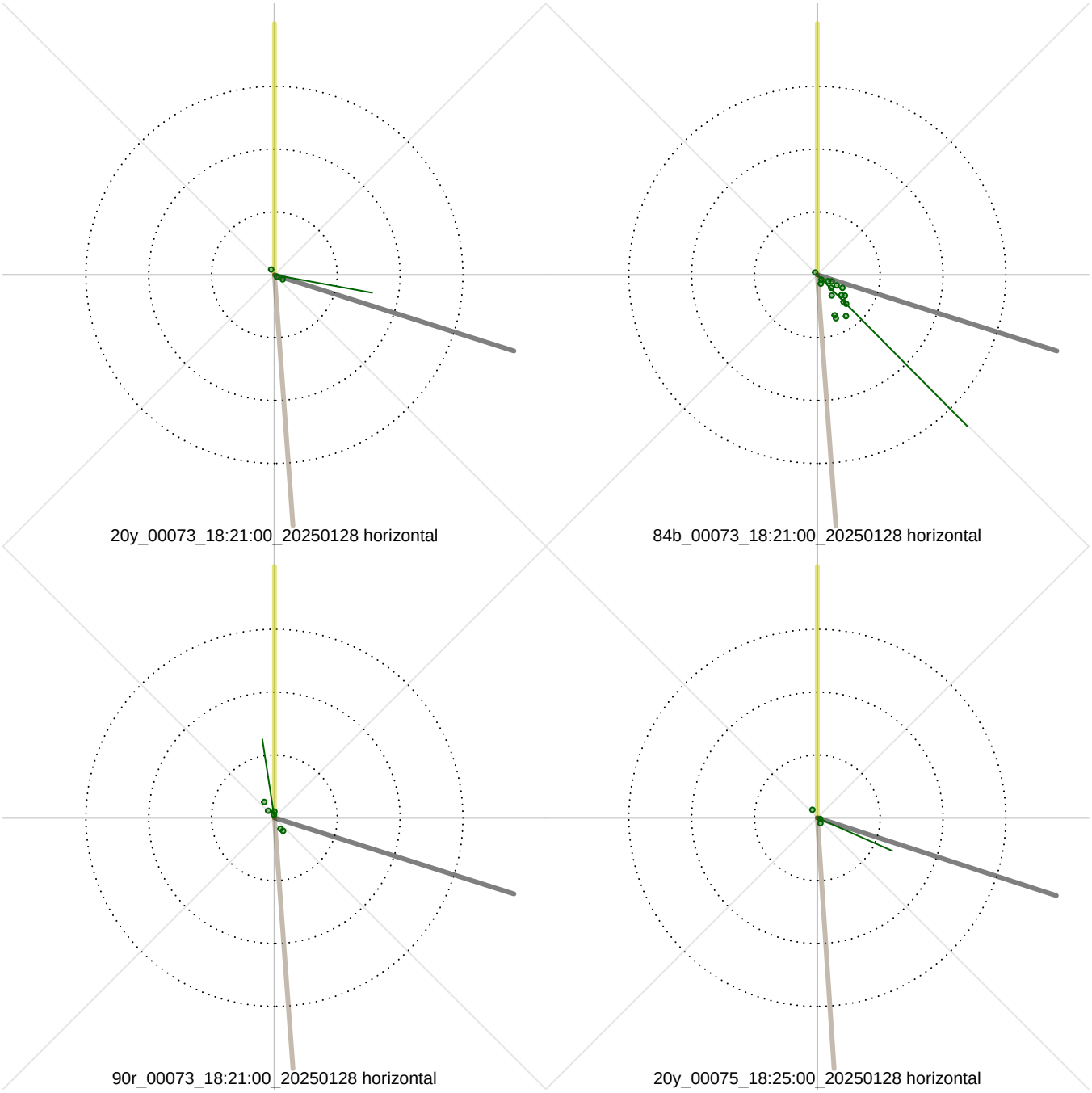
90r_00068_18:16:00_20250128 horizontal

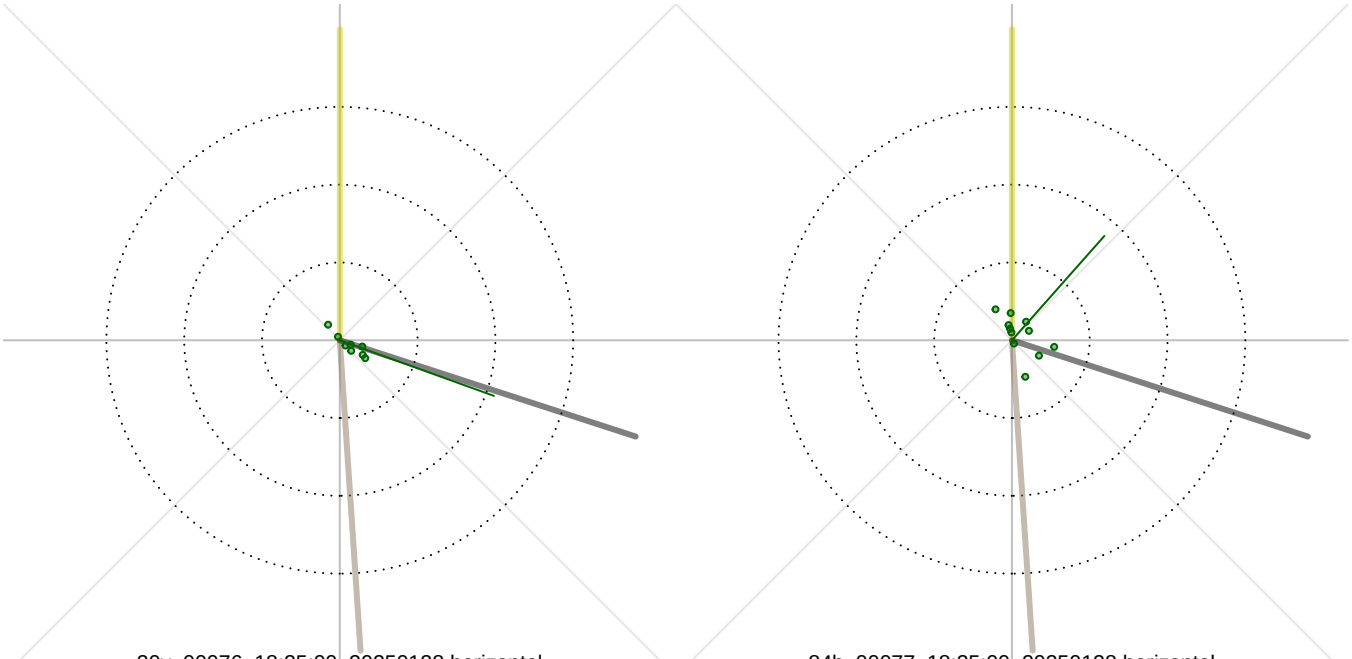
95r_00071_18:16:00_20250128 horizontal



95r_00072_18:16:00_20250128 horizontal

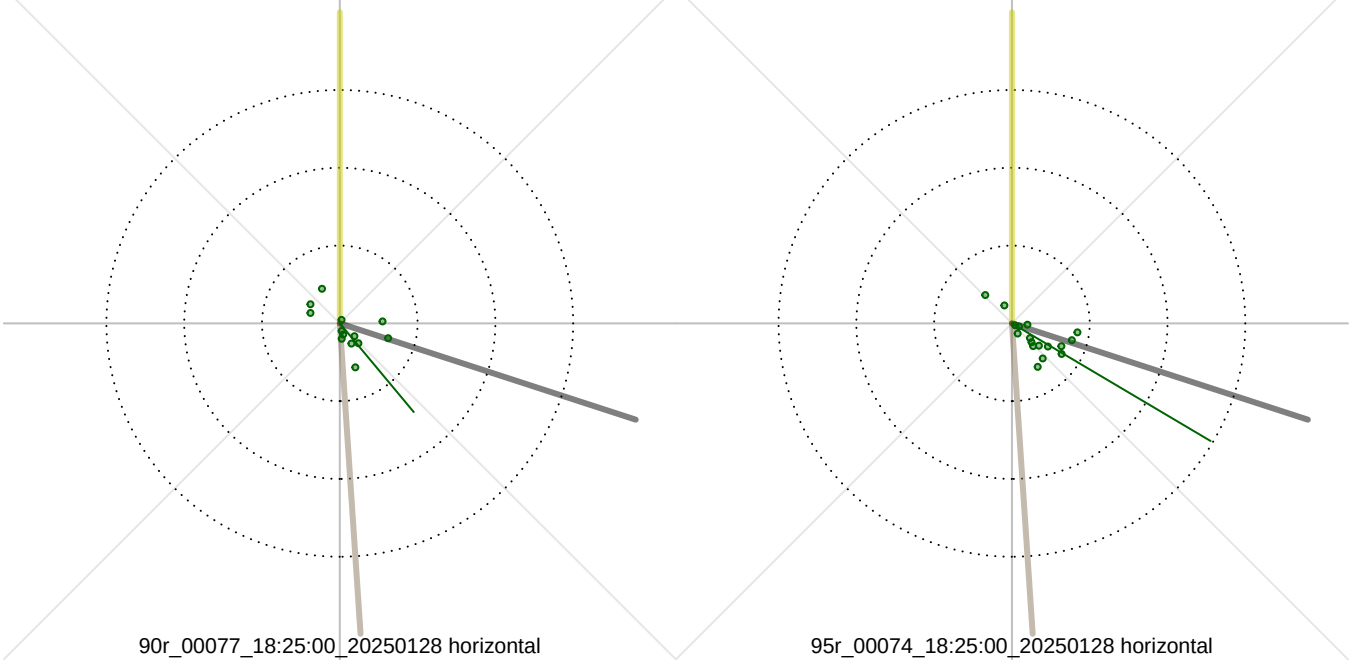
17y_00073_18:21:00_20250128 horizontal


20y_00073_18:21:00_20250128 horizontal
84b_00073_18:21:00_20250128 horizontal
90r_00073_18:21:00_20250128 horizontal
20y_00075_18:25:00_20250128 horizontal



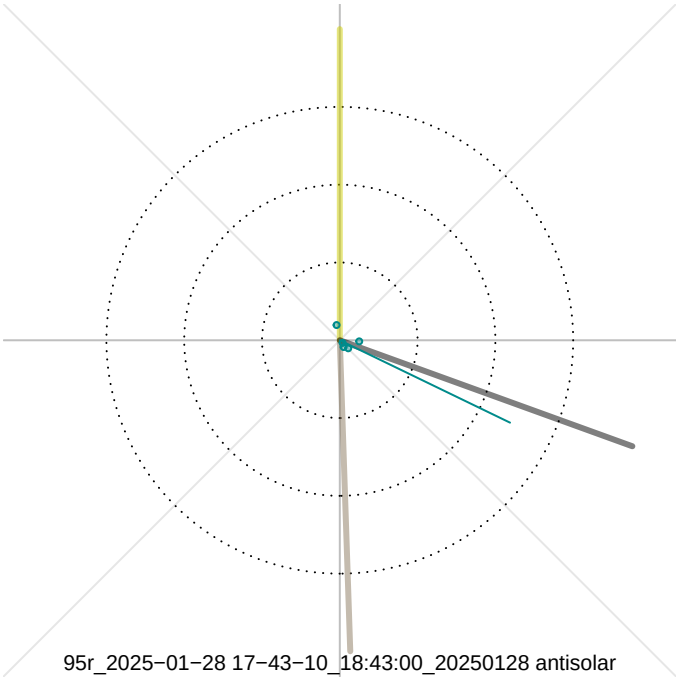
20y_00076_18:25:00_20250128 horizontal

84b_00077_18:25:00_20250128 horizontal



90r_00077_18:25:00_20250128 horizontal

95r_00074_18:25:00_20250128 horizontal


95r_2025−01−28 17−43−10_18:43:00_20250128 antisolar